

Operations Management

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Manuel Patton



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Table of Contents

Preface

xv

Chapter 1	Introduction	1
	The Concept of Management	1
	Functional Concept	2
	‘Getting Things Done through Others’ Concept	2
	Leadership and Decision-making Concept	3
	Productivity Concept	3
	Universality Concept	3
	Nature of Management	4
	Universality	4
	Purposeful	4
	Social Process	4
	Coordinating Force	4
	Intangible	4
	Continuous Process	5
	Composite Process	5
	Creative Organ	5
	Scope of Management	5
	The Process of Management	6
	Significance of Management	6
	Managerial Roles (Mintzberg)	7
	Interpersonal Roles	7
	Informational Roles	8
	Decisional Roles	9
	Mintzberg’s Ten Management Roles	10

Functional Areas of Management	12
Production Management	12
Marketing Management	12
Finance and Accounting Management	12
Personnel Management	13
Development of Management Thought	13
Taylor's Four Fundamental Principles	13
Frank and Lillian Gilbreth: Lean Production	14
Henri Fayol's General Principles of Management	14
Tenants of Modern Bureaucracy	15
Hawthorne Studies: Birth of the Neo-Classical School	16
The Human Factor in Management	16
Sources of Managerial Authority	17
Abraham H. Maslow and the Behaviouralist School	17
Douglas McGregor	18
Frederick Herzberg	18
External Environment	20
Classical and Neo-Classical Theories of Management	20
Classical Management Theory	20
Evolution of Classical Approach to Management	20
Neoclassical Theory of Management	21
A Framework for Managing Operations	23
Planning	23
Objectives of Operations Management	25
Managing Global Operations	26
Scope of Production and Operations Management	27
Location of Facilities	27
Plant Layout and Material Handling	28
Product Design	28
Process Design	29
Production Planning and Control	29
Quality Control	30
Materials Management	30
Maintenance Management	31
Functions of Production Management Department	31
Place of Production Management Department in the Organisation	33
Organisation Chart for P.P.C. Department	34
Types of Production Systems	34
Chapter 2 Operational Planning Management	37
Business Plan	38
Audience	38
Revisiting the Business Plan	40

Legal and Liability Issues	40
Within Corporations	41
Concept	42
Planning	42
Objectives and Policies	44
Planning Basics	45
Applications	45
Control	46
Capacity Planning	55
Long-term Capacity Planning	56
Short-term Capacity Planning	56
Capacity-planning Techniques	57
Manufacturing Planning and Control.....	58
Framework.....	59
Aspects of Planning.....	60
Event Planning	60
Financial Plan	62
Marketing Plan	62
Strategic Planning	69
Aggregate Production Planning	74
Management Options to Meet Fluctuating Demand	75
Kinds of Costs Involved	75
Master Production Scheduling	75
Powerful Scheduling Tools	76
Planner's Workbench	76
Complete Planning Visibility	76
"What If" Simulation	76
Scheduling Cycles	76
Instant Available - to - Promise Information	76
Performance Measurement.....	76
Job Shop Scheduling	77
Problem Variations	77
Major Results	77
Shop Floor Control.....	77
Empower Shop Floor Personnel for Maximum Effectiveness and Output ..	77
Closed-Loop Feedback	78
Work-in-Process Visibility	78
Shop Floor Empowerment	78
Move Ticket Reporting	78
On-Screen Dispatching	78
Shop Floor Control.....	79
User-Controlled Operation Scheduling.....	79
Work Order Dispatching	79

Move Ticket Recording	80
Demand Management	80
Water Demand	81
Tools and Techniques	81
Utility Programmes	82
Chapter 3 Facility and Layouts Management	83
Facility Management	83
Types of Layouts	84
Fixed Location Layout	87
Flexible Manufacturing Systems	87
Layout Problem	87
Input Data	88
Distance	88
Criterion for Comparison	90
Finding a Layout with the CRAFT Method	90
Facility Location	91
Example	92
Solution	93
Concentrator	96
Importance of Facility Location	98
Facility Layout	98
Facility Layout and Design	99
Factors in Determining Layout and Design	99
Differences Between Office and Factory Layouts	101
Chapter 4 Project Management	103
Introduction	103
Planning Large Projects and Programmes	104
Kotter's 8-Step Change Model	104
Key Points	108
Stakeholder Analysis	108
How to Use the Tool	109
Influence Maps	111
Influence Mapping	111
Elements of an Influence Map	112
Creating an Influence Map	113
Programme Evaluation and Review Technique	113
Conventions	114
Terminology	114
Advantages	115
Disadvantages	115
Uncertainty in Project Scheduling	116

Resource Leveling	116
Schedule	117
Process Mix	118
Product Mix	119
Product Mix Decisions	119
Product-mix Management and Responsibilities	119
General Management Workflow	120
Product-mix Analysis	121
Critical Path Analysis and PERT Charts	123
How to Use the Tool	123
Drawing a Critical Path Analysis Chart	124
Crash Action	126
PERT	127
Key Points	127
Logframes, and the Logical Framework Approach	128
Logical Framework Approach and the Logframe	128
Using a Logframe	129
Key Points	132
Estimating Time Accurately	132
How to Use the Tool	132
Key Points	133
Risk Impact/Probability Chart	133
Use the Risk Impact/Probability Chart	134
Key Points	135
Scheduling Simple Projects	135
Use	136
Key Points	136
Gantt Charts	136
Sequential and Parallel Activities	136
Drawing a Gantt Chart	137
Key Points	139
Chapter 5 Production and Operations Management	140
Introduction	140
Product	140
Production	141
Management	142
Criteria of Performance	142
Definition of Production Management	143
Objective of Production Management	143
Scope of Production Management	143
Benefits Derived From Efficient Production Management	145
Production Cycle	145

Production Design	146
Product Development	147
Product Development Process	147
Documentation	147
Research	148
Making a Prototype	148
Filing a Patent	149
Marketing Your Invention	149
Chapter 6 Operation of Pricing Decisions and Strategies Management	150
Questions Involved in Pricing	150
What a Price should Do	151
Definitions	152
Reference Price Effect	154
Difficult Comparison Effect	154
Switching Costs Effect	154
Price-Quality Effect	154
Expenditure Effect	154
End-Benefit Effect	154
Shared-cost Effect	154
Fairness Effect	155
The Framing Effect	155
Approaches	155
Tactics	155
Pricing Mistakes	155
Chapter 7 Quality Management	156
Concept	156
Quality Assurance	156
Quality Control	157
Good Laboratory Practice	158
Quality Control	159
Accuracy of Quality Assurance	160
Steps for Quality Assurance Process	160
Quality Assurance Versus Quality Control	160
QA in Software Development	162
Company Quality	162
Using Contractors and/or Consultants	163
Quality Control	164
Statistical Process Control	165
General	166
How to Use SPC?	167
Process Capability Improvement	167

Process Capability	167
Process Improvement	169
Benchmarking	169
Business Process Improvement	170
Business Process Reengineering	175
The Impact of BPR on Organizational Performance	176
How to Implement a BPR Project	176
Tools to Support BPR	176
Capability Maturity Model Integration	177
CMMI Topics	178
Inspection	181
Total Quality Control	181
Quality Circles	182
Genesis of Quality Circles	182
Definition	182
Philosophy	183
Concept	183
Objective	183
Organisational Structure	183
Launching Quality Circles	184
Training	185
Process of Operation	185
Basic Problem Solving Techniques	185
Benefits and Limitations of Quality Circles	186
Applicability in Government Organisations	186
Kaizen	188
Implementation	190
Quality Assurance	190
Early Efforts to Control the Quality of Production	191
Wartime Production	192
Postwar	192
Concept of Quality and Manufacturing	193
Definitions	193
Market Sector Perspectives	194
Quality Management Techniques	195
Inspection	196
Chapter 8 Purchasing Management	197
Objectives	198
Functions	198
Methods of Purchase	200
Purchasing Procedure	201
Purchase Request	201
Selection of Companies for Estimate Submission	201

Estimate Request	201
Submission Estimate	201
Negotiation	201
Contract Conclusion	202
Delivery and Inspection	202
Payment	202
Value Analysis	202
Concept of Value	202
Introduction to Value Analysis	202
Value Analysis Method	203
Value Analysis Process	204
Function Analysis System Technique	205
Creating a Fast Model	206
Integrating QFD With Fast	209
Value Improvement Process	211
Stock Control Systems	212
Fixed Re-order Stock Level	212
Fixed Time Reordering	213
Economic Order Quantity	213
Just-in-time Production	213
Definition of Purchasing	214
Acquisition Process	215
Selection of Bidders	216
Bidding Process	217
Technical Evaluation	218
Commercial Evaluation	218
Negotiating	219
Post-Award Administration	219
Chapter 9 Risk Management and Business Continuity	220
Risk Communication	220
Communicate your Ideas Effectively	221
Risk Assessment	222
Explanation	223
Risk Assessment in Public Health	223
Risk Assessment in Auditing	225
Risk Assessment in Information Security	225
Criticisms of Quantitative Risk Assessment	226
Principles of Risk Management	226
Risk Management Process	227
How Teams use Risk Management?	228
Limitations of Risk Assessment	229
Areas in Risk Management	230

Chapter 10 Strategic Planning Knowledge Management	232
Using Knowledge Management Systems to Make Strategic Planning more Effective	232
Short- To Long-range Strategic Planning in a Changing World	233
Utilization of Knowledge in Executive Visioning	234
Implementing Successful Knowledge	235
Essential Focus of Knowledge Management Systems	236
Redesign May be Necessary for Changing Knowledge Work Processes	236
Cost Justification	238
Successful Knowledge Management Systems include	
Selecting Appropriate Software	238
Continuing Support of Knowledge Management Systems	240
Real-world Case Studies of Successful Knowledge Management Systems	241
Software Useful in Knowledge Management Systems	243
Knowledge Extraction Tools	243
Methodologies for Developing Successful Knowledge Management Systems	253
Steps in Implementing Successful Knowledge Management Systems ..	258
Strategic Planning Principles Based on Knowledge of Total Business Operations	263
Important Factors	267
Knowledge Management Systems in Strategic Planning	276
Areas that Lend Themselves to Discovering Strategic Planning Knowledge	277
Long-range Strategic Planning	278
Short-range And Medium-range Strategic Planning	284
Global Best Practices	289
Chapter 11 Elements of Strategic Management	291
Mission	291
Levels of Strategy	293
Societal Strategy	295
Evolution of Strategic Management	301
Deliberateness of Strategy	302
Strategic Thinking	305
Evolution of Strategic Thinking	305
Becoming a Strategic Thinker	308
Elements of Strategic Thinking	308
 <i>Bibliography</i>	310
 <i>Index</i>	312

Preface

Operations management is a field of business concerned with the administration of business practices to maximize efficiency within an organization. It involves planning, organizing, and overseeing the organization's processes to balance revenues and costs and achieve the highest possible operating profit. An operations manager is tasked with ensuring that the organization successfully converts inputs such as materials, labor, and technology into outputs in an efficient manner.

Operations management is an area of business concerned with the production of goods and services, and involves the responsibility of ensuring that business operations are efficient in terms of using as little resource as needed, and effective in terms of meeting customer requirements. Operations management involves utilizing resources from staff, materials, equipment, and technology. Operations managers acquire, develop, and deliver goods to clients based on client needs and the abilities of the company.

Operations management handles various strategic issues, including determining the size of manufacturing plants and project management methods and implementing the structure of information technology networks. Other operational issues include the management of inventory levels, including work-in-process levels and raw materials acquisition, quality control, materials handling, and maintenance policies.

An operations management professional understands local and global trends, customer demand, and the available resources for production. Operations management approaches the acquisition of materials and the use of labor in a timely, cost-effective manner to deliver customer expectations. Inventory levels are monitored to ensure excessive quantities are on hand. Operations management is responsible for finding vendors that supply the appropriate goods at reasonable prices and have the ability to deliver the product when needed.

An operations manager needs to have good coordination by knowing how to integrate resources, activities, and time to ensure proper use of the resources toward the achievement of the organization's goals. Coordination involves carrying out specific activities simultaneously and switching between the activities with ease. It also involves dealing with interruptions, obstacles, and crises, and efficiently going back to the normal routine functions to prevent further interruptions.

Operations is one of the three strategic functions of any organization. This means that it is a vital part of accomplishing the organization's strategy and ensuring its long-term survival. The other two areas of strategic importance to the organization are marketing and finance. The operations strategy should support the overall organization strategy. Many companies prepare a 5-year proforma to assist in their operation planning. The pro forma uses information from past and current financial statements in an effort to predict future events such as sales, and capital investments.

The Subject of Production Management is studied under different Headings-such as Production Planning and control, Production and Inventory control, production and operations control and many more. What ever may be the title of the subject, the contents of the subject are more or less one and the same. Before we discuss about production management, let us discuss about product, production and management. This will give us a rough idea about production Management and with what a production manager has to deal with.

This book is concerned with the fundamental activity of organizations - how they provide goods and services. One of the earliest branches of business and management studies, the increase in international competition has seen a resurgence of interest in the development of the operation management field.

– *Author*

Introduction

Operations management is an area of business concerned with the production of goods and services, and involves the responsibility of ensuring that business operations are efficient in terms of using as little resource as needed, and effective in terms of meeting customer requirements. It is concerned with managing the process that converts inputs into outputs. Operations traditionally refers to the production of goods and services separately, although the distinction between these two main types of operations is increasingly difficult to make as manufacturers tend to merge product and service offerings. More generally, Operations Management aims to increase the content of value-added activities in any given process. Fundamentally, these value-adding creative activities should be aligned with market opportunity for optimal enterprise performance.

The U.S. Department of Education, Operations Management, the physical and/or technical functions of a firm or organization, particularly those relating to development, production, and manufacturing. Instruction in principles of general management, manufacturing and production systems, plant management, equipment maintenance management, production control, industrial labour relations and skilled trades supervision, strategic manufacturing policy, systems analysis, productivity analysis and cost control, and materials planning.

THE CONCEPT OF MANAGEMENT

The concept of management is not fixed. It has changing just as to time and circumstances. The concept of management has been used in integration and authority,

etc. Different authors on management have given different concepts of management. The main concepts of management are as follows:

FUNCTIONAL CONCEPT

This concept 'management is what a manager does'. The main followers of this concept are Louis Allen, George R. Terry, Henry Fayol, E.F.L. Brech, James L. Lundy, Koontz and O. Donnell, G.E. Milward, McFarland, etc. The functional concept as given by some of the authors is given below:

- Louis Allen, "Management is what a manager does."
- James L. Lundy, "Management is principally the task of planning, coordinating, motivating and controlling the effort of others towards a specific objective. Management is what management does. It is the task of planning, executing and controlling."
- George R. Terry, "Management is a distinct process consisting of planning, organizing, activating and controlling performed to determine and accomplish the objective by the use of human beings and other resources."
- Howard M. Carlisle, "Management is defined as the process by which the elements of a group are integrated, coordinated and/or utilized so as to effectively and efficiently achieve organizational objectives."
- Henry Fayol, "To manage is to forecast, and plan, to organize, to command, to coordinate and to control."

'GETTING THINGS DONE THROUGH OTHERS' CONCEPT

This concept, '*Management is the art of getting things done through others*'. It is very narrow and traditional concept of management. The followers of this concept are Koontz and O. Donnell, Mooney and Railey, Lawrence A. Appley, S. George, Mary Parker Follet, etc. Under this concept, the workers are treated as a factor of production only and the work of the manager is confined to taking work from the workers. He need not do any work himself. Modern management experts do not agree with this concept of management.

Some of these authors have explained this concept in the following words:

- Mary Parker Follet, "Management is the art of getting things done through others."
- Harold Koontz, "Management is the art of getting things done through and with people in formally organized groups. It is the art of creating and environment in which people can perform as individuals and yet cooperate towards attaining of group goals."
- J.D. Mooney and A.C. Railey, "Management is the art of directing and inspiring people."

LEADERSHIP AND DECISION-MAKING CONCEPT

This concept, “management is an art and science of decision-making and leadership.” Most of the time of managers is consumed in taking decisions. Achievement of objects depends on the quality of decisions. Similarly, production and productivity both can be increased by efficient leadership only. Leadership provides efficiency, coordination and continuity in an organization.

Leadership and decision-making concept as given by some authors is given below:

- Donald J. Clough, “Management is the art and science of decision-making and leadership”.
- Ralph, C. Davix, “Management is the function of executive leadership anywhere.”
- Association of Mechanical Engineers, U.S.A., “Management is the art and science of preparing, organizing and directing human efforts applied to control the forces and utilize the materials of nature for the benefit to man.”
- F.W. Taylor, “Management implies substitution of exact scientific investigation and knowledge for the old individual judgement or opinion, in all matters in the establishment.”

PRODUCTIVITY CONCEPT

This concept, “*management is an art of increasing productivity.*” Economists treat management as an important factor of production.

According to them, “*Management is also a factor of production like land, labour, capital and enterprise.*” The main followers of this concept of management are John F. Mee, Marry Cushing Niles, F.W. Taylor, etc.

The productivity concept, as given by the authors is given below:

- Jon, F. Mee, “Management may be defined as the art of securing maximum prosperity with a minimum of effort so as to secure maximum prosperity and happiness for both employer and employee and give the public the best possible service.”
- F. W. Taylor, “Management is the art of knowing what you want to do in the best and cheapest way.”
- Marry Cushing Niles, “Good management achieves a social objectives with the best use of human and material energy and time and with satisfaction of the participants and the public.

UNIVERSALITY CONCEPT

According to this concept, “*Management is universal*”. Management is universal in the sense that it is applicable anywhere whether social, religious or business and industrial. The followers of this concept are Henry Fayol, Lawrence A. Appley, F.W. Taylor, Theo Haimann, etc. According to:

- Henry Fayol, “*Management is an universal activity which is equally applicable in all types of organization whether social, religious or business and industrial*”.
- Megginson, “*Management is management, whether it is in Lisbon, or in London or in Los Angeles.*”
- Theo Haimann, “*Management principles are universal. It may be applied to any kind of enterprises, where the human efforts are coordinated.*”

NATURE OF MANAGEMENT

UNIVERSALITY

Management is an universal phenomenon in the sense that it is common and essential element in all enterprises. Managers perform more or less the same functions irrespective of their position or nature of the organization. The basic principles of management can be applied in all managerial situations regardless of the size, nature and location of the organization. Universality of managerial tasks and principles also implies that managerial skills are transferable and managers can be trained and developed.

PURPOSEFUL

Management is always aimed at achieving organizational goals and purposes. The success of management is measured by the extent to which the desired objectives are attained. In both economic and non-economic enterprises, the tasks of management are directed towards effectiveness (*i.e.*, attainment of organizational goals) and efficiency (*i.e.*, goal attainment with economy of resource use).

SOCIAL PROCESS

Management essentially involves managing people organized in work groups. It includes retaining, Developing and motivating people at work, as well as taking care of their satisfaction as social beings. All these interpersonal relations and interactions makes the management as asocial process.

COORDINATING FORCE

Management coordinates the efforts of organization members through orderly arrangement of interrelated activities so as to avoid duplication and overlapping. Management reconciles the individual goals with the organizational goals and integrates human and physical resources.

INTANGIBLE

Management is intangible. It is an unseen force. Its presence can be felt everywhere by the results of its effort which comes in the form of orderliness, adequate work output, satisfactory working climate, employees satisfaction, etc.

CONTINUOUS PROCESS

Management is a dynamic and an on-going process. The cycle of management continues to operate so long as there is organised action for the achievement of group goals.

COMPOSITE PROCESS

Functions of management cannot be undertaken sequentially, independent of each other. Management is a composite process made up of individual ingredients. All the functions are performed by involving several ingredients. Therefore, the whole process is integrative and performed in a network fashion.

CREATIVE ORGAN

Management creates energetics effect by producing results which are more than the sum of individual efforts of the group members. It provides sequence to operations, matches jobs to goals, connects work to physical and financial resources. It provides creative ideas, new imaginations and visions to group efforts. It is not a passive force adopting to external environment but a dynamic life giving element in every organization.

SCOPE OF MANAGEMENT

- We will now delve deeper in order to understand what are all the scope of the activities performed by managers in organizations.
- Managers must make decisions to establish the purpose of the organization and to perform a variety of activities to make the goal a reality.
- All organizations have collective or shared goals, over and above the individual goals of their members, for the achievement of which the organization is responsible to its owners and stakeholders.
- The collective goals of the organization can be achieved by controlling the activities being performed by the individual members of the organization. This controlled performance means that:
 - The collective goals are known and understood by all members—planning;
 - The necessary resources are obtained and utilized efficiently—implementation;
 - The performance can be controlled and measured to determine the extent of reaching the goals - control. In order to implement the plan by utilizing obtaining and utilizing the necessary
- Let us now classify managers in the organization.
- Managers are customarily classified vertically in organizations as top, middle, or first-line managers.

- First-line managers are usually called supervisors. They are responsible for directing the day-to-day activities of operative employees.
- Middle managers manage other managers—and possibly some operative employees—and are typically responsible for translating the goals set by top management into specific details that lower-level managers can perform.
- Top managers are responsible for making decisions about the direction of the organization and establishing policies that affect all organizational members.
- The other major difference in management jobs occurs horizontally across the organization:
- Functional managers are responsible for departments that perform a single functional task and have employees with similar training and skills. Functional departments include manufacturing, marketing, finance, and human resources.
- Line managers are responsible for the manufacturing and marketing departments that make or sell the product or service.
- Staff managers are in charge of departments such as finance and human resources that support line departments.
- General managers are responsible for several departments that perform different functions. Project managers also have general management responsibility, because they coordinate people across several departments to accomplish a specific project.

THE PROCESS OF MANAGEMENT

Management is considered a process because it involves a series of interrelated functions. The management process includes planning, organizing, staffing, directing and controlling functions.

Management as a process has the following implications:

- *Management as social process:* Management process involves interaction among people. Goals can be achieved only when relations between people are productive.
- *Management as integrated process:* Management process brings together human, physical and financial resources. Management process also integrates human efforts so as to maintain harmony among them.
- *Management as iterative process:* All managerial functions are contained within each other. For example, when a manager prepares plans, he is also laying down standards for control
- *Management as continuous process:* Management involves continuous identifying and solving problems. It is repeated again and again.

SIGNIFICANCE OF MANAGEMENT

Management is a function, a discipline, a task to be done, and managers practice this discipline, carry out the functions and discharge these tasks—Peter F. Drucker. Analysis

of business failures made over many years shows that a high percentage of these failures were due to unqualified or inexperienced management. The Bank of America has said, in its publication “Small Business Reporter”. “In the final analysis, more than 90% of business failures are due to managerial incompetence and inexperience.”

The significance of management is much more in case of developing countries. In recent years development specialists has shown in their research that availability of money or technology does not bring development. The limiting factor in almost every case has been the lack of quality on the part of managers. There are some people or groups who believe that there is no need of management. In fact, they feel that people would work together better and with more personal satisfaction if there were no managers.

They refer ‘team effort’ as the ideal group operation. Perhaps they do not realise that whenever an individual plays a game, he has clear group goals as well as some personal goals also. In fact, every group effort designed to attain goals with minimum cost of time, money and materials, adopts some basic principles and techniques of management. Management is critical in the nation’s economic and social development also. Efficient management helps a developing country to make better use of resources for the achievement of economic development.

MANAGERIAL ROLES (MINTZBERG)

Managers must wear many different hats in formulating and implementing task activities related to their positions. In an attempt to understand the diversity of hats managers must wear, Henry Mintzberg examined managerial activities on a daily basis. His study enabled him to identify ten different but, coordinated sets of behaviour, or roles, that managers assume. These ten roles can be separated into three general groupings: interpersonal roles, informational roles, and decisional roles.

INTERPERSONAL ROLES

Three of the manager’s roles come into play when the manager must engage in interpersonal relationships. The three roles of figurehead, leader, and liaison are each necessary under differing circumstances. Adopting one or another of the three interpersonal roles is made easier by the formal authority the manager obtains from the organization.

The figurehead role is enacted when activity of a ceremonial nature is required within the organization. A baseball manager attending a minor league all-star game, the head chef of a prominent restaurant greeting customers at the door, and the president of a bank congratulating a new group of trainees are all examples of the figurehead role. While the figurehead role is routine, with little serious communication and no important decision making, its importance should not be overlooked. At the interpersonal level, it provides members and non-members alike with a sense of what the organization is

about and the type of people the organization recruits. The second interpersonal role, the leader role, involves the coordination and control of the work of the manager's subordinates. The leader role may be exercised in a direct or an indirect manner. Hiring, training, and motivating may all require direct contact with subordinates. However, establishing expectations regarding work quality, decision-making responsibility, or time commitments to the job are all outcomes of the leader role that are indirectly related to subordinates.

Quite often, managers are required to obtain information or resources outside their authority. The liaison role is enacted when managers make contact with other individuals, who may or may not reside in the organization, in order to complete the work performed by their departments or work units. An auto assembly plant supervisor may telephone a tire supplier to determine the amount of inventory available for next week; a prosecuting attorney may meet with the presiding judge and defence attorney to discuss the use of motions and evidence in a libel trial; or a college professor may meet with professors in a separate department on campus to obtain information on a prospective doctoral student. Ultimately, the liaison role enables a manager to develop a network for obtaining external information which can be useful for completing current and future work activities.

INFORMATIONAL ROLES

Monitor, disseminator, and spokesperson are the three informational roles that a manager may assume. These informational roles are created as a result of enacting the set of interpersonal roles already described. A network of interpersonal contacts with both subordinates and individuals outside the work unit serves to establish the manager as an informational nerve center of the unit, responsible for gathering, receiving, and transmitting information that concerns members of the work unit.

A manager assumes the monitor role by continually scanning the environment for information or activities and events that may identify opportunities or threats to the functioning of the work unit. Much of the manager's gathering of information is achieved through the network of contacts that has been established through the interpersonal roles. Hearing small talk at a banquet about a competitor's planned marketing programme, learning through casual conversation at a ball game about the negative medical evaluation of an unsigned ball player, or daily reading of a business periodical are all examples of the kinds of information gathering involved in the monitor role.

The information a manager gathers as a monitor must be evaluated and transmitted as appropriate to members of the organization. The transmittal of information by a manager activates the disseminator role. Privileged information may be disseminated to subordinates, peers, or superiors in the organization. The manager may inform the marketing vice-president about the specific marketing strategy a competitor is planning to implement. A baseball manager may inform the team owner that an impending trade should be cancelled because of the unfavourable medical report on one of the players. Or reading *The Wall Street Journal* may inform the manager that a shipping strike is

looming and thus enable her to inform subordinates that temporary layoffs may occur next month. Occasionally, a manager must assume the spokesperson role by speaking on behalf of the work unit to people inside or outside the organization. This might involve lobbying for critical resources or appealing to individuals who have influence on activities that affect the work unit. A top manager asking the board of directors to keep the work unit together during a reorganization period or a corporate president speaking to a college audience on the role the company plays in education would both constitute engaging in the spokesperson role.

DECISIONAL ROLES

Both interpersonal and informational roles are really preludes to what are often considered to be a manager's most important set of roles: the decisional roles of entrepreneur, disturbance handler, resource allocator, and negotiator.

The entrepreneur role comes into action when the manager seeks to improve the work unit. This can be accomplished by adapting new techniques to fit a particular situation or modifying old techniques to improve individual or group activity.

Managers usually learn of new or innovative methods through information gathered in the monitor role. As a result, a supervisor purchases a new kiln which will shorten the drying process for ceramic tiles; a director of a youth club trains staff in the use of personal computers to increase file access; or a president establishes a new pension plan to improve employee morale. Whereas the entrepreneur role establishes the manager as the initiator of change, the disturbance handler role establishes the manager as a responder to change. Organizations, unfortunately, do not run so smoothly that managers are never called upon to respond to unwelcome pressures. In these cases, the manager is required to act quickly to bring stability back to the organization. A law partner must settle a disagreement among associates in the firm on who will present a case before a judge; a personnel director must negotiate with striking employees dissatisfied with the procedures for laying off employees; or a cannery first-line manager must respond to a sudden shortage of cans used to package perishable fruit because the supplier has reneged on a contract.

When a manager is placed in the position of having to decide to whom and in what quantity resources will be dispensed, the resource allocator role is assumed. Resources may include money, time, power, equipment, or people. During periods of resource abundance, this role can be easily performed by a manager. In most cases, however, organizations operate under conditions of resource scarcity; thus, decisions on the allocation of resources can be critical for the success of the work unit, division, or organization.

As a decision maker, the manager must strive not only to appropriately match resources with subordinates but also to ensure that the distribution of resources is coordinated to effectively complete the task to be performed. An office manager must provide secretaries with appropriate equipment to generate and duplicate documents. A

manager of a fast-food restaurant must coordinate work shifts to have the maximum number of employees working during the lunch hour. Corporate presidents may provide their administrative assistants with decision-making responsibility for day-to-day matters.

In addition to decisions concerning organizational changes, disturbances, and resources, the manager must enact a negotiator role. The process of negotiation is possible only when an individual has the authority to commit organizational resources. Hence, as managers move up the managerial hierarchy and obtain control over more resources, they become more involved in the negotiator role. For example, the president of a record company may be called in to discuss terms of a possible contract with a major rock group; a production manager must negotiate with the personnel department to obtain employees with specialized skills; or a college dean must negotiate with department heads over course offerings and the number of faculty to be hired.

The relative emphasis a manager places on these ten roles is highly dependent on the manager's authority and status in the organization. Length of time on the job, position in the management hierarchy, goals of the subunit to be achieved, and skills the manager possesses all play a part in determining which roles are more prominent than others at any given time. For instance, a marketing manager is more likely to emphasize the interpersonal roles because of the importance of personal contact in the marketing process. A financial manager, charged with responsibility for the economic efficiency of the organization, will probably focus on the decisional roles. A staff manager, or a manager who performs in an advisory capacity, is likely to be more heavily involved in the informational roles. Regardless of the differences that may occur, however, all managers enact interpersonal, informational, and decisional roles while performing their tasks. Effectively managing an organization is a demanding task. Managers not only must develop skills related to the functional areas of management but also must learn how to integrate these activities. What makes this process demanding is that events and activities external and internal to an organization can radically change the techniques and methods managers must use in order to arrive at successful outcomes. Managers cannot afford to be limited in their view of management, nor can they simply rely on how things were done in the past. Even the most seasoned and successful managers are prone to mistakes. However, a more complete knowledge of the managerial process can reduce the chances of mistakes that will have dire consequences for an organization. Such knowledge may help managers to better plan, organize and staff, direct, and control organization activities within the context of their organization.

MINTZBERG'S TEN MANAGEMENT ROLES

Mintzberg's Ten Management Roles are a complete set of behaviours or roles within a business environment. Each role is different, thus spanning the variety of all identified management behaviours. When collected together as an integrated whole (gestalt), the capabilities and competencies of a manager can be further evaluated in a role-specific way.

The Ten Management Roles

The ten roles explored in this theory have extensive explanations which are briefly developed here:

1. *Figurehead*: All social, inspiration, legal and ceremonial obligations. In this light, the manager is seen as a symbol of status and authority.
2. *Leader*: Duties are at the heart of the manager-subordinate relationship and include structuring and motivating subordinates, overseeing their progress, promoting and encouraging their development, and balancing effectiveness.
3. *Liaison*: Describes the information and communication obligations of a manager. One must network and engage in information exchange to gain access to knowledge bases.
4. *Monitor*: Duties include assessing internal operations, a department's success and the problems and opportunities which may arise. All the information gained in this capacity must be stored and maintained.
5. *Disseminator*: Highlights factual or value based external views into the organisation and to subordinates. This requires both filtering and delegation skills.
6. *Spokesman*: Serves in a PR capacity by informing and lobbying others to keep key stakeholders updated about the operations of the organisation.
7. *Entrepreneur*: Roles encourage managers to create improvement projects and work to delegate, empower and supervise teams in the development process.
8. *Disturbance handler*: A generalist role that takes charge when an organisation is unexpectedly upset or transformed and requires calming and support.
9. *Resource Allocator*: Describes the responsibility of allocating and overseeing financial, material and personnel resources.
10. *Negotiator*: Is a specific task which is integral for the spokesman, figurehead and resource allocator roles.

As a secondary filtering, Mintzberg distinguishes these roles by their responsibilities towards information. Interpersonal roles, categorised as the figurehead, leader and liaison, provide information.

Informational roles link all managerial work together by processing information. These roles include the monitor, the disseminator and the spokesperson. All the remaining roles are decisional, in that they use information and make decisions on how information is delivered to secondary parties.

Generalist and Specialist Management

The core of Mintzberg's Ten Managerial Roles is that managers need to be both organisational generalists and specialists.

This is due to three reasons:

1. External frustrations including operational imperfections and environmental pressures.

2. Authority disputes which upset even basic routines.
3. The expected fallibility of the individual and human, manager.

Mintzberg's summary statement may be that the role of a manager is quite varied and contradictory in its demands, and that it is therefore not always the lack of managerial prowess, but the complexity of individual situations demanding a variety of roles, which troubles today's manager.

FUNCTIONAL AREAS OF MANAGEMENT

PRODUCTION MANAGEMENT

Production means creation of utilities by converting raw material in to final product by various scientific methods and regulations. It is very important field of management.

Various sub-areas of the production department are as follows:

- *Plant lay out and location:* This area deals with designing of plant layout, decide about the plant location for various products and providing various plant utilities
- *Production planning:* Managers has to plan about various production policies and production methods.
- *Material management:* This area deals with purchase, storage, issue and control of the material required for production department.
- *Research and Development:* This area deals with research and developmental activities of manufacturing department. Refinement in existing product line or develop a new product are the major activities.
- *Quality Control:* Quality control department works for production of quality product by doing various tests which ensure the customer satisfaction.

MARKETING MANAGEMENT

Marketing management involves distribution of the product to the buyers.

It may need number of steps. Sub areas are as follows:

- *Advertising:* This area deals with advertising of product, introducing new product in market by various means and encourage the customer to buy thee products.
- *Sales management:* Sales management deals with fixation of prices, actual transfer of products to the customer after fulfilling certain formalities and after sales services.
- *Market research:* It involves in collection of data related to product demand and performance by research and analysis of market.

FINANCE AND ACCOUNTING MANAGEMENT

Financial and accounting management deals with managerial activities related to procurement and utilization of fund for business purpose.

Its sub areas are as follows:

- *Financial accounting:* It relates to record keeping of various financial transactions their classification and preparation of financial statements to show the financial position of the organization.
- *Management accounting:* It deals with analysis and interpretation of financial record so that management can take certain decisions on investment plans, return to investors and dividend policy
- *Taxation:* This area deals with various direct and indirect taxes which organization has to pay.
- *Costing:* Costing deals with recording of costs, their classification, analysis and cost control.

PERSONNEL MANAGEMENT

Personnel management is the phase of management which deals with effective use and control of manpower.

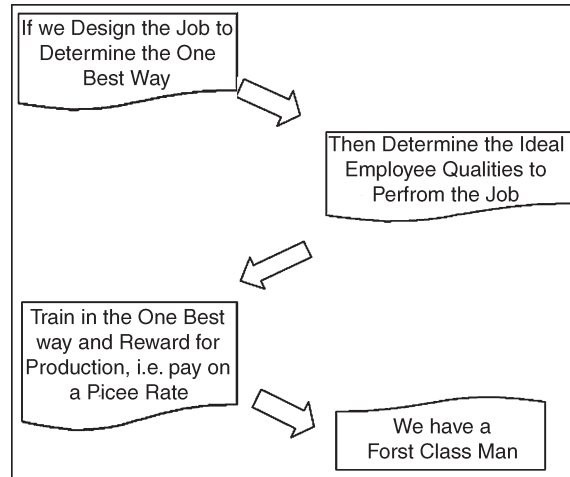
Following are the sub areas of Personnel management:

- *Personnel planning:* This deals with preparation inventory of available manpower and actual requirement of workers in organization.
- *Recruitment and selection:* This deals with hiring and employing human being for various positions as required.
- *Training and development:* Training and development deals with process of making the employees more efficient and effective by arranging training programmes. It helps in making team of competent employees which work for growth of organisation.
- *Wage administration:* It deals in job evaluation, merit rating of jobs and making wage and incentive policy for employees.
- *Industrial relation:* It deals with maintenance of overall employee relation, providing good working conditions and welfare services to employees.

DEVELOPMENT OF MANAGEMENT THOUGHT

TAYLOR'S FOUR FUNDAMENTAL PRINCIPLES

1. Develop one best way to do each job, based on careful job analysis.
2. Scientifically select, train, and develop workers to perform the job in the "one best way."
3. Provide a culture between managers and workers in which workers closely follow the manager's guidance, and managers work to improve processes.
4. Equal division of work between management and workers; it is the responsibility of workers to diligently carry out jobs in the one best way and management must work to continuously improve the firm's performance.



FRANK AND LILLIAN GILBRETH: LEAN PRODUCTION

- *Similarities with Taylor's work:* Both viewed careful job analysis and design as the starting point of efficiency gains. Both sought to find “standard” high efficiency approaches to completing a job.
- *Differences:* The Gilbreth's saw the initial job design scheme as the starting point at which employees could be looked to for further improvement. Taylor saw his “one best way” as sort of the end of the process, and really did not see workers making contributions to enhance performance. The Gilbreth's and Toyota which embraced their approach saw workers as sources of continuous improvement.

HENRI FAYOL'S GENERAL PRINCIPLES OF MANAGEMENT

- *Division of Work:* More and better work with the same effort results from allowing individual task specialization.
- *Authority and Responsibility:* Authority is the right to utilize organizational resources, give orders, and to exact obedience. Responsibility is the obligation to use authority to effectively and efficiently achieve organizational objectives. Authority and responsibility go hand in hand and flows from the top down.
- *Discipline:* Conformance with directives and instructions. Without discipline no enterprise could prosper.
- *Unity of Command:* For any action whatsoever, an employee should receive orders from one superior only.
- *Unity of Direction:* One head and one plan for a group of activities having the same objective.
- *Subordination of Individual Interest to General Interest:* The interest of one employee or group of employees should not prevail over that of the enterprise.

- *Remuneration of Personnel:* Remuneration of personnel is the price of services rendered.
- *Centralization:* The optimum degree of centralization must be found for each enterprise.
- *Scalar Chain:* The scalar chain is the chain of superiors ranging from the ultimate authority to the lowest ranks. It represents an unbroken series of authority-accountability relationships
- *Order:* A place for everyone and everyone in his place.
- *Equity:* The head of the enterprise should strive to instill a sense of equity throughout all levels of the scalar chain.
- *Stability of Tenure of Personnel:* It takes a long time for managers to settle in. Excessive turnover is undesirable.
- *Initiative:* Initiative is the power of thinking out a plan and ensuring its success. The initiative of all is a great source of strength for the enterprise.
- *Esprit de Corps:* In union there is strength.

TENANTS OF MODERN BUREAUCRACY

- The organization is governed by written policies, procedures, and directives.
- There should be a hierarchy of formal authority and responsibility.
- Individuals, both employees and clients, are treated impersonally.
- Authority is invested in the office, not the person holding the office.
- Written directives and instructions are provided (a job description).
- Positions are filled on the basis of pre-defined qualifications and promotions are based on merit.

Pros and Cons of Bureaucracy

Pros:

- Insures (or attempts to insure) that organizations treat each client and employee equitably and consistently.
- Insures uniformity and consistency across the organization.
- Seeks to reduce graft and corruption.

Cons:

- Very inflexible—does not deal with exceptions and unique situations well.
- The rules tend to be difficult to change, and therefore slow to change, which creates problems when the environment is dynamic.

According to Weber, Bureaucratization Resulted from Trelated Causes

1. Competition among capitalist firms in the marketplace.
2. Competition among states, increasing rulers' need to control their staff and citizenry.

3. *Bourgeois demands for equal protection under the law*: “It is primarily the capitalist market economy which demands that the official business of administration be discharged precisely, unambiguously, continuously, and with as much speed as possible. Normally, the very large, modern capitalist enterprises are themselves unequally models of strict bureaucratic organization.”

HAWTHORNE STUDIES: BIRTH OF THE NEO-CLASSICAL SCHOOL

- Series of experiments that took place at the Hawthorne plant of the Western Electric Company. The original purpose of the experiments was to look at the effects of working conditions on efficiency, but they became much more complex.
- Key Researchers were Elton Mayo and Fritz Roethlisberger.
- Most famous experiment was the Illumination Study—Researchers thought that by lowering the lighting in the workroom, productivity would go down; however, he found just the opposite.
- Assessment—these studies pointed out that workers’ feelings, attitudes, motivations, and needs had an impact on their productivity. Management realised for the first time that they needed to be concerned about workers in order to make the organization more efficient and productive. Hawthorne Studies also pointed out the importance of the informal organization created by the social relationships formed by employees.
- Not many concrete suggestions for improving management flowed from the studies.

THE HUMAN FACTOR IN MANAGEMENT

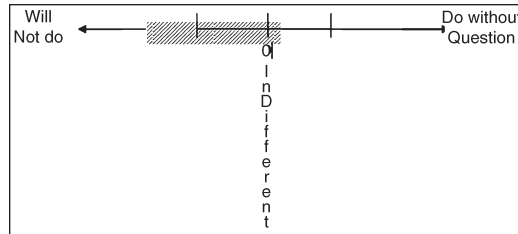
- *Mary Parker Follett*: Great works are accomplished in groups. Shared power and group accomplishment should be important goals of management. She would now be talking about subordinate empowerment
- *Chester Bernard, Acceptance View of Authority*: Authority flows from the bottom up, through the Scalar Chain. In effect, the right to lead is granted by those who are led—i.e., the consent of the governed.

In order for a subordinate to likely comply with an order the following must be true:

- They understand the order
- They feel the order is consistent with the purpose of the organization.
- The order does not conflict with their personal beliefs, and is considered a reasonable request (within the zone of indifference).
- They are able to perform the task as directed.

Zone Indifference

The psychological zone that encompasses acceptable behaviours employees do not mind following.



SOURCES OF MANAGERIAL AUTHORITY

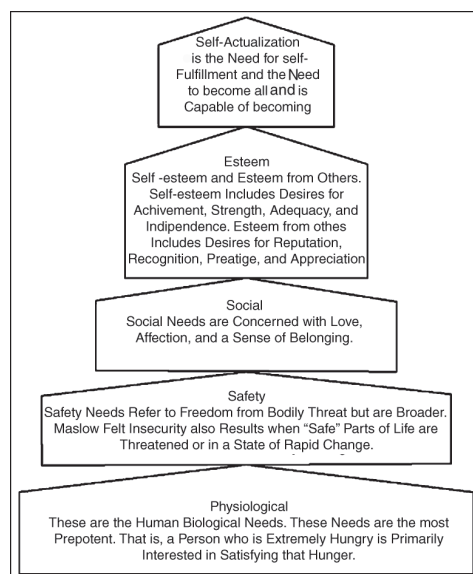
The Classical School vs. the Neo-Classical view:

- According to the classical view, authority flows from the top of an organization to the bottom. It is based on the rights of property owners and is a logical, but legalistic, view of the source of a manager's formal authority.
- According to the acceptance view, authority flows from the bottom of an organization to the top; that is, managers get their authority from their subordinates. Chester Barnard suggested there is a zone of indifference within which subordinates will accept orders from their superiors more or less without thinking about them.

ABRAHAM H. MASLOW AND THE BEHAVIOURALIST SCHOOL

Hierarchy of Human Needs

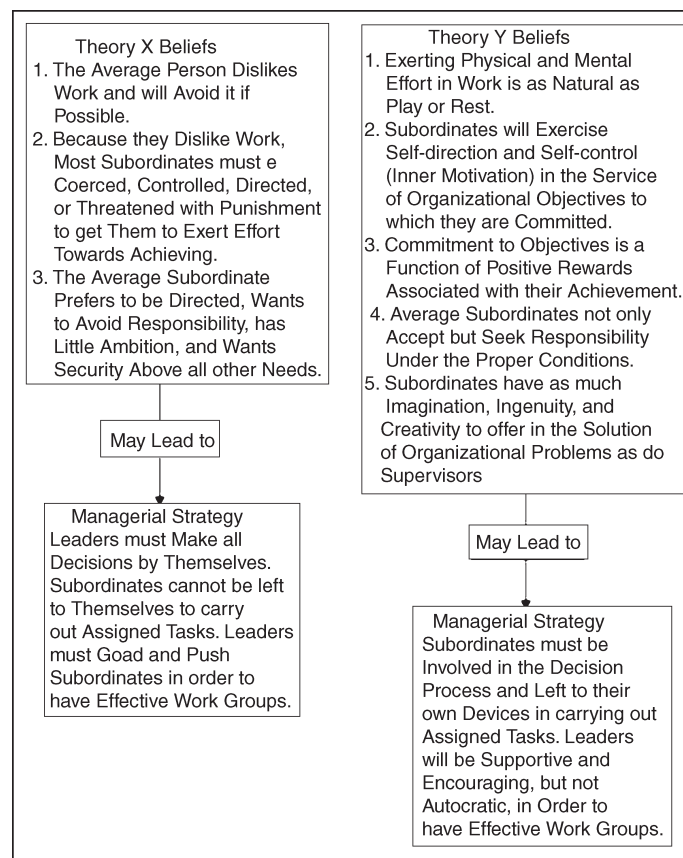
Maslow's idea of a "hierarchy" means that needs at one level must be relatively satisfied before the needs at the next higher level can motivate behaviour. Once a need is relatively satisfied it ceases to motivate behaviour. In Maslow's view, only needs that are relatively unsatisfied are capable of motivating people.



This is known as Maslow's prepotency principle—the need that is most powerful in motivating behaviour is the need at the next higher level of the hierarchy above the level that is currently satisfied.

DOUGLAS MCGREGOR

Theory X and Theory Y



FREDERICK HERZBERG

Hygiene Factors and Motivating Factors

Primary needs represent the need for basic survival and the need to avoid pain and discomfort in work situations. These primary needs are satisfied by what Herzberg called hygiene factors in the work environment—factors such as money and working conditions. Herzberg considers money the most important hygiene factor. In motivator-hygiene theory, hygienics have a unique role. They prevent employees from being dissatisfied with their jobs, but they largely do not affect employee satisfaction.

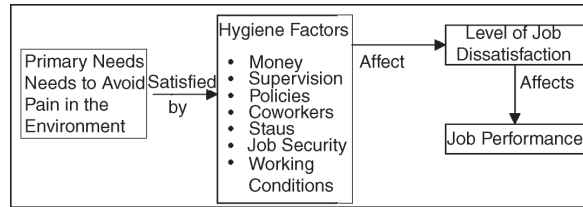


Fig. Role of Hygienes in Motivator-Hygiene Theory.

In addition to primary needs, people at work have growth needs. These needs are satisfied by motivator factors—factors such as recognition and achievement. Herzberg considers achievement the most important motivator. Motivators help make people satisfied with their jobs.

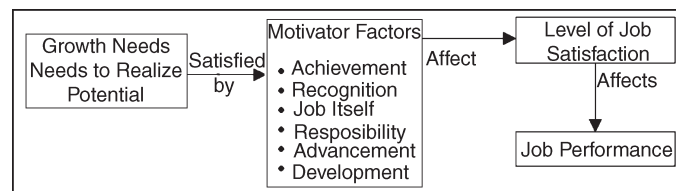
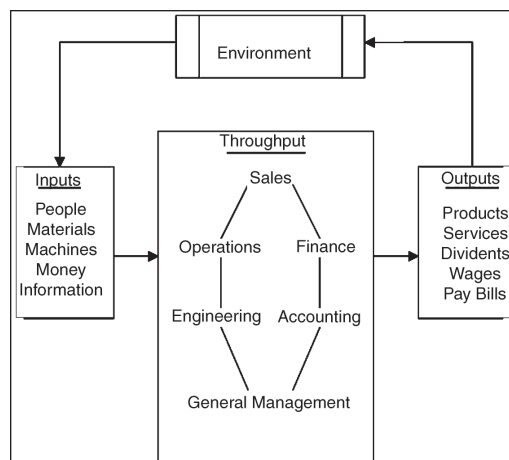
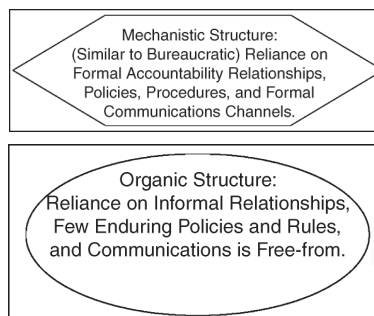


Fig. Role of Motivators in Motivator-Hygiene Theory.

The Business Firm as an Open System:



Organization Structure:



EXTERNAL ENVIRONMENT

- *Burns and Stalker Study*: Research on the type of organizational structure that is most successful in a given type of external environment. A highly centralized and bureaucratic organization (mechanistic structure) is more successful in a stable environment while an organization that is more flexible and decentralized (organic structure) works best in a dynamic environment of change.
- *Lawrence and Lorsch Study*: Looked at environmental uncertainty—important to match structure of organization to type of environment.

CLASSICAL AND NEO-CLASSICAL THEORIES OF MANAGEMENT

CLASSICAL MANAGEMENT THEORY

There are three well-established theories of classical management: Taylor's Theory of Scientific Management, Fayol's Administrative Theory, Weber's Theory of Bureaucracy. Although these schools, or theories, developed historical sequence, later ideas have not replaced earlier ones. Instead, each new school has tended to complement or coexist with previous ones.

Theory recognizing the role that management plays in an organization. The importance of the function of management was first recognized by French industrialist Henri Fayol in the early 1900s. In contrast to the purely scientific examination of work and organizations conducted by F W Taylor, Fayol proposed that any industrial undertaking had six functions: technical; commercial; financial; security; accounting; and managerial. Of these, he believed the managerial function, 'to forecast and plan, to organize, to command, to coordinate, and control', to be quite distinct from the other five. Fayol also identified general principles of management: division of work; authority and responsibility; discipline; unity of command; unity of direction; subordination of individual interest to general interest; remuneration of personnel; centralization; scalar chain of authority; order; equity; stability of tenure of personnel; initiative; and esprit de corps. Fayol's views on management remained popular throughout a large part of the 20th century.

EVOLUTION OF CLASSICAL APPROACH TO MANAGEMENT

Traditional process of learning is either through observation and experiment. Nature or environment is considered uniform and when we observe certain phenomenon or events uniformly leading to the same result or results, we conclude a cause and effect relationship between the two. This is learning by observation or in other words by

experience. Earlier thinkers on management followed this approach in developing theories of management. Learning principally is through empirical process and through analysis of the data collected through observation. Draw the principles of management by looking at and analyzing the jobs that all managers commonly do. This approach served as a starting point for pioneers on management science to verify the validity and improve the applicability of the principles and practices of management. Analysis of observed data is what constitutes a case study. The observational method of case study helps arriving at logical conclusions about past experience and to test the same as standards for future events.

The German socialists, Max Weber followed the classical approach and developed his theory of Bureaucracy, which portrays the structure and design of organization characterized by a hierarchy of authority, formalized rules and regulations that serve to guide the coordinated functioning of an organization.

Basic Postulates of the Classical Approach by Max Weber:

- Management of an organization is considered as a chain of interrelated functions. The study of the scope and features of these functions, the sequence through which these are performed and their interrelationship leads one to draw principles of management suitable for universal application
- Learning principles of management is done through the past experiences of actual practicing managers
- As business environment consists of uniform cycles exhibiting an underlying unity of realities, functions and principles of management derived through process of empirical reasoning are suitable for universal application
- Emerging new managers through formal education and case study can develop skill and competency in management concepts and practices
- The classical approach also recognized the importance of economic efficiency and formal organizational structure as guiding pillars of management effectiveness.
- Business activity is based on economic benefit. Organizations should therefore control economic incentives

NEOCLASSICAL THEORY OF MANAGEMENT

Human Relations Theory

Explains the modern advancement of Human Relations Management theory which takes into account human factors like the employer-employee relationship. Human relations theory is largely seen to have been born as a result of the Hawthorne experiments which Elton Mayo conducted at the Western Electrical Company. The important strand in the development of modern management was the increase in attention to the human factors, which has become known as the 'human relations school of management.' The

core aspect of Human Relations Theory is that, when workers were being observed and included in the research, they felt more important and valued by the company. As a result, their productivity levels went up significantly. This represented a significant departure from many of the classical theories, particularly Fordism, as it went against the notion that management needed to control workers, and remove their autonomy at every step. Instead, it showed that by engaging with workers and considering their requirements and needs, company's could benefit from increased productivity.

Behavioural Theory

The behavioural management theory is often called the human relations movement because it addresses the human dimension of work. Behavioural theorists believed that a better understanding of human behaviour at work, such as motivation, conflict, expectations, and group dynamics, improved productivity.

The theorists who contributed to this school viewed employees as individuals, resources, and assets to be developed and worked with—not as machines, as in the past. Several individuals and experiments contributed to this theory.

Social Systems Theory

Developed by Niklas Luhmann is an option for the theoretical foundation of Human Resource Management (HRM). After clarifying the advantages of using a grand (social) theory as the basic theoretical perspective, the roots of this social systems theory - the deterministic view of systems as machines, the open systems approach and non-linear systems theory - are addressed. Based on the view of social systems as autopoietically closed systems, five major contributions to a theoretical foundation of HRM are identified:

- The conceptualization of organizing and managing human resources as social processes, thus overcoming an individualistic angle;
- The new importance of individuals as essential element in the system's environment;
- the abstention from far reaching or highly unrealistic assumptions about the 'nature' of human beings;
- The interaction between various levels and units of analysis built into the theory which is essential for comprehensive and in-depth analyses of HR phenomena and
- The openness for additional theories for which social systems theory provides the overall framework.

George Elton Mayo was in charge of certain experiments on human behaviour carried out at the Hawthorne Works of the Western Electric company in Chicago between 1924 and 1927. His research findings have contributed to organizational development in terms of human relations and motivation theory.

Elton Mayo's contributions came as part of the Hawthorne studies, a series of experiments that rigorously applied classical management theory only to reveal its shortcomings. The Hawthorne experiments consisted of two studies conducted at the Hawthorne Works of the Western Electric Company in Chicago from 1924 to 1932. The first study was conducted by a group of engineers seeking to determine the relationship of lighting levels to worker productivity. Surprisingly enough, they discovered that worker productivity increased as the lighting levels decreased—that is, until the employees were unable to see what they were doing, after which performance naturally declined. A few years later, a second group of experiments began. Harvard researchers Mayo and F. J. Roethlisberger supervised a group of five women in a bank wiring room. They gave the women special privileges, such as the right to leave their workstations without permission, take rest periods, enjoy free lunches, and have variations in pay levels and workdays. This experiment also resulted in significantly increased rates of productivity.

In this case, Mayo and Roethlisberger concluded that the increase in productivity resulted from the supervisory arrangement rather than the changes in lighting or other associated worker benefits. Because the experimenters became the primary supervisors of the employees, the intense interest they displayed for the workers was the basis for the increased motivation and resulting productivity. Essentially, the experimenters became a part of the study and influenced its outcome. This is the origin of the term Hawthorne effect, which describes the special attention researchers give to a study's subjects and the impact that attention has on the study's findings.

A FRAMEWORK FOR MANAGING OPERATIONS

Managing operations can be enclosed in a frame of general management function. Operation managers are concerned with planning, organizing, and controlling the activities which affect human behaviour through models.

PLANNING

Activities that establishes a course of action and guide future decision-making is planning. The operations manager defines the objectives for the operations subsystem of the organization, and the policies, and procedures for achieving the objectives. This stage includes clarifying the role and focus of operations in the organization's overall strategy. It also involves product planning, facility designing and using the conversion process.

Organizing

Activities that establishes a structure of tasks and authority. Operation managers establish a structure of roles and the flow of information within the operations subsystem.

They determine the activities required to achieve the goals and assign authority and responsibility for carrying them out.

Controlling

Activities that assure the actual performance in accordance with planned performance. To ensure that the plans for the operations subsystems are accomplished, the operations manager must exercise control by measuring actual outputs and comparing them to planned operations management. Controlling costs, quality, and schedules are the important functions here.

Behaviour

Operation managers are concerned with how their efforts to plan, organize, and control affect human behaviour. They also want to know how the behaviour of subordinates can affect management's planning, organizing, and controlling actions. Their interest lies in decision-making behaviour.

Models

As operation managers plan, organise, and control the conversion process, they encounter many problems and must make many decisions. They can simplify their difficulties using models like *aggregate planning models* for examining how best to use existing capacity in short-term, *break even analysis* to identify break even volumes, *linear programming and computer simulation* for capacity utilisation, *decision tree analysis* for long-term capacity problem of facility expansion, *simple median model* for determining best locations of facilities, etc.

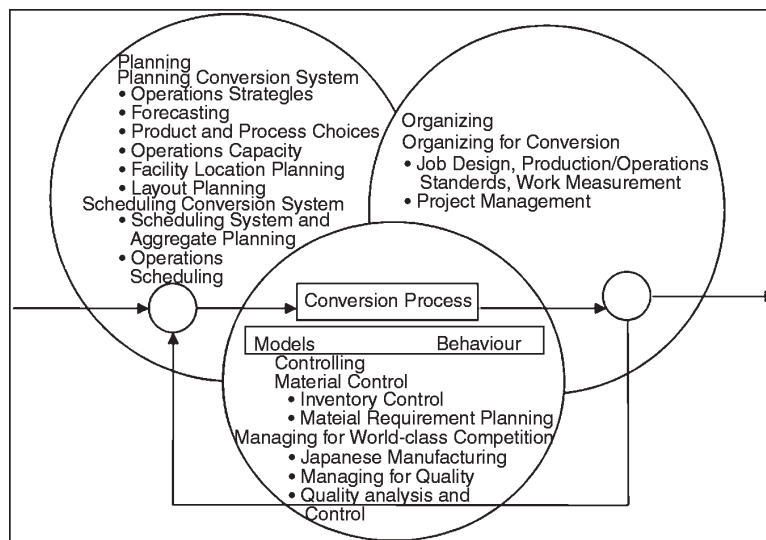


Fig. General Model for Managing Operations.

OBJECTIVES OF OPERATIONS MANAGEMENT

Objectives of operations management can be categorised into customer service and resource utilisation.

Customer Service

The first objective of operating systems is the customer service to the satisfaction of customer wants. Therefore, customer service is a key objective of operations management.

The operating system must provide something to a specification which can satisfy the customer in terms of cost and timing. Thus, primary objective can be satisfied by providing the 'right thing at a right price at the right time'.

These aspects of customer service—specification, cost and timing—are described for four functions in Table. They are the principal sources of customer satisfaction and must, therefore, be the principal dimension of the customer service objective for operations managers.

Principal Function	Principal Customer Wants	
	Primary Considerations	Other Considerations
Manufacture	Goods of a given, Requested or Acceptable Specification	Cost, i.e., Purchase Price or Cost of Obtaining Goods. Timing, i.e., Delivery Delay from Order or Request to Receipt of Goods.
Transport	Management of a Given, Requested or Acceptable Specification	Cost, i.e., Cost of Movement. Timing, i.e., 1. Duration or Time to Move. 2. Wait or Delay from Requesting to its Commencement.
Supply	Goods of a Given, Requested or Acceptable Specification	Cost, i.e., Delivery Delay From Order or Request to Receipt of Goods. Timing, i.e., Delivery Delay from Order or Request to receipt of Goods.
Service	Treatment of a given, Requested or Acceptable Specification	Cost, i.e., Cost of Movements. Timing, i.e., 1. Duration or time Required for Treatment 2. Wait or Delay From requesting Treatment to its Commencement

Fig. Aspects of Customer Service.

Generally an organization will aim reliably and consistently to achieve certain standards and operations manager will be influential in attempting to achieve these standards. Hence, this objective will influence the operations manager's decisions to achieve the required customer service.

Resource Utilisation

Another major objective of operating systems is to utilise resources for the satisfaction of customer wants effectively, *i.e.*, customer service must be provided with the achievement of effective operations through efficient use of resources. Inefficient use of resources or inadequate customer service leads to commercial failure of an operating system.

Operations management is concerned essentially with the utilisation of resources, *i.e.*, obtaining maximum effect from resources or minimising their loss, under utilisation or waste. The extent of the utilisation of the resources' potential might be expressed in terms of the proportion of available time used or occupied, space utilisation, levels of activity, etc. Each measure indicates the extent to which the potential or capacity of such resources is utilised. This is referred as the objective of resource utilisation.

Operations management is also concerned with the achievement of both satisfactory customer service and resource utilisation. An improvement in one will often give rise to deterioration in the other. Often both cannot be maximised, and hence a satisfactory performance must be achieved on both objectives.

All the activities of operations management must be tackled with these two objectives in mind, and many of the problems will be faced by operations managers because of this conflict. Hence, operations managers must attempt to balance these basic objectives.

Table. The Twin Objectives of Operations Management.

The customer service objective.	The resource utilisation objective.
To provide agreed/adequate levels of customer service (and hence customer satisfaction) by providing goods or services with the right specification, at the right cost and at the right time	To achieve adequate levels of resource utilisation (or productivity) <i>e.g.</i> , to achieve agreed levels of utilisation of materials, machines and labour.

MANAGING GLOBAL OPERATIONS

The term 'globalization' describes businesses' deployment of facilities and operations around the world. Globalization can be defined as a process in which geographic distance becomes a factor of diminishing importance in the establishment and maintenance of cross border economic, political and socio-cultural relations.

It can also be defined as worldwide drive towards a globalized economic system dominated by supranational corporate trade and banking institutions that are not accountable to democratic processes or national governments.

There are four developments, which have spurred the trend towards globalization.

These are:

1. Improved transportation and communication technologies;
2. Opened financial systems;
3. Increased demand for imports; and
4. Reduced import quotas and other trade barriers.

When a firm sets up facilities abroad it involve some added complexities in its operation. Global markets impose new standards on quality and time. Managers should not think about domestic markets first and then global markets later, rather it could be

think globally and performance locally. Also, they must have a good understanding of their competitors. Some other important challenges of managing multinational operations include other languages and customs, different management style, unfamiliar laws and regulations, and different costs.

Managing global operations would focus on the following key issues:

- To acquire and properly utilize the following concepts and those related to global operations, supply chain, logistics, etc.
- To associate global historical events to key drivers in global operations from different perspectives.
- To develop criteria for conceptualization and evaluation of different global operations.
- To associate success and failure cases of global operations to political, social, economical and technological environments.
- To envision trends in global operations.
- To develop an understanding of the world vision regardless of their country of origin, residence or studies in a respectful way of perspectives of people from different races, studies, preferences, religion, politic affiliation, place of origin, etc.

SCOPE OF PRODUCTION AND OPERATIONS MANAGEMENT

Production and operations management concern with the conversion of inputs into outputs, using physical resources, so as to provide the desired utilities to the customer while meeting the other organizational objectives of effectiveness, efficiency and adoptability. It distinguishes itself from other functions such as personnel, marketing, finance, etc., by its primary concern for 'conversion by using physical resources.'

Following are the activities which are listed under production and operations management functions:

- Location of facilities
- Plant layouts and material handling
- Product design
- Process design
- Production and planning control
- Quality control
- Materials management
- Maintenance management.

LOCATION OF FACILITIES

Location of facilities for operations is a long-term capacity decision which involves a long term commitment about the geographically static factors that affect a business

organization. It is an important strategic level decision-making for an organization. It deals with the questions such as ‘where our main operations should be based?’

The selection of location is a key-decision as large investment is made in building plant and machinery. An improper location of plant may lead to waste of all the investments made in plant and machinery equipments. Hence, location of plant should be based on the company’s expansion plan and policy, diversification plan for the products, changing sources of raw materials and many other factors. The purpose of the location study is to find the optimal location that will results in the greatest advantage to the organization.

PLANT LAYOUT AND MATERIAL HANDLING

Plant layout refers to the physical arrangement of facilities. It is the configuration of departments, work centres and equipment in the conversion process. The overall objective of the plant layout is to design a physical arrangement that meets the required output quality and quantity most economically.

“Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipments and all other supporting services along with the design of best structure to contain all these facilities”. ‘Material Handling’ refers to the ‘moving of materials from the store room to the machine and from one machine to the next during the process of manufacture’. It is also defined as the ‘art and science of moving, packing and storing of products in any form’.

It is a specialised activity for a modern manufacturing concern, with 50 to 75% of the cost of production. Operation and maintenance of material handling devices.

Material handling devices increases the output, improves quality, speeds up the deliveries and decreases the cost of production. Hence, material handling is a prime consideration in the designing new plant and several existing plants.

PRODUCT DESIGN

Product design deals with conversion of ideas into reality. Every business organization have to design, develop and introduce new products as a survival and growth strategy. Developing the new products and launching them in the market is the biggest challenge faced by the organizations.

The entire process of need identification to physical manufactures of product involves three functions: marketing, product development, manufacturing. Product development translates the needs of customers given by marketing into technical specifications and designing the various features into the product to these specifications.

Manufacturing has the responsibility of selecting the processes by which the product can be manufactured. Product design and development provides link between marketing, customer needs and expectations and the activities required to manufacture the product.

PROCESS DESIGN

Process design is a macroscopic decision-making of an overall process route for converting the raw material into finished goods. These decisions encompass the selection of a process, choice of technology, process flow analysis and layout of the facilities.

Hence, the important decisions in process design are to analyse the workflow for converting raw material into finished product and to select the workstation for each included in the workflow.

PRODUCTION PLANNING AND CONTROL

Production planning and control can be defined as the process of planning the production in advance, setting the exact route of each item, fixing the starting and finishing dates for each item, to give production orders to shops and to follow up the progress of products according to orders.

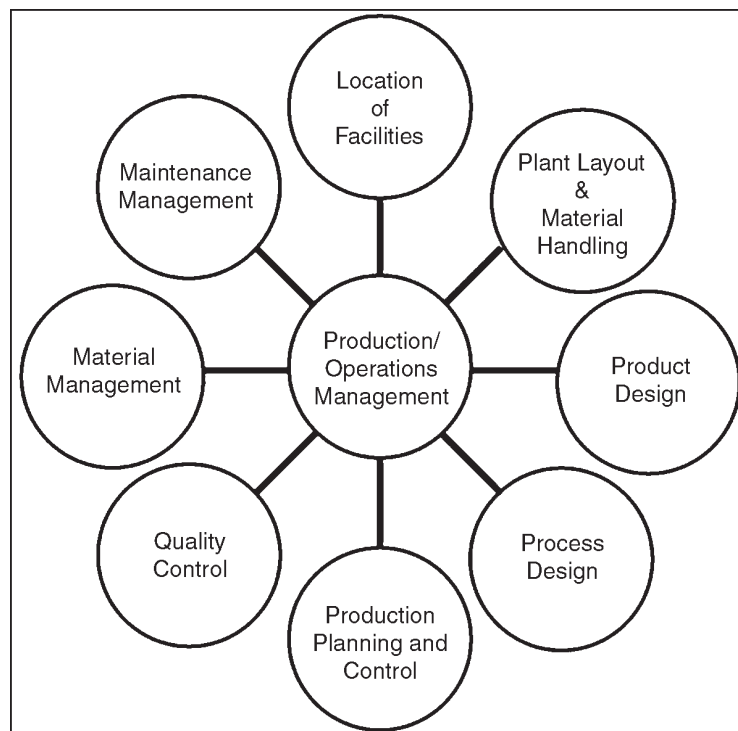


Fig. Scope of Production and Operations Management.

The principle of production planning and control lies in the statement ‘First Plan Your Work and then Work on Your Plan’. Main functions of production planning and control includes planning, routing, scheduling, dispatching and follow-up.

Planning is deciding in advance what to do, how to do it, when to do it and who is to do it. Planning bridges the gap from where we are, to where we want to go. It makes it

possible for things to occur which would not otherwise happen. Routing may be defined as the selection of path which each part of the product will follow, which being transformed from raw material to finished products. Routing determines the most advantageous path to be followed from department to department and machine to machine till raw material gets its final shape.

Scheduling determines the programme for the operations. Scheduling may be defined as 'the fixation of time and date for each operation' as well as it determines the sequence of operations to be followed.

Dispatching is concerned with the starting the processes. It gives necessary authority so as to start a particular work, which has already been planned under 'Routing' and 'Scheduling'. Therefore, dispatching is 'release of orders and instruction for the starting of production for any item in acceptance with the route sheet and schedule charts'. The function of follow-up is to report daily the progress of work in each shop in a prescribed proforma and to investigate the causes of deviations from the planned performance.

QUALITY CONTROL

Quality Control may be defined as 'a system that is used to maintain a desired level of quality in a product or service'. It is a systematic control of various factors that affect the quality of the product. Quality control aims at prevention of defects at the source, relies on effective feed back system and corrective action procedure.

Quality control can also be defined as 'that industrial management technique by means of which product of uniform acceptable quality is manufactured'. It is the entire collection of activities which ensures that the operation will produce the optimum quality products at minimum cost.

The main objectives of quality control are:

- To improve the companies income by making the production more acceptable to the customers *i.e.*, by providing long life, greater usefulness, maintainability, etc.
- To reduce companies cost through reduction of losses due to defects.
- To achieve interchangeability of manufacture in large scale production.
- To produce optimal quality at reduced price.
- To ensure satisfaction of customers with productions or services or high quality level, to build customer goodwill, confidence and reputation of manufacturer.
- To make inspection prompt to ensure quality control.
- To check the variation during manufacturing.

MATERIALS MANAGEMENT

Materials management is that aspect of management function which is primarily concerned with the acquisition, control and use of materials needed and flow of goods and services connected with the production process having some predetermined objectives in view.

The main objectives of materials management are:

- To minimise material cost.
- To purchase, receive, transport and store materials efficiently and to reduce the related cost.
- To cut down costs through simplification, standardisation, value analysis, import substitution, etc.
- To trace new sources of supply and to develop cordial relations with them in order to ensure continuous supply at reasonable rates.
- To reduce investment tied in the inventories for use in other productive purposes and to develop high inventory turnover ratios.

MAINTENANCE MANAGEMENT

In modern industry, equipment and machinery are a very important part of the total productive effort. Therefore, their idleness or downtime becomes are very expensive. Hence, it is very important that the plant machinery should be properly maintained.

The main objectives of maintenance management are:

- To achieve minimum breakdown and to keep the plant in good working condition at the lowest possible cost.
- To keep the machines and other facilities in such a condition that permits them to be used at their optimal capacity without interruption.
- To ensure the availability of the machines, buildings and services required by other sections of the factory for the performance of their functions at optimal return on investment.

FUNCTIONS OF PRODUCTION MANAGEMENT DEPARTMENT

The functions of Production Management depend upon the size of the firm. In small firms the production Manager may have to look after production planning and control along with Personnel, Marketing, Finance and Purchase functions. In medium sized firms, there may be separate managers for Personnel, marketing and Finance functions.

But the production planning and control and Purchase and stores may be under the control of Production management department. In large sized firms the activities of Production Management is confined to the management of production activities only.

As such, there are no hard and fast rule or guidelines to specify the function of Production Management, but in the academic interest we can mention some of the functions, which are looked after by the Production Management department.

They are:

- *Materials:* The selection of materials for the product. Production manager must have sound Knowledge of materials and their properties, so that he can select appropriate materials for his product. Research on materials is necessary to find alternatives to satisfy the changing needs of the design in the product and availability of material resumes.
- *Methods:* Finding the best method for the process, to search for the methods to suit the available resources, identifying the sequence of process are some of the activities of Production Management.
- *Machines and Equipment:* Selection of suitable machinery for the process desired, designing the maintenance policy and design of layout of machines are taken care of by the Production Management department.
- *Estimating:* To fix up the Production targets and delivery dates and to keep the production costs at minimum, production management department does a thorough estimation of Production times and production costs. In competitive situation this will help the management to decide what should be done in arresting the costs at desired level.
- *Loading and Scheduling:* The Production Management department has to draw the time table for various production activities, specifying when to start and when to finish the process required. It also has to draw the timings of materials movement and plan the activities of manpower. The scheduling is to be done keeping in mind the loads on hand and capacities of facilities available.
- *Routing:* This is the most important function of Production Management department. The Routing consists of fixing the flow lines for various raw materials, components, etc., from the stores to the packing of finished product, so that all concerned knows what exactly is happening on the shop floor.
- *Despatching:* The Production Management department has to prepare various documents such as Job Cards, Route sheets, Move Cards, Inspection Cards for each and every component of the product. These are prepared in a set of five copies. These documents are to be released from Production Management department to give green signal for starting the production. The activities of the shop floor will follow the instructions given in these documents. Activity of releasing the document is known as dispatching.
- *Expediting or Follow up:* Once the documents are dispatched, the management wants to know whether the activities are being carried out as per the plans or not. Expediting engineers go round the production floor along with the plans, compare the actual with the plan and feed back the progress of the work to the management. This will help the management to evaluate the plans.
- *Inspection:* Here inspection is generally concerned with the inspection activities during production, but a separate quality control department does the quality inspection, which is not under the control of Production Management. This is

true because, if the quality inspection is given to production Management, then there is a chance of qualifying the defective products also. For example Teaching and examining of students is given to the same person, then there is a possibility of passing all the students in the first grade. To avoid this situation an external person does correction of answer scripts, so that the quality of answers are correctly judged.

- *Evaluation:* The Production department must evaluate itself and its contribution in fulfilling the corporate objectives and the departmental objectives. This is necessary for setting up the standards for future. What ever may be the size of the firm; Production management department alone must do Routing, Scheduling, Loading, Dispatching and expediting.

This is because this department knows very well regarding materials, Methods, and available resources, etc. If the firms are small, all the functions (i to x) are to be carried out by Production Management Department. In medium sized firms in addition to Routing, Scheduling and Loading, Dispatching and expediting, some more functions like Methods, Machines may be under the control of Production Management Department. In large firms, there will be Separate departments for Methods, Machines, Materials and others but routing, loading and scheduling are the sole functions of Production Management. All the ten functions are categorized in three stage, that is Preplanning, Planning and control stages as shown in figure.

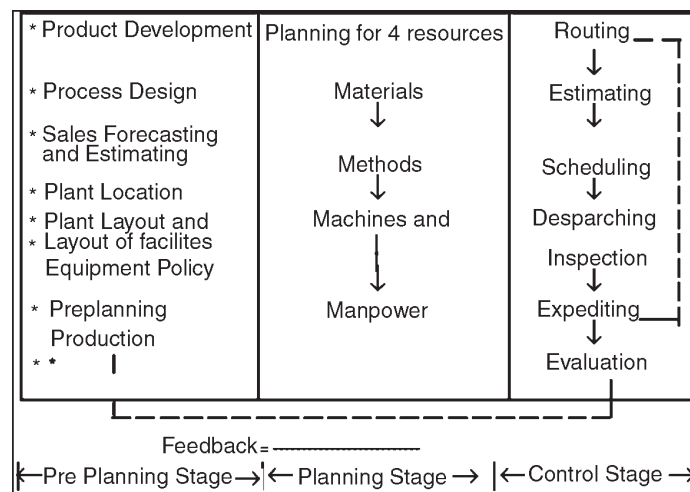


Fig. Production Function.

PLACE OF PRODUCTION MANAGEMENT DEPARTMENT IN THE ORGANISATION

Production is the centre of all activities of an organization. This is to say all the activities of an organization, such as: Finance, Personnel, Marketing...etc., are exists in

an organization because of production activity. Hence the position of Production Management in an organization is very important.

Whether it should be a line function or Staff function, more or less depends upon the corporate management policy. In small organization, Production Management is whole and sole of it. In large industries, generally it is advisable to have Production as line function, because, the decision taken by the line manager and the advices given by the Staff personnel will be based on the Production activities.

The Production Manager, directly report to General Manager, who in turn report to the Board. The figure shows a typical organization structure showing the position of Production Management. Figure shows an organization chart for Production and Operations Management department.

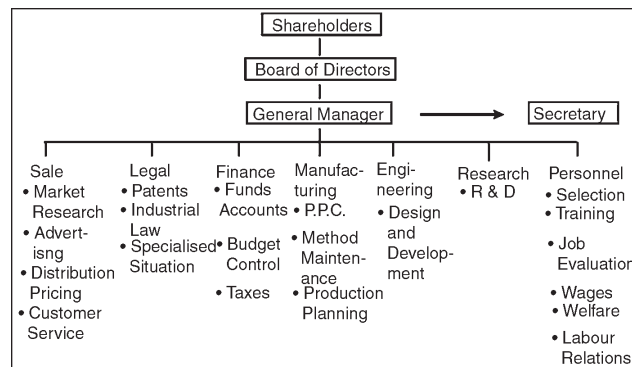


Fig. Typical Organisation Chart for an Organisation.

ORGANISATION CHART FOR P.P.C. DEPARTMENT

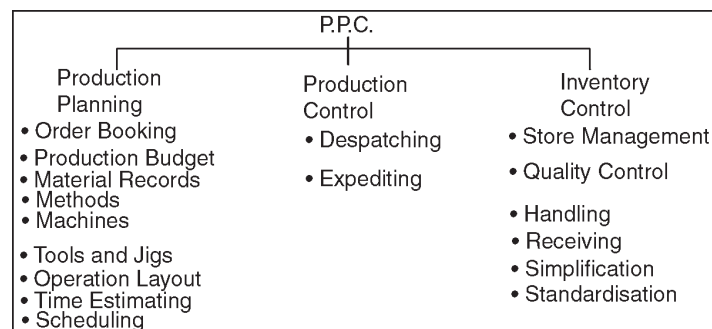


Fig. An Organization Chart for Production Management Department.

TYPES OF PRODUCTION SYSTEMS

The organization of manufacturing systems, also planning and control of production greatly depends on type of product type of the product line. Basic principles that guide the formation of planning policy and its execution may be the same for all the

manufacturing concerns. But emphasis on a particular aspect of production management in fulfilling of specific requirement of the plant and the management approach to the problems of inventory, machine selection, machine setting, tooling, routing, scheduling, loading, follow up and general control will differ depending on the type of production system.

Three main factors generally determine this aspect are:

1. Type of production *i.e.*, quantities of finished products and regularity of manufacture. For example whether Job production or Batch Production or Continuous Production.
2. Size of the Plant *i.e.*, Small Industry, Medium sized Industry or Large Industry.
3. Type of Production: In general there are three classifications in types of Production system..
 - *Job Production*: In this system Products are manufactured to meet the requirements of a specific order. The quality involved is small and the manufacturing of the product will take place as per the specifications given by the customer. This system may be further classified as.
 - a. *The Job produced only once*: Here the customer visit the firm and book his order. After the completion of the product, he takes delivery of the product and leaves the firm. He may not visit the firm to book the order for the same product. The firm has to plan for material, process and manpower only after receiving the order from the customer. The firms have no scope for pre-planning the production of the product.
 - b. *The job produced at irregular intervals*: Here the customer visits the firm to place orders for the same type of the product at irregular intervals. The firm will not have any idea of customer's visit. Here also planning for materials, process and manpower will start only after taking the order from the Customer. In case the firm maintains the record of the Jobs Produced by it, it can refer to the previous plans, when the customer arrives at the firm to book the order.
 - c. *The Jobs Produced periodically at regular intervals*: In this system, the customer arrives at the firm to place orders for the same type of product at regular intervals. Here firm knows very well that the customer visits at regular intervals, it can plan for materials, and process and manpower and have them in a master file. As soon as the customer visits and books the order, the firm can start production. If the volume of the order is considerably large and the number of regularly visiting customers are large in number, the Job Production system slowly transform into Batch Production system.
 - *Batch Production*: Batch Production is the manufacture of number of identical products either to meet the specific order or to satisfy the demand. When the Production of plant and equipment is terminated, the

plant and equipment can be used for producing similar products. This system also can be classified under three categories.

- a. *A batch produced only once*: Here customer places order with the firm for the product of his specification. The size of the order is greater than that of job production order. The firm has to plan for the resources after taking the order from the customer.
- b. *A Batch produced at irregular intervals as per Customer order or when the need arises*: As the frequency is irregular, the firm can maintain a file of its detailed plans and it can refer to its previous files and start production.
- c. *A Batch Produced periodically at known Intervals*: Here the firm either receives order from the customer at regular intervals or it may produce the product to satisfy the demand. It can have well designed file of its plans, material requirement and instructions for the ready reference. It can also purchase materials required in bulk in advance. As the frequency of regular orders goes on increasing the Batch Production system becomes Mass Production System. Here also, incase the demand for a particular product ceases, the plant and machinery can be used for producing other products with slight modification in layout or in machinery and equipment.

Operational Planning Management

An operational planning is a subset of strategic work plan. It describes short-term ways of achieving milestones and explains how, or what portion of, a strategic plan will be put into operation during a given operational period, in the case of commercial application, a fiscal year or another given budgetary term. An operational plan is the basis for, and justification of an annual operating budget request. Therefore, a five-year strategic plan would need five operational plans funded by five operating budgets.

Operational plans should establish the activities and budgets for each part of the organisation for the next 1 - 3 years. They link the strategic plan with the activities the organization will deliver and the resources required to deliver them.

An operational plan draws directly from agency and programme strategic plans to describe agency and programme missions and goals, programme objectives, and programme activities.

Like a strategic plan, an operational plan addresses four questions:

1. Where are we now?
2. Where do we want to be?
3. How do we get there?
4. How do we measure our progress?

The OP is both the first and the last step in preparing an operating budget request. As the first step, the OP provides a plan for resource allocation; as the last step, the OP may be modified to reflect policy decisions or financial changes made during the budget development process.

Operational plans should be prepared by the people who will be involved in implementation. There is often a need for significant cross-departmental dialogue as plans created by one part of the organisation inevitably have implications for other parts.

Operational plans should contain:

- Clear objectives
- Activities to be delivered
- Quality standards
- Desired outcomes
- Staffing and resource requirements
- Implementation timetables
- A process for monitoring progress.

BUSINESS PLAN

A business plan is a formal statement of a set of business goals, the reasons why they are believed attainable, and the plan for reaching those goals. It may also contain background information about the organization or team attempting to reach those goals. The business goals may be defined for for-profit or for non-profit organizations. For-profit business plans typically focus on financial goals, such as profit or creation of wealth. Non-profit and government agency business plans tend to focus on organizational mission which is the basis for their governmental status or their non-profit, tax-exempt status, respectively—although non-profits may also focus on optimizing revenue.

In non-profit organizations, creative tensions may develop in the effort to balance mission with “margin”. Business plans may also target changes in perception and branding by the customer, client, tax-payer, or larger community. A business plan having changes in perception and branding as its primary goals is called a marketing plan.

AUDIENCE

Business plans may be internally or externally focused. Externally focused plans target goals that are important to external stakeholders, particularly financial stakeholders. They typically have detailed information about the organization or team attempting to reach the goals. With for-profit entities, external stakeholders include investors and customers. External stakeholders of non-profits include donors and the clients of the non-profit’s services. For government agencies, external stakeholders include tax-payers, higher-level government agencies, and international lending bodies such as the IMF, the World Bank, various economic agencies of the UN, and development banks. Internally focused business plans target intermediate goals required to reach the external goals. They may cover the development of a new product, a new service, a new IT system, a restructuring of finance, the refurbishing of a factory or a restructuring of the organization.

An internal business plan is often developed in conjunction with a balanced scorecard or a list of critical success factors. This allows success of the plan to be measured using non-financial measures. Business plans that identify and target internal goals, but provide only general guidance on how they will be met are called strategic plans. Operational plans describe the goals of an internal organization, working group or department. Project plans, sometimes known as project frameworks, describe the goals of a particular project. They may also address the project's place within the organization's larger strategic goals.

Presentation Formats

The format of a business plan depends on its presentation context.

It is not uncommon for businesses, especially start-ups to have three or four formats for the same business plan:

1. *An "elevator pitch":* A three minute summary of the business plan's executive summary. This is often used as a teaser to awaken the interest of potential funders, customers, or strategic partners.
2. *An oral presentation:* A hopefully entertaining slide show and oral narrative that is meant to trigger discussion and interest potential investors in reading the written presentation. The content of the presentation is usually limited to the executive summary and a few key graphs showing financial trends and key decision making benchmarks. If a new product is being proposed and time permits, a demonstration of the product may also be included.
3. *A written presentation for external stakeholders:* A detailed, well written, and pleasingly formatted plan targeted at external stakeholders.
4. *An internal operational plan:* A detailed plan describing planning details that are needed by management but may not be of interest to external stakeholders. Such plans have a somewhat higher degree of candour and informality than the version targeted at external stakeholders.

Typical structure for a business plan for a start up venture:

- Cover page and table of contents
- Executive summary
- Business description
- Business environment analysis
- Industry background
- Competitive analysis
- Market analysis
- Marketing plan
- Operations plan
- Management summary
- Financial plan
- Attachments and milestones

REVISITING THE BUSINESS PLAN

Cost Overruns and Revenue Shortfalls

Cost and revenue estimates are central to any business plan for deciding the viability of the planned venture. But costs are often underestimated and revenues overestimated resulting in later cost overruns, revenue shortfalls, and possibly non-viability. During the dot-com bubble 1997-2001 this was a problem for many technology start-ups.

However, the problem is not limited to technology or the private sector; public works projects also routinely suffer from cost overruns and/or revenue shortfalls. The main causes of cost overruns and revenue shortfalls are optimism bias and strategic misrepresentation. Reference class forecasting has been developed to reduce the risks of cost overruns and revenue shortfalls.

LEGAL AND LIABILITY ISSUES

Disclosure Requirements

An externally targeted business plan should list all legal concerns and financial liabilities that might negatively affect investors. Depending on the amount of funds being raised and the audience to whom the plan is presented, failure to do this may have severe legal consequences.

Limitations on Content and Audience

Non disclosure agreements with third parties, non-compete agreements, conflicts of interest, privacy concerns, and the protection of one's trade secrets may severely limit the audience to which one might show the business plan. Alternatively, they may require each party receiving the business plan to sign a contract accepting special clauses and conditions.

This situation is complicated by the fact that many venture capitalists will refuse to sign an NDA before looking at a business plan, lest it put them in the untenable position of looking at two independently developed look-alike business plans, both claiming originality. In such situations one may need to develop two versions of the business plan: a stripped down plan that can be used to develop a relationship and a detail plan that is only shown when investors have sufficient interest and trust to sign an NDA.

Open Business Plans

Traditionally business plans have been highly confidential and quite limited in audience. The business plan itself is generally regarded as secret. However the emergence of free software and open source has opened the model and made the notion of an open business plan possible.

An open business plan is a business plan with unlimited audience. The business plan is typically web published and made available to all. In the free software and open source business model, trade secrets, copyright and patents can no longer be used as effective locking mechanisms to provide sustainable advantages to a particular business and therefore a secret business plan is less relevant in those models. While the origin of the open business plan model is in the free software and Libre services arena, the concept is likely applicable to other domains.

Uses

Venture capital:

- *Business plan contests:* Provides a way for venture capitals to find promising projects
- *Venture capital assessment of business plans:* Focus on qualitative factors such as team.

Public offerings:

- In a public offering, potential investors can evaluate perspectives of issuing company

WITHIN CORPORATIONS

Fundraising

Fundraising is the primary purpose for many business plans, since they are related to the inherent probable success/failure of the company risk.

Total Quality Management

Total quality management is a business management strategy aimed at embedding awareness of quality in all organizational processes. TQM has been widely used in manufacturing, education, call centres, government, and service industries, as well as NASA space and science programmes.

Management by Objective

Management by objectives is a process of agreeing upon objectives within an organization so that management and employees agree to the objectives and understand what they are in the organization.

Strategic Planning

Strategic planning is an organization's process of defining its strategy, or direction, and making decisions on allocating its resources to pursue this strategy, including its capital and people. Various business analysis techniques can be used in strategic planning,

including SWOT analysis and PEST analysis or STEER analysis involving Socio-cultural, Technological, Economic, Ecological, and Regulatory factors and EPISTELS

Satires

The business plan is the subject of many satires. Satires are used both to express cynicism about business plans and as an educational tool to improve the quality of business plans.

For example:

- Five Criteria for a successful business plan in biotech uses Dilbert comic strips to remind people of what *not* to do when researching and writing a business plan for a biotech start-up. Scott Adams, the author of Dilbert, is an MBA graduate who sees humour as a critical tool that can improve the behaviour of businesses and their managers. He has written numerous critiques of business practices, including business planning. Games has a mission statement generator that satirizes the wording often found in mission statements. His book *The Dilbert Principle – A Cubicle’s Eye View of Bosses, Meetings, Management Fads and Other Workplace Afflictions* discusses the foibles of management and their plans as depicted in the Dilbert comic strips by Scott Adams.
- “South Park’s” Investing Lesson, The Motley Fool columnist “Fool on the Hill” uses the Underpants Gnomes to illustrate the fallacy of focusing on goals without a clear implementation strategy. The Underpants Gnomes episode satirizes the business plans of the Dot-com era.

CONCEPT

PLANNING

Planning in organizations and public policy is both the organizational process of creating and maintaining a plan; and the psychological process of thinking about the activities required to create a desired goal on some scale. As such, it is a fundamental property of intelligent behaviour.

This thought process is essential to the creation and refinement of a plan, or integration of it with other plans, that is, it combines forecasting of developments with the preparation of scenarios of how to react to them. An important, albeit often ignored aspect of planning, is the relationship it holds with forecasting. Forecasting can be described as predicting what the future will look like, whereas planning predicts what the future should look like. The term is also used to describe the formal procedures used in such an endeavour, such as the creation of documents, diagrams, or meetings to discuss the important issues to be addressed, the objectives to be met, and the strategy to be followed. Beyond this,

planning has a different meaning depending on the political or economic context in which it is used. Two attitudes to planning need to be held in tension: on the one hand we need to be prepared for what may lie ahead, which may mean contingencies and flexible processes. On the other hand, our future is shaped by consequences of our own planning and actions.

An Overview

Planning is a process for accomplishing purpose. It is a blue print of business growth and a road map of development. It helps in deciding objectives both in quantitative and qualitative terms. It is setting of goals on the basis of objectives and keeping in view the resources.

What should a Plan be?

A plan should be a realistic view of the expectations. Depending upon the activities, a plan can be long range, intermediate range or short range. It is the framework within which it must operate. For management seeking external support, the plan is the most important document and key to growth. Preparation of a comprehensive plan will not guarantee success, but lack of a sound plan will almost certainly ensure failure.

Purpose of a Plan

Just as no two organizations are alike, so also their plans. It is therefore important to prepare a plan keeping in view the necessities of the enterprise. A plan is an important aspect of business.

It serves the following three critical functions:

1. Helps management to clarify, focus, and research their business's or project's development and prospects.
2. Provides a considered and logical framework within which a business can develop and pursue business strategies over the next three to five years.
3. Offers a benchmark against which actual performance can be measured and reviewed.

Importance of the Planning Process

A plan can play a vital role in helping to avoid mistakes or recognize hidden opportunities. Preparing a satisfactory plan of the organization is essential.

The planning know the business and that they have thought through its development in terms of products, management, finances, and most importantly, markets and competition. Planning helps in forecasting the future, makes the future visible to some extent. It bridges between where we are and where we want to go. Planning is looking ahead.

Types of Plans or Planning

- Architectural planning
- Business plan
- Comprehensive planning
- Enterprise Architecture Planning
- Event Planning and Production
- Family planning
- Financial planning
- Infrastructure planning
- Land use planning
- Life planning
- Marketing plan
- Network resource planning
- Strategic planning
- Urban planning
- Operational planning

OBJECTIVES AND POLICIES

Objectives

The objectives are general parts of the planning process. They are the end-results towards which all business activities are directed. They are needed in every aspect where performance and result directly and vitally affect the survival and success of the firm. In other words, the objective of the firm justifies its existence. Robert C. Appley, "Objectives are goals; they are aims which management and administration wish the organization to achieve." In other words, goals, aims and purposes are also used to signify objectives. Newman and Summer stated, "For managerial purposes, it is useful to think of objectives as the results we want to achieve. Objective covers firm's long-range plans specific departmental goals and short-term individual assignment also."

Policies

George R. Ferry, "Policy is a verbal, written or implied overall guide setting up boundaries that supply the general units and directions in which managerial action will take place." Policies are specific guidelines and constraints for managerial thinking on decision-making and action. Policies provide the framework within which decision-makers are expected to operate while making organizational decisions. They are the basic guides to be consistent in decision-making.

PLANNING BASICS

Essentials of Planning

Planning is not done off hand. It is prepared after careful and extensive research.

For a comprehensive business plan, management has to:

- Clearly define the target/goal in writing.
 - It should be set by a person having authority.
 - The goal should be realistic.
 - It should be specific.
 - Acceptability
 - Easily measurable
- Identify all the main issues which need to be addressed.
- Review past performance.
- Decide budgetary requirement.
- Focus on matters of strategic importance.
- What are requirements and how will they be met?
- What will be the likely length of the plan and its structure?
- Identify shortcomings in the concept and gaps.
- Strategies for implementation.
- Review periodically.

APPLICATIONS

In Organizations

Planning is also a management process, concerned with defining goals for future organizational performance and deciding on the tasks and resources to be used in order to attain those goals. To meet the goals, managers may develop plans such as a business plan or a marketing plan. Planning always has a purpose.

The purpose may be achievement of certain goals or targets. The planning helps to achieve these goals or target by using the available time and resources. To minimize the timing and resources also require proper planning. The concept of planning is to identify what the organization wants to do by using the four questions which are “where are we today in terms of our business or strategy planning? Where are we going? Where do we want to go? How are we going to get there?...”

In Public Policy

Planning refers to the practice and the profession associated with the idea of planning an idea yourself. In many countries, the operation of a town and country planning system is often referred to as “planning” and the professionals which operate the system are

known as “planners”. It is a conscious as well as sub-conscious activity. It is “an anticipatory decision making process” that helps in coping with complexities. It is deciding future course of action from amongst alternatives. It is a process that involves making and evaluating each set of interrelated decisions.

It is selection of missions, objectives and “translation of knowledge into action.” A planned performance brings better results compared to an unplanned one. A manager’s job is planning, monitoring and controlling. Planning and goal setting are important traits of an organization. It is done at all levels of the organization.

Planning includes the plan, the thought process, action, and implementation. Planning gives more power over the future. Planning is deciding in advance what to do, how to do it, when to do it, and who should do it. It bridges the gap from where the organization is to where it wants to be. The planning function involves establishing goals and arranging them in logical order.

CONTROL

Control is one of the managerial functions like planning, organizing, staffing and directing. It is an important function because it helps to check the errors and to take the corrective action so that deviation from standards are minimized and stated goals of the organization are achieved in desired manner.

Modern concepts, control is a foreseeing action whereas earlier concept of control was used only when errors were detected. Control in management means setting standards, measuring actual performance and taking corrective action. Thus, control comprises these three main activities.

Characteristics of Control

- Control is a continuous process
- Control is a management process
- Control is embedded in each level of organizational hierarchy
- Control is forward looking
- Control is closely linked with planning
- Control is a tool for achieving organizational activities

Elements of Control

The four basic elements in a control system:

1. The characteristic or condition to be controlled,
2. The sensor,
3. The comparator, and
4. *The activator*: Occur in the same sequence and maintain a consistent relationship to each other in every system.

The first element is the characteristic or condition of the operating system which is to be measured. We select a specific characteristic because a correlation exists between

it and how the system is performing. The characteristic may be the output of the system during any stage of processing or it may be a condition that has resulted from the output of the system. For example, it may be the heat energy produced by the furnace or the temperature in the room which has changed because of the heat generated by the furnace. In an elementary school system, the hours a teacher works or the gain in knowledge demonstrated by the students on a national examination are examples of characteristics that may be selected for measurement, or control. The second element of control, the sensor, is a means for measuring the characteristic or condition. The control subsystem must be designed to include a sensory device or method of measurement. In a home heating system this device would be the thermostat, and in a quality-control system this measurement might be performed by a visual inspection of the product.

The third element of control, the comparator, determines the need for correction by comparing what is occurring with what has been planned. Some deviation from plan is usual and expected, but when variations are beyond those considered acceptable, corrective action is required. It is often possible to identify trends in performance and to take action before an unacceptable variation from the norm occurs. This sort of preventative action indicates that good control is being achieved.

The fourth element of control, the activator, is the corrective action taken to return the system to expected output. The actual person, device, or method used to direct corrective inputs into the operating system may take a variety of forms. It may be a hydraulic controller positioned by a solenoid or electric motor in response to an electronic error signal, an employee directed to rework the parts that failed to pass quality inspection, or a school principal who decides to buy additional books to provide for an increased number of students. As long as a plan is performed within allowable limits, corrective action is not necessary; this seldom occurs in practice, however. Information is the medium of control, because the flow of sensory data and later the flow of corrective information allow a characteristic or condition of the system to be controlled. To illustrate how information flow facilitates control, let us review the elements of control in the context of information.

Relationship between the Elements of Control and Information

Controlled Characteristic or Condition The primary requirement of a control system is that it maintain the level and kind of output necessary to achieve the system's objectives. It is usually impractical to control every feature and condition associated with the system's output. Therefore, the choice of the controlled item is extremely important. There should be a direct correlation between the controlled item and the system's operation. In other words, control of the selected characteristic should have a direct relationship to the goal or objective of the system.

Sensor

After the characteristic is sensed, or measured, information pertinent to control is fed back. Exactly what information needs to be transmitted and also the language that

will best facilitate the communication process and reduce the possibility of distortion in transmission must be carefully considered. Information that is to be compared with the standard, or plan, should be expressed in the same terms or language as in the original plan to facilitate decision making. Using machine methods may require extensive translation of the information.

Since optimal languages for computation and for human review are not always the same, the relative ease of translation may be a significant factor in selecting the units of measurement or the language unit in the sensing element. In many instances, the measurement may be sampled rather than providing a complete and continuous feedback of information about the operation. A sampling procedure suggests measuring some segment or portion of the operation that will represent the total.

Comparison with Standard

In a social system, the norms of acceptable behaviour become the standard against which so-called deviant behaviour may be judged. Regulations and laws provide a more formal collection of information for society. Social norms change, but very slowly. In contrast, the standards outlined by a formal law can be changed from one day to the next through revision, discontinuation, or replacement by another. Information about deviant behaviour becomes the basis for controlling social activity. Output information is compared with the standard or norm and significant deviations are noted. In an industrial example, frequency distribution may be used to show the average quality, the spread, and the comparison of output with a standard.

If there is a significant and uncorrectable difference between output and plan, the system is “out of control.” This means that the objectives of the system are not feasible in relation to the capabilities of the present design. Either the objectives must be reevaluated or the system redesigned to add new capacity or capability. For example, the traffic in drugs has been increasing in some cities at an alarming rate. The citizens must decide whether to revise the police system so as to regain control, or whether to modify the law to reflect a different norm of acceptable behaviour.

Implimenter

The activator unit responds to the information received from the comparator and initiates corrective action. If the system is a machine-to-machine system, the corrective inputs are designed into the network.

When the control relates to a man-to-machine or man-to-man system, however, the individual(s) in charge must evaluate:

- The accuracy of the feedback information,
- The significance of the variation, and
- What corrective inputs will restore the system to a reasonable degree of stability. Once the decision has been made to direct new inputs into the system,

the actual process may be relatively easy. A small amount of energy can change the operation of jet airplanes, automatic steel mills, and hydroelectric power plants. The pilot presses a button, and the landing gear of the airplane goes up or down; the operator of a steel mill pushes a lever, and a ribbon of white-hot steel races through the plant; a worker at a control board directs the flow of electrical energy throughout a regional network of stations and substations. It takes but a small amount of control energy to release or stop large quantities of input.

The comparator may be located far from the operating system, although at least some of the elements must be in close proximity to operations. For example, the measurement is usually at the point of operations. The measurement information can be transmitted to a distant point for comparison with the standard and when deviations occur, the correcting input can be released from the distant point. However, the input will be located at the operating system. This ability to control from afar means that aircraft can be flown by remote control, dangerous manufacturing processes can be operated from a safe distance, and national organizations can be directed from centralized headquarters.

Process of Controlling

- Setting performance standards.
- Measurement of actual performance.
- Comparing actual performance with standards.
- Analysing deviations.
- Correcting deviations.

Kinds of Control

Control may be grouped according to three general classifications:

1. The nature of the information flow designed into the system,
2. The kind of components included in the design, and
3. The relationship of control to the decision process.

Open- and Closed-Loop Control

A street-lighting system controlled by a timing device is an example of an open-loop system. At a certain time each evening, a mechanical device closes the circuit and energy flows through the electric lines to light the lamps. Note, however, that the timing mechanism is an independent unit and is not measuring the objective function of the lighting system. If the lights should be needed on a dark, stormy day the timing device would not recognize this need and therefore would not activate energy inputs. Corrective properties may sometimes be built into the controller but this would not close the loop. In another instance, the sensing, comparison, or adjustment may be made through action

taken by an individual who is not part of the system. For example, the lights may be turned on by someone who happens to pass by and recognizes the need for additional light. If control is exercised as a result of the operation rather than because of outside or predetermined arrangements, it is a closed-loop system. The home thermostat is the classic example of a control device in a closed-loop system. When the room temperature drops below the desired point, the control mechanism closes the circuit to start the furnace and the temperature rises. The furnace-activating circuit is turned off as the temperature reaches the preselected level. The significant difference between this type of system and an open-loop system is that the control device is an element of the system it serves and measures the performance of the system. In other words, all four control elements are integral to the specific system.

An essential part of a closed-loop system is feedback; that is, the output of the system is measured continually through the item controlled, and the input is modified to reduce any difference or error towards zero. Many of the patterns of information flow in organizations are found to have the nature of closed loops, which use feedback. The reason for such a condition is apparent when one recognizes that any system, if it is to achieve a predetermined goal, must have available to it at all times an indication of its degree of attainment. In general, every goal-seeking system employs feedback. ‘ ==

Man and Machine Control

The elements of control are easy to identify in machine systems. For example, the characteristic to be controlled might be some variable like speed or temperature, and the sensing device could be a speedometer or a thermometer. An expectation of precision exists because the characteristic is quantifiable and the standard and the normal variation to be expected can be described in exact terms. In automatic machine systems, inputs of information are used in a process of continual adjustment to achieve output specifications. When even a small variation from the standard occurs, the correction process begins. The automatic system is highly structured, designed to accept certain kinds of input and produce specific output, and programmed to regulate the transformation of inputs within a narrow range of variation.

For an illustration of mechanical control, as the load on a steam engine increases and the engine starts to slow down, the regulator reacts by opening a valve that releases additional inputs of steam energy. This new input returns the engine to the desired number of revolutions per minute. This type of mechanical control is crude in comparison to the more sophisticated electronic control systems in everyday use. Consider the complex missile-guidance systems that measure the actual course according to predetermined mathematical calculations and make almost instantaneous corrections to direct the missile to its target.

Machine systems can be complex because of the sophisticated technology, whereas control of people is complex because the elements of control are difficult to determine. In human control systems, the relationship between objectives and associated

characteristics is often vague; the measurement of the characteristic may be extremely subjective; the expected standard is difficult to define; and the amount of new inputs required is impossible to quantify. To illustrate, let us refer once more to a formalized social system in which deviant behaviour is controlled through a process of observed violation of the existing law, court hearings and trials, incarceration when the accused is found guilty and release from custody after rehabilitation of the prisoner has occurred. The speed limit established for freeway driving is one standard of performance that is quantifiable, but even in this instance, the degree of permissible variation and the amount of the actual variation are often a subject of disagreement between the patrolman and the suspected violator. The complexity of our society is reflected in many of our laws and regulations, which establish the general standards for economic, political, and social operations. A citizen may not know or understand the law and consequently would not know whether or not he was guilty of a violation.

Most organized systems are some combination of man and machine; some elements of control may be performed by machine whereas others are accomplished by man. In addition, some standards may be precisely structured whereas others may be little more than general guidelines with wide variations expected in output. Man must perform as the controller when measurement is subjective and judgement is required. Machines such as computers are incapable of making exceptions from the specified control criteria regardless of how much a particular case might warrant special consideration. A pilot acts in conjunction with computers and automatic pilots to fly large jets. In the event of unexpected weather changes, or possible collision with another plane, he must intercede and assume direct control.

Organizational and Operational Control

The concept of organizational control is implicit in the bureaucratic theory of Max Weber. Associated with this theory are such concepts as “span of control”, “closeness of supervision”, and “hierarchical authority”. Weber’s view tends to include all levels or types of organizational control as being the same. More recently, writers have tended to differentiate the control process between that which emphasizes the nature of the organizational or systems design and that which deals with daily operations.

To illustrate the difference, we “evaluate” the performance of a system to see how effective and efficient the design proved to be or to discover why it failed. In contrast, we operate and “control” the system with respect to the daily inputs of material, information, and energy. In both instances, the elements of feedback are present, but organizational control tends to review and evaluate the nature and arrangement of components in the system, whereas operational control tends to adjust the daily inputs. The direction for organizational control comes from the goals and strategic plans of the organization. General plans are translated into specific performance measures such as share of the market, earnings, return on investment, and budgets.

The process of organizational control is to review and evaluate the performance of the system against these established norms. Rewards for meeting or exceeding standards may range from special recognition to salary increases or promotions. On the other hand, a failure to meet expectations may signal the need to reorganize or redesign. In organizational control, the approach used in the programme of review and evaluation depends on the reason for the evaluation — that is, is it because the system is not effective (accomplishing its objectives)? Is the system failing to achieve an expected standard of efficiency? Is the evaluation being conducted because of a breakdown or failure in operations? Is it merely a periodic audit-and-review process?

When a system has failed or is in great difficulty, special diagnostic techniques may be required to isolate the trouble areas and to identify the causes of the difficulty. It is appropriate to investigate areas that have been troublesome before or areas where some measure of performance can be quickly identified. For example, if an organization's output backlog builds rapidly, it is logical to check first to see if the problem is due to such readily obtainable measures as increased demand or to a drop in available man hours. When a more detailed analysis is necessary, a systematic procedure should be followed.

In contrast to organizational control, operational control serves to regulate the day-to-day output relative to schedules, specifications, and costs. Is the output of product or service the proper quality and is it available as scheduled? Are inventories of raw materials, goods-in-process, and finished products being purchased and produced in the desired quantities? Are the costs associated with the transformation process in line with cost estimates? Is the information needed in the transformation process available in the right form and at the right time? Is the energy resource being utilized efficiently?

The most difficult task of management concerns monitoring the behaviour of individuals, comparing performance to some standard, and providing rewards or punishment as indicated. Sometimes this control over people relates entirely to their output.

For example, a manager might not be concerned with the behaviour of a salesman as long as sales were as high as expected. In other instances, close supervision of the salesman might be appropriate if achieving customer satisfaction were one of the sales organization's main objectives.

The larger the unit, the more likely that the control characteristic will be related to some output goal. It also follows that if it is difficult or impossible to identify the actual output of individuals, it is better to measure the performance of the entire group. This means that individuals' levels of motivation and the measurement of their performance become subjective judgements made by the supervisor. Controlling output also suggests the difficulty of controlling individuals' performance and relating this to the total system's objectives.

Problems of Control

The perfect plan could be outlined if every possible variation of input could be anticipated and if the system would operate as predicted. This kind of planning is neither realistic, economical, nor feasible for most business systems. If it were feasible, planning requirements would be so complex that the system would be out of date before it could be operated. Therefore, we design control into systems.

This requires more thought in the systems design but allows more flexibility of operations and makes it possible to operate a system using unpredictable components and undetermined input. Still, the design and effective operation of control are not without problems. The objective of the system is to perform some specified function. The purpose of organizational control is to see that the specified function is achieved; the objective of operational control is to ensure that variations in daily output are maintained within prescribed limits. It is one thing to design a system that contains all of the elements of control, and quite another to make it operate true to the best objectives of design.

Operating “in control” or “with plan” does not guarantee optimum performance. For example, the plan may not make the best use of the inputs of materials, energy, or information — in other words, the system may not be designed to operate efficiently. Some of the more typical problems relating to control include the difficulty of measurement, the problem of timing information flow, and the setting of proper standards.

Measurement of Output

When objectives are not limited to quantitative output, the measurement of system effectiveness is difficult to make and subsequently perplexing to evaluate. Many of the characteristics pertaining to output do not lend themselves to quantitative measurement. This is true particularly when inputs of human energy cannot be related directly to output. The same situation applies to machines and other equipment associated with human involvement, when output is not in specific units.

In evaluating man-machine or human-oriented systems, psychological and sociological factors obviously do not easily translate into quantifiable terms. For example, how does mental fatigue affect the quality or quantity of output? And, if it does, is mental fatigue a function of the lack of a challenging assignment or the fear of a potential injury? Subjective inputs may be transferred into numerical data, but there is always the danger of an incorrect appraisal and transfer, and the danger that the analyst may assume undue confidence in such data after they have been quantified. Let us suppose, for example, that the decisions made by an executive are rated from 1 to 10, 10 being the perfect decision. After determining the ranking for each decision, adding these, and dividing by the total number of decisions made, the average ranking would indicate a particular executive's score in his decision-making role. On the basis of this score, judgements — which could be quite erroneous — might be made about his decision-

making effectiveness. One executive with a ranking of 6.75 might be considered more effective than another who had a ranking of 6.25, and yet the two managers may have made decisions under different circumstances and conditions. External factors over which neither executive had any control may have influenced the difference in “effectiveness”.

Quantifying human behaviour, despite its extreme difficulty, subjectivity, and imprecision in relation to measuring physical characteristics is the most prevalent and important measurement made in large systems. The behaviour of individuals ultimately dictates the success or failure of every man-made system.

Information Flow

Another problem of control relates to the improper timing of information introduced into the feedback channel. Improper timing can occur in both computerized and human control systems, either by mistakes in measurement or in judgement. The more rapid the system's response to an error signal, the more likely it is that the system could overadjust; yet the need for prompt action is important because any delay in providing corrective input could also be crucial. A system generating feedback inconsistent with current need will tend to fluctuate and will not adjust in the desired manner.

The most serious problem in information flow arises when the delay in feedback is exactly one-half cycle, for then the corrective action is superimposed on a variation from norm which, at that moment, is in the same direction as that of the correction. This causes the system to overcorrect, and then if the reverse adjustment is made out of cycle, to correct too much in the other direction, and so on until the system fluctuates out of control. If, at Point A, the trend below standard is recognized and new inputs are added, but not until Point B, the system will overreact and go beyond the allowable limits. Again, if this is recognized at Point C, but inputs are not withdrawn until Point D, it will cause the system to drop below the lower limit of allowable variation.

One solution to this problem rests in anticipation, which involves measuring not only the change but also the rate of change. The correction is outlined as a factor of the type and rate of the error. The difficulty also might be overcome by reducing the time lag between the measurement of the output and the adjustment to input.

If a trend can be indicated, a time lead can be introduced to compensate for the time lag, bringing about consistency between the need for correction and the type and magnitude of the indicated action. It is usually more effective for an organization to maintain continuous measurement of its performance and to make small adjustments in operations constantly. Information feedback, consequently, should be timely and correct to be effective. That is, the information should provide an accurate indication of the status of the system.

Setting Standards

Setting the proper standards or control limits is a problem in many systems. Parents are confronted with this dilemma in expressing what they expect of their children, and

business managers face the same issue in establishing standards that will be acceptable to employees. Some theorists have proposed that workers be allowed to set their own standards, on the assumption that when people establish their own goals, they are more apt to accept and achieve them.

Standards should be as precise as possible and communicated to all persons concerned. Moreover, communication alone is not sufficient; understanding is necessary. In human systems, standards tend to be poorly defined and the allowable range of deviation from standard also indefinite. For example, how many hours each day should a professor be expected to be available for student consultation? Or, what kind of behaviour should be expected by students in the classroom? Discretion and personal judgement play a large part in such systems, to determine whether corrective action should be taken. Perhaps the most difficult problem in human systems is the unresponsiveness of individuals to indicated correction. This may take the form of opposition and subversion to control, or it may be related to the lack of defined responsibility or authority to take action. Leadership and positive motivation then become vital ingredients in achieving the proper response to input requirements. Most control problems relate to design; thus the solution to these problems must start at that point. Automatic control systems, provided that human intervention is possible to handle exceptions, offer the greatest promise. There is a danger, however, that we may measure characteristics that do not represent effective performance or that improper information may be communicated.

CAPACITY PLANNING

Capacity planning is the process of determining the production capacity needed by an organization to meet changing demands for its products. In the context of capacity planning, “capacity” is the maximum amount of work that an organization is capable of completing in a given period of time. The phrase is also used in business computing as a synonym for Capacity Management.

A discrepancy between the capacity of an organization and the demands of its customers results in inefficiency, either in under-utilized resources or unfulfilled customers. The goal of capacity planning is to minimize this discrepancy. Demand for an organization’s capacity varies based on changes in production output, such as increasing or decreasing the production quantity of an existing product, or producing new products. Better utilization of existing capacity can be accomplished through improvements in overall equipment effectiveness. Capacity can be increased through introducing new techniques, equipment and materials, increasing the number of workers or machines, increasing the number of shifts, or acquiring additional production facilities.

Capacity is calculated: (number of machines or workers) × (number of shifts) × (utilization) × (efficiency).

The broad classes of capacity planning are lead strategy, lag strategy, and match strategy:

- Lead strategy is adding capacity in anticipation of an increase in demand. Lead strategy is an aggressive strategy with the goal of luring customers away from the company's competitors. The possible disadvantage to this strategy is that it often results in excess inventory, which is costly and often wasteful.
- Lag strategy refers to adding capacity only after the organization is running at full capacity or beyond due to increase in demand. This is a more conservative strategy. It decreases the risk of waste, but it may result in the loss of possible customers.
- Match strategy is adding capacity in small amounts in response to changing demand in the market. This is a more moderate strategy.

In the context of systems engineering, capacity planning is used during system design and system performance monitoring. Capacity planning is long-term decision that establishes a firm's overall level of resources.

It extends over time horizon long enough to obtain resources. Capacity decisions affect the production lead time, customer responsiveness, operating cost and company ability to compete. Inadequate capacity planning can lead to the loss of the customer and business. Excess capacity can drain the company's resources and prevent investments into more lucrative ventures. The question of when capacity should be increased and by how much are the critical decisions.

LONG-TERM CAPACITY PLANNING

Over the long term, capacity planning relates primarily to strategic issues involving the firm's major production facilities. In addition, long-term capacity issues are interrelated with location decisions. Technology and transferability of the process to other products is also intertwined with long-term capacity planning.

Long-term capacity planning may evolve when short-term changes in capacity are insufficient. For example, if the firm's addition of a third shift to its current two-shift plan still does not produce enough output, and subcontracting arrangements cannot be made, one feasible alternative is to add capital equipment and modify the layout of the plant. It may even be desirable to add additional plant space or to construct a new facility.

SHORT-TERM CAPACITY PLANNING

In the short term, capacity planning concerns issues of scheduling, labour shifts, and balancing resource capacities. The goal of short-term capacity planning is to handle unexpected shifts in demand in an efficient economic manner. The time frame for short-term planning is frequently only a few days but may run as long as six months.

Alternatives for making short-term changes in capacity are fairly numerous and can even include the decision to not meet demand at all. The easiest and most commonly-

used method to increase capacity in the short term is working overtime. This is a flexible and inexpensive alternative. While the firm has to pay one and one half times the normal labour rate, it foregoes the expense of hiring, training, and paying additional benefits.

When not used abusively, most workers appreciate the opportunity to earn extra wages. If overtime does not provide enough short-term capacity, other resource-increasing alternatives are available. These include adding shifts, employing casual or part-time workers, the use of floating workers, leasing workers, and facilities subcontracting.

Firms may also increase capacity by improving the use of their resources. The most common alternatives in this category are worker cross training and overlapping or staggering shifts. Most manufacturing firms inventory some output ahead of demand so that any need for a capacity change is absorbed by the inventory buffer. From a technical perspective, firms may initiate a process design intended to increase productivity at work stations.

Manufacturers can also shift demand to avoid capacity requirement fluctuation by backlogging, queuing demand, or lengthening the firm's lead times. Service firms accomplish the same results through scheduling appointments and reservations. A more creative approach is to modify the output. Standardizing the output or offering complimentary services are examples. In services, one might allow customers to do some of the process work themselves. Another alternative—reducing quality—is an undesirable yet viable tactic.

Finally, the firm may attempt to modify demand. Changing the price and promoting the product are common. Another alternative is to partition demand by initiating a yield or revenue management system. Utilities also report success in shifting demand by the use of “off-peak” pricing.

CAPACITY-PLANNING TECHNIQUES

There are four procedures for capacity planning; capacity planning using overall factors, capacity bills, resource profiles, and capacity requirements planning. The first three are rough-cut approaches that can be used with or without manufacturing resource planning systems. CRP is used in conjunction with MRP systems. Capacity using overall factors is a simple, manual approach to capacity planning that is based on the master production schedule and production standards that convert required units of finished goods into historical loads on each work centre.

Bills of capacity is a procedure based on the MPS. Instead of using historical ratios, however, it utilizes the bills of material and routing sheet. Capacity requirements can then be determined by multiplying the number of units required by the MPS by the time needed to produce each. Resource profiles are the same as bills of capacity, except lead times are included so that workloads fall into the correct periods.

Capacity requirements planning is only applicable in firms using MRP or MRP II. CRP uses the information from one of the previous rough-cut methods, plus MRP outputs on existing inventories and lot sizing. The result is a tabular load report for each work centre or a graphical load profile for helping plan-production requirements. This will indicate where capacity is inadequate or idle, allowing for imbalances to be corrected by shifts in personnel or equipment or the use of overtime or added shifts. Finite capacity scheduling is an extension of CRP that simulates job order stopping and starting to produce a detailed schedule that provides a set of start and finish dates for each operation at each work centre.

A failure to understand the critical nature of managing capacity can lead to chaos and serious customer service problems. If there is a mismatch between available and required capacity, adjustments should be made. However, it should be noted that firms cannot have perfectly-balanced material and capacity plans that easily accommodate emergency orders. If flexibility is the firm's competitive priority, excess capacity would be appropriate.

MANUFACTURING PLANNING AND CONTROL

Manufacturing planning and control address decisions on the acquisition, utilization and allocation of production resources to satisfy customer requirements in the most efficient and effective way. Typical decisions include workforce level, production lot sizes, assignment of overtime and sequencing of production runs. Optimization models are widely applicable for providing decision support in this context.

We focus on optimization models for production planning for discrete-parts, batch manufacturing environments. We do not cover detailed scheduling or sequencing models, nor do we address production planning for continuous processes. We consider only discrete-time models, and do not include continuous-time models such as developed by Hackman and Leachman.

Our intent is to provide an overview of applicable optimization models; we present the most generic formulations and briefly describe how these models are solved. There is an enormous range of problem contexts and model formulations, as well as solution methods. We make no effort to be exhaustive in the treatment herein.

Rather, we have made choices of what to include based on personal judgement and preferences. We have organized the substance into four major parts. In the first part we present a framework for the decisions, issues and tradeoffs involved in implementing an optimization model for discrete-part production planning. The remaining three parts present and discuss three distinct types of models. In the second part we discuss linear programming models for production planning, in which we have linear costs.

This category is of great practical interest, as many important problem features can be captured with these models and powerful solution methods for linear programmes

are readily available. In the third part, we present a production-planning model for a single aggregate product with quadratic costs; this model is of historical significance as it represents one of the earliest applications of optimization to manufacturing planning. In the final part we introduce the multi-item capacitated lot-size problem, which is modelled as a mixed integer linear programme. This is an important model as it introduces economies of scale in production, due to the presence of production setups.

FRAMEWORK

There are a variety of considerations that go into the development and implementation of an optimization model for manufacturing planning and control. In this part we highlight and comment upon a number of key issues and questions that should be addressed. Excellent general references on production planning are Thomas and McClain, Shapiro and Silver *et al.*

Any planning problem starts with a specification of customer demand that is to be met by the production plan. In most contexts, future demand is at best only partially known, and often is not known at all. Consequently, one relies on a forecast for the future demand. To the extent that any forecast is inevitably inaccurate, one must decide how to account for or react to this demand uncertainty. The optimization models described in this substance treat demand as being known; as such they must be periodically revised and rerun to account for forecast updates. A key choice is what planning decisions to include in the model. By definition, production-planning models include decisions on production and inventory quantities. But in addition, there might be resource acquisition and allocation decision, such as adding to the workforce and upgrading the training of the current workforce. In many planning contexts, an important construct is to set a planning hierarchy. Namely, one structures the planning process in a hierarchical way by ordering the decisions according to their relative importance.

Hax and Meal introduced the notion of hierarchical production planning and provide a specific framework for this, whereby there is an optimization model with each level of the hierarchy. Each optimization model imposes a constraint on the model at the next level of the hierarchy. Bitran and Tirupati provide a comprehensive survey of hierarchical planning methods and models. The identification of the relevant costs is also an important issue. For production planning, one typically needs to determine the variable production costs, including setup-related costs, inventory holding costs, and any relevant resource acquisition costs. There might also be costs associated with imperfect customer service, such as when demand is backordered.

A planning problem exists because there are limited production resources that cannot be stored from period to period. Choices must be made as to which resources to include and how to model their capacity and behaviour, and their costs. Also, there may be uncertainty associated with the production function, such as uncertain yields or lead times. One might only include the most critical or limiting resource in the planning problem, *e.g.*, a bottleneck. Alternatively, when there is not a dominant resource, then

one must model the resources that could limit production. We describe in this substance two types of production functions.

The first assumes a linear relationship between the production quantity and the resource consumption. The second assumes that there is a required fixed charge or setup to initiate production and then a linear relationship between the production quantity and resource usage. Related to these choices is the selection of the time period and planning horizon. The planning literature distinguishes between “big bucket” and “small bucket” time periods. A time period is a big bucket if multiple items are typically produced within a time period; a small bucket is such that at most one item would be produced in the time period. For big bucket models, one has to worry about how to schedule or sequence the production runs assigned to any time period. The choice of planning horizon is dictated by the lead times to enact production and resource-related decisions, as well as the quality of knowledge about future demand.

Planning is typically done in a rolling horizon fashion. A plan is created for the planning horizon, but only the decisions in the first few periods are implemented before a revised plan is issued. Indeed the plan must be periodically revised due to the uncertainties in the demand forecasts and production. For instance a firm might plan for the next 26 weeks, but then revise this once a month to incorporate new information on demand and production.

Production planning is usually done at an aggregate level, for both products and resources. Distinct but similar products are combined into aggregate product families that can be planned together so as to reduce planning complexity. Similarly production resources, such as distinct machines or labour pools, are aggregated into an aggregate machine or labour resource. Care is required when specifying these aggregates to assure that the resulting aggregate plan can be reasonably disaggregated into feasible production schedules. Finally for complex products, one must decide the level and extent of the product structure to include in the planning process. For instance, in some contexts it is sufficient to just plan the production of end items; the production plan for components and subassemblies is subservient to the master production schedule for end items.

In other contexts, planning just the end items is suboptimal, as there are critical resource constraints applicable to multiple levels of the product structure. In this instance, a multistage planning model allows for the simultaneous planning of end items and components or subassemblies. Of course, this produces a much larger model.

ASPECTS OF PLANNING

EVENT PLANNING

Event planning is the process of planning a festival, ceremony, competition, party, or convention. Event planning includes budgeting, establishing dates and alternate dates, selecting and reserving the event site, acquiring permits, and coordinating transportation

and parking. Event planning also includes some or all of the following, depending on the event: developing a theme or motif for the event, arranging for speakers and alternate speakers, coordinating location support, arranging decor, tables, chairs, tents, event support and security, catering, police, fire, portable toilets, parking, signage, emergency plans, health care professionals, and cleanup.

Event Planning is a relatively new career field. There is now training that helps one trying to break into the career field. There must be training for an event planner to handle all the pressure and work efficiently. This career deals with a lot of communication and organization aspects. There are many different names for an event planner such as a conference coordinator, a convention planner, a special event coordinator, and a meeting manager.

Steps to Planning an Event

The first step to planning an event is determining its purpose, whether it is for a wedding, company, birthday, festival, graduation or any other event requiring extensive planning. From this the event planner needs to choose entertainment, location, guest list, speakers, and content. The location for events is endless, but with event planning they would likely be held at hotels, convention centres, reception halls, or outdoors depending on the event. Once the location is set the coordinator/planner needs to prepare the event with staff, set up the entertainment, and keep contact with the client. After all this is set the event planner has all the smaller details to address like set up of the event such as food, drinks, music, guest list, budget, advertising and marketing, decorations, all this preparation is what is needed for an event to run smoothly. An event planner needs to be able to manage their time wisely for the event, and the length of preparation needed for each event so it is a success.

Working Conditions

Event planners work is considered either stressful or energizing. This line of work is also considered fast paced and demanding. Planners face deadlines and communicating with multiple people at one time. Planners spend most of their time in offices, but when meeting with clients the work is usually on-site at the location where the event is taking place. Some physical activity is required such as carrying boxes of materials and decorations or supplies needed for the event. Also, long working hours can be a part of the job. The day the event is taking place could start as early as 5:00 a.m. and then work until midnight. Working on weekends is sometimes required, which is when many events take place.

Publications and Resources

Many business-to-business trade publications exist to help event planning and production professionals become educated about the issues and trends in their industry.

Many are controlled circulation publications available at no cost to qualified event professionals. Qualification is based on multiple variables like job title, company type, industry segment or geographic region, and is at the publisher's discretion.

FINANCIAL PLAN

In general usage, a financial plan can be a budget, a plan for spending and saving future income. This plan allocates future income to various types of expenses, such as rent or utilities, and also reserves some income for short-term and long-term savings. A financial plan can also be an investment plan, which allocates savings to various assets or projects expected to produce future income, such as a new business or product line, shares in an existing business, or real estate.

In business, a financial plan can refer to the three primary financial statements created within a business plan. Financial forecast or financial plan can also refer to an annual projection of income and expenses for a company, division or department. A financial plan can also be an estimation of cash needs and a decision on how to raise the cash, such as through borrowing or issuing additional shares in a company. While a financial plan refers to estimating future income, expenses and assets, a financing plan or finance plan usually refers to the means by which cash will be acquired to cover future expenses, for instance through earning, borrowing or using saved cash.

MARKETING PLAN

A marketing plan is a written document that details the necessary actions to achieve one or more marketing objectives. It can be for a product or service, a brand, or a product line. Marketing plans cover between one and five years. A marketing plan may be part of an overall business plan. Solid marketing strategy is the foundation of a well-written marketing plan. While a marketing plan contains a list of actions, a marketing plan without a sound strategic foundation is of little use.

The Marketing Planning Process

Marketing process can be realised by the marketing mix in step 4. The last step in the process is the marketing controlling. In most organizations, "strategic planning" is an annual process, typically covering just the year ahead. Occasionally, a few organizations may look at a practical plan which stretches three or more years ahead.

To be most effective, the plan has to be formalized, usually in written form, as a formal "marketing plan." The essence of the process is that it moves from the general to the specific, from the overall objectives of the organization down to the individual action plan for a part of one marketing programme. It is also an interactive process, so that the draft output of each stage is checked to see what impact it has on the earlier stages, and is amended.

Marketing Planning Aims and Objectives

Behind the corporate objectives, which in themselves offer the main context for the marketing plan, will lie the “corporate mission,” which in turn provides the context for these corporate objectives. In a sales-oriented organization, the marketing planning function designs incentive pay plans to not only motivate and reward frontline staff fairly but also to align marketing activities with corporate mission.

This “corporate mission” can be thought of as a definition of what the organization is, of what it does: “Our business is ...”. This definition should not be too narrow, or it will constrict the development of the organization; a too rigorous concentration on the view that “We are in the business of making meat-scales,” as IBM was during the early 1900s, might have limited its subsequent development into other areas. On the other hand, it should not be too wide or it will become meaningless; “We want to make a profit” is not too helpful in developing specific plans. Abell suggested that the definition should cover three dimensions: “customer groups” to be served, “customer needs” to be served, and “technologies” to be utilized. Thus, the definition of IBM’s “corporate mission” in the 1940s might well have been: “We are in the business of handling accounting information [customer need] for the larger US organizations [customer group] by means of punched cards [technology].”

Perhaps the most important factor in successful marketing is the “corporate vision.” Surprisingly, it is largely neglected by marketing textbooks, although not by the popular exponents of corporate strategy - indeed, it was perhaps the main theme of the book by Peters and Waterman, in the form of their “Superordinate Goals.” “In Search of Excellence” said: “Nothing drives progress like the imagination. The idea precedes the deed.” If the organization in general, and its chief executive in particular, has a strong vision of where its future lies, then there is a good chance that the organization will achieve a strong position in its markets. This will be not least because its strategies will be consistent and will be supported by its staff at all levels. In this context, all of IBM’s marketing activities were underpinned by its philosophy of “customer service,” a vision originally promoted by the charismatic Watson dynasty. The emphasis at this stage is on obtaining a complete and accurate picture.

A “traditional” - albeit product-based - format for a “brand reference book” was suggested by Godley more than three decades ago:

1. *Financial data:* Facts for this part will come from management accounting, costing and finance parts.
2. *Product data:* From production, research and development.
3. *Sales and distribution data:* Sales, packaging, distribution parts.
4. *Advertising, sales promotion, merchandising data:* Information from these departments.
5. *Market data and miscellany:* From market research, who would in most cases performance as a source for this information. His sources of data, however, assume the resources of a very large organization. In most organizations they would be obtained from a much smaller set of people.

It is apparent that a marketing audit can be a complex process, but the aim is simple: “it is only to identify those existing factors which will have a significant impact on the future plans of the company.” It is clear that the basic material to be input to the marketing audit should be comprehensive.

Accordingly, the best approach is to accumulate this material continuously, as and when it becomes available; since this avoids the otherwise heavy workload involved in collecting it as part of the regular, typically annual, planning process itself - when time is usually at a premium.

Even so, the first task of this annual process should be to check that the material held in the current facts book or facts files actually is comprehensive and accurate, and can form a sound basis for the marketing audit itself. The structure of the facts book will be designed to match the specific needs of the organization, but one simple format - suggested by Malcolm McDonald - may be applicable in many cases.

This splits the material into three groups:

1. *Review of the marketing environment:* A study of the organization’s markets, customers, competitors and the overall economic, political, cultural and technical environment; covering developing trends, as well as the current situation.
2. *Review of the detailed marketing activity:* A study of the company’s marketing mix; in terms of the 7 Ps -
3. *Review of the marketing system:* A study of the marketing organization, marketing research systems and the current marketing objectives and strategies. The last of these is too frequently ignored. The marketing system itself needs to be regularly questioned, because the validity of the whole marketing plan is reliant upon the accuracy of the input from this system, and ‘garbage in, garbage out’ applies with a vengeance.
 - Portfolio planning. In addition, the coordinated planning of the individual products and services can contribute towards the balanced portfolio.
 - 80:20 rule. To achieve the maximum impact, the marketing plan must be clear, concise and simple. It needs to concentrate on the 20 per cent of products or services, and on the 20 per cent of customers, which will account for 80 per cent of the volume and 80 per cent of the profit.
 - 7 P’s: Product, Place, Price and Promotion, Physical Environment, People, Process. The 7 P’s can sometimes divert attention from the customer, but the framework they offer can be very useful in building the action plans.

It is only at this stage that the active part of the marketing planning process begins. This next stage in marketing planning is indeed the key to the whole marketing process. The “marketing objectives” state just where the company intends to be at some specific time in the future. James Quinn succinctly defined objectives in general as: Goals state what is to be achieved and when results are to be accomplished, but they do not state

'how' the results are to be achieved. They typically relate to what products will be where in what markets. They are essentially about the match between those "products" and "markets." Objectives for pricing, distribution, advertising and so on are at a lower level, and should not be confused with marketing objectives.

They are part of the marketing strategy needed to achieve marketing objectives. To be most effective, objectives should be capable of measurement and therefore "quantifiable." This measurement may be in terms of sales volume, money value, market share, percentage penetration of distribution outlets and so on. An example of such a measurable marketing objective might be "to enter the market with product Y and capture 10 per cent of the market by value within one year." As it is quantified it can, within limits, be unequivocally monitored, and corrective action taken as necessary.

The marketing objectives must usually be based on the organization's financial objectives; converting these financial measurements into the related marketing measurements. He went on to explain his view of the role of "policies," with which strategy is most often confused: "Policies are rules or guidelines that express the 'limits' within which action should occur." Simplifying somewhat, marketing strategies can be seen as the means, or "game plan," by which marketing objectives will be achieved and, in the framework that we have chosen to use, are generally concerned with the 8 P's.

Examples are:

1. *Price:* The amount of money needed to buy products
2. *Product:* The actual product
3. *Promotion:* Getting the product known
4. *Placement:* Where the product is located
5. *People:* Represent the business
6. *Physical environment:* The ambiance, mood, or tone of the environment
7. *Process -* How do people obtain your product
8. *Packaging:* How the product will be protected

In principle, these strategies describe how the objectives will be achieved. The 7 P's are a useful framework for deciding how the company's resources will be manipulated to achieve the objectives. It should be noted, however, that they are not the only framework, and may divert attention from the real issues. The focus of the strategies must be the objectives to be achieved - not the process of planning itself. Only if it fits the needs of these objectives should you choose, as we have done, to use the framework of the 7 P's.

The strategy statement can take the form of a purely verbal description of the strategic options which have been chosen. Alternatively, and perhaps more positively, it might include a structured list of the major options chosen. One aspect of strategy which is often overlooked is that of "timing."

Exactly when it is the best time for each element of the strategy to be implemented is often critical. Taking the right action at the wrong time can sometimes be almost as

bad as taking the wrong action at the right time. Timing is, therefore, an essential part of any plan; and should normally appear as a schedule of planned activities. Having completed this crucial stage of the planning process, you will need to re-check the feasibility of your objectives and strategies in terms of the market share, sales, costs, profits and so on which these demand in practice.

As in the rest of the marketing discipline, you will need to employ judgement, experience, market research or anything else which helps you to look at your conclusions from all possible angles.

Detailed Plans and Programmes

At this stage, you will need to develop your overall marketing strategies into detailed plans and programme. Although these detailed plans may cover each of the 7 P's, the focus will vary, depending upon your organization's specific strategies. A product-oriented company will focus its plans for the 7 P's around each of its products. A market or geographically oriented company will concentrate on each market or geographical area. Each will base its plans upon the detailed needs of its customers, and on the strategies chosen to satisfy these needs.

Again, the most important element is, indeed, that of the detailed plans, which spell out exactly what programmes and individual activities will take place over the period of the plan. Without these specified - and preferably quantified - activities the plan cannot be monitored, even in terms of success in meeting its objectives. It is these programmes and activities which will then constitute the "marketing" of the organization over the period. As a result, these detailed marketing programmes are the most important, practical outcome of the whole planning process.

These plans must therefore be:

- *Clear:* They should be an unambiguous statement of 'exactly' what is to be done.
- *Quantified:* The predicted outcome of each activity should be, as far as possible, quantified, so that its performance can be monitored.
- *Focused:* The temptation to proliferate activities beyond the numbers which can be realistically controlled should be avoided. The 80:20 Rule applies in this context too.
- *Realistic:* They should be achievable.
- *Agreed:* Those who are to implement them should be committed to them, and agree that they are achievable. The resulting plans should become a working document which will guide the campaigns taking place throughout the organization over the period of the plan. If the marketing plan is to work, every exception to it must be questioned; and the lessons learned, to be incorporated in the next year's planning.

Content of the Marketing Plan

A marketing plan for a small business typically includes Small Business Administration Description of competitors, including the level of demand for the product or service and the strengths and weaknesses of competitors

- Description of the product or service, including special features
- Marketing budget, including the advertising and promotional plan
- Description of the business location, including advantages and disadvantages for marketing
- Pricing strategy
- Market Segmentation

Measurement of Progress

The final stage of any marketing planning process is to establish targets so that progress can be monitored. Accordingly, it is important to put both quantities and timescales into the marketing objectives and into the corresponding strategies. Changes in the environment mean that the forecasts often have to be changed. Along with these, the related plans may well also need to be changed. Continuous monitoring of performance, against predetermined targets, represents a most important aspect of this. However, perhaps even more important is the enforced discipline of a regular formal review. Again, as with forecasts, in many cases the best planning cycle will revolve around a quarterly review. Best of all, at least in terms of the quantifiable aspects of the plans, if not the wealth of backing detail, is probably a quarterly rolling review - planning one full year ahead each new quarter. Of course, this does absorb more planning resource; but it also ensures that the plans embody the latest information, and - with attention focused on them so regularly - forces both the plans and their implementation to be realistic. Plans only have validity if they are actually used to control the progress of a company: their success lies in their implementation, not in the writing'.

Performance Analysis

The most important elements of marketing performance, which are normally tracked, are:

Sales Analysis

Most organizations track their sales results; or, in non-profit organizations for example, the number of clients. The more sophisticated track them in terms of 'sales variance' - the deviation from the target figures - which allows a more immediate picture of deviations to become evident. 'Micro-analysis', which is a nicely pseudo-scientific term for the normal management process of investigating detailed problems, then investigates the individual elements which are failing to meet targets.

Market Share Analysis

Few organizations track market share though it is often an important metric. Though absolute sales might grow in an expanding market, a firm's share of the market can decrease which bodes ill for future sales when the market starts to drop.

Where such market share is tracked, there may be a number of aspects which will be followed:

- Overall market share
- Segment share - that in the specific, targeted segment
- Relative share -in relation to the market leaders
- Annual fluctuation rate of market share

Expense Analysis

The key ratio to watch in this area is usually the 'marketing expense to sales ratio'; although this may be broken down into other elements.

Financial Analysis

The "bottom line" of marketing activities should at least in theory, be the net profit.

There are a number of separate performance figures and key ratios which need to be tracked:

- Gross contribution<>net profit
- Gross profit<>return on investment
- Net contribution<>profit on sales

There can be considerable benefit in comparing these figures with those achieved by other organizations; using, for instance, the figures which can be obtained from 'The Centre for Interfirm Comparison'. The most sophisticated use of this approach, however, is typically by those making use of PIMS, initiated by the General Electric Company and then developed by Harvard Business School, but now run by the Strategic Planning Institute.

The performance analyses concentrate on the quantitative measures which are directly related to short-term performance. But there are a number of indirect measures, essentially tracking customer attitudes, which can also indicate the organization's performance in terms of its longer-term marketing strengths and may accordingly be even more important indicators.

Some useful measures are:

- *Market research:* Including customer panels
- *lost business:* The orders which were lost because, for example, the stock was not available or the product did not meet the customer's exact requirements
- *Customer complaints:* How many customers complain about the products or services, or the organization itself, and about what

Use of Marketing Plans

A formal, written marketing plan is essential; in that it provides an unambiguous reference point for activities throughout the planning period. However, perhaps the most important benefit of these plans is the planning process itself. This typically offers a unique opportunity, a forum, for information-rich and productively focused discussions between the various managers involved. The plan, together with the associated discussions, then provides an agreed context for their subsequent management activities, even for those not described in the plan itself.

Budgets as Managerial Tools

The classic quantification of a marketing plan appears in the form of budgets. Because these are so rigorously quantified, they are particularly important. They should, thus, represent an unequivocal projection of actions and expected results. What is more, they should be capable of being monitored accurately; and, indeed, performance against budget is the main management review process. The purpose of a marketing budget is, thus, to pull together all the revenues and costs involved in marketing into one comprehensive document. It is a managerial tool that balances what is needed to be spent against what can be afforded, and helps make choices about priorities. It is then used in monitoring performance in practice.

The marketing budget is usually the most powerful tool by which you think through the relationship between desired results and available means. Its starting point should be the marketing strategies and plans, which have already been formulated in the marketing plan itself; although, in practice, the two will run in parallel and will interact. At the very least, the rigorous, highly quantified, budgets may cause a rethink of some of the more optimistic elements of the plans.

STRATEGIC PLANNING

Strategic planning is an organization's process of defining its strategy, or direction, and making decisions on allocating its resources to pursue this strategy, including its capital and people. Various business analysis techniques can be used in strategic planning, including SWOT analysis, PEST analysis, STEER analysis, and EPISTEL. Strategic planning is the formal consideration of an organization's future course.

All strategic planning deals with at least one of three key questions:

1. "What do we do?"
2. "For whom do we do it?"
3. "How do we excel?"

In business strategic planning, the third question is better phrased "How can we beat or avoid competition?". In many organizations, this is viewed as a process for determining where an organization is going over the next year or more -typically 3 to 5 years, although some extend their vision to 20 years. In order to determine where it is going, the

organization needs to know exactly where it stands, then determine where it wants to go and how it will get there. The resulting document is called the “strategic plan.” It is also true that strategic planning may be a tool for effectively plotting the direction of a company; however, strategic planning itself cannot foretell exactly how the market will evolve and what issues will surface in the coming days in order to plan your organizational strategy. Therefore, strategic innovation and tinkering with the ‘strategic plan’ have to be a cornerstone strategy for an organization to survive the turbulent business climate.

Mission, Vision and Values

- *Mission*: Defines the fundamental purpose of an organization or an enterprise, succinctly describing why it exists and what it does to achieve its Vision.
- *Vision*: Defines the desired or intended future state of an organization or enterprise in terms of its fundamental objective and/or strategic direction. Vision is a long term view, sometimes describing how the organization would like the world in which it operates to be. For example a charity working with the poor might have a vision statement which read “A world without poverty” It is sometimes used to set out a ‘picture’ of the organization in the future. A vision statement provides inspiration, the basis for all the organization’s planning. It could answer the question: “Where do we want to go?”
- *Values*: Beliefs that are shared among the stakeholders of an organization. Values drive an organization’s culture and priorities.
- *Strategy*: Strategy narrowly defined, means “the art of the general”. A combination of the ends for which the firm is striving and the means by which it is seeking to get there.

Mission Statements and Vision Statements

Organizations sometimes summarize goals and objectives into a mission statement and/or a vision statement. Others begin with a vision and mission and use them to formulate goals and objectives. While the existence of a shared mission is extremely useful, many strategy specialists question the requirement for a written mission statement.

However, there are many models of strategic planning that start with mission statements, so it is useful to examine them here:

- A Mission statement tells you the fundamental purpose of the organization. It defines the customer and the critical processes. It informs you of the desired level of performance.
- A Vision statement outlines what the organization wants to be, or how it wants the world in which it operates to be. It concentrates on the future. It is a source of inspiration. It provides clear decision-making criteria.

An advantage of having a statement is that it creates value for those who get exposed to the statement, and those prospects are managers, employees and sometimes even

customers. Statements create a sense of direction and opportunity. They both are an essential part of the strategy-making process. Many people mistake vision statement for mission statement, and sometimes one is simply used as a longer term version of the other.

The Vision should describe why it is important to achieve the Mission. A Vision statement defines the purpose or broader goal for being in existence or in the business and can remain the same for decades if crafted well. A Mission statement is more specific to what the enterprise can achieve itself. Vision should describe what will be achieved in the wider sphere if the organization and others are successful in achieving their individual missions.

A mission statement can resemble a vision statement in a few companies, but that can be a grave mistake. It can confuse people. The mission statement can galvanize the people to achieve defined objectives, even if they are stretch objectives, provided it can be elucidated in SMART terms. A mission statement provides a path to realise the vision in line with its values. These statements have a direct bearing on the bottom line and success of the organization. Which comes first? The mission statement or the vision statement? That depends. If you have a new start up business, new programme or plan to re engineer your current services, then the vision will guide the mission statement and the rest of the strategic plan. If you have an established business where the mission is established, then many times, the mission guides the vision statement and the rest of the strategic plan.

Either way, you need to know your fundamental purpose - the mission, your current situation in terms of internal resources and capabilities and external conditions, and where you want to go - the vision for the future. It's important that you keep the end or desired result in sight from the start..

Features of an effective vision statement include:

- Clarity and lack of
- Vivid and clear picture
- Description of a bright future
- Memorable and engaging wording
- Realistic aspirations
- Alignment with organizational values and culture

To become really effective, an organizational vision statement must become assimilated into the organization's culture. Leaders have the responsibility of communicating the vision regularly, creating narratives that illustrate the vision, acting as role-models by embodying the vision, creating short-term objectives compatible with the vision, and encouraging others to craft their own personal vision compatible with the organization's overall vision. In addition, mission statements need to be subjected to an internal assessment and an external assessment.

The internal assessment should focus on how members inside the organization interpret their mission statement. The external assessment — which includes all of the

businesses stakeholders — is valuable since it offers a different perspective. These discrepancies between these two assessments can give insight on the organization's mission statement effectiveness. Another approach to defining Vision and Mission is to pose two questions. Firstly, "What aspirations does the organization have for the world in which it operates and has some influence over?", and following on from this, "What can the organization do or contribute to fulfil those aspirations?". The succinct answer to the first question provides the basis of the Vision Statement. The answer to the second question determines the Mission Statement.

Methodologies

There are many approaches to strategic planning but typically a three-step process may be used:

1. *Situation*: Evaluate the current situation and how it came about.
2. *Target*: Define goals and/or objectives
3. *Path*: Map a possible route to the goals/objectives

One alternative approach is called Draw-See-Think:

- *Draw*: What is the ideal image or the desired end state?
- *See*: What is today's situation? What is the gap from ideal and why?
- *Think*: What specific actions must be taken to close the gap between today's situation and the ideal state?
- *Plan*: What resources are required to execute the activities?

An alternative to the Draw-See-Think approach is called See-Think-Draw:

- *See*: What is today's situation?
- *Think*: Define goals/objectives
- *Draw*: Map a route to achieving the goals/objectives

In other terms strategic planning can be as follows:

- *Vision*: Define the vision and set a mission statement with hierarchy of goals and objectives
- *SWOT*: Analysis conducted according to the desired goals
- *Formulate*: Formulate actions and processes to be taken to attain these goals
- *Implement*: Implementation of the agreed upon processes
- *Control*: Monitor and get feedback from implemented processes to fully control the operation

Situational Analysis

When developing strategies, analysis of the organization and its environment as it is at the moment and how it may develop in the future, is important. The analysis has to be executed at an internal level as well as an external level to identify all opportunities and threats of the external environment as well as the strengths and weaknesses of the organizations.

There are several factors to assess in the external situation analysis:

- Markets
- Competition
- Technology
- Supplier markets
- Labour markets
- The economy
- The regulatory environment

It is rare to find all seven of these factors having critical importance. It is also uncommon to find that the first two - markets and competition - are not of critical importance.

Analysis of the external environment normally focuses on the customer. Management should be visionary in formulating customer strategy, and should do so by thinking about market environment shifts, how these could impact customer sets, and whether those customer sets are the ones the company wishes to serve. Analysis of the competitive environment is also performed, many times based on the framework suggested by Michael Porter.

Goals, Objectives and Targets

Strategic planning is a very important business activity. It is also important in the public sector areas such as education. It is practiced widely informally and formally. Strategic planning and decision processes should end with objectives and a roadmap of ways to achieve those objectives.

One of the core goals when drafting a strategic plan is to develop it in a way that is easily translatable into action plans. Most strategic plans address high level initiatives and over-arching goals, but don't get articulated into day-to-day projects and tasks that will be required to achieve the plan. Terminology or word choice, as well as the level a plan is written, are both examples of easy ways to fail at translating your strategic plan in a way that makes sense and is executable to others. Often, plans are filled with conceptual terms which don't tie into day-to-day realities for the staff expected to carry out the plan. The following terms have been used in strategic planning: desired end states, plans, policies, goals, objectives, strategies, tactics and actions.

Definitions vary, overlap and fail to achieve clarity. The most common of these concepts are specific, time bound statements of intended future results and general and continuing statements of intended future results, which most models refer to as either goals or objectives. One model of organizing objectives uses hierarchies.

The items listed may be organized in a hierarchy of means and ends and numbered as follows: Top Rank Objective, Second Rank Objective, Third Rank Objective, etc. From any rank, the objective in a lower rank answers to the question "How?" and the objective in a higher rank answers to the question "Why?" The exception is the Top Rank Objective: there is no answer to the "Why?" question. That is how the TRO is

defined. People typically have several goals at the same time. "Goal congruency" refers to how well the goals combine with each other. Does goal A appear compatible with goal B? Do they fit together to form a unified strategy? "Goal hierarchy" consists of the nesting of one or more goals within other goals. One approach recommends having short-term goals, medium-term goals, and long-term goals. In this model, one can expect to attain short-term goals fairly easily: they stand just slightly one's reach. At the other extreme, long-term goals appear very difficult, almost impossible to attain.

Strategic management jargon sometimes refers to "Big Hairy Audacious Goals" in this context. Using one goal as a stepping-stone to the next involves goal sequencing. A person or group starts by attaining the easy short-term goals, then steps up to the medium-term, then to the long-term goals. Goal sequencing can create a "goal stairway". In an organizational setting, the organization may co-ordinate goals so that they do not conflict with each other. The goals of one part of the organization should mesh compatibly with those of other parts of the organization.

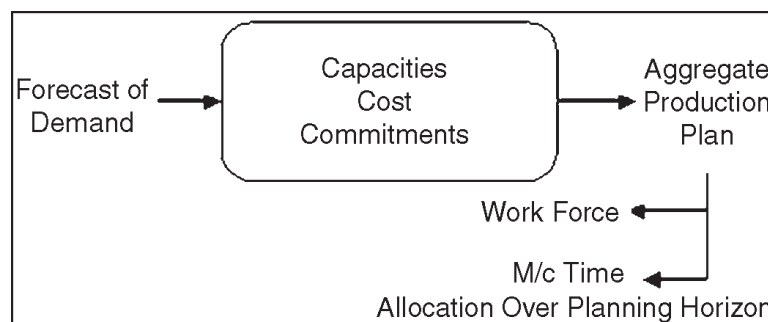
AGGREGATE PRODUCTION PLANNING

Aggregate production planning refers to the process of deciding the overall quantities of products to be manufactured or produced in a plant or other manufacturing facility during a medium term planning period such as a month, or a quarter.

The aggregate plan output consist of the total quantities of each product or a group of product to be manufactured in the plan period of going into details of scheduling of different manufacturing activities required to achieve the planned production levels. The aggregate production will also not specify details such as the dates when material ordered against individual customer order will be ready for delivery.

The aggregate production plan is prepared as a means of setting overall production targets and as input for planning availability of other inputs and supporting activities to meet the production targets.

The aggregate plans then form the basis of more detailed production, including such as daily and weekly production schedules and customer delivery schedules. Such production plan are further detailed out as machine loading schedules.



MANAGEMENT OPTIONS TO MEET FLUCTUATING DEMAND

- Build inventories in slack periods in anticipation of higher demands later in planning horizon.
- Carry backorders or tolerate lost sales during peak periods.
- Use over time in peak periods, under time in slack periods to vary output, while holding workforce and facilities constant.
- Vary capacity by changing size of workforce through hiring and firing.
- Vary capacity through changes in plant and equipment
- Subcontracting may be resolved. Each option involves cost. Aim in aggregate production planning is to choose best option.

KINDS OF COSTS INVOLVED

- Procurement Costs
- Production Costs
- Inventory holding Costs
- Shortage losses associated with backorders and lost sales
- Costs of increasing/decreasing workforce
- Cost of overtime/under time
- Cost of changing production rates

MASTER PRODUCTION SCHEDULING

The master production scheduling process translates a business plan into a dynamic and comprehensive product manufacturing schedule. Master Production Scheduling helps your management team balance and integrate the needs of marketing, manufacturing, finance and most importantly your customers.

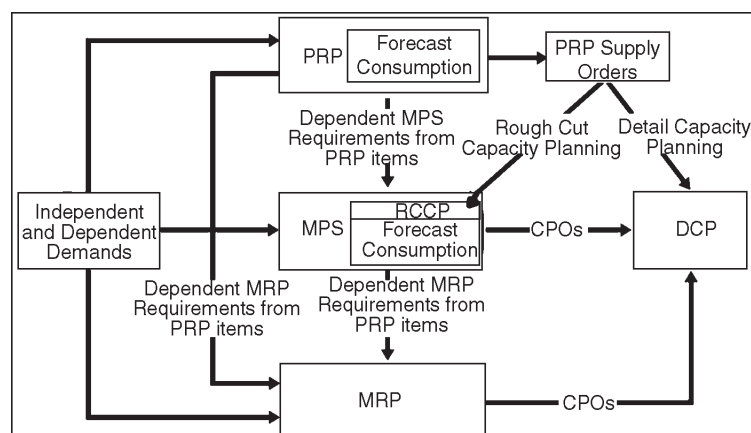


Fig. The Planning Hierarchy.

POWERFUL SCHEDULING TOOLS

Master scheduling requires ongoing analysis, measurement and adjustment to achieve revenue goals and ensure profitability through the careful allocation of materials and resources. MPS gives you a set of powerful tools for resource planning and scheduling, analysis, and performance measurement.

PLANNER'S WORKBENCH

A Planner's Workbench provides the means for consolidating all planning and scheduling functions onto one easy-to-use screen.

The workbench gives you immediate access to information in all areas necessary for successful planning/scheduling including sales demand, committed production, and purchased and planned supply. This allows planners, analysts, and programme managers to react quickly and easily to any exceptions that occur.

COMPLETE PLANNING VISIBILITY

To help resolve any conflicts, MPS takes a variety of variables into account, including available inventory, forecasted sales, actual backlog, and firm planned production, among others.

"WHAT IF" SIMULATION

A "What If" simulation capability ensures the best possible manufacturing scenario by allowing you to analyse the effects of alternate scenarios, using either current or simulated data.

SCHEDULING CYCLES

And because no single view of your supply and demand data can satisfy all of your planning needs, MPS enables you to create and maintain an unlimited number of planning scenarios. One for near-term deliveries, another for long-range production capacities, and so on. Each cycle provides a unique view of the information necessary to help you develop the correct priorities for your factory.

INSTANT AVAILABLE - TO - PROMISE INFORMATION

Your customer has an emergency requirement for an item that is in critical demand. Can you make the shipment? Our online Available-To-Promise feature allows you to instantly investigate the current product availability. You'll have up-to-the-minute item information at hand to make critical delivery commitments.

PERFORMANCE MEASUREMENT

MPS can give you an online "snapshot" estimation of the profit or loss that your build plan will yield. Performance analysis can be based on current or simulated data. This helps you focus on areas of maximum payback and set priorities before committing critical resources.

JOB SHOP SCHEDULING

Job Shop Scheduling is an optimization problem in computer science in which ideal jobs are assigned to resources at particular times. The most basic version is as follows:

We are given n jobs $J_1, J_2, \dots, J_n$ of varying sizes, which need to be scheduled on m identical machines, while trying to minimize the total length of the schedule. Mostly, the problem is presented as an online problem, that is, each job is presented, and the online algorithm needs to make a decision about that job before the next job is presented.

This problem is one of the most well known online problems, and was the first problem for which competitive analysis was presented, by Graham in 1966. Best problem instances for basic model with makespan objective are due to Taillard.

PROBLEM VARIATIONS

Many variations of the problem exist, which can be summarized as follows:

- Machines can be related, independent, equal
- Machines can require a certain gap between jobs
- Machines can have sequence-dependent setups
- Objective function can be to minimize the makespan, the L_p norm, etc
- Jobs may have constraints, for example a job i needs to finish before job j can be started
- Jobs and machines have mutual constraints, for example, certain jobs can be scheduled on some machines only

MAJOR RESULTS

Graham had already provided the List scheduling algorithm in 1966, which is $(2 - 1/m)$ -competitive, where m is the number of machines. Also, it was proved that List scheduling is optimum online algorithm for 2 and 3 machines. In 1992, Bartal, Fiat, Karloff and Vohra presented an algorithm that is 1.986 competitive. In 1992, Albers provided a different algorithm that is 1.923-competitive. Currently, the best known result is an algorithm given by Fleischer and Wahl, which achieves a competitive ratio of 1.9201. Currently, best known lower bound is 1.88. Taillard instances has an important role in developing job shop scheduling with makespan objective.

SHOP FLOOR CONTROL

EMPOWER SHOP FLOOR PERSONNEL FOR MAXIMUM EFFECTIVENESS AND OUTPUT

To focus on the highest priority jobs, your factory needs the latest shop floor information. Glovia G2 Shop Floor Control provides this information along with a set of easy-to-use tools for achieving maximum productivity.

CLOSED-LOOP FEEDBACK

The final phase in the closed-loop process is the tracking and feedback of actual shop floor activity. The successful execution of your master production schedule and material requirements planning depends on rapid and accurate dispatch of daily production schedules. Shop Floor Control is designed to integrate these tasks fully.

WORK-IN-PROCESS VISIBILITY

Shop Floor Control provides shop personnel with a set of easy-to-use transactions for reporting work order operation completions and quantities of items produced. Shop status provides the added visibility necessary to stay on top of the current work-in-process.

SHOP FLOOR EMPOWERMENT

Shop Floor Control gives you the tools you need to tailor the detail schedules and workflows to best match the people and machines available at each work centre. This helps you optimize schedules and anticipate needs for critical resources, as well as helps the people on the factory floor actively contribute to the success of your production plan.

MOVE TICKET REPORTING

Move ticket provide an easy method for reporting shop floor progress when separate shop floor documents are needed for tracking physical movement from one work centre to the next.

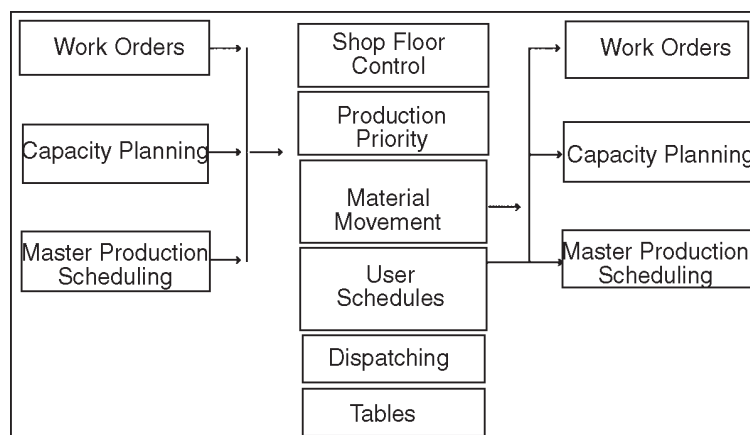


Fig. Shop Floor Control Integration.

ON-SCREEN DISPATCHING

On-screen dispatching enquiries and reports provide factory supervisors with the information they need to control the production floor. This up-to-the-minute input

provides a powerful tool for monitoring currently scheduled operations, work centre loading and priority sequencing. It also provides the visibility into what needs to be completed in order to ensure a smooth and coordinated production flow.

SHOP FLOOR CONTROL

- View all production jobs by work centre
- Select workload by date range and shift
- Update operation completions
- Report completions by quantity or percentage
- Review operation start and end dates
- Work centre dispatch list
- List of work by priority, order, line and type
- Pay points automatically back-fill prior operations
- Labour and machine load hours
- Setup and run times
- Completion status
- Outside processing status
- Next or prior work centre
- Labour and machine capacities with effectivity
- Efficiency factors, queue times and cost centres
- Work order shortages

USER-CONTROLLED OPERATION SCHEDULING

- Modify detail scheduling and loading of shop orders
- Reassign shift codes
- Assign order priority code
- Select schedule code: backward, forward and user scheduling
- Set operation start and end dates
- Define overlapped or parallel operation sequencing
- Create user-defined status codes for shop floor control

WORK ORDER DISPATCHING

- View selected work order lines
- Display part numbers and order quantities
- Display and update work order operations
- Review scheduled operation start and end dates
- Report completed quantities or percentages
- Identify Next and prior work centre
- Assign Shifts

- View pay point operations
- View operation routing with function description
- Review setup times, run times, crew sized and move times
- Notify of bill of materials component shortages

MOVE TICKET RECORDING

- Work order activity and progress
- Operation completions using quantities or percentages
- Assign posting date
- Report completed activity types
- Labour setup and run, machine set up and run, outside processing

DEMAND MANAGEMENT

Demand management is the purposeful and beneficial manipulation of the level and timing of water usage. Demand management deploys various techniques for conserving water and improving the efficient use of water by end users. Improvements to economic efficiency are achieved whenever the total benefits of a measure are outweighed by the total costs of implementation.

Demand management evolved in the context of least-cost or integrated resource planning, which balanced supply and demand management considerations. Managing demand can complement or supplant traditional and emerging supply-management options for water utilities.

Demand management involves measures that promote the efficient use of water, including load management and load reduction or conservation. Water conservation also can be understood as the economically and/or socially beneficial reduction of water withdrawals, water use, or water waste.

Conservation can forestall future supply-capacity needs; it can be implemented on the supply side as well as the demand side; and it can consist of both temporary measures used during emergencies and permanent measures used to improve long-term efficiency. Reductions in water usage can be beneficial to both water utilities and wastewater utilities in terms of flow reduction.

Demand management or strategic load management complements supply management because controlling the level and timing of demand can improve overall efficiency of system operations and help eliminate, reduce, or defer the need for an investment in new capacity by the water utility. Reductions in peak and off-peak demand affect the total capacity requirements of the utility system and thus the total cost of providing water service.

All demand management activities that decrease the demand for utility services tend to affect supply management since existing system capacity is released for other customers and other uses. That is, the freed or redirected utility capacity can be compared

to that provided by more traditional means. Thus, the benefits of demand management can be measured in terms of avoided costs, or the incremental savings associated with not having to produce additional units of water or water service. Avoided cost can be used to compare demand management and supply management options and encourage utilities to seek out least-cost alternatives for meeting future water needs.

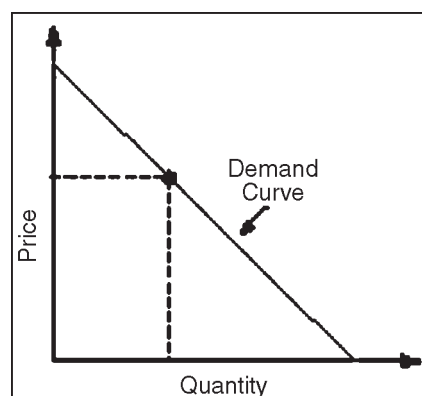
Although demand management should not be equated with drought management, the experience of water utilities and customers in implementing efficiency practices can be beneficial during periods of water shortage. Some of the basic demand management techniques can be accelerated during supply emergencies or droughts.

A sign warns residents that they may no longer water their lawns because of drought conditions. Lack of rain in conjunction with high temperatures can lead communities and even states to enact water-use restrictions in an attempt to control water demand.

WATER DEMAND

Several factors influence the residential and nonresidential demand for water. In the aggregate, per capita water demand is very stable. Residential water usage is largely a function of basic demographics, particularly household size, property size, and income. Nonresidential water varies substantially according to type of industry. The production of some goods is highly water intensive.

The demand for water, like the demand for other goods and services, can be represented by a downward-sloping curve where price is represented on the y-axis and quantity demanded on the x-axis. However, compared with many other goods and services, water demand is relatively price inelastic; that is, changes in price generally do not induce large changes in water use. Nonresidential (large-volume) water use and outdoor residential water use (such as summer lawn watering) are generally more price responsive than indoor water use.



TOOLS AND TECHNIQUES

Demand management techniques include conservation-oriented pricing, water-fixture plumbing standards and retrofitting, water-efficient landscaping, changes in water-use

practices, and public education. Combining demand and management measures can prove to be effective. For example, a public education programme can enhance the effectiveness of a conservation-oriented rate structure. Despite water's relative price inelasticity, price is considered an essential tool of demand management because it sends customers an essential signal about the value of water. Water metering is a basic demand management measure because it allows the utility to charge a variable price (that is, a per unit rate) so that a customer's bill varies with the level of water usage.

Some methods of rate design are particularly efficiency oriented. These include increasing-block rates (where unit rates are increased at higher levels of usage), seasonal rates (where unit rates are higher during peak usage periods), and water budget and excess-use rates.

UTILITY PROGRAMMES

Some demand management measures can be implemented by consumers on their own, while others can be implemented through utility-sponsored programmes. High water and energy prices can induce customers to invest in water-efficient fixtures and appliances as well as to change water-use behaviour.

Utility programmes can help consumers provide more informed choices as well as provide specific incentives for engaging in demand management. Some programmes have helped utilities reduce operating costs (water, energy, and chemicals) and postpone or avoid capital costs (treatment plants and other facilities). Both water and wastewater systems can benefit from utility sponsored demand management programmes.

Utility demand management programmes can range from very passive to very active. Passive approaches include the distribution of educational materials and low-cost conservation devices (such as leak detection tablets and faucet aerators). More active methods include water audits and rebates to households that purchase replace fixtures (such as toilets).

An even more active approach is for the water utility to directly perform onsite retrofits. Programme elements can be designed specifically for the needs of residential and nonresidential customers. Utilities can work directly with large-volume users in identifying methods for reducing or shifting loads. Programme design and implementation are critical to the success of a conservation programme. Attention must be paid to the characteristics, needs, and preferences of water users. Utilities can partner with other local utilities, government agencies, or community groups. A pilot programme can be useful for developing an effective programme. Utilities often collect data throughout the course of implementation in order to assess programme effectiveness. The programme can be refined over time to ensure that programme goals are met.

Facility and Layouts Management

FACILITY MANAGEMENT

Facility management is an interdisciplinary field primarily devoted to the maintenance and care of commercial or institutional buildings, such as hospitals, clinics, hotels, resorts, schools, office complexes, sports arenas or convention centres. Duties may include the care of air conditioning, electric power, plumbing and lighting systems; cleaning; decoration; groundskeeping and security. Some or all of these duties can be assisted by computer programmes.

These duties can be thought of as non-core or support services, because they are not the primary business of the owner organization. It is the role of the facility management function to coordinate and oversee the safe, secure, and environmentally-sound operations and maintenance of these assets in a cost effective manner aimed at long-term preservation of the asset value, and also other janitorial duties such as making sure the environment is properly cleaned and sanitized for its tenants.

In those cases where the operation of the facility directly involves the occupants and/or customers of the owner organization, the satisfactory delivery of facility-related services to these people will be an important consideration too; hence, the term "end-user satisfaction" is often used both as a goal and a measure of performance. The term facility management is similar to property management although not exactly the same. While both manage the day to day operations of a facility the property such as cleaning, maintenance and security, similar to Janitors, one must not confuse it with such a title.

The property manager has an expanded role which includes leasing and marketing activities whereas the facility manager role focuses on existing tenants who usually are owner occupants.

An important feature of facility management is that it takes account of human needs of its tenants in the use of buildings and other constructed facilities. These softer factors complement the harder factors associated with the maintenance and care of engineering services installations. Atkin and Brooks, an important concept in the facility management field is that of outsourcing, where the owner enters into an arrangement with external organizations to provide one or more services in preference to their being provided through internal arrangements.

The reasons for this action can vary, including lack of in-house resources, lack of expertise and pressure to reduce costs. Unfortunately, confusion can exist because of the close association that facility management has with outsourcing. The two concepts are not synonymous; rather, outsourcing is one means for providing facility-related services to the owner organization.

Facility management is performed during the operational phase of a building's life cycle, which normally extends over many decades. As such, it will represent a continuous process of service provision to support the owner's core business and one where improvement will be sought on a continuous basis. It is essential that decision-making in the preceding design and construction phases is therefore properly informed about operational requirements if the facility is to provide optimal support to the owner's business.

In this connection, facility management can be seen as an integral part of a coordinated and controlled process of design, engineering, construction and operations. Where a facility is provided on a turnkey basis, for example design-build-finance-operate, the consortium responsible for the delivery of the physical asset and then operating the core service will need to understand implicitly the day-to-day demands in managing that facility. Under such arrangements - typically public-private partnerships- owner-operators must fully integrate operational thinking into early design decision-making. A major challenge facing facility owners is reducing demand for energy for economic reasons, but also because energy consumption goes hand-in-hand with carbon emissions. Reducing energy during the operational phase of a facility's life similarly reduces carbon emissions. When considering that 30-40% of a country's total carbon emissions is attributable to buildings and other constructed facilities, it is clear that operations and, hence, facility management have a significant role to play.

TYPES OF LAYOUTS

There are several alternative layout types that are appropriate for different product mixes and production volumes. Determination of the layout type is a major design decision because it impacts on so many other aspects of the production system. In the

Job Shop Layout, machines are grouped according to function into machine centres. Orders for individual products are routed through the various machine centres to obtain the required processing.

This layout may be appropriate when there are many different products, each with a low volume of production. Machines are general purpose, within their general function area, so that a wide variety of products can be handled. Because the expense of automation may be too great to be justified by the low volume, the machines in this arrangement will probably be at a relatively low level of automation. Workers will be highly skilled.

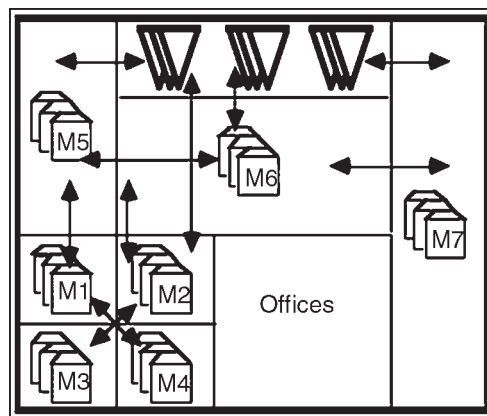


Fig. Job Shop Layout.

Production scheduling is difficult with this type of arrangement because the level and type of work is highly variable. This results in large amounts of work-in-process, long product lead times, and high levels of management interaction. Typically there is a high degree of product movement required by the long and variable routes of individual products through the system. The costs for setting up machines to produce the various products will be high because of the variety of different products and small lot sizes. The arrangement can adapt readily to changes in product volume and design because of its inherent flexibility.

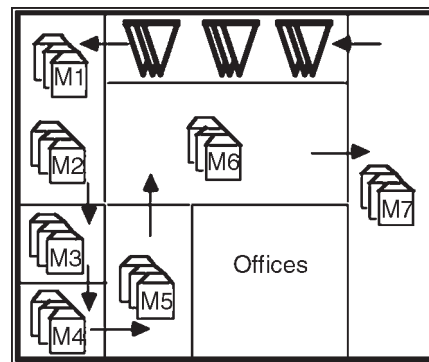


Fig. Assembly Line or Flow Shop Layout.

Here the product follows a fixed path through the production resources. The resources are arranged to minimize the material movement required. This type of layout is typical for an assembly line where a single product, or a few very similar products, passes through the line in a continuous fashion. Because of the high volume of production, the machines on the line can be designed with a high level of fixed automation, with very little manual labour. Direct labour will be much less than for the job shop, but there will be high costs for maintenance. Setup costs and work-in-progress will be low for this arrangement.

The line, in general, is not flexible to product or volume changes. It is very sensitive to failures that cause the entire line to shut down. The arrangement is also appropriate for a flow shop that may have a number of products that all pass through the machine centres in the same order. In this case, the machines implementing the system may or may not be automated depending on the product mix and volume, but one would expect a higher level of automation than for the job shop.

The product mix appropriate for this arrangement is similar to that of the job shop. Products are grouped into classes that have some similarity with respect to processing. A manufacturing cell is designed for each group consisting of machines particularly adapted to the processing required. The figure shows the cells as collections of dissimilar machines. Because the range of products manufactured by each cell is less than that for the job shop, the machines and workers can be more specialized.

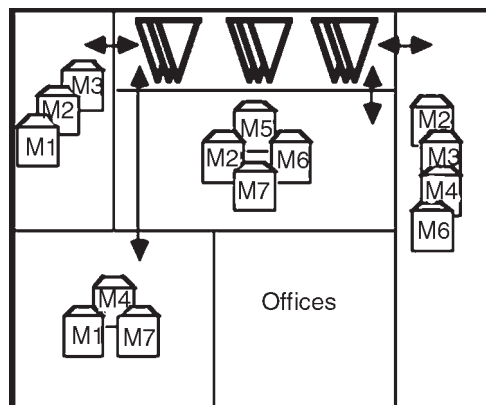


Fig. Group Technology Layout.

Typically, the workers in a cell are given more of the responsibility for production scheduling of a product class. This, together with the start-to-finish nature of the processing, results in more interesting jobs for the workers. The group technology arrangement requires less setup time and cost than the job shop because of the greater specialization of function. It is compatible with the just-in-time concept of manufacture, so prevalent today, because of the smaller lot sizes made possible by the low setup costs.

Often the level of automation with group technology is low, indicating the dependence of the concept on the skill of the labour force. Many companies have, however, introduced

highly automated flexible manufacturing cells into the system. Because the cell has a smaller range of products than the entire plant, it is easier to design the automation to handle the set of products in a group.

The group technology approach is more sensitive to changes in product mix and volume than the job shop, again because of the specialization introduced because of the manufacturing cell approach. When a product requires processing in more than one cell, problems similar to those of the job shop are introduced.

FIXED LOCATION LAYOUT

For tasks on large objects such as the manufacture of an electrical generator, the construction of a building, or the repair of a large airplane, the machines implementing the operation must come to the product, rather than the product moving to the machines. Here the question is more often the scheduling of operations rather than the layout of machines. For tasks on large objects such as the manufacture of an electrical generator, the construction of a building, or the repair of a large airplane, the machines implementing the operation must come to the product, rather than the product moving to the machines. Here the question is more often the scheduling of operations rather than the layout of machines.

FLEXIBLE MANUFACTURING SYSTEMS

The FMS is a system with automated material handling moving individual units of product between automated processors. Robotic manipulators often handle material. Using computer controlled movement and processing, a wide variety of products can be manufactured. All of this is to be accomplished with very low setup time, great flexibility of function, and very little manual labour. The diversity of possible FMS, and the rapidly changing technologies, makes detailed consideration of the design of FMS beyond the scope of this text. Certainly many of the classical questions associated with facility design are no longer relevant for the FMS. In this text we consider the FMS as just another kind of machine, with perhaps a very broad range of capabilities.

LAYOUT PROBLEM

The layout problem is to arrange the physical spaces required for several departments in a given space provided for the departments. In practice the facility layout problem is often solved by intuition, using the artistic and spatial skills of the human designer; however, when there are quantitative considerations associated with the layout problem, the human is at a disadvantage as compared to the computer. We concentrate on computerized procedures for solving the layout problem. There are a variety of problems regarding layout one might encounter. In this part we explain the problem by specifying the data and describing the decisions.

INPUT DATA

Here we are considering the problem of arranging several departments on a plant with a single floor and fixed dimensions.

Certain data is necessary to describe the layout problem.

- Number of departments, n ,
- Physical area of each department, A_i for $i = 1 \dots n$
- Physical dimensions of the plant in which the departments are to be placed: Length, L , and Width, W .
- Product flow between every pair of departments: f_{ij} for $i = 1 \dots n$ and $j = 1 \dots n$.
- Material handling cost between every pair of departments measured in dollars per unit-foot: c_{ij} for $i = 1 \dots n$ and $j = 1 \dots n$.

DISTANCE

Our models involve the distance from one department to another. The distance depends on the layout. To illustrate consider a problem with ten departments with each department having an area of 100 square feet. The ten departments are to be placed in an area that is 50 feet long by 20 feet wide. One layout is shown in Figure.

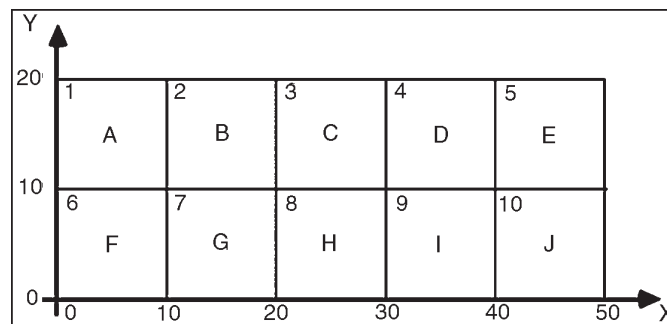


Fig. A Layout of Departments.

This is only one of many possible layouts. If we assume that the departments must maintain a square shape, every permutation of the letters A through J is a different layout. There are $n!$ permutations. The matrix as in Fig. describes the flow between the departments. This is called the *From-To* matrix because an element (i, j) contains f_{ij} , the flow from department i to department j .

The criterion for the layout problem involves the distance between departments. First we must prescribe the end points for the distance measurement. Here we assume that distances are measured between the centroids or centres of gravity of departments. Second, we must specify the route of travel. One possibility is that flow will follow a straight-line path. This is the Euclidean measure. More common in layout analysis, is to assume that flow travels via paths that are parallel to the axes of the layout. This is the rectilinear measure.

From \ To	A	B	C	D	E	F	G	H	I	J
A					5				1	9
B				3	4		1			
C				1			9			2
D							6		1	
E						8	3			
F							2	5		
G								6		
H									8	
I										
J										

Fig. From-to Matrix for the Office Example.

The centroids are specified in terms of the coordinate system as

$x(i)$ = x -coordinate of the centroid of department i , and

$y(i)$ = y -coordinate of the centroid of department i .

The centroid is the same as the centre of the area when the department is rectangular. For a more general shape, the centroid is the centre of gravity of area. In the example case of Fig. we have the centroids as follows.

Department A: $x(A) = 5$, $y(A) = 15$.

Department B: $x(B) = 15$, $y(B) = 15$.

Department C: $x(C) = 25$, $y(C) = 15$.

The distance between two departments by a rectilinear measure is,

$$d_{ij} = |x(i) - x(j)| + |y(i) - y(j)|.$$

Here the vertical lines indicate absolute value. Fig. shows both the flow and distance between all pairs of departments on the from-to chart. The flow appears the diagonals and the distance appears.

From \ To	A	B	C	D	E	F	G	H	I	J	Sum
A					5				1	9	690
B				3	4		1				190
C				1			9			2	250
D							6		1		190
E						8	3				520
F							2	5			120
G								6			60
H									8		80
I											0
J											0

Fig. Computation of the Total Distance Travelled.

CRITERION FOR COMPARISON

The flow multiplied by the distance and summed over all cells of the chart. We compute the cost for the flow from i to j as the product of the material handling cost, the flow and the distance between the departments. The cost of the layout is the sum of the flow cost.

$$z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} f_{ij} d_{ij}$$

The cost associated with each row of the chart is shown at the right of Figure. The cost of the layout is the sum of the row costs. The factors c_{ij} and f_{ij} are given as data, but the factor d_{ij} depends on the layout.

FINDING A LAYOUT WITH THE CRAFT METHOD

The plant layout problem is to find an arrangement of departments that minimizes the total distance travelled. This measure is called the “cost” in the following. The problem is difficult for the human because of the large number of calculations required to evaluate the cost of an alternative. The computer can greatly aid in the layout determination, but the computer also has its weaknesses.

The problem of finding the optimum arrangement is a very difficult one for mathematical optimization. Rather than attempt to obtain the optimum, we use a heuristic approach that tries to better a given solution by switching pairs of departments. At each iteration, every pair of departments is considered for switching locations. The pair that gives the greatest savings in cost will be interchanged and the process will continue by looking for another pair to switch. If no pair results in a positive savings the process stops.

This is commonly called the CRAFT* method. Only the centroids of the two departments are affected. For example, if department A is switched with department H, the centroid of A becomes, while the centroid of H becomes. All other departments remain fixed. The only effect on the cost of the layout is related to the flows entering or leaving the two departments.

The cost of the flow between the other departments is unaffected because the distances between these departments have not changed. This is the logic used by the solution method. The cost savings associated with switching two departments is determined by calculating the effect of interchanging the centroids of the two departments. There are several reasons why the CRAFT method may not yield the optimum solution. First, it may be necessary to switch more than two departments to find a better solution. More complex implementations of the method do allow switching three or more departments, but this increases considerably the computation time.

Problems for which the department areas are not equal cause several new difficulties. If two departments are not of equal area, switching their locations may not be equivalent to switching the centroids. In fact, if two unequally sized departments are not physically adjacent it will not be possible to interchange them.

The method allows departments of different areas to be switched only if they are physically adjacent. Allowing departments with different areas to switch causes another problem. When a small department is interchanged with a larger department there are a number of possible arrangements of the two areas with a corresponding variety of centroids.

It is impossible to enforce a requirement that areas remain rectangular, so often the interchange of such departments causes irregularities in shape that grow worse as the algorithm progresses. Because the change in cost is not exactly represented by the interchange of centroids, the estimate of the cost reduction may not be accurate.

It is possible that the programme will switch two areas and have the total cost increase. To guard against cycling, the method stops if a certain number of successive iterations do not result in an improvement. Since the CRAFT method does not guarantee an optimum solution, it should be run several times with different initial layouts.

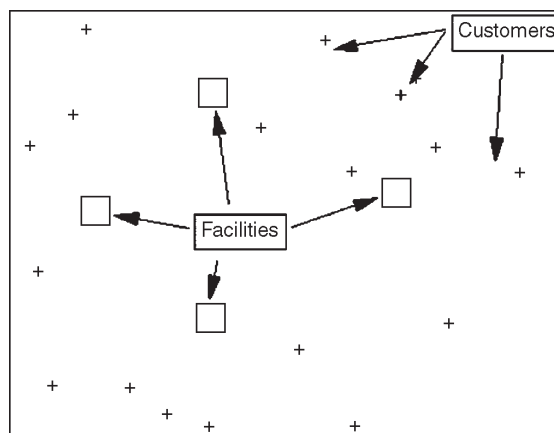
FACILITY LOCATION

The general facility location problem is: given a set of facility locations and a set of customers who are served from the facilities then:

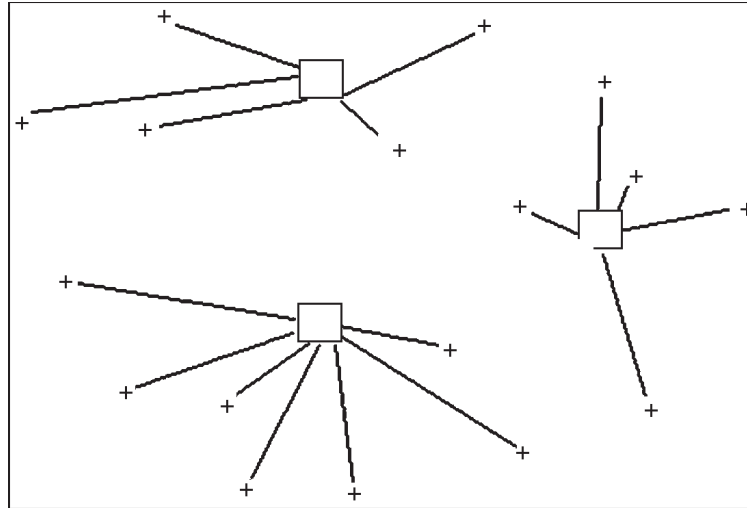
- Which facilities should be used
- Which customers should be served from which facilities so as to minimise the total cost of serving all the customers.

Typically here facilities are regarded as “open” or “closed” and there is a fixed cost which is incurred if a facility is open. Which facilities to have open and which closed is our decision.

We show a graphical representation of the problem.



One possible solution is.



Other factors often encountered here:

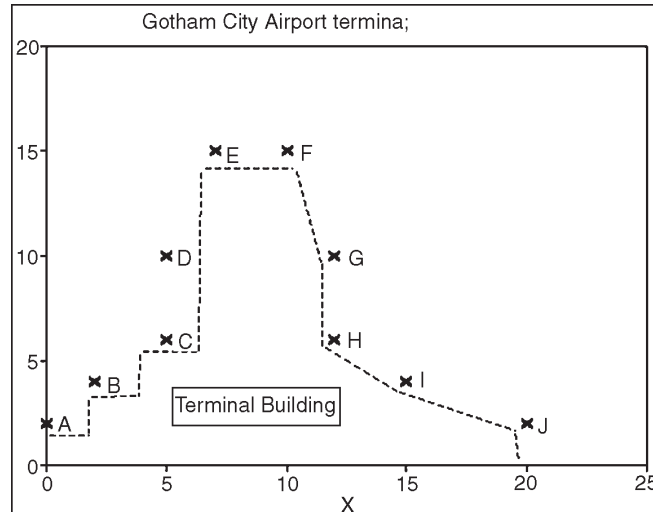
- Customers have an associated demand with capacities on the total customer demand that can be served from a facility
- Customers being served by more than one facility.

EXAMPLE

The problems of facility size and facility location are very closely linked and should be considered *simultaneously*. In fact the package used here disregards the problem of facility size and deals only with facility location. We shall illustrate the problem of facility location by means of an example. At Gotham City airport terminal there are 10 arrival gates. A pictorial representation of the terminal is given with the location of the gates being:

Gate	x coordinate	y coordinate
A	0	2
B	2	4
C	5	6
D	5	10
E	7	15
F	10	15
G	12	10
H	12	6
I	15	4
J	20	2

Luggage from arriving flights is unloaded at these gates and moved to a passenger luggage pick-up point. It is estimated that the number of pieces of luggage arriving per day at each gate is: 3600, 2500, 1800, 2200, 1000, 4500, 5600, 1400, 1800 and 3000 respectively. Where should the passenger luggage pick-up point be located in order to minimize movement of luggage?



SOLUTION

In order to logically locate the passenger luggage pick-up point we need to make use of the amount of luggage flowing from the gates to the pick-up point. Logically a gate from which there is a large flow should be nearer to the pick-up point than a gate with a small flow.

Informally therefore we would like to position the pick-up point so as to *minimise* the sum over all gates g . This approach to choosing the location of the pick-up point is precisely the approach used by the facility location module in the package.

Typically in such location problems we are concerned with a *load-distance* score. The initial input to the package for this problem.

Problem Specification	
Problem Type ☒ Facility Location ☐ Functional Layout ☐ Line Balancing	Objective Criterion ☒ Minimization ☐ Maximization
Problem Title: Gotham City Terminal	
Number of Existing Facilities: 10	
Number of Planned New Facilities: 1	
Number of Coordinate Dimensions (2/3): 2	
ok	Cancel
Help	

Note here the package terminology is somewhat peculiar:

- *Existing facilities:* Are points where we know in advance exactly where they are and they are fixed in position
- *New facilities:* Are points where we do not know where they are and their location is what we have to determine

In order to enter the data relating to the gates we can ignore the columns in the package input that are concerned with flows between existing facilities - here all flows are from the existing facilities to the new facility.

Facility Number	Facility Name		To New 1 Flow/Unit Cost	Location X Axis	Location Y Axis
Existing 1	A		3600	0	2
Existing 2	B		2500	2	4
Existing 3	C		1800	5	6
Existing 4	D		2200	5	10
Existing 5	E		1000	7	15
Existing 6	F		4500	10	15
Existing 7	G		5600	12	10
Existing 8	H		1400	12	6
Existing 8	I		1800	15	4
Existing 10	J		3000	20	2
New 1	New				

Note here that in solving the problem we need to specify the appropriate *distance model*. This is because as we do not yet know where the new facility is to be we cannot specify the distance between it and the gates *without* a general expression for calculating the distance between two locations.

If (x_i, y_i) and (x_j, y_j) represent the coordinates of two locations i and j then the distance model measures can be:

- *Rectilinear:* Distance between i and j is: $|x_i - x_j| + |y_i - y_j|$
- *Euclidean:* Distance between i and j is: $[(x_i - x_j)^2 + (y_i - y_j)^2]^{0.5}$
- *Squared Euclidean:* Distance between i and j is: $(x_i - x_j)^2 + (y_i - y_j)^2$

The rectilinear distance measure is often used for factories, American cities, etc which are laid out in the form of a rectangular grid.

For this reason it is sometimes called the Manhattan distance measure. The Euclidean distance measure is used where genuine straight line travel is possible. The squared Euclidean distance measure is used where straight line travel is possible but where we wish to discourage excessive distances.

The output from the package for each of the distance measures is:

Note here that the initial choice of distance measure is often between rectilinear and Euclidean. If the luggage flows in an Euclidean manner then choosing to use squared Euclidean, rather than Euclidean, penalises excessively long distances.

One point to note about the picture is that we are not really interested in determining positions down to the nearest millimetre. Instead we are using the package to get an *indication* of the approximate region where it would be sensible to site a luggage pick-up point. Note too that the package uses a “centre of gravity” approach to decide where to locate a single pick-up point.

11-09-2000 14:33:37	Facility Name	X Axis	Y Axis	Flow to All Facilities	Cost to All facilities
1	A	0	2	3600	50400
2	B	2	4	2500	25000
3	C	5	6	1800	9000
4	D	5	10	2200	19800
5	E	7	15	1000	12000
6	F	10	15	4500	40500
7	G	12	10	5600	33600
8	H	12	6	1400	2800
9	I	15	4	1800	12600
10	J	20	2	3000	42000
11	New	10	6	0	0
Total				27400	247700
Distance		Measure: Rectilinear			

Fig. Rectilinear.

11-09-2000 14:33:37	Facility Name	X Axis	Y Axis	Flow to All Facilities	Cost to All facilities
1	A	0	2	3600	44,266.76
2	B	2	4	2500	23,819.75
3	C	5	6	1800	10,667.54
4	D	5	10	2200	11,483.13
5	E	7	15	1000	6,776.21
6	F	10	15	4500	27,073.98
7	G	12	10	5600	11,965.45
8	H	12	6	1400	4,938.54
9	I	15	4	1800	12,556.59
10	J	20	2	3000	36,299.11
11	New	9.05	7.67	0	0
Total				27400	189,847.06
Distance		Measure: Euclidean			

Fig. Euclidean.

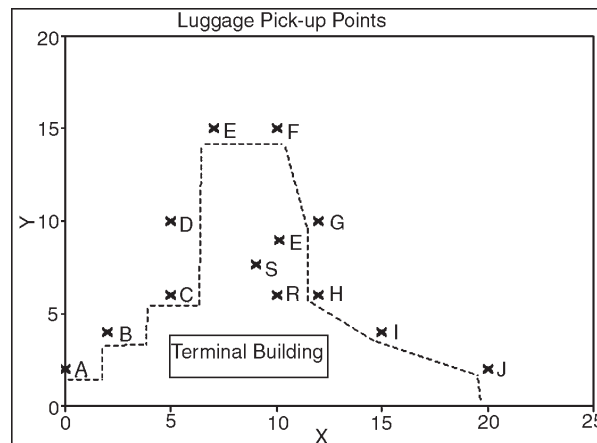
11-09-2000 14:33:37	Facility Name	X Axis	Y Axis	Flow to All Facilities	Cost to All facilities
1	A	0	2	3600	410570
2	B	2	4	2500	157,928.28
3	C	5	6	1800	34,547.77
4	D	5	10	2200	48,070.31
5	E	7	15	1000	57,966.92
6	F	10	15	4500	245,971.59
7	G	12	10	5600	79,154.93
8	H	12	6	1400	16,069.03
9	I	15	4	1800	87,917.11
10	J	20	2	3000	456,010.25
11	New	9.05	7.67	0	0
Total				27400	1,594,206.25
Distance		Measure: Squared Euclidean			

Fig. Squared Euclidean.

From the package output we can see that the location for the luggage pick-up point should be:

Distance measure	x coordinate	y coordinate
Rectilinear	10	6
Euclidean	10.12	8.98
Squared Euclidean	9.05	7.67

The picture shows these three possible locations with respect to the terminal.



CONCENTRATOR

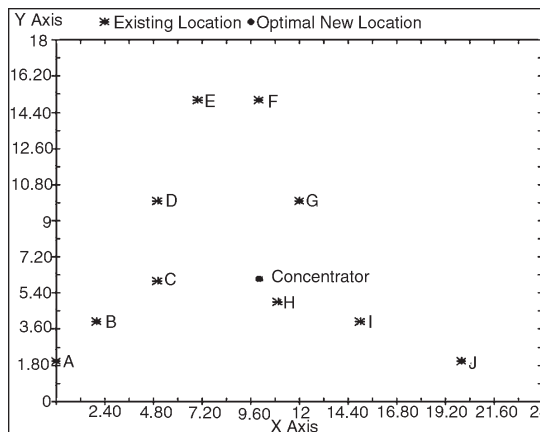
Suppose now that in the example we just considered the luggage pick-up point for gates C to H is merely acting as a concentrator, concentrating luggage before passing it to the other pick-up point, which collects luggage from gates A, B, I and J directly. In this system all passengers see is a single pick-up point at which they collect their luggage. To locate this concentrator we make use of flows between new facilities. Here the total amount of luggage collected from C to H will flow to the final pick-up point from this intermediate concentrator. This will total $1800+2200+1000+4500+5600+140 = 16500$ pieces of luggage. Hence the input to the package is:

Facility Number	Facility Name	To New 1 Flow/Unit Cost	To New 2 Flow/Unit Cost	Location X Axis	Location Y Axis
Existing 1	A	0	3600	0	2
Existing 2	B	0	2500	2	4
Existing 3	C	1800		5	6
Existing 4	D	2200		5	10
Existing 5	E	1000		7	15
Existing 6	F	4500		10	15
Existing 7	G	5600		12	10
Existing 8	H	1400		12	6
Existing 9	I	0	1800	15	4
Existing 10	J	0	3000	20	2
New 1	Concentrator		16500		
New 2	Final				

The solutions are:

11-12-2000	New Facility	X Axis	Y Axis
1	Concentrator	10.00	6.12
2	Final	10.00	6.12
Total	Flow to&From	New Location	= 43900
Total	Cost to&From	New Location	= 247,818.11
(by	Rectilinear	Distance)	

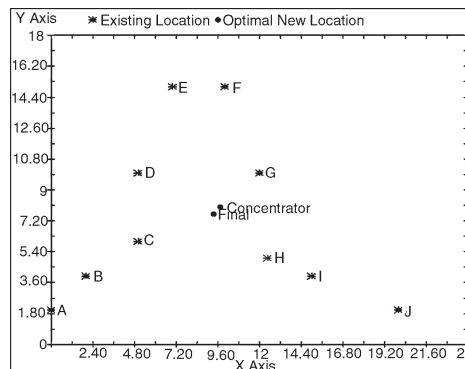
Fig. Rectilinear.



So in this case the final and concentrator locations coincide.

11-12-2000	New Facility	X Axis	Y Axis
1	Concentrator	9.46	6.12
2	Final	8.94	6.12
Total	Flow to&From	New Location	= 43900
Total	Cost to&From	New Location	= 198,276.50
(by	Euclidean	Distance)	

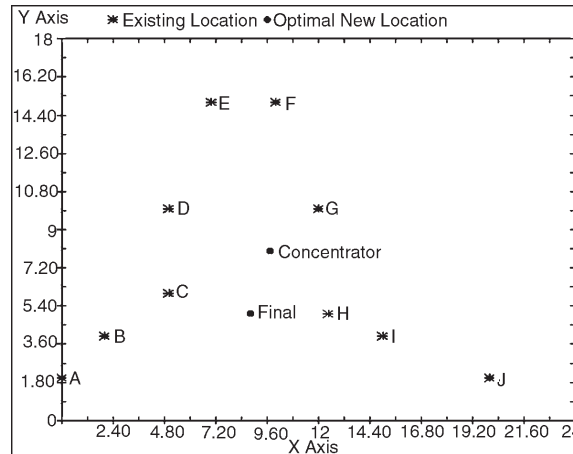
Fig. Euclidean.



So in this case the final and concentrator locations are near to each other.

11-12-2000	New Facility	X Axis	Y Axis
1	Concentrator	9.17	8.59
2	Final	8.88	6.28
Total	Flow to&From	New Location	= 43900
Total	Cost to&From	New Location	= 1,469,664.25
(by	Squared Euclidean	Distance)	

Fig. Squared Euclidean.



So in this case the final and concentrator locations are not particularly close to each other.

IMPORTANCE OF FACILITY LOCATION

- Location of facility decides the production technology and cost structure. Like location in under developed country will have labour intensive technology.
- Location depends on size and nature of business. A large scale industry needs huge investment and option for extension and can not be shifted in future.
- Location affected the firms ability to serve the customers quickly and efficiently.
- Location affects the competitive advantage.
- Optimum location reduces transporation cost, labour cost, taxes, etc.

FACILITY LAYOUT

A typical manufacturing plant has a number of diverse activities interacting with each other. Thus, raw materials arrive at a shipping dock, they are unpacked and checked in a quality control area, they may then be processed through several processing areas, and finally the finished product again passes through the shipping dock.

In addition to areas specifically related to production, there must be dressing rooms, lunch rooms, and restrooms for employees; offices for supervision, design, and production control; and space for inventory and aisles. In fact, a plant may be viewed as a large number of finite geometric areas arranged on the floor space of the building.

The problem of arranging these areas in an effective manner is the facility layout problem. Clearly, the layout problem has relevance in many areas of facility and equipment design, from the layout of the rooms in a home to the layout of chips on an electronic circuit board.

Although the facility layout problem may arise in many contexts, in this part we assume we are dealing with a plant manufacturing products for sale.

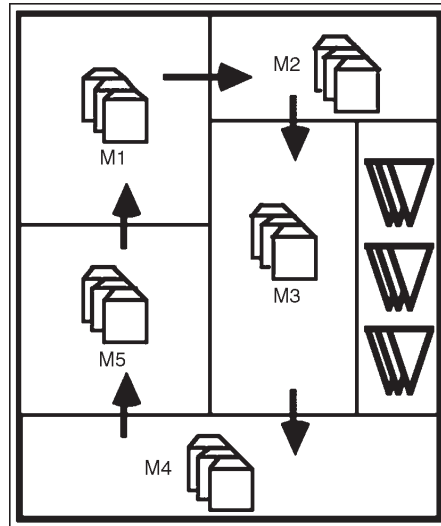


Fig. Facility layout.

FACILITY LAYOUT AND DESIGN

Facility layout and design is an important component of a business's overall operations, both in terms of maximizing the effectiveness of production processes and meeting employee needs and/or desires. Writing in *Production and Operations Management*, Howard J. Weiss and Mark E. Gershon defined facility layout as “the physical arrangement of everything needed for the product or service, including machines, personnel, raw materials, and finished goods. The criteria for a good layout necessarily relate to people, materials, machines, and their interactions.”

FACTORS IN DETERMINING LAYOUT AND DESIGN

Small business owners need to consider many operational factors when building or renovating a facility for maximum layout effectiveness.

These criteria include the following:

- *Ease of future expansion or change:* Facilities should be designed so that they can be easily expanded or adjusted to meet changing production needs. “Although redesigning a facility is a major, expensive undertaking not to be done lightly, there is always the possibility that a redesign will be necessary,” said Weiss and Gershon. “Therefore, any design should be flexible.... Flexible manufacturing systems most often are highly automated facilities having intermediate-volume production of a variety of products. Their goal is to minimize

changeover or setup times for producing the different products while still achieving close to assembly line production rates.”

- *Flow of movement:* The facility design should reflect a recognition of the importance of smooth process flow. In the case of factory facilities, the editors of *How to Run a Small Business* state that “ideally, the plan will show the raw materials entering your plant at one end and the finished product emerging at the other. The flow need not be a straight line. Parallel flows, U-shaped patterns, or even a zig-zag that ends up with the finished product back at the shipping and receiving bays can be functional. However, backtracking is to be avoided in whatever pattern is chosen. When parts and materials move against or across the overall flow, personnel and paperwork become confused, parts become lost, and the attainment of coordination becomes complicated.”
- *Materials handling:* Small business owners should make certain that the facility layout makes it possible to handle materials. In an orderly, efficient—and preferably simple—manner.
- *Output needs:* The facility should be laid out in a way that is conducive to helping the business meet its production needs.
- *Space utilization:* This aspect of facility design includes everything from making sure that traffic lanes are wide enough to making certain that inventory storage warehouses or rooms utilize as much vertical space as possible.
- *Shipping and receiving:* The J.K. Lasser Institute counseled small business owners to leave ample room for this aspect of operations. “While space does tend to fill itself up, receiving and shipping rarely get enough space for the work to be done effectively,” it said in *How to Run a Small Business*.
- *Ease of communication and support:* Facilities should be laid out so that communication within various areas of the business and interactions with vendors and customers can be done in an easy and effective manner. Similarly, support areas should be stationed in areas that help them to serve operating areas.
- *Impact on employee morale and job satisfaction:* Since countless studies have indicated that employee morale has a major impact on productivity, Weiss and Gershon counsel owners and managers to heed this factor when pondering facility design alternatives: “Some ways layout design can increase morale are obvious, such as providing for light-coloured walls, windows, space. Other ways are less obvious and not directly related to the production process. Some examples are including a cafeteria or even a gymnasium in the facility design. Again, though, there are costs to be traded off. That is, does the increase in morale due to a cafeteria increase productivity to the extent that the increased productivity covers the cost of building and staffing the cafeteria.”
- *Promotional value:* If the business commonly receives visitors in the form of customers, vendors, investors, etc., the small business owner may want to

make sure that the facility layout is an attractive one that further burnishes the company's reputation. Design factors that can influence the degree of attractiveness of a facility include not only the design of the production area itself, but the impact that it has on, for instance, ease of fulfilling maintenance/cleaning tasks.

- *Safety*: The facility layout should enable the business to effectively operate in accordance with Occupational Safety and Health Association guidelines and other legal restrictions.

"Facility layout must be considered very carefully because we do not want to constantly redesign the facility," summarized Weiss and Gershon. "Some of the goals in designing the facility are to ensure a minimum amount of materials handling, to avoid bottlenecks, to minimize machine interference, to ensure high employee morale and safety, and to ensure flexibility.

Essentially, there are two distinct types of layout. *Product layout* is synonymous with assembly line and is oriented towards the products that are being made. *Process layout* is oriented around the processes that are used to make the products.

Generally, product layout is applicable for high-volume repetitive operations, while process layout is applicable for low-volume custom-made goods."

DIFFERENCES BETWEEN OFFICE AND FACTORY LAYOUTS

Offices and manufacturing facilities are typically designed in much different ways—a reflection of the disparate products that the two entities make. "A factory produces things," wrote Stephan Konz in *Facility Design*.

"These things are moved with conveyors and lift trucks; factory utilities include gas, water, compressed air, waste disposal, and large amounts of power as well as telephones and computer networks. A layout criterion is minimization of transportation cost." Konz pointed out, however, that the mandate of business offices is to produce information, whether disseminated in physical, electronic or oral form. "Office layout criteria, although hard to quantify, are minimization of communication cost and maximization of employee productivity," wrote Konz.

Layout requirements can also differ dramatically by industry. The needs of service-oriented businesses, for instance, are often predicated on whether customers receive their services at the physical location of the business or whether the business goes to the customer's home or place of business to provide the service. In the latter instances, these businesses will likely have facility layouts that emphasize storage space for equipment, chemicals, and paperwork rather than spacious customer waiting areas.

Manufacturers may also have significantly different facility layouts, depending on the unique needs that they have. After all, the production challenges associated with producing jars of varnish or mountaineering equipment are apt to be considerably different than those of making truck chassis or foam beach toys. Retail outlets comprise yet another business sector that have unique facility layout needs. Such establishments

typically emphasize sales floor space, inventory logistics, foot traffic issues, and overall store attractiveness when studying facility layout issues.

Konz also observed that differences in factory and office layouts can often be traced to user expectations. “Historically, office workers have been much more concerned with status and aesthetics than factory workers,” he noted. “A key consideration in many office layouts is ‘Who will get the best window location?’

To show their status, executives expect, in addition to preferred locations, to have larger amounts of space. Rank expects more privacy and more plush physical surroundings.” In addition, he stated, “Offices are designed to be ‘tasteful’ and to ‘reflect the organization’s approach to business dealings.’ “ Conversely, in the factory setting, aesthetic elements take a back seat to utility.

Given these emphases, it is not surprising that, as a general rule, office workers will enjoy advantages over their material production brethren in such areas as ventilation, lighting, acoustics, and climate control.

Project Management

INTRODUCTION

Project management is the discipline of planning, organizing, and managing resources to bring about the successful completion of specific project goals and objectives. It is sometimes conflated with programme management, however technically a programme is actually a higher level construct: a group of related and somehow interdependent projects.

A project is a temporary endeavour, having a defined beginning and end undertaken to meet unique goals and objectives, usually to bring about beneficial change or added value. The temporary nature of projects stands in contrast to business as usual, which are repetitive, permanent or semi-permanent functional work to produce products or services.

In practice, the management of these two systems is often found to be quite different, and as such requires the development of distinct technical skills and the adoption of separate management.

The primary challenge of project management is to achieve all of the project goals and objectives while honouring the preconceived project constraints. Typical constraints are scope, time, and budget. The secondary—and more ambitious—challenge is to optimize the allocation and integration of inputs necessary to meet pre-defined objectives.

PLANNING LARGE PROJECTS AND PROGRAMMES

However, this approach will only scale up to a certain extent – as projects get larger, they can reach a level of complexity where *ad hoc* approaches to project management become wasteful and inefficient. For these projects, project management becomes a technical discipline in its own right. To run such projects efficiently, project managers use formal project management methodologies such as PMBOK or PRINCE2.

PMBOK, which stands for the “Project Management Body Of Knowledge”, was first published in 1996 as a manual called *A Guide to the Project Management Body of Knowledge*. It is now in its fourth edition. It is a standard that represents generally-recognized good practice, and is published by the Project Management Institute. This is a not-for-profit membership association for the project management profession, which administers the associated Project Management Professional qualifications. PMBOK is the dominant project management methodology used in North America.

PRINCE2, which stands for “PRojects IN Controlled Environments”, is widely used in the UK and in English-speaking countries outside North America, and was originally devised as a “best practice” standard to be used for managing UK government information systems projects. Since then, it has become increasingly widely used for projects of all kinds, in the private as well as the public sector, and although copyright in it is retained by the Crown and managed by the Office of Government Commerce, its methods are in the public domain and are free for organizations to use.

Various accredited training organizations offer PRINCE2 qualifications at Foundation level, for those who need to be familiar with its terminology, and at Practitioner level, which is aimed at project and programme managers.

PRINCE2 is powerful in that it completely clarifies people’s roles in projects, ensures that lines of communication are clear, makes sure that project risk is actively managed, sets up appropriate controls, establishes baseline costs, schedule and scope, and so on. In this, it embodies and codifies much of project management best practice.

KOTTER’S 8-STEP CHANGE MODEL

What was true more than two thousand years ago is just as true today. We live in a world where “business as usual” IS change. New initiatives, project-based working, technology improvements, staying ahead of the competition - these things come together to drive ongoing changes to the way we work.

Whether you’re considering a small change to one or two processes, or a systemwide change to an organization, it’s common to feel uneasy and intimidated by the scale of the challenge.

You know that the change needs to happen, but you don’t really know how to go about doing delivering it. Where do you start? Whom do you involve? How do you see

it through to the end? There are many theories about how to “do” change. Many originate with leadership and change management guru, John Kotter. A professor at Harvard Business School and world-renowned change expert, Kotter introduced his eight-step change process in his 1995.

“We look at his eight steps for leading change”:

- *Step One: Create Urgency:* For change to happen, it helps if the whole company really wants it. Develop a sense of urgency around the need for change. This may help you spark the initial motivation to get things moving. This isn’t simply a matter of showing people poor sales statistics or talking about increased competition. Open an honest and convincing dialogue about what’s happening in the marketplace and with your competition. If many people start talking about the change you propose, the urgency can build and feed on itself.

What you can do:

- Identify potential threats, and develop scenarios showing what could happen in the future.
 - Examine opportunities that should be, or could be, exploited.
 - Start honest discussions, and give dynamic and convincing reasons to get people talking and thinking.
 - Request support from customers, outside stakeholders and industry people to strengthen your argument.
- *Step Two: Form a Powerful Coalition:* Convince people that change is necessary. This often takes strong leadership and visible support from key people within your organization. Managing change isn’t enough - you have to lead it. You can find effective change leaders throughout your organization - they don’t necessarily follow the traditional company hierarchy. To lead change, you need to bring together a coalition, or team, of influential people whose power comes from a variety of sources, including job title, status, expertise, and political importance. Once formed, your “change coalition” needs to work as a team, continuing to build urgency and momentum around the need for change.

What you can do:

- Identify the true leaders in your organization.
 - Ask for an emotional commitment from these key people.
 - Work on team building within your change coalition.
 - Check your team for weak areas, and ensure that you have a good mix of people from different departments and different levels within your company.
- *Step Three: Create a Vision for Change:* When you first start thinking about change, there will probably be many great ideas and solutions floating around. Link these concepts to an overall vision that people can grasp easily and

remember. A clear vision can help everyone understand why you're asking them to do something. When people see for themselves what you're trying to achieve, then the directives they're given tend to make more sense.

What you can do:

- Determine the values that are central to the change.
 - Develop a short summary that captures what you “see” as the future of your organization.
 - Create a strategy to execute that vision.
 - Ensure that your change coalition can describe the vision in five minutes or less.
 - Practice your “vision speech” often.
- *Step Four: Communicate the Vision:* What you do with your vision after you create it will determine your success. Your message will probably have strong competition from other day-to-day communications within the company, so you need to communicate it frequently and powerfully, and embed it within everything that you do. Don't just call special meetings to communicate your vision. Instead, talk about it every chance you get. Use the vision daily to make decisions and solve problems. When you keep it fresh on everyone's minds, they'll remember it and respond to it. It's also important to “walk the talk.” What you do is far more important - and believable - than what you say. Demonstrate the kind of behaviour that you want from others.

What you can do:

- Talk often about your change vision.
 - Openly and honestly address peoples' concerns and anxieties.
 - Apply your vision to all aspects of operations - from training to performance reviews. Tie everything back to the vision.
 - Lead by example.
- *Step Five: Remove Obstacles:* If you follow these steps and reach this point in the change process, you've been talking about your vision and building buy-in from all levels of the organization. Hopefully, your staff wants to get busy and achieve the benefits that you've been promoting. But is anyone resisting the change? And are there processes or structures that are getting in its way? Put in place the structure for change, and continually check for barriers to it. Removing obstacles can empower the people you need to execute your vision, and it can help the change move forward.

What you can do:

- Identify, or hire, change leaders whose main roles are to deliver the change.
- Look at your organizational structure, job descriptions, and performance and compensation systems to ensure they're in line with your vision.
- Recognize and reward people for making change happen.

- Identify people who are resisting the change, and help them see what's needed.
- Take action to quickly remove barriers.
- *Step Six: Create Short-term Wins:* Nothing motivates more than success. Give your company a taste of victory early in the change process. Within a short time frame, you'll want to have results that your staff can see. Without this, critics and negative thinkers might hurt your progress. Create short-term targets - not just one long-term goal. You want each smaller target to be achievable, with little room for failure. Your change team may have to work very hard to come up with these targets, but each "win" that you produce can further motivate the entire staff.

What you can do:

- Look for sure-fire projects that you can implement without help from any strong critics of the change.
- Don't choose early targets that are expensive. You want to be able to justify the investment in each project.
- Thoroughly analyse the potential pros and cons of your targets. If you don't succeed with an early goal, it can hurt your entire change initiative.
- Reward the people who help you meet the targets.
- *Step Seven: Build on the Change:* Kotter argues that many change projects fail because victory is declared too early. Real change runs deep. Quick wins are only the beginning of what needs to be done to achieve long-term change. Launching one new product using a new system is great. But if you can launch 10 products, that means the new system is working. To reach that 10th success, you need to keep looking for improvements. Each success provides an opportunity to build on what went right and identify what you can improve.

What you can do:

- After every win, analyse what went right and what needs improving.
- Set goals to continue building on the momentum you've achieved.
- Learn about kaizen, the idea of continuous improvement.
- Keep ideas fresh by bringing in new change agents and leaders for your change coalition.
- *Step Eight: Anchor the Changes in Corporate Culture:* Finally, to make any change stick, it should become part of the core of your organization. Your corporate culture often determines what gets done, so the values behind your vision must show in day-to-day work. Make continuous efforts to ensure that the change is seen in every aspect of your organization. This will help give that change a solid place in your organization's culture. It's also important that your company's leaders continue to support the change. This includes existing staff and new leaders who are brought in. If you lose the support of these people, you might end up back where you started.

What you can do:

- Talk about progress every chance you get. Tell success stories about the change process, and repeat other stories that you hear.
- Include the change ideals and values when hiring and training new staff.
- Publicly recognize key members of your original change coalition, and make sure the rest of the staff - new and old - remembers their contributions.
- Create plans to replace key leaders of change as they move on. This will help ensure that their legacy is not lost or forgotten.

KEY POINTS

You have to work hard to change an organization successfully. When you plan carefully and build the proper foundation, implementing change can be much easier, and you'll improve the chances of success. If you're too impatient, and if you expect too many results too soon, your plans for change are more likely to fail. Create a sense of urgency, recruit powerful change leaders, build a vision and effectively communicate it, remove obstacles, create quick wins, and build on your momentum. If you do these things, you can help make the change part of your organizational culture. That's when you can declare a true victory. Then sit back and enjoy the change that you envisioned so long ago.

STAKEHOLDER ANALYSIS

"Stakeholder management is critical to the success of every project in every organization I have ever worked with. By engaging the right people in the right way in your project, you can make a big difference to its success... and to your career."

– Rachel Thompson, Experienced Project Manager

As you become more successful in your career, the actions you take and the projects you run will affect more and more people. The more people you affect, the more likely it is that your actions will impact people who have power and influence over your projects. These people could be strong supporters of your work – or they could block it.

Stakeholder Management is an important discipline that successful people use to win support from others. It helps them ensure that their projects succeed where others fail.

There are two major elements to Stakeholder Management: Stakeholder Analysis and Stakeholder Planning. Stakeholder Analysis is the technique used to identify the key people who have to be won over. You then use Stakeholder Planning to build the support that helps you succeed.

The benefits of using a stakeholder-based approach are that:

- You can use the opinions of the most powerful stakeholders to shape your projects at an early stage. Not only does this make it more likely that they will support you, their input can also improve the quality of your project.
- Gaining support from powerful stakeholders can help you to win more resources – this makes it more likely that your projects will be successful.

- By communicating with stakeholders early and often, you can ensure that they know what you are doing and fully understand the benefits of your project – this means they can support you actively when necessary.
- You can anticipate what people's reaction to your project may be, and build into your plan the actions that will win people's support.

HOW TO USE THE TOOL

The first step in Stakeholder Analysis is to identify who your stakeholders are. The next step is to work out their power, influence and interest, so you know who you should focus on. The final step is to develop a good understanding of the most important stakeholders so that you can work out how to win their support. You record this analysis on a stakeholder map. After you have used this tool and created a stakeholder map, you can use the stakeholder planning tool to plan how you will communicate with each stakeholder.

The steps of Stakeholder Analysis are:

- *Identifying Your Stakeholders:* The first step in your stakeholder analysis is to brainstorm who your stakeholders are. As part of this, think of all the people who are affected by your work, who have influence or power over it, or have an interest in its successful or unsuccessful conclusion. The table shows some of the people who might be stakeholders in your job or in your projects:

Your boss	Shareholders	Government
Senior executives	Alliance partners	Trades associations
Your coworkers	Suppliers	The press
Your team	Lenders	Interest groups
Customers	Analysts	The public
Prospective customers	Future recruits	The community
Your family		

Remember that although stakeholders may be both organizations and people, ultimately you is shown in figure, and classify them by their power over your work and by their interest in your work.

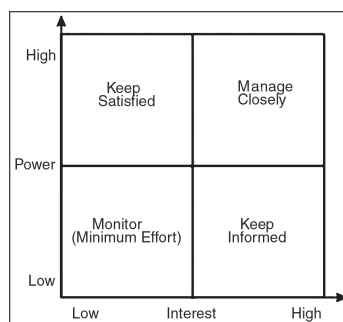


Fig. Power/Interest Grid for Stakeholder Prioritization.

For example, your boss is likely to have high power and influence over your projects and high interest. Your family may have high interest, but are unlikely to have power over it.

Someone's position on the grid shows you the actions you have to take with them:

- *High power, interested people:* These are the people you must fully engage with, and make the greatest efforts to satisfy.
- *High power, less interested people:* Put enough work in with these people to keep them satisfied, but not so much that they become bored with your message.
- *Low power, interested people:* Keep these people adequately informed, and talk to them to ensure that no major issues are arising. These people can often be very helpful with the detail of your project.
- *Low power, less interested people:* Again, monitor these people, but do not bore them with excessive communication.
- *Understanding your key stakeholders:* You now need to know more about your key stakeholders. You need to know how they are likely to feel about and react to your project. You also need to know how best to engage them in your project and how best to communicate with them. Key questions that can help you understand your stakeholders are:
 - What financial or emotional interest do they have in the outcome of your work? Is it positive or negative?
 - What motivates them most of all?
 - What information do they want from you?
 - How do they want to receive information from you? What is the best way of communicating your message to them?
 - What is their current opinion of your work? Is it based on good information?
 - Who influences their opinions generally, and who influences their opinion of you? Do some of these influencers therefore become important stakeholders in their own right?
 - If they are not likely to be positive, what will win them around to support your project?
 - If you don't think you will be able to win them around, how will you manage their opposition?
 - Who else might be influenced by their opinions? Do these people become stakeholders in their own right?

A very good way of answering these questions is to talk to your stakeholders directly – people are often quite open about their views, and asking people's opinions is often the first step in building a successful relationship with them.

You can summarize the understanding you have gained on the stakeholder map, so that you can easily see which stakeholders are expected to be blockers or critics, and which stakeholders are likely to be advocates and supporters of your project. A good

way of doing this is by colour coding: showing advocates and supporters in green, blockers and critics in red, and others who are neutral in orange.

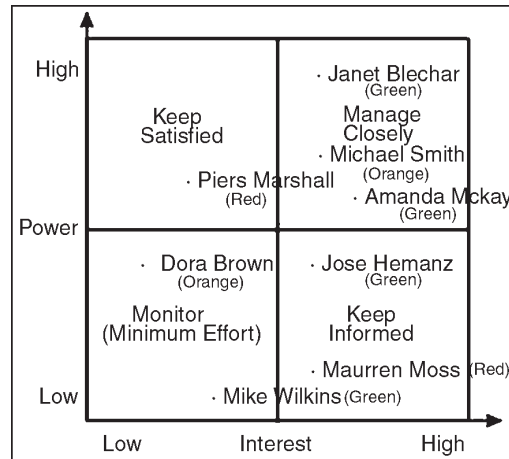


Fig. Example Power/Interest Grid with Stakeholders Marked.

Figure shows an example of this - in this example, you can see that a lot of effort needs to be put into persuading Piers and Michael of the benefits of the project – Janet and Amanda also need to managed well as powerful supporters.

INFLUENCE MAPS

Many people can have influence over your projects. Some influencers are obvious and easy to spot. Others are less obvious, but are no less significant. If you fail to recognize and “manage” these influencers, you’ll most-likely experience unexpected resistance to your projects, and sometimes bewildering failure. This is increasingly the case as you run large projects, and as the number of people affected by your projects increases.

People within your organization, at least, are supposed to work together openly and willingly. However, even here, your boss, your teammates, your customers, your boss’s boss - even the CEO’s nephew in the mailroom - can all impact you, given certain sets of circumstances.

However people outside your organization have all sorts of interests and motivations that you can’t control. Here, knowing who influences who can be critical if you want to get anything done at all.

INFLUENCE MAPPING

So do you understand who has influence over your projects? Do you know the nature, direction, and strength of these influences? After all, using the normal “chain of command” may not always be the best way to advance your objectives: Knowing who the real influencers are can help you determine where you should put your effort if you

really want to succeed. This is what influence mapping is all about - discovering your project's true stakeholders and the influence relationships between them. This helps you target the key influencers so that you can win the resources and support you need to reach your goal.

ELEMENTS OF AN INFLUENCE MAP

An influence map is a visual model showing the people who influence and make decisions about your project. The map helps you understand how stakeholders relate to one-another, so that you can quickly see the way in which influence flows.

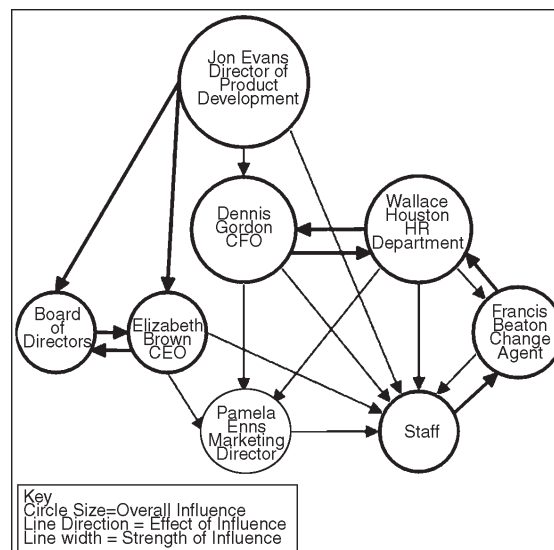
Remember that even the most powerful people rarely perform alone. Top executives and other people in authority rely on advisers. Find out who the advisers are, and understand how they operate. This can be vital to your project's success.

There are three main considerations when you construct an influence map:

1. The importance or weight of a stakeholder's overall influence.
2. The relationships between stakeholders.
3. The amount of influence stakeholders have over others.

Your completed influence map shows the stakeholders with the most influence as individuals with the largest circles. Lines drawn to other stakeholders indicate the presence and strength of influence.

The resulting influence map looks something like this:



This influence map clearly shows how important Jon Evans is to the success of your restructuring plan. It also indicates that you should spend energy on gaining support from Wallace Houston and Dennis Gordon before moving on to other executives. Before you thought about stakeholder influences, you might have assumed that the CEO and

CFO had the most influence on organization-wide change. But the influence map shows you that this is probably not for the case in this situation.

CREATING AN INFLUENCE MAP

Follow these steps to construct an influence map:

- *Step One: Prepare a stakeholder analysis:* This helps you identify, prioritize, and understand your key stakeholders.
- *Step Two: For each stakeholder, find out the following:*
 - Whom does he or she influence, and who influences him or her?
 - How strong is that influence?
 - What is the history of each relationship? How does this impact overall influence?
 - What role does hierarchy play in the amount of influence?
- *Step Three:* Map the importance of influence using the size and position of the circles. The largest circles belong to stakeholders with the most influence.
- *Step Four:* Map the direction of influence by drawing arrows to link the stakeholders..
- *Step Five:* Map the strength of influence by using thicker lines to indicate stronger influence.
- *Step Six:* Study the map, and identify stakeholders with the most overall influence. Form a stakeholder management plan that will allow you to communicate with, and hopefully influence, these important influencers.
- *Step Seven:* Map these influence relationships on a regular basis. This way, you'll better understand the dynamics of decision making relating to your project.

PROGRAMME EVALUATION AND REVIEW TECHNIQUE

The Programme Evaluation and Review Technique, commonly abbreviated PERT, is a model for project management designed to analyse and represent the tasks involved in completing a given project. It is commonly used in conjunction with the critical path method or CPM.

PERT is a method to analyse the involved tasks in completing a given project, especially the time needed to complete each task, and identifying the minimum time needed to complete the total project.

PERT was developed primarily to simplify the planning and scheduling of large and complex projects. It was developed by Bill Pocock of Booz Allen Hamilton and Gordon Perhson of the U.S. Navy Special Projects Office in 1957 to support the U.S. Navy's Polaris nuclear submarine project. It was able to incorporate uncertainty by making it

possible to schedule a project while not knowing precisely the details and durations of all the activities. It is more of an event-oriented technique rather than start- and completion-oriented, and is used more in projects where time, rather than cost, is the major factor. It is applied to very large-scale, one-time, complex, non-routine infrastructure and Research and Development projects.

This project model was the first of its kind, a revival for scientific management, founded by Frederick Taylor and later refined by Henry Ford. DuPont corporation's critical path method was invented at roughly the same time as PERT.

CONVENTIONS

- A PERT chart is a tool that facilitates decision making; The first draft of a PERT chart will number its events sequentially in 10s to allow the later insertion of additional events.
- Two consecutive events in a PERT chart are linked by activities, which are conventionally represented as arrows.
- The events are presented in a logical sequence and no activity can commence until its immediately preceding event is completed.
- The planner decides which milestones should be PERT events and also decides their "proper" sequence.
- A PERT chart may have multiple pages with many sub-tasks.

PERT is valuable to manage where multiple tasks are going simultaneously to reduce the redundancy

TERMINOLOGY

- *PERT event*: A point that marks the start or completion of one or more activities. It consumes no time, and uses no resources. When it marks the completion of one or more tasks, it is not "reached" until all of the activities leading to that event have been completed.
- *Predecessor event*: An event that immediately precedes some other event without any other events intervening. An event can have multiple predecessor events and can be the predecessor of multiple events.
- *Successor event*: An event that immediately follows some other event without any other intervening events. An event can have multiple successor events and can be the successor of multiple events.
- *PERT activity*: The actual performance of a task which consumes time and requires resources. It can be understood as representing the time, effort, and resources required to move from one event to another. A PERT activity cannot be performed until the predecessor event has occurred.
- *Optimistic time*: The minimum possible time required to accomplish a task, assuming everything proceeds better than is normally expected
- *Pessimistic time*: The maximum possible time required to accomplish a task, assuming everything goes wrong.

- *Most likely time*: The best estimate of the time required to accomplish a task, assuming everything proceeds as normal.
- *Expected time*: The best estimate of the time required to accomplish a task, assuming everything proceeds as normal.

$$T_E = (O + 4M + P) \div 6$$

- Float or Slack is the amount of time that a task in a project network can be delayed without causing a delay - Subsequent tasks –or Project Completion – (total float)
- *Critical Path*: The longest possible continuous pathway taken from the initial event to the terminal event. It determines the total calendar time required for the project; and, therefore, any time delays along the critical path will delay the reaching of the terminal event by at least the same amount.
- *Critical Activity*: An activity that has total float equal to zero. Activity with zero float does not mean it is on the critical path.
- *Lead time*: The time by which a predecessor event must be completed in order to allow sufficient time for the activities that must elapse before a specific PERT event reaches completion.
- *Lag time*: The earliest time by which a successor event can follow a specific PERT event.
- *Slack*: The slack of an event is a measure of the excess time and resources available in achieving this event. Positive slack would indicate ahead of schedule; negative slack would indicate behind schedule; and zero slack would indicate on schedule.
- *Fast tracking*: Performing more critical activities in parallel
- *Crashing critical path*: Shortening duration of critical activities

ADVANTAGES

- PERT chart explicitly defines and makes visible dependencies between the WBS elements
- PERT facilitates identification of the critical path and makes this visible
- PERT facilitates identification of early start, late start, and slack for each activity,
- PERT provides for potentially reduced project duration due to better understanding of dependencies leading to improved overlapping of activities and tasks where feasible.

DISADVANTAGES

- There can be potentially hundreds or thousands of activities and individual dependency relationships
- The network charts tend to be large and unwieldy requiring several pages to print and requiring special size paper

- The lack of a timeframe on most PERT/CPM charts makes it harder to show status although colours can help
- When the PERT/CPM charts become unwieldy, they are no longer used to manage the project.

UNCERTAINTY IN PROJECT SCHEDULING

During project execution, however, a real-life project will never execute exactly as it was planned due to uncertainty. It can be ambiguity resulting from subjective estimates that are prone to human errors or it can be variability arising from unexpected events or risks. And Project Evaluation and Review Technique may provide inaccurate information about the project completion time for main reason uncertainty. This inaccuracy is large enough to render such estimates as not helpful.

One possibility to maximize solution robustness is to include safety in the baseline schedule in order to absorb the anticipated disruptions. This is called proactive scheduling. A pure proactive scheduling is an utopia, incorporating safety in a baseline schedule that allows to cope with every possible disruption would lead to a baseline schedule with a very large makespan. A second approach, reactive scheduling, consists of defining a procedure to react to disruptions that cannot be absorbed by the baseline schedule.

RESOURCE LEVELLING

Resource levelling is a project management process used to examine unbalanced use of resources over time, and for resolving over-allocations or conflicts.

When performing project planning activities, the manager will attempt to schedule certain tasks simultaneously. When more resources such as machines or people are needed than are available, or perhaps a specific person is needed in both tasks, the tasks will have to be rescheduled concurrently or even sequentially to manage the constraint. Project planning resource levelling is the process of resolving these conflicts. It can also be used to balance the workload of primary resources over the course of the project[s], usually at the expense of one of the traditional triple constraints.

When using specially designed project software, levelling typically means resolving conflicts or over allocations in the project plan by allowing the software to calculate delays and update tasks automatically. Project management software levelling requires delaying tasks until resources are available. In more complex environments, resources could be allocated across multiple, concurrent projects thus requiring the process of resource levelling to be performed at company level. In either definition, levelling could result in a later project finish date if the tasks affected are in the critical path.

Resource Levelling is also useful in the world of maintenance management. Many organizations have maintenance backlogs. These backlogs consist of work orders. In a

“planned state” these work orders have estimates such as 2 electricians for 8 hours. These work orders have other attributes such as report date, priority, asset operational requirements, and safety concerns. These same organizations have a need to create weekly schedules. Resource-levelling can take the “work demand” and balance it against the resource pool availability for the given week. The goal is to create this weekly schedule in advance of performing the work. Without resource-levelling the organization is most likely performing subjective selection. For the most part, when it comes to maintenance scheduling, there are very few logic ties and therefore no need to calculate critical path and total float.

SCHEDULE

In project management, a schedule consists of a list of a project’s terminal elements with intended start and finish dates. Terminal elements are the lowest element in a schedule, which is not further subdivided. Those items are often estimated in terms of resource requirements, budget and duration, linked by dependencies and scheduled.

Before a project schedule can be created, a project manager should typically have a work breakdown structure, an effort estimate for each task, and a resource list with availability for each resource. If these are not yet available, it may be possible to create something that looks like a schedule, but it will essentially be a work of fiction. They can be created using a consensus-driven estimation method like Wideband Delphi. The reason for this is that a schedule itself is an estimate: each date in the schedule is estimated, and if those dates do not have the buy-in of the people who are going to do the work, the schedule will be inaccurate. In many industries, such as engineering and construction, the development and maintenance of the project schedule is the responsibility of a full time scheduler or team of schedulers, depending on the size of the project. And though the techniques of scheduling are well developed, they are inconsistently applied throughout industry.

Standardization and promotion of scheduling best practices are being pursued by the Association for the Advancement of Cost Engineering, the Project Management Institute. In some large corporations, scheduling, as well as cost, estimating, and risk management are organized under the department of project controls. Many project scheduling software products exist which can do much of the tedious work of calculating the schedule automatically, and plenty of books and tutorials dedicated to teaching people how to use them. However, before a project manager can use these tools, he or she should understand the concepts behind the WBS, dependencies, resource allocation, critical paths, Gantt charts and earned value. These are the real keys to planning a successful project.

In order for a project schedule to be healthy, the following criteria must be met:

- The schedule must be constantly updated.

- The EAC value must be equal to the baseline value.
- The remaining effort must be appropriately distributed among team members, taking into consideration vacations.

PROCESS MIX

Process is another element of the extended marketing mix, or 7P's. There are a number of perceptions of the concept of process within the business and marketing literature. Some see processes as a means to achieve an outcome, for example - to achieve a 30% market share a company implements a marketing planning process. Another view is that marketing has a number of processes that integrate together to create an overall marketing process, for example - telemarketing and Internet marketing can be integrated. A further view is that marketing processes are used to control the marketing mix, *i.e.*, processes that measure the achievement marketing objectives. All views are understandable, but not particularly customer focused.

For the purposes of the marketing mix, process is an element of service that sees the customer experiencing an organisation's offering. It's best viewed as something that your customer participates in at different points in time. Here are some examples to help you build a picture of marketing process, from the customer's point of view.

Going on a cruise - from the moment that you arrive at the dockside, you are greeted; your baggage is taken to your room. You have two weeks of services from restaurants and evening entertainment, to casinos and shopping. Finally, you arrive at your destination, and your baggage is delivered to you. This is a highly focused marketing process

Booking a flight on the Internet - the process begins with you visiting an airline's web site. You enter details of your flights and book them. Your ticket/booking reference arrive by e-mail or post. You catch your flight on time, and arrive refreshed at your destination. This is all part of the marketing process.

At each stage of the process, markets:

- Deliver value through all elements of the marketing mix. Process, physical evidence and people enhance services.
- Feedback can be taken and the mix can be altered.
- Customers are retained, and other serves or products are extended and marked to them.
- The process itself can be tailored to the needs of different individuals, experiencing a similar service at the same time.

Processes essentially have inputs, throughputs and outputs. Marketing adds value to each of the stages. Take a look at the lesson on value chain analysis to consider a series of processes at work.

PRODUCT MIX

Product mix is a combination of products manufactured or traded by the same business house to reinforce their presence in the market, increase market share and increase the turnover for more profitability. Normally the product mix is within the synergy of other products for a medium size organization. However large groups of Industries may have diversified products within core competency. *Larsen and Toubro Ltd, Godrej, Reliance in India are some of the examples.*

One of the realities of business is that most firms deal with multi-products. This helps a firm diffuse its risk across different product groups/Also it enables the firm to appeal to a much larger group of customers or to different needs of the same customer group. So when Videocon chose to diversify into other consumer durables like music systems, washing machines and refrigerators, it sought to satisfy the needs of the middle and upper middle income group of consumers.

Likewise, Bajaj Electricals. A household name in India, has almost ninety products in its portfolio ranging from low value items like bulbs to high priced consumer durables like mixers and luminaires and lighting projects. The number of products carried by a firm at a given point of time is called its product mix. This product mix contains product lines and product items. In other words it's a composite of products offered for sale by a firm.

PRODUCT MIX DECISIONS

Often firms take decisions to change their product mix. These decisions are dictated by the factors and also by the changes occurring in the market place. Like the changing lifestyles of Indian consumers led BPL-Sanyo to launch an entire range of white goods like refrigerators, washing machines, and microwave ovens. It also motivate the firm to launch other entertainment electronics.

Rahejas, a well-known builders firm in Bombay, took a major decision to convert one of its theatre buildings in the western suburbs of Bombay into a large garments and accessories store for men, women and children, perhaps the first of its kind in India to have almost all products required by these customer groups Competition from low priced washing powders forced Hindustan Levers to launch different brands of detergent powder at different price levels positioned at different market segments.

Customer preferences for herbs, mainly shikakai motivated Lever to launch black Sunsilk Shampoo, which has shikakai. Also, low purchasing power and cultural bias against shampoo market made Hindustan Lever consider smaller packaging mainly sachets, for single use. So, it is the changes or anticipated changes in the market place that motivates a firm to consider changes in its product mix.

PRODUCT-MIX MANAGEMENT AND RESPONSIBILITIES

It is extremely important for any organization to have a well-managed product mix. Most organizations break down managing the product mix, product line, and actual

product into three different levels. Product-mix decisions are concerned with the combination of product lines offered by the company. Management of the companies' product mix is the responsibility of top management.

Some basic product-mix decisions include:

- Reviewing the mix of existing product lines;
- Adding new lines to and deleting existing lines from the product mix;
- Determining the relative emphasis on new versus existing product lines in the mix;
- Determining the appropriate emphasis on internal development versus external acquisition in the product mix;
- Gauging the effects of adding or deleting a product line in relationship to other lines in the product mix; and
- Forecasting the effects of future external change on the company's product mix.

Product-line decisions are concerned with the combination of individual products offered within a given line. The product-line manager supervises several product managers who are responsible for individual products in the line. Decisions about a product line are usually incorporated into a marketing plan at the divisional level. Such a plan specifies changes in the product lines and allocations to products in each line.

Generally, product-line managers have the following responsibilities:

- Considering expansion of a given product line;
- Considering candidates for deletion from the product line;
- Evaluating the effects of product additions and deletions on the profitability of other items in the line; and
- Allocating resources to individual products in the line on the basis of marketing strategies recommended by product managers.

Decisions at the first level of product management involve the marketing mix for an individual brand/product. These decisions are the responsibility of a brand manager. Decisions regarding the marketing mix for a brand are represented in the product's marketing plan. The plan for a new brand would specify price level, advertising expenditures for the coming year, coupons, trade discounts, distribution facilities, and a five-year statement of projected sales and earnings.

The plan for an existing product would focus on any changes in the marketing strategy. Some of these changes might include the product's target market, advertising and promotional expenditures, product characteristics, price level, and recommended distribution strategy.

GENERAL MANAGEMENT WORKFLOW

Top management formulates corporate objectives that become the basis for planning the product line. Product-line managers formulate objectives for their line to guide brand managers in developing the marketing mix for individual brands. Brand strategies

are then formulated and incorporated into the product-line plan, which is in turn incorporated into the corporate plan. The corporate plan details changes in the firm's product lines and specifies strategies for growth. Once plans have been formulated, financial allocations flow from top management to product line and then to brand management for implementation.

Implementation of the plan requires tracking performance and providing data from brand to product line to top management for evaluation and control. Evaluation of the current plan then becomes the first step in the next planning cycle, since it provides a basis for examining the company's current offerings and recommending modifications as a result of past performance.

PRODUCT-MIX ANALYSIS

Since top management is ultimately responsible for the product mix and the resulting profits or losses, they often analyse the company product mix. The first assessment involves the area of opportunity in a particular industry or market.

Opportunity is generally defined in terms of current industry growth or potential attractiveness as an investment. The second criterion is the company's ability to exploit opportunity, which is based on its current or potential position in the industry.

The company's position can be measured in terms of market share if it is currently in the market, or in terms of its resources if it is considering entering the market. These two factors—opportunity and the company's ability to exploit it—provide four different options for a company to follow.

- High opportunity and ability to exploit it result in the firm's introducing new products or expanding markets for existing products to ensure future growth.
- Low opportunity but a strong current market position will generally result in the company's attempting to maintain its position to ensure current profitability.
- High opportunity but a lack of ability to exploit it results in either
 - Attempting to acquire the necessary resources or
 - Deciding not to further pursue opportunity in these markets.
- Low opportunity and a weak market position will result in either
 - Avoiding these markets or
 - Divesting existing products in them.

These options provide a basis for the firm to evaluate new and existing products in an attempt to achieve balance between current and future growth. This analysis may cause the product mix to change, depending on what management decides.

The most widely used approach to product portfolio analysis is the model developed by the Boston Consulting Group. The BCG analysis emphasizes two main criteria in evaluating the firm's product mix: the market growth rate and the product's relative market share.

BCG uses these two criteria because they are closely related to profitability, which is why top management often uses the BCG analysis. Proper analysis and conclusions may lead to significant changes to the company's product mix, product line, and product offerings.

The market growth rate represents the products' category position in the product life cycle. Products in the introductory and growth phases require more investment because of research and development and initial marketing costs for advertising, selling, and distribution.

This category is also regarded as a high-growth area. Relative market share represents the company's competitive strength. Market share is compared to that of the leading competitor.

Once the analysis has been done using the market growth rate and relative market share, products are placed into one of four categories:

1. *Stars:* Products with high growth and market share are known as stars. Because these products have high potential for profitability, they should be given top priority in financing, advertising, product positioning, and distribution. As a result, they need significant amounts of cash to finance rapid growth and frequently show an initial negative cash flow.
2. *Cash cows:* Products with a high relative market share but in a low growth position are cash cows. These are profitable products that generate more cash than is required to produce and market them. Excess cash should be used to finance high-opportunity areas. Strategies for cash cows should be designed to sustain current market share rather than to expand it. An expansion strategy would require additional investment, thus decreasing the existing positive cash flow.
3. *Problem children:* These products have low relative market share but are in a high-growth situation. They are called "problem children" because their eventual direction is not yet clear. The firm should invest heavily in those that sales forecasts indicate might have a reasonable chance to become stars. Otherwise divestment is the best course, since problem children may become dogs and thereby candidates for deletion.
4. *Dogs:* Products in the category are clearly candidates for deletion. Such products have low market shares and unlike problem children, have no real prospect for growth. Eliminating a dog is not always necessary, since there are strategies for dogs that could make them profitable in the short term. These strategies involve "harvesting" these products by eliminating marketing support and selling the product only to intensely loyal consumers who will buy in the absence of advertising. However, over the long term companies will seek to eliminate dogs.

As can be seen from the description of the four BCG alternatives, products are evaluated as producers or users of cash. Products with a positive cash flow will finance

high-opportunity products that need cash. The emphasis on cash flow stems from management's belief that it is better to finance new entries and to support existing products with internally produced funds than to increase debt or equity in the company. Based on this belief, companies will normally take money from cash cows and divert it to stars and to some problem children.

The hope is that the stars will turn into cash cows and the problem children will turn into stars. The dogs will continue to receive lower funding and eventually be dropped.

CRITICAL PATH ANALYSIS AND PERT CHARTS

Critical Path Analysis and PERT are powerful tools that help you to schedule and manage complex projects. They were developed in the 1950s to control large defence projects, and have been used routinely since then.

As with Gantt Charts, Critical Path Analysis or the Critical Path Method helps you to plan all tasks that must be completed as part of a project. They performance as the basis both for preparation of a schedule, and of resource planning. During management of a project, they allow you to monitor achievement of project goals. They help you to see where remedial action needs to be taken to get a project back on course.

Within a project it is likely that you will display your final project plan as a Gantt Chart. The benefit of using CPA within the planning process is to help you develop and test your plan to ensure that it is robust. Critical Path Analysis formally identifies tasks which must be completed on time for the whole project to be completed on time. It also identifies which tasks can be delayed if resource needs to be reallocated to catch up on missed or overrunning tasks. The disadvantage of CPA, if you use it as the technique by which your project plans are communicated and managed against, is that the relation of tasks to time is not as immediately obvious as with Gantt Charts. This can make them more difficult to understand. A further benefit of Critical Path Analysis is that it helps you to identify the minimum length of time needed to complete a project. Where you need to run an accelerated project, it helps you to identify which project steps you should accelerate to complete the project within the available time.

HOW TO USE THE TOOL

As with Gantt Charts, the essential concept behind Critical Path Analysis is that you cannot start some activities until others are finished. These activities need to be completed in a sequence, with each stage being more-or-less completed before the next stage can begin.

These are 'sequential' activities. Other activities are not dependent on completion of any other tasks. You can do these at any time before or after a particular stage is reached. These are non-dependent or 'parallel' tasks.

DRAWING A CRITICAL PATH ANALYSIS CHART

Use the following steps to draw a CPA Chart:

- *Step 1. List all activities in the plan:* For each activity, show the earliest start date, estimated length of time it will take, and whether it is parallel or sequential. If tasks are sequential, show which stage they depend on.
- *Step 2. Plot the activities as a circle and arrow diagram:* Critical Path Analyses are presented using circle and arrow diagrams. In these, circles show events within the project, such as the start and finish of tasks. The number shown in the left hand half of the circle allows you to identify each one easily. Circles are sometimes known as nodes. An arrow running between two event circles shows the activity needed to complete that task. A description of the task is written underneath the arrow. The length of the task is shown on it. By convention, all arrows run left to right. Arrows are also sometimes called arcs.

An example of a very simple diagram is shown:

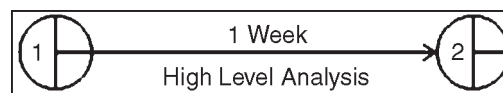


Fig. Simple Circle and Arrow Diagram.

This shows the start event and the completion of the ‘High Level Analysis’ task. The arrow between them shows the activity of carrying out the High Level Analysis. This activity should take 1 week. Where one activity cannot start until another has been completed, we start the arrow for the dependent activity at the completion event circle of the previous activity. An example of this is shown:

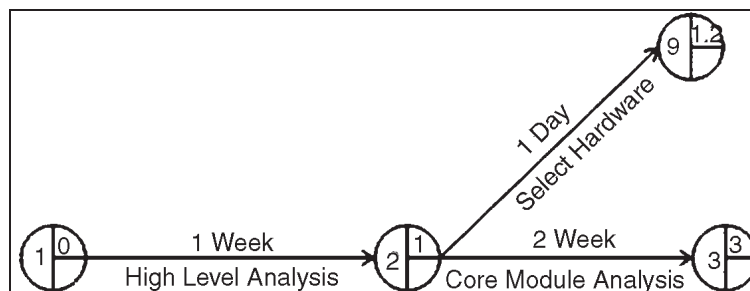


Fig. Circle and Arrow Diagram Showing Two Activities that cannot be Started until the First Activity has been Completed.

Here the activities of ‘Select Hardware’ and ‘Core Module Analysis’ cannot be started until ‘High Level Analysis’ has been completed.

This diagram also brings out a number of other important points:

- Within Critical Path Analysis, we refer to activities by the numbers in the circles at each end. For example, the task ‘Core Module Analysis’ would be called activity 2 to 3. ‘Select Hardware’ would be activity 2 to 9.

- Activities are not drawn to scale. In the diagram, activities are 1 week long, 2 weeks long, and 1 day long. Arrows in this case are all the same length.
- In the example, you can see a second number in the top, right hand quadrant of each circle. This shows the earliest start time for the following activity. It is conventional to start at 0. Here units are whole weeks.

A different case is shown:

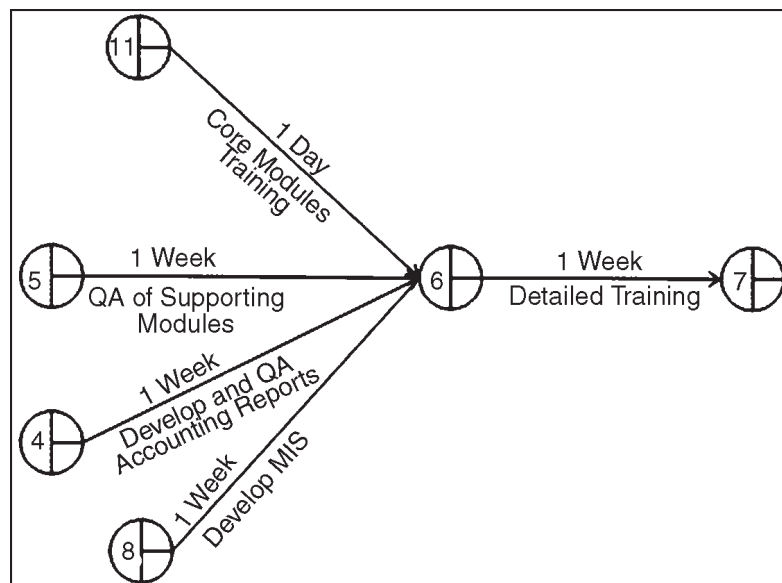


Fig. Circle and Arrow Diagram Showing an Activity (6to7) that cannot start until other Activities have been Completed.

Here activity 6 to 7 cannot start until the other four activities have been completed. The full circle and arrow diagram for the computer project we are using as an example. This shows all the activities that will take place as part of the project. Notice that each event circle also has a figure in the bottom, right hand quadrant.

This shows the latest finish time that's permissible for the preceding activity if the project is to be completed in the minimum time possible. You can calculate this by starting at the last event and working backwards.

The latest finish time of the preceding event and the earliest start time of the following event will be the same for circles on the critical path.

You can see that event M can start any time between weeks 6 and 8. The timing of this event is not critical. Events 1 to 2, 2 to 3, 3 to 4, 4 to 5, 5 to 6 and 6 to 7 must be started and completed on time if the project is to be completed in 10 weeks.

This is the 'critical path' – these activities must be very closely managed to ensure that activities are completed on time. If jobs on the critical path slip, immediate action should be taken to get the project back on schedule. Otherwise completion of the whole project will slip.

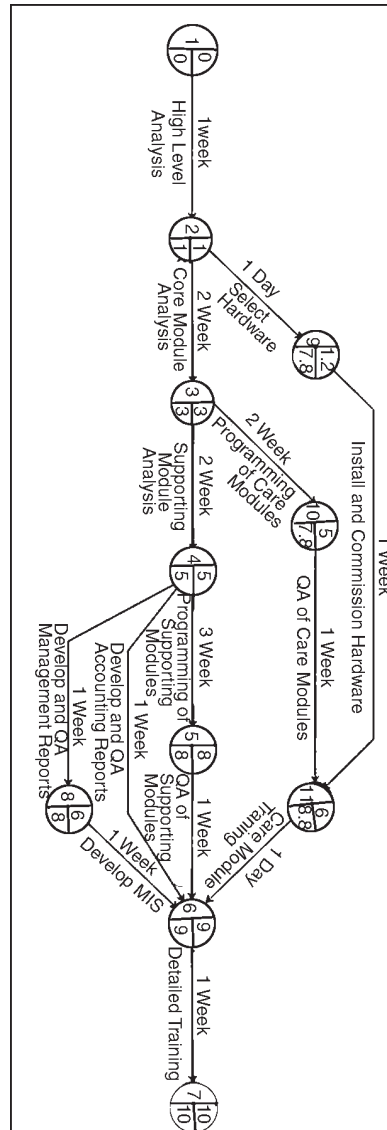


Fig. Critical Path Analysis for Exapmle Computer Project.

CRASH ACTION

You may find that you need to complete a project earlier than your Critical Path Analysis says is possible. In this case you need to re-plan your project. You have a number of options and would need to assess the impact of each on the project's cost, quality and time required to complete it.

For example, you could increase resource available for each project activity to bring down time spent on each but the impact of some of this would be insignificant and a

more efficient way of doing this would be to look only at activities on the critical path. As an example, it may be necessary to complete the computer project in Figure weeks rather than 10 weeks.

In this case you could look at using two analysts in activities 2 to 3 and 3 to 4. This would shorten the project by two weeks, but may raise the project cost – doubling resources at any stage may only improve productivity by, say, 50% as additional time may need to be spent getting the team members up to speed on what is required, coordinating tasks split between them, integrating their contributions, etc.

In some situations, shortening the original critical path of a project can lead to a different series of activities becoming the critical path. For example, if activity 4 to 5 were reduced to 1 week, activities 4 to 8 and 8 to 6 would come onto the critical path.

As with Gantt Charts, in practice project managers use software tools like Microsoft Project to create CPA Charts. Not only do these ease make them easier to draw, they also make modification of plans easier and provide facilities for monitoring progress against plans.

PERT

PERT is a variation on Critical Path Analysis that takes a slightly more sceptical view of time estimates made for each project stage. To use it, estimate the shortest possible time each activity will take, the most likely length of time, and the longest time that might be taken if the activity takes longer than expected.

Use the formula to calculate the time to use for each project stage:

$$\frac{\text{shortest time} + 4 \times \text{likely time} + \text{longest time}}{6}$$

This helps to bias time estimates away from the unrealistically short time-scales normally assumed.

KEY POINTS

Critical Path Analysis is an effective and powerful method of assessing:

- What tasks must be carried out.
- Where parallel activity can be performed.
- The shortest time in which you can complete a project.
- Resources needed to execute a project.
- The sequence of activities, scheduling and timings involved.
- Task priorities.
- The most efficient way of shortening time on urgent projects.

An effective Critical Path Analysis can make the difference between success and failure on complex projects. It can be very useful for assessing the importance of problems faced during the implementation of the plan. PERT is a variant of Critical Path Analysis that takes a more sceptical view of the time needed to complete each project stage.

LOGFRAMES, AND THE LOGICAL FRAMEWORK APPROACH

In practice, even the best project managers can find it difficult to plan major projects without missing important activities, and without failing to spot all significant risks and issues. What's more, once you're immersed in the detail of project planning, it's hard to keep site of the big picture: What are you trying to achieve and why? What are the risks and assumptions? And how you can tell whether the project is a success once it's implemented?

The Logical Framework Approach is a useful technique for helping you do these things, thereby making your projects more robust and coherent – and more successful. The Logical Framework Approach was developed in the 1970s as a tool for strategic planning, using the ideas of Management by Objectives. It's a tool of choice used by development agencies and in the international donor community.

Large aid organizations throughout the world use the LFA for planning, approving, evaluating and monitoring their projects. That said, this is a powerful and useful technique, and is one that richly deserves much wider application than in international development alone.

LOGICAL FRAMEWORK APPROACH AND THE LOGFRAME

The Logical Framework Approach elegantly weaves together top-down and bottom-up approaches to project management. It brings together the classical, top-down, “waterfall approach” for identifying the activities in a project, with a rigorous bottom-up checking process to make sure that these activity lists are comprehensive. It then reinforces this with a rigorous risks and assumptions analysis, which is again thoroughly checked. And it concludes by identifying the controls needed to monitor and manage the project through to successful conclusion. It does this within the framework of the Logframe Matrix, shown in figure.

This cross-references seven key areas of the project to ensure that the key questions are asked:

- *Goal:* What results do we expect?
- *Purpose:* Why are we doing this?
- *Outputs:* What are the deliverables?
- *Activities:* what will we do to deliver the outputs?
- *Indicators of Achievement:* How will we know we've been successful?
- *Means of Verification:* How will we check our reported results?
- *Risks and Assumptions:* What assumptions underlie the structure of our project and what is the risk they will not prevail?

The answers to these questions are put into a Logical Framework Matrix and become the output of the Logical Framework Analysis exercise. The Logframe is a four by four matrix:

Table. The Logframe Matrix.

Project Achievement	Indicators of Verification	Means of Risks and	Important	Summary Assumptions
Goal:				
Purpose:				
Outputs:				
Activities:				

The process has significant value for any size of project. It helps identify the big picture and allows you to see how other items cascade down from it. As well, it helps flesh out the core assumptions that are used in the project development process.

USING A LOGFRAME

Carry out the following steps in consultation with your stakeholders, after you've completed a thorough analysis of the situation. By involving stakeholders, you'll end up with a much more robust analysis of the project than you would on your own.

- *Step 1: Identifying Outputs and Activities:* The first step is to brainstorm the outputs and activities required by the project, starting with the project goal. Do this in the Project Summary column of the Logframe. Start by defining the Goal and Purpose of the project and, from these, identify the outputs and the activities required:
 - *Goal:* What is the “to be” state of the project? What are you trying to achieve?
 - *Purpose:* What good will you do by achieving the goal? Who are the beneficiaries? What is the underlying motivation for starting the project in the first place?
 - *Outputs:* What specific things will be delivered as a result of this project? In order for the project to be considered a success, what changes must be made, and what will the result be?
 - *Activities:* What will actually be done in order to deliver the intended outputs? The Logframe is not intended as an implementation guide, so this part is typically presented in bullet point form.
- *Step 2: Verify the Vertical Logic* Next, we take a bottom-up approach to checking that this list of activities will deliver the desired results - after all, it's possible that activities have been missed, or that the actual results of these activities may not be the ones wanted. This checking process is an important part of making sure that your project plan is robust. Column one shows a hierarchy of objectives, so it is important to check that actions identified deliver the results wanted. Check the logic in column one by using an if/then test as follows. Starting with your activities, ensure that:
 - *IF you complete the activity, Then the outputs will occur.* You want to make sure your activities and outputs are directly linked.

- *IF your outputs are achieved, Then the purpose of your project will be satisfied.* Are the planned outputs closely tied to your purpose? Make sure the beneficiaries you identified in your purpose actually receive the beneficial outcome desired.
- *IF your purpose is satisfied, Then the goal of the project is achieved.* Examine your purpose and goal to make sure that the purpose fully incorporates the intent within the goal. If, in this step, you find that activities and outputs are missing or are wrong, add or adjust them appropriately. And bear in mind that if you identify issues with elements higher up in this hierarchy, you'll need to go back to Step 1 and identify appropriate outcomes and activities for those elements.
- *Step 3: Identify the Risks and Assumptions of your plan* We now cross over to the other side of the Logframe to identify risks associated with the project, and possible false assumptions that may undermine it. There are any number of external factors that can throw projects off course. In the planning and design phase, it is prudent to identify the major assumptions you've used and the degree or risk associated with them. For each of the points in the project's structure, identify the assumptions you're making and look at the associated risks. To define your assumptions, ask "What actions or variables must exist for the project to start and proceed as planned?" Start at the bottom and work up.
 - *Activity Assumptions:* What do you need to happen for your activities to be completed successfully? And what conditions and resources are you assuming will be in place?
 - *Output Assumptions:* What factors outside of your control must be present to achieve the outputs you need?
 - *Purpose Assumptions:* To achieve the purpose, what external factors do you need to have in place?
 - *Goal Assumptions:* What are the necessary conditions for long-term viability of the project goal?
 - Clarify these assumptions with stakeholders immediately, if you can. If you can't, make sure you have early activities in place within your project plan to confirm that your assumptions are correct.

Next, repeat this process looking at risks. Make sure you plan in all of the activities needed to manage or eliminate risk, and if risk can neither be managed or eliminated, make sure that it's clearly identified so that it can be evaluated in the next step.
- *Step 4: Verify the Logic of the Risks and Assumptions* Once you have identified assumptions and risks, you need to check them to determine:
 - Whether your assumptions will link one level of the project to the next; and

- Whether risks are too large. First of all, check that your assumptions are logical using an if/and/then analysis. Start at the bottom and work up to ensure:
 - IF the activity is completed successfully, AND the assumptions underlying it are true, THEN the output will be delivered.
 - IF the output is delivered, AND the assumptions underlying it are true, THEN the purpose will be achieved.
 - IF the purpose is achieved, AND the assumptions underlying it are true, THEN the goal will be achieved. Then, check some additional points related to your risk and assumption analysis:
 - a. Make sure you have identified as many assumptions and risks as possible. Have you talked to everyone involved? Have you looked at the project from all angles?
 - b. Make sure your assumptions are stated specifically and are not too vague. You can't assess risk accurately if you are working with generalities.
 - c. Do you have plans at each level to manage the risks you have identified?
 - d. If the risks you're not able to manage are too high, consider redesigning the project or, if you still can't reduce these to sensible levels, reconsider the project's viability. Again, where this process exposes issues with your Logframe, update it appropriately.
- *Step 5: Determine the Indicators of Achievement and Means of Verification*

When you are satisfied with the structure of the Logframe so far, and are comfortable that you can manage the risks related to your assumptions, you can move on to think about how you will monitor progress towards success. Performance indicators are the specific measures used to monitor this progress. Here are the criteria for a good indicator of achievement:

 - *Valid*: It must measure the intended result.
 - *Reliable*: The measure must be consistently attained over time.
 - *Sensitive*: The measure should respond to changes, and should sufficiently-quickly identify if things are going wrong.
 - *Simple*: The measure should be easy to collect or perform.
 - *Useful*: It must help with decision making or provide information for future learning.
 - *Affordable*: You need to be able to afford the financial and time costs involved in taking the measurement on a regular basis.

Using these criteria, for each goal, purpose, output and activity, indicate what will be used to determine whether it was successfully achieved. Also note who will be responsible for setting these targets. Then indicate exactly how you will verify that achievement. What sources of data will you use? How will you collect the data? How often? Make sure that appropriate activities are in place within your plan to set up and manage these monitoring systems.

KEY POINTS

The Logical Framework Approach is a great technique for making sure that your project plan is robust and coherent. By using it, you significantly increase the likelihood that your project will be successful. Firstly, it provides a useful framework for working through the design of your project with key stakeholders, making sure that you can take full advantage of their knowledge, insights and experience.

Secondly, it provides a useful process for testing and checking your project plan, making sure that it contains all the necessary activities, is based on sound assumptions, and fairly weighs and manages the risks inherent within the project. Thirdly, it helps you ensure that appropriate control measures are embedded within the project, meaning that you can quickly identify where things are going wrong, and take appropriate corrective action.

ESTIMATING TIME ACCURATELY

Accurate time estimation is a skill essential to good project management.

It is important to get time estimates right for two main reasons:

1. Time estimates drive the setting of deadlines for delivery of projects, and hence peoples' assessments of your reliability
2. They often determine the pricing of contracts and hence their profitability.

Usually people vastly underestimate the amount of time needed to implement projects. This is true particularly when they are not familiar with the task to be carried out. They forget to take into account unexpected events or unscheduled high priority work. People also often simply fail to allow for the full complexity involved with a job.

This part discusses how to estimate time on small projects. Time estimates are important inputs into the other techniques used to organize and structure medium and large sized projects. Both of these techniques reduce large projects down into a set of small projects.

HOW TO USE THE TOOL

The first stage in estimating time accurately is to fully understand what you need to achieve. This involves reviewing the task in detail so that there are no unknowns. Inevitably it is the difficult-to-understand, tricky problems that take the greatest amount

of time to solve. The best way to review the job is to list all tasks in full detail. Simple techniques such as Drill-Down are useful for this.

Once you have a detailed list of all the tasks that you must achieve, make your best guess at how long each task will take to complete. Ensure that within your estimate you also allow time for project management, detailed project planning, liaison with outside bodies, meetings, quality assurance and any supporting documentation necessary.

Also make sure that you have allowed time for:

- Other high urgency tasks to be carried out which will have priority over this one
- Accidents and emergencies
- Internal meetings
- Holidays and sickness in essential staff
- Contact with other customers, perhaps to arrange the next job
- Breakdowns in equipment
- Missed deliveries by suppliers
- Interruptions
- Quality control rejections

These factors may double the length of time needed to complete a project. If the accuracy of time estimates is critical, you may find it effective to develop a systematic approach to including these factors. If possible, base this on past experience.

KEY POINTS

You can lose a great deal of credibility by underestimating the length of time needed to implement a project. If you underestimate time, not only do you miss deadlines, you also put other project workers under unnecessary stress. Projects will become seriously unprofitable, and other tasks cannot be started.

The first step towards making good time estimates is to fully understand the problem to be solved. You can then prepare a detailed list of tasks that must be achieved. This list should include all the administrative tasks and meetings you need to carry out as well as the work itself. Finally, allow time for all the expected and unexpected disruptions and delays to work that will inevitably happen.

RISK IMPACT/PROBABILITY CHART

Risk management is an important function in organizations today. Companies undertake increasingly complex and ambitious projects, and those projects must be executed successfully, in an uncertain and often risky environment.

As a responsible manager, you need to be aware of these risks. Does this mean that you should try to address each and every risk that your project might face? Probably not - in all but the most critical environments, this can be much too expensive, both in time and resources.

Instead, you need to prioritize risks. If you do this effectively, you can focus the majority of your time and effort on the most important risks. The Risk Impact/Probability Chart provides a useful framework that helps you decide which risks need your attention.

USE THE RISK IMPACT/PROBABILITY CHART

The Risk Impact/Probability Chart is based on the principle that a risk has two primary dimensions:

1. *Probability:* A risk is an event that “may” occur. The probability of it occurring can range anywhere from just above 0% to just below 100%.
2. *Impact:* A risk, by its very nature, always has a negative impact. However, the size of the impact varies in terms of cost and impact on health, human life, or some other critical factor.

The chart allows you to rate potential risks on these two dimensions. The probability that a risk will occur is represented on one axis of the chart - and the impact of the risk, if it occurs, on the other. You use these two measures to plot the risk on the chart. This gives you a quick, clear view of the priority that you need to give to each. You can then decide what resources you will allocate to managing that particular risk. The basic form of the Risk Impact/Probability Chart is shown.

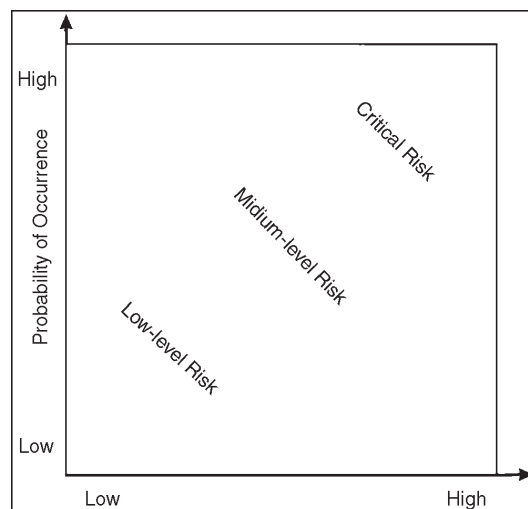


Fig. The Risk Impact/Probability Chart.

The corners of the chart have these characteristics:

- *Low impact/Low probability:* Risks in the bottom left corner are low level, and you can often ignore them.
- *Low impact/High probability:* Risks in the top left corner are of moderate importance - if these things happen, you can cope with them and move on. However, you should try to reduce the likelihood that they'll occur.
- *High impact/Low probability:* Risks in the bottom right corner are of high importance if they do occur, but they're very unlikely to happen. For these,

however, you should do what you can to reduce the impact they'll have if they do occur, and you should have contingency plans in place just in case they do.

- *High impact/High probability:* Risks towards the top right corner are of critical importance. These are your top priorities, and are risks that you must pay close attention to.

To use the Risk Impact/Probability Chart, print this free worksheet, and then follow these steps:

- List all of the likely risks that your project faces. Make the list as comprehensive as possible.
- Assess the probability of each risk occurring, and assign it a rating. For example, you could use a scale of 1 to 10. Assign a score of 1 when a risk is extremely unlikely to occur, and use a score of 10 when the risk is extremely likely to occur.
- Estimate the impact on the project if the risk occurs. Again, do this for each and every risk on your list. Using your 1-10 scale, assign it a 1 for little impact and a 10 for a huge, catastrophic impact.
- Map out the ratings on the Risk Impact/Probability Chart.
- Develop a response to each risk, according to its position in the chart. Remember, risks in the bottom left corner can often be ignored, while those in the top right corner need a great deal of time and attention. Read Risk Analysis and Risk Management for detailed strategies on developing a risk response plan.

KEY POINTS

To successfully implement a project, you must identify and focus your attention on middle and high-priority risks - otherwise you risk spreading your efforts too thinly, and you'll waste resources on unnecessary risk management. With the Risk Impact/Probability Chart, you map out each risk - and its position determines its priority. High-probability/high-impact risks are the most critical, and you should put a great deal of effort into managing these. The low-probability/high-impact risks and high-probability/low-impact risks are next in priority, though you may want to adopt different strategies for each. Low-probability/low-impact risks can often be ignored.

SCHEDULING SIMPLE PROJECTS

Simple projects involve only one or a few people over a short time. Typically, simple projects will have few tasks dependent on other tasks, and will be relatively simple and easy to coordinate. Examples might be coordinating delivery of resources for a workshop session, implementing a small marketing plan, or delivering a simple software enhancement. With simple projects, tools like Gantt Charts and Critical Path Diagrams may overcomplicate project scheduling and communication. Unless project team

members are trained in their use, they can often 'blind people with science', leading to poor communication and muddled projects.

USE

Appropriate Timetables and Action Plans are often sufficient to coordinate and implement simple projects. These should be explained and negotiated with project staff to improve the plans and get staff understanding, input and buy-in. It will often be enough to create a workback schedule, starting from the date by which the project must be completed, and listing all of the tasks in reverse order with due dates for each. Whatever the size of your project, ensure that you have agreed its scope with its sponsor before you start planning. This will help you to resist changes to its scope, which will seriously affect your plans, once you have started working.

KEY POINTS

Simple projects are often best run using simple Timetables and Action Plans. These should be prepared and negotiated with project staff to improve plans and get buy-in. During the project these will contain sufficient control points and deliveries to monitor project progress and take any appropriate remedial action.

GANTT CHARTS

Gantt Charts are useful tools for analyzing and planning more complex projects.

- Help you to plan out the tasks that need to be completed
- Give you a basis for scheduling when these tasks will be carried out
- Allow you to plan the allocation of resources needed to complete the project, and
- Help you to work out the critical path for a project where you must complete it by a particular date.

When a project is under way, Gantt Charts help you to monitor whether the project is on schedule. If it is not, it allows you to pinpoint the remedial action necessary to put it back on schedule.

SEQUENTIAL AND PARALLEL ACTIVITIES

An essential concept behind project planning is that some activities are dependent on other activities being completed first. As a shallow example, it is not a good idea to start building a bridge before you have designed it! These dependent activities need to be completed in a sequence, with each stage being more-or-less completed before the next activity can begin. We can call dependent activities 'sequential' or 'linear'. Other activities are not dependent on completion of any other tasks. These may be done at any time before or after a particular stage is reached. These are nondependent or 'parallel' tasks.

DRAWING A GANTT CHART

To draw up a Gantt diagram, follow these steps:

- *Step 1. List all activities in the plan:* For each task, show the earliest start date, estimated length of time it will take, and whether it is parallel or sequential. If tasks are sequential, show which stages they depend on. You will end up with a task list like the one in figure. This will allow you to compare the results of the two approaches. Figure. Gantt Chart *Example: Planning a custom-written computer project*

Task	Earliest start	Length	Type	Dependent on...
A. High level analysis	Week 0	1 week	Sequential	
B. Selection of hardware platform	Week 1	1 day	Sequential	A
C. Installation and commissioning of hardware	Week 1.2	2 weeks	Parallel	B
D. Detailed analysis of core modules	Week 1	2 weeks	Sequential	A
E. Detailed analysis of supporting modules	Week 3	2 weeks	Sequential	D
F. Programming of core modules	Week 3	2 weeks	Sequential	D
G. Programming of supporting modules	Week 5	3 weeks	Sequential	E
H. Quality assurance of core modules	Week 5	1 week	Sequential	F
I. Quality assurance of supporting modules	Week 8	1 week	Sequential	G
J. Core module training	Week 6	1 day	Parallel	C,H
K. Development and QA of accounting reporting	Week 5	1 week	Parallel	E
L. Development and QA of management reporting	Week 5	1 week	Parallel	E
M. Development of Management Information System	Week 6	1 week	Sequential	L
N. Detailed training	Week 9	1 week	Sequential	I, J, K, M

- *Step 2.* Head up graph paper with the days or weeks through to task completion
- *Step 3. Plot the tasks onto the graph paper:* Next draw up a rough draft of the Gantt Chart. Plot each task on the graph paper, showing it starting on the earliest possible date. Draw it as a bar, with the length of the bar being the length of the task. The task bars, mark the time taken to complete them. Schedule them in such a way that sequential actions are carried out in the

required sequence. Ensure that dependent activities do not start until the activities they depend on have been completed.

This will produce an untidy diagram like the one:

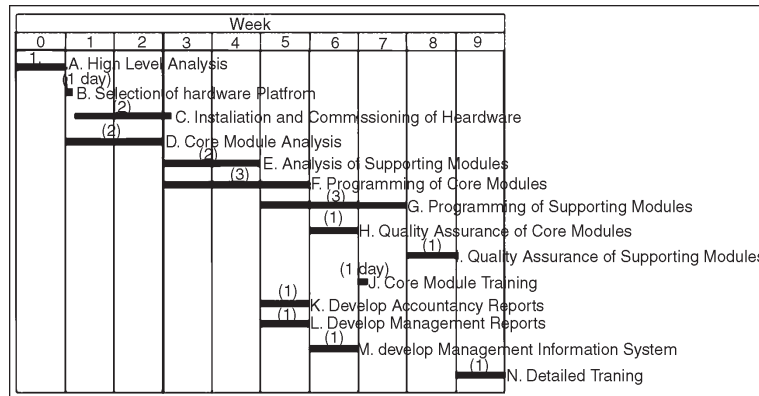


Fig. Draft Gantt Chart: Exampe Computer Project.

- *Step 4. Presenting the analysis:* The last stage in this process is to prepare a final version of the Gantt Chart. This shows how the sets sequential activities link together, and the identifies critical path activities. At this stage you also need to check the resourcing of the various activities. While scheduling, ensure that you make best use of the resources you have available, and do not over-commit resource. You can also use colour to represent the different resource types that you need to use such as programmers, or analysts.

A redrawn version of the example project is shown:

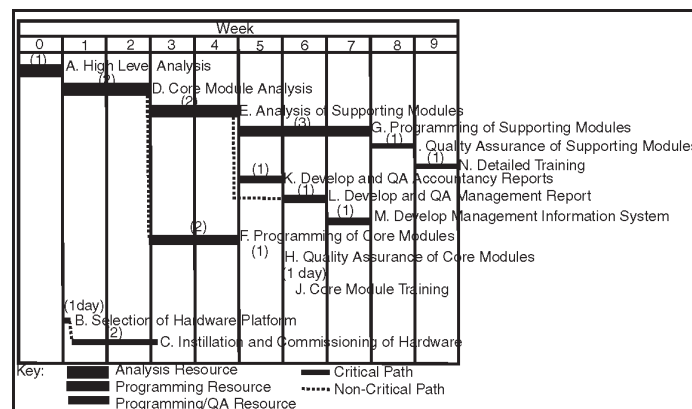


Fig. Critical Path Analysis: Activities Scheduled on a Gantt Chart.

By drawing this example Gantt Chart, you can see that:

- If all goes well, the project can be completed in 10 weeks.
- If you want to complete the task as rapidly as possible, you need:
 - 1 analyst for the first 5 weeks.

- 1 programmer for 5 weeks starting week 4.
- 1 programmer/QA expert for 3 weeks starting week 6. Note: Activities L and M have been moved back a week. This does not affect the critical path, but it does mean that a single programming/QA resource can carry out all three of activities K, L and M.
- Analysis, development and testing of supporting modules are essential activities that must be completed on time.
- Hardware installation and commissioning is not time-critical as long as it is completed before the Core Module Training starts.

While this part describes how to draw a Gantt Chart manually, in practice project managers use software tools like Microsoft Project to create Gantt Charts. Not only do these ease the drawing of Gantt Charts, they also make modification of plans easier and provide facilities for monitoring progress against plans, as well as generating resource histograms.

KEY POINTS

Gantt charts are useful tools for planning and scheduling projects. They allow you to assess how long a project should take, determine the resources needed, and lay out the order in which tasks need to be carried out. They are useful in managing the dependencies between tasks. When a project is under way, Gantt charts are useful for monitoring its progress. You can immediately see what should have been achieved at a point in time, and can therefore take remedial action to bring the project back on course. This can be essential for the successful and profitable implementation of the project.

Production and Operations Management

INTRODUCTION

The Subject of Production Management is studied under different Headings-such as Production Planning and control, Production and Inventory control, production and operations control and many more. What ever may be the title of the subject, the contents of the subject are more or less one and the same. Before we discuss about production management, let us discuss about product, production and management. This will give us a rough idea about production Management and with what a production manager has to deal with.

PRODUCT

Though many authors define the product with Consumer orientation, it is better for us to deal with different angles, because it will be helpful for us to understand the subject of production and Operation Management:

- *For a Consumer:* The product is a combination of or optimal mix of potential utilities. This is because every consumer expects some use or uses from the product. Hence he/she always identifies the product in terms of the uses. Say for example-Soap can be identified by complexion, cleanliness of body,

freshness, fragrance or health..., etc. Because of this, many producers advertise that they are selling health, or they are selling Cine star Complexion or they are selling freshness and so on.

- *For a Production Manager:* Product is the combination of various surfaces and processes. This is because the production Manager is solely responsible for producing the product. He has to think of the various surfaces by which the product is made of, so that he can plan for processes by which a particular surface can be made and plan for required capacity of the facility by which the surface is produced. While planning he has to see that the required surface is produced by the best and cheapest method, so as to make the product to face competition in the market.
- *For a Financial Manager:* For him the product is a mix of various cost elements as he is responsible for the profitability of the product.
- *For a Personnel Manager:* For him the product is a mix of various skills, as he is the person who selects and trains the personnel to meet the demand of the skill to produce the product. In general we can define the product as a bundle of tangible and intangible attributes, which along with the service is meant to satisfy the customer wants.

PRODUCTION

Production means application of processes. The raw material to add the use and economic values to arrive at desired product by the best method, without sacrificing the desired quality.

We have three ways of Production, they are:

- *Production by Disintegration:* By separating the contents of Crude oil or a mixture the desired products are produced. For example the crude oil is disintegrated into various fuel oils. Similarly salt production is also an example for product produced by disintegrated. We can use Mechanical or Chemical or both technologies to get the desired product, so that it will have desired use value.
- *Production by Integration:* In this type of Production various Components of the products are assembled together to get the desired product. In this process, Physical and Chemical Properties of the materials used may change. The examples are: Assembly of Two wheelers, Four wheelers and so on.
- *Production by Service:* Here the Chemical and Mechanical Properties of materials are improved without any physical change. The example for this is Heat Treatment of metals. In real world, a combination of methods is used. In general production is the use of any process or procedure designed to transform a set of input elements into a set of output elements, which have use value and economic value.

MANAGEMENT

Management can be explained as an art or science, of getting things done by the people, by planning, coordinating, organizing, directing and controlling the activities to meet specified goals, with in the frame work of agreed policies. The explanation put emphasis on getting things done, Planning, Organizing, Coordinating, and controlling and specific objectives and agreed policies. Today's manager needs scientific base as well as personal tactics to manage the people under him to achieve the desired goals. Production and management will help us to understand what exactly the Production Management or Production, and Operations Management.

CRITERIA OF PERFORMANCE

Three aims of performance of the Production and Operations Management system may be identified.

They are,:

- Effectiveness,
- Customer satisfaction,
- Efficiency.

The case of Efficiency is productive utilization of resources is clear. Whether the organization is in 'private sector' or in the 'public sector', is a 'manufacturing or 'non-manufacturing' organization or a 'profit' or a 'non-profit' organization, the optimal utilization of resource inputs is always a desired objective. The effectiveness has more dimensions to it. It involves optimality in the fulfilment of multiple objectives with a possible prioritization within the objectives. Modern production and operations management has to serve the target customers, the people working within, as also the region, country or society at large. Thus Production/Operations Management system, has not only to be profitable and/or efficient, but must necessarily satisfy many more customers. This effectiveness has to be again viewed in terms of the short and long-term horizons depending upon the operations system.

Optimum, Good, Better operations management can improve:

- Efficiency of operation system to do things right and broader concept.
- Effectiveness of operation system refers to doing right things that is seven rights, they are: Right operation, Right Quantity, Right Quality, Right Supplier or Right Vendor, Right Time, Right Place and Right Price.

Basically, efficiency and effectiveness of the operations system can be measured by four dimensions, they are:

1. Cost,
2. Quality,
3. Dependability and
4. Reliability.

In fact these directly relate to the competitiveness of the organization, both nationally and internationally. Modern developments in better tools and techniques, methods and systems like Automation, Flexible manufacturing, CAD, CAM, CIM at management, CADD, CIMS, Use of Robotics, TQM, OR Techniques etc, are taking place to achieve improvements in Cost, Quality, Dependability, Reliability and Flexibility and thus to help for better management.

DEFINITION OF PRODUCTION MANAGEMENT

It may be defined as:

- The performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system.
- It is the processes of effectively planning, coordinating and controlling the production, that is the operations of that part of an enterprise, it means to say that production and operations Management is responsible for the actual transformation of raw materials into finished products.
- Production management is a function of Management, related to planning, coordinating and controlling the resources required for production to produce specified product by specified methods, by optimal utilization of resources.
- Production management is defined as management function which plans, organizes, coordinates, directs and controls the material supply and Processing activities of an enterprise, so that specified products are produced by specified methods to meet an approved sales programme. These activities are being carried out in such a manner that Labour, Plant and Capital available are used to the best advantage of the organization.

OBJECTIVE OF PRODUCTION MANAGEMENT

The objective of Production Management is to produce the desired product or specified product by specified methods so that the optimal utilization of available resources is met with. Hence the production management is responsible to produce the desired product, which has marketability at the cheapest price by proper planning, the manpower, material and processes. Production management must see that it will deliver right goods of right quantity at right place and at right price. When the objective is achieved, we say that we have effective Production Management system.

SCOPE OF PRODUCTION MANAGEMENT

In fact, we apply Principles of Management; and functions of Management in our day-to-day life. We all know, from morning till night, we plan our activities; we coordinate available resources and control our activities to achieve certain goals. So also any organization must follow the Principles of Management for its survival and growth. The same is applicable to production Management also. Reading and learning Production Management will enable one to be capable of solving the problems of the organization, may be an Educational Institution, Production Shop, Hospital, Departmental shop or even a barber shop. The problems a manager face in various organizations are more or less similar to that of Production department but smaller in magnitude. Hence the knowledge of Production Management will help any professional Manager to tackle the problems of his business easily.

For example: The Production Management consists of Planning, selection of materials, planning of processes, Routing, Scheduling and controlling the activities, etc., Take the example of an Educational Institution/University. Here also selection of raw students, Planning of the Course Work, Educating the students and conducting the examination. Therefore this knowledge will enable one to apply the principles of Production Management to any field of life without restriction. Here, we have to remember that this is also applicable to the management of a service organization and the management of a Project.

Here it is better to distinguish between product, Service and Project, so as to help the reader to know on which particular aspect of Production Management to put much emphasis, in managing a service organization or a project.

- *Product:* Manufacturing system often produces standardized products in large volumes. The plant and machinery have a finite capacity. The facilities constitute fixed costs, which are allocated to the products produced. Variable costs, such as, labour cost and materials costs. While manufacturing the product use value and economic values are added to the product. Hence the product is a store of values added during manufacture. Because the input costs and output costs are measurable, the productivity can be measured with certain degree of accuracy. Product can be transported to the markets and stored physically until it is sold.
- *Service:* Service system present more uncertainty with respect to capacity and costs. Services are produced and consumed in the presence of the customer. We cannot store the service physically. Because of this the service organizations, such as Hotels, Hospitals, Transport Organizations and many other service organizations the capacity must be sufficiently or consciously managed to accommodate a highly variable demand. Sometimes services like legal practice and medical practice involve Professional or intellectual

judgements, which cannot be easily standardized. Because of this the calculation of cost and productivity is difficult.

- *Project*: Project system does not produce standardized products. The Plant, Machinery, Men and Materials are often brought to project site and the project is completed. The project is of big size and remains in the site itself after completion. As the costs can be calculated and allocated to the project with considerable accuracy, Productivity can be measured. Once the project is completed, all the resources are removed from site.

BENEFITS DERIVED FROM EFFICIENT PRODUCTION MANAGEMENT

The efficient Production Management will give benefits to the various parts of the society.

They are:

- Consumer benefits from improved industrial Productivity, increased use value in the product. Products are available to him at right place, at right price, at right time, in desired quantity and of desired quality.
- *Investors*: They get increased security for their investments, adequate market returns, and creditability and good image in the society.
- Employee gets adequate Wages, Job security, improved working conditions and increased Personal and Job satisfaction.
- *Suppliers*: Will get confidence in management and their bills can be realised with out any delay.
- *Community*: community enjoys Benefits from economic and social stability.
- The Nation will achieve prospects and security because of increased Productivity and healthy industrial atmosphere.

PRODUCTION CYCLE

The production cycle starts from Market Research. Market research reveals consumer preferences and needs. The marketing department will transfer this information to the design department.

The design department basing on the information received from marketing department designs the product to fulfil consumer needs and supplies design specifications and drawings to production department. The production department verifies whether the product can be manufactured with the technology and skill available in the firm. If yes it will give the acceptance. Otherwise the Production Manager, Design engineer and Marketing Manager, discuss together and make alterations in the product without sacrificing the customer needs. Once this is done the design department conduct the market survey and analyse the demand and submits its sales programme to the top

management. The top management after going through the proposals sanctions the budget and gives green signal for production. Production department produces a trial run and sales department release the product in the test markets to get consumer acceptance sells the products. Meanwhile, the production department prepares the detailed production plans and production scheduling. After getting the acceptance from the marketing department, actual production of product starts to meet the marketing programme.

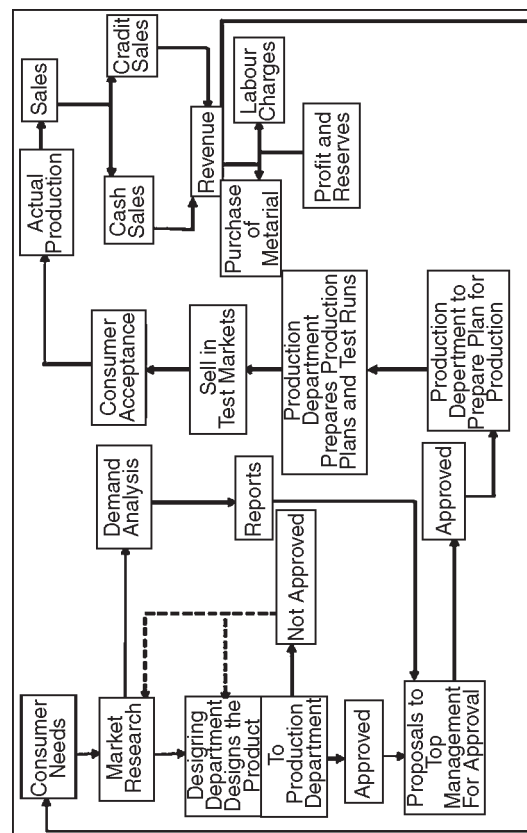


Fig. Production cycle.

PRODUCTION DESIGN

Production design is the creation and organization of the physical world surrounding a film story. The term was coined by producer David O. Selznick to describe the greater-than-normal contribution of designer William Cameron Menzies to *Gone with the Wind*, but the exact responsibilities of a production designer inevitably vary from film to film. In some cases, the production designer is almost completely responsible for the overall look of a film; in others, particularly when working with directors with strong visual styles, a designer's contribution tends to be much more limited. Art direction and

production design often overlap, although credit for production design is seen as more inclusive. During the studio era, production designers, as opposed to art directors, were the exception. The production designer's primary, though by no means exclusive, responsibility is the design of the sets. Exact responsibility varies from one film industry to another. In the United States, for example, production design and costume design are usually two separate professions. In other major film industries, the two responsibilities are often held by a single person. Before designing anything, the designer develops a "design concept," an overarching metaphor for the film's appearance that governs individual choices. This "concept" may or may not be established in conjunction with the director. Once settled upon, however, it structures all decisions made, helping the art staff to give an individual film visual distinction.

PRODUCT DEVELOPMENT

Product development is the process of designing, creating, and marketing an idea or product. The product can either be one that is new to the marketplace or one that is new to your particular company, or, an existing product that has been improved. In many instances a product will be labelled new and improved when substantial changes have been made.

PRODUCT DEVELOPMENT PROCESS

All product development goes through a similar planning process. Although the process is a continuous one, it is crucial that companies stand back after each step and evaluate whether the new product is worth the investment to continue. That evaluation should be based on a specific set of objective criteria, not someone's gut feeling. Even if the product is wonderful, if no one buys it the company will not make a profit. Brainstorming and developing a concept is the first step in product development. Once an idea is generated, it is important to determine whether there is a market for the product, what the target market is, and whether the idea will be profitable, as well as whether it is feasible from an engineering and financial standpoint. Once the product is determined to be feasible, the idea or concept is tested on a small sample of customers within the target market to see what their reactions are.

DOCUMENTATION

Simply having an "idea" is worthless--you need to have proof of when you came up with the idea for your invention. Write down everything you can think of that relates to your invention, from what it is and how it works to how you'll make and market it. This is the first step to patenting your idea and keeping it from being stolen. You've probably heard about the "poor man's patent"--writing your idea down and mailing it to yourself in a sealed envelope so you have dated proof of your invention's conception. This is

unreliable and unlikely to hold up in court. Write your idea down in an inventor's journal and have it signed by a witness. This journal will become your bible throughout the patent process. An inventor's journal can be any bound notebook whose pages are numbered consecutively and can't be removed or reinserted.

You can find specially designed inventor's journals at bookstores or you can save money and purchase a generic notebook anywhere they're sold, such as the grocery store, office supply store, stationery store, etc.

RESEARCH

You will need to research your idea from a legal and business standpoint.

Before you file a patent, you should:

- *Complete an initial patent search:* Just because you haven't seen your invention doesn't mean it doesn't already exist. Before you hire a patent attorney or agent. You should also complete a non-patent "prior art" search. If you find any sort of artwork or design related to your idea, you cannot patent it--regardless of whether a prior patent has been filed.
- *Research your market:* Sure, your brother thinks your idea for a new lawn sprinkler is a great idea, but that doesn't mean your neighbour would buy one. More than 95 per cent of all patents never make money for the inventor. Before you invest too much time and money into patenting your invention, do some preliminary research of your target market. Is this something people will actually buy? Once you know there's a market, make sure your product can be manufactured and distributed at a low enough cost so that your retail price is reasonable. You can determine these costs by comparing those of similar products currently on the market. This will also help you size up your competition--which you will have, no matter how unique you think your invention is.

MAKING A PROTOTYPE

A prototype is a model of your invention that puts into practice all of the things you have written in your inventor's journal. This will demonstrate the design of your invention when you present it to potential lenders and licensees. Do not file a patent before you have made a prototype. You will almost always discover a flaw in your original design or think of a new feature you would like to add. If you patent your idea before you work out these kinks, it will be too late to include them in the patent and you will risk losing the patent rights of the new design to someone else.

Here are some general rules of thumb when prototyping your invention:

- *Begin with a drawing:* Before you begin the prototyping phase, sketch out all of your ideas into your inventor's journal.
- Create a concept mockup out of any material that will allow you to create a 3-D model of your design.

- Once you're satisfied with the mockup, create a full-working model of your idea. There are many books and kits that can help you create prototypes. If your invention is something that will cost a lot of money or is unreasonable to prototype, consider using a computer-animated virtual prototype.

FILING A PATENT

Now that you have all of the kinks worked out of your design, it's finally time to file a patent. There are two main patents you will have to choose from: a utility patent or a design patent. You can write the patent and fill out the application yourself, but do not file it yourself until you have had a skilled patent professional look it over first. If the invention is really valuable, someone will infringe on it. If you do not have a strong patent written by a patent attorney or agent, you will be pulling your hair out later when a competitor finds a loophole that allows them to copy your idea. It's best to get the legal help now to avoid any legal problems in the future. When searching for a patent attorney or agent, remember one thing: If you see them advertised on TV, run away!

Once you are far, far away, follow these steps to choosing the best patent professional:

- *Do your homework:* Have your inventor's journal, prototype and notes with you. This will save them time, and you money. This will also help persuade them to work with you.
- *Make sure they are registered with the U.S. Patent and Trademark Office.*
- *Ask them what their technical background is:* If your invention is electronic, find a patent professional who is also an electrical engineer.
- *Discuss fees:* Keep your focus on smaller patent firms. They are less expensive and will work more closely with you. Agree to the estimated total cost before hiring your patent professional.

MARKETING YOUR INVENTION

Now it's time to figure out how you're going to bring your product to market. Create a business plan: How will you get money? Where will you manufacture the product? How will you sell it? Now is a good time to decide if you will manufacture and sell the product yourself, or license it for sale through another company. When you license your product you will probably only receive two per cent to five per cent in royalty fees. This often scares away inventors who feel they deserve more. But consider the upside: You will not have the financial burden associated with maintaining a business. This could end up making you more money in the long run. Following these five steps will ensure an easy road to patenting your invention. Just remember that an easy road doesn't necessarily mean a short one. From the time you conceive your idea to the time you see your product on the shelf is a very long process. Most inventions take years to come to fruition. Have patience and follow due diligence in your steps to patenting your invention and your years of hard work will finally pay off.

Operation of Pricing Decisions and Strategies Management

Pricing is a fundamental aspect of financial modelling, and is one of the four Ps of the marketing mix. The other three aspects are product, promotion, and place. It is also a key variable in microeconomic price allocation theory.

Price is the only revenue generating element amongst the four Ps, the rest being cost centres. Pricing is the manual or automatic process of applying prices to purchase and sales orders, based on factors such as: a fixed amount, quantity break, promotion or sales campaign, specific vendor quote, price prevailing on entry, shipment or invoice date, combination of multiple orders or lines, and many others. Automated systems require more setup and maintenance but may prevent pricing errors. The needs of the consumer can be converted into demand only if the consumer has the willingness and capacity to buy the product. Thus pricing is very important in marketing.

QUESTIONS INVOLVED IN PRICING

Pricing involves asking questions like:

- *How much to charge for a product or service?* This question is that a typical starting point for discussions about pricing, however, a better question for a vendor to ask is-How much do customers *value* the products, services, and other intangibles that the vendor provides.

- What are the pricing objectives?
- Do we use profit maximization pricing?
- How to set the price?: (cost-plus pricing, demand based or value-based pricing, rate of return pricing, or competitor indexing)
- Should there be a single price or multiple pricing?
- Should prices change in various geographical areas, referred to as zone pricing?
- Should there be quantity discounts?
- What prices are competitors charging?
- Do you use a price skimming strategy or a penetration pricing strategy?
- What image do you want the price to convey?
- Do you use psychological pricing?
- How important are customer price sensitivity (*e.g.*, “sticker shock”) and elasticity issues?
- Can real-time pricing be used?
- Is price discrimination or yield management appropriate?
- Are there legal restrictions on retail price maintenance, price collusion, or price discrimination?
- Do price points already exist for the product category?
- How flexible can we be in pricing? : The more competitive the industry, the less flexibility we have.
 - The price floor is determined by production factors like costs (often only variable costs are taken into account), economies of scale, marginal cost, and degree of operating leverage
 - The price ceiling is determined by demand factors like price elasticity and price points
- Are there transfer pricing considerations?
- What is the chance of getting involved in a price war?
- How visible should the price be? Should the price be neutral? (*i.e.*: not an important differentiating factor), should it be highly visible? (To help promote a low priced economy product, or to reinforce the prestige image of a quality product), or should it be hidden? (So as to allow marketers to generate interest in the product unhindered by price considerations).
- Are there joint product pricing considerations?
- What are the non-price costs of purchasing the product? (*e.g.*: travel time to the store, wait time in the store, disagreeable elements associated with the product purchase-dentist-> pain, fishmarket-> smells)
- What sort of payments should be accepted? Pricing

WHAT A PRICE SHOULD DO

A well chosen price should do three things:

- Achieve the financial goals of the company (*e.g.*, profitability)
- Fit the realities of the marketplace (Will customers buy at that price?)

- Support a product's positioning and be consistent with the other variables in the marketing mix
 - Price is influenced by the type of distribution channel used, the type of promotions used, and the quality of the product
 - Price will usually need to be relatively high if manufacturing is expensive, distribution is exclusive, and the product is supported by extensive advertising and promotional campaigns
 - A low price can be a viable substitute for product quality, effective promotions, or an energetic selling effort by distributors

From the marketer's point of view, an efficient price is a price that is very close to the maximum that customers are prepared to pay. In economic terms, it is a price that shifts most of the consumer surplus to the producer. A good pricing strategy would be the one which could balance between the price floor (the price below which the organization ends up in losses) and the price ceiling (the price beyond which the organization experiences a no demand situation).

DEFINITIONS

Pricing is the process of determining what a company will receive in exchange for its products. Pricing factors are manufacturing cost, market place, competition, market condition, Quality of product.

The effective price is the price the company receives after accounting for discounts, promotions, and other incentives.

Price lining is the use of a limited number of prices for all your product offerings. This is a tradition started in the old five and dime stores in which everything cost either 5 or 10 cents. Its underlying rationale is that these amounts are seen as suitable price points for a whole range of products by prospective customers. It has the advantage of ease of administering, but the disadvantage of inflexibility, particularly in times of inflation or unstable prices.

A loss leader is a product that has a price set below the operating margin. This results in a loss to the enterprise on that particular item, but this is done in the hope that it will draw customers into the store and that some of those customers will buy other, higher margin items.

Promotional pricing refers to an instance where pricing is the key element of the marketing mix. The price/quality relationship refers to the perception by most consumers that a relatively high price is a sign of good quality. The belief in this relationship is most important with complex products that are hard to test, and experiential products that cannot be tested until used (such as most services). The greater the uncertainty surrounding a product, the more consumers depend on the price/quality hypothesis and the more of a premium they are prepared to pay. The classic example of this is the pricing of the snack cake Twinkies, which were perceived as low quality when the price was lowered. Note, however, that excessive reliance on the price/quantity

relationship by consumers may lead to the raising of prices on all products and services, even those of low quality, which in turn causes the price/quality relationship to no longer apply.

Premium pricing (also called prestige pricing) is the strategy of consistently pricing at, or near, the high end of the possible price range to help attract status-conscious consumers. A few examples of companies which partake in premium pricing in the marketplace include Rolex and Bentley. People will buy a premium priced product because:

1. They believe the high price is an indication of good quality;
2. They believe it to be a sign of self worth “They are worth it”. It authenticates their success and status-It is a signal to others that they are a member of an exclusive group;
3. They require flawless performance in this application-The cost of product malfunction is too high to buy anything but the best-example : heart pacemaker.

The term Goldilocks pricing is commonly used to describe the practice of providing a “gold-plated” version of a product at a premium price in order to make the next-lower priced option look more reasonably priced; for example, encouraging customers to see business-class airline seats as good value for money by offering an even higher priced first-class option. Similarly, third-class railway carriages in Victorian England are said to have been built without windows, not so much to punish third-class customers (for which there was no economic incentive), as to motivate those who could afford second-class seats to pay for them instead of taking the cheaper option. This is also known as a potential result of price discrimination.

The name derives from the Goldilocks story, in which Goldilocks chose neither the hottest nor the coldest porridge, but instead the one that was “just right”. More technically, this form of pricing exploits the general cognitive bias of aversion to extremes. This practice is known academically as “framing”. By providing three options (*i.e.*, small, medium, and large; first, business, and coach classes) you can manipulate the consumer into choosing the middle choice and thus, the middle choice should yield the most profit to the seller, since it is the most chosen option.

Demand-based pricing is any pricing method that uses consumer demand-based on perceived value-as the central element. These include : price skimming, price discrimination and yield management, price points, psychological pricing, bundle pricing, penetration pricing, price lining, value-based pricing, geo and premium pricing. Pricing factors are manufacturing cost, market place, competition, market condition, quality of product.

Multidimensional pricing is the pricing of a product or service using multiple numbers. In this practice, price no longer consists of a single monetary amount (*e.g.*, sticker price of a car), but rather consists of various dimensions (*e.g.*, monthly payments, number of payments, and a downpayment). Research has shown that this practice can significantly influence consumers’ ability to understand and process price information.

The 9 Laws of Price Sensitivity

In their book, “The Strategy and Tactics of Pricing”, Thomas Nagle and Reed Holden outline 9 laws or factors that influence a buyer’s price sensitivity with respect to a given purchase:

REFERENCE PRICE EFFECT

Buyer’s price sensitivity for a given product increases the higher the product’s price relative to perceived alternatives. Perceived alternatives can vary by buyer segment, by occasion, and other factors.

DIFFICULT COMPARISON EFFECT

Buyers are less sensitive to the price of a known/more reputable product when they have difficulty comparing it to potential alternatives.

SWITCHING COSTS EFFECT

The higher the product-specific investment a buyer must make to switch suppliers, the less price sensitive that buyer is when choosing between alternatives.

PRICE-QUALITY EFFECT

Buyers are less sensitive to price the more that higher prices signal higher quality. Products for which this effect is particularly relevant include: image products, exclusive products, and products with minimal cues for quality.

EXPENDITURE EFFECT

Buyers are more price sensitive when the expense accounts for a large percentage of buyers’ available income or budget.

END-BENEFIT EFFECT

The effect refers to the relationship a given purchase has to a larger overall benefit, and is divided into two parts:

Derived demand: The more sensitive buyers are to the price of the end benefit, the more sensitive they will be to the prices of those products that contribute to that benefit.

Price proportion cost: The price proportion cost refers to the percent of the total cost of the end benefit accounted for by a given component that helps to produce the end benefit (*e.g.*, think CPU and PCs). The smaller the given components share of the total cost of the end benefit, the less sensitive buyers will be to the component’s price.

SHARED-COST EFFECT

The smaller the portion of the purchase price buyers must pay for themselves, the less price sensitive they will be.

FAIRNESS EFFECT

Buyers are more sensitive to the price of a product when the price is outside the range they perceive as “fair” or “reasonable” given the purchase context.

THE FRAMING EFFECT

Buyers are more price sensitive when they perceive the price as a loss rather than a forgone gain, and they have greater price sensitivity when the price is paid separately rather than as part of a bundle.

APPROACHES

Pricing as the most effective profit lever. Pricing can be approached at three levels. The industry, market, and transaction level.

Pricing at the industry level focuses on the overall economics of the industry, including supplier price changes and customer demand changes. Pricing at the market level focuses on the competitive position of the price in comparison to the value differential of the product to that of comparative competing products.

Pricing at the transaction level focuses on managing the implementation of discounts away from the reference, or list price, which occur both on and off the invoice or receipt.

TACTICS

Micromarketing is the practice of tailoring products, brands (microbrands), and promotions to meet the needs and wants of microsegments within a market. It is a type of market customization that deals with pricing of customer/product combinations at the store or individual level.

PRICING MISTAKES

Many companies make common pricing mistakes. Bernstein’s article “Supplier Pricing Mistakes” outlines several which include:

- Weak controls on discounting
- Inadequate systems for tracking competitor selling prices and market share
- Cost-Up pricing
- Price increases poorly executed
- Worldwide price inconsistencies
- Paying sales reps on dollar volume vs. addition of profitability measures.

Quality Management

CONCEPT

Quality Management is the assembly and management of all activities aimed at the production of quality by organizations of various kinds. In the present case this implies the introduction and proper running of a “Quality System” in laboratories. A statement of objectives and policy to produce quality should be made for the organization or department concerned. This statement also identifies the internal organization and responsibilities for the effective operation of the Quality System. Quality Management can be considered a somewhat wider interpretation of the concept of “Good Laboratory Practice” (GLP). Therefore, inevitably the basics of the present Guidelines largely coincide with those of GLP.

Note. An even wider concept of quality management is presently coming into vogue: “Total Quality Management” (TQM). This concept includes additional aspects such as leadership style, ethics of the work, social aspects, relation to society, etc. For an introduction to TQM the reader is referred to Parkany.

QUALITY ASSURANCE

Proper Quality Management implies consequent implementation of the next level: Quality Assurance. The ISO definition reads: “the assembly of all planned and systematic actions necessary to provide adequate confidence that a product, process, or service

will satisfy given quality requirements.” The result of these actions aimed at the production of quality, should ideally be checked by someone independent of the work: the Quality Assurance Officer. If no QA officer is available, then usually the Head of Laboratory performs this job as part of his quality management task. In case of special projects, customers may require special quality assurance measures or a Quality Plan.

QUALITY CONTROL

A major part of the quality assurance is the Quality Control defined by ISO as “the operational techniques and activities that are used to satisfy quality requirements. “ An important part of the quality control is the Quality Assessment: the system of activities to verify if the quality control activities are effective, in other words: an evaluation of the products themselves. Quality control is primarily aimed at the prevention of errors. Yet, despite all efforts, it remains inevitable that errors are made. Therefore, the control system should have checks to detect them.

When errors or mistakes are suspected or discovered it is essential that the “Five Ws” are trailed:

1. What error was made?
2. Where was it made?
3. When was it made?
4. Who made it?
5. Why was it made?

Only when all these questions are answered, proper action can be taken to correct the error and prevent the same mistake being repeated.

The techniques and activities involved in Quality Control can be divided into four levels of operation:

1. First-line control: Instrument performance check.
2. Second-line control: Check of calibration or standardization.
3. Third-line control: Batch control.
4. Fourth-line control: Overall check.

Because the first two control levels both apply to the correct functioning of the instruments they are often taken together and then only three levels are distinguished.

This designation is used throughout the present Guidelines:

- *First-line control:* Instrument check/calibration.
- *Second-line control:* Batch control
- *Third-line control:* External check

It will be clear that producing quality in the laboratory is a major enterprise requiring a continuous human effort and input of money. The rule-of-fist is that 10-20% of the total costs of analysis should be spent on quality control.

Therefore, for quality work at least four conditions should be fulfilled:

1. Means are available
2. Efficient use of time and means

3. Expertise is available
4. Upholding and improving level of output

In quality work, management aspects and technical aspects are inherently cobbled together and for a clear insight and proper functioning of the laboratory these aspects have to be broken down into their components.

GOOD LABORATORY PRACTICE

Quality Management in the present context can be considered a modern version of the hitherto much used concept “Good Laboratory Practice” with a somewhat wider interpretation. The OECD Document defines GLP as follows: “Good Laboratory Practice is concerned with the organizational process and the conditions under which laboratory studies are planned, performed, monitored, recorded, and reported.”

Thus, GLP prescribes a laboratory to work according to a system of procedures and protocols. This implies the organization of the activities and the conditions under which these take place are controlled, reported and filed. GLP is a policy for all aspects of the laboratory which influence the quality of the analytical work.

When properly applied, GLP should then:

- Allow better laboratory management
- Improve efficiency
- Minimize errors
- Allow quality control
- Stimulate and motivate all personnel
- Improve safety
- Improve communication possibilities, both internally and externally.

The result of GLP is that the performance of a laboratory is improved and its working effectively controlled. An important aspect is also that the standards of quality are documented and can be demonstrated to authorities and clients. This results in an improved reputation for the laboratory.

In short, the message is:

- Say what you do
- Do what you say
- Do it better
- Be able to show what you have done

The basic rule is that all relevant plans, activities, conditions and situations are recorded and that these records are safely filed and can be produced or retrieved when necessary. These aspects differ strongly in character and need to be attended to individually.

As an assembly, the involved documents constitute a so-called Quality Manual.

This comprises then all relevant information on:

- Organization and Personnel
- Facilities

- Equipment and Working materials
- Analytical or testing systems
- Quality control
- Reporting and filing of results.

Since institutions having a laboratory are of divergent natures, there is no standard format and each has to make its own Quality Manual. The present Guidelines contain examples of forms, protocols, procedures and artificial situations. They need at least to be adapted and many new ones will have to be made according to the specific needs, but all have to fulfil the basic requirement of usefulness and verifiability. As already indicated, the guidelines for Quality Management given here are mainly based on the principles of Good Laboratory Practice as they are laid down in various relevant documents such as ISO and ISO/IEC guides, ISO 9000 series, OECD and CEN documents, national standards, as well as a number of text books. The consulted documents are listed in the Literature.

Use is also made of documents developed by institutes which have obtained accreditation or are working towards this. This concerns mainly so-called Standard Operating Procedures and Protocols. Sometimes these documents are hard to acquire as they are classified information for reasons of competitiveness. The institutes and persons which cooperated in the development of these Guidelines are listed in the Acknowledgements.

QUALITY CONTROL

Quality control is a process employed to ensure a certain level of quality in a product or service. It may include whatever actions a business deems necessary to provide for the control and verification of certain characteristics of a product or service. The basic goal of quality control is to ensure that the products, services, or processes provided meet specific requirements and are dependable, satisfactory, and fiscally sound. Essentially, quality control involves the examination of a product, service, or process for certain minimum levels of quality. The goal of a quality control team is to identify products or services that do not meet a company's specified standards of quality.

If a problem is identified, the job of a quality control team or professional may involve stopping production temporarily. Depending on the particular service or product, as well as the type of problem identified, production or implementation may not cease entirely. Usually, it is not the job of a quality control team or professional to correct quality issues. Typically, other individuals are involved in the process of discovering the cause of quality issues and fixing them.

Once such problems are overcome, the product, service, or process continues production or implementation as usual. Quality control can cover not just products, services, and processes, but also people. Employees are an important part of any company. If a company has employees that don't have adequate skills or training, have trouble

understanding directions, or are misinformed, quality may be severely diminished. When quality control is considered in terms of human beings, it concerns correctable issues. However, it should not be confused with human resource issues.

Often, quality control is confused with quality assurance. Though the two are very similar, there are some basic differences. Quality control is concerned with the product, while quality assurance is process-oriented. Even with such a clear-cut difference defined, identifying the differences between the two can be hard. Basically, quality control involves evaluating a product, activity, process, or service. By contrast, quality assurance is designed to make sure processes are sufficient to meet objectives. Simply put, quality assurance ensures a product or service is manufactured, implemented, created, or produced in the right way; while quality control evaluates whether or not the end result is satisfactory.

ACCURACY OF QUALITY ASSURANCE

There is an extensive process of trial and error in order to ensure quality assurance. By the end of the trials you arrive at an acceptable process that helps you decide the reliability and efficiency of the sample. The process involves meeting specifications such as performance measures and depends on environment operation. The costs of resulting in failure is very high which result in the process of Quality Assurance to be extensive and it is acceptable to delay production until all tests are done and double-checked.

STEPS FOR QUALITY ASSURANCE PROCESS

- Plan to improve
- Design to include improvements and requirements
- Manufacture with improvements
- Review new item and improvements
- Test new item

The process for Quality Assurance is very rigorous and requires a lot of testing and planning. The team or firm has to comply with previous requirements, implement new requirements and improve the old item. Other than following requirements, the team or firm has to comply with consumers needs.

QUALITY ASSURANCE VERSUS QUALITY CONTROL

Quality control emphasizes testing of products to uncover defects, and reporting to management who make the decision to allow or deny the release, whereas quality assurance attempts to improve and stabilize production, and associated processes, to avoid, or at least minimize, issues that led to the defects in the first place. On applying Quality Assurance in Education the gross purpose of applying quality assurance is served. To prevent mistakes from arising, several QA methodologies are used. However, QA

does not eliminate the need for QC: some product parameters are so critical that testing is still essential. QC activities are treated as one of the overall QA processes.

Failure Testing

A valuable process to perform on a whole consumer product is failure testing or stress testing. In mechanical terms this is the operation of a product until it fails, often under stresses such as increasing vibration, temperature, and humidity.

This exposes many unanticipated weaknesses in a product, and the data are used to drive engineering and manufacturing process improvements. Often quite simple changes can dramatically improve product service, such as changing to mold-resistant paint or adding lock-washer placement to the training for new assembly personnel.

Statistical Control

Many organizations use statistical process control to bring the organization to Six Sigma levels of quality, in other words, so that the likelihood of an unexpected failure is confined to six standard deviations on the normal distribution. This probability is less than four one-millionths. Items controlled often include clerical tasks such as order-entry as well as conventional manufacturing tasks. Traditional statistical process controls in manufacturing operations usually proceed by randomly sampling and testing a fraction of the output. Variances in critical tolerances are continuously tracked and where necessary corrected before bad parts are produced.

Total Quality Management

Invariably, the Quality of output is directly dependent upon that of the participating constituents, some of which are sustainably and effectively controlled while others are [not]. The fluid state spells lack of Quality control, and the process(es) which are properly managed for Quality such that Quality is assured, pertain to Total Quality Management.

The major problem which leads to a decrease in sales was that the specifications did not include the most important factor, "What the specifications have to state in order to satisfy the customer requirements?".

The major characteristics, ignored during the search to improve manufacture and overall business performance were:

- Reliability
- Maintainability
- Safety
- Strength

As the most important factor had been ignored, a few refinements had to be introduced:

- Marketing had to carry out their work properly and define the customer's specifications.
- Specifications had to be defined to conform to these requirements.

- Conformance to specifications *i.e.*, drawings, standards and other relevant documents, were introduced during manufacturing, planning and control.
- Management had to confirm all operators are equal to the work imposed on them and holidays, celebrations and disputes did not affect any of the quality levels.
- Inspections and tests were carried out, and all components and materials, bought in or otherwise, conformed to the specifications, and the measuring equipment was accurate, this is the responsibility of the QA/QC department.
- Any complaints received from the customers were satisfactorily dealt with in a timely manner.
- Feedback from the user/customer is used to review designs.
- Consistent data recording and assessment and documentation integrity.
- Product and/or process change management and notification.

If the specification does not reflect the true quality requirements, the product's quality cannot be guaranteed. For instance, the parameters for a pressure vessel should cover not only the material and dimensions but operating, environmental, safety, reliability and maintainability requirements.

QA IN SOFTWARE DEVELOPMENT

The following are examples of QA models relating to the software development process.

Models and Standards

ISO 17025 is an international standard that specifies the general requirements for the competence to carry out tests and or calibrations. There are 15 management requirements and 10 technical requirements. These requirements outline what a laboratory must do to become accredited. Management system refers to the organization's structure for managing its processes or activities that transform inputs of resources into a product or service which meets the organization's objectives, such as satisfying the customer's quality requirements, complying with regulations, or meeting environmental objectives. The CMMI (Capability Maturity Model Integration) model is widely used to implement Quality Assurance (PPQA) in an organization. The CMMI maturity levels can be divided into 5 steps, which a company can achieve by performing specific activities within the organization.

COMPANY QUALITY

During the 1980s, the concept of "company quality" with the focus on management and people came to the fore. It was realized that, if all departments approached quality with an open mind, success was possible if the management led the quality improvement process.

The company-wide quality approach places an emphasis on four aspects:

1. Elements such as controls, job management, adequate processes, performance and integrity criteria and identification of records
2. Competence such as knowledge, skills, experience, qualifications
3. Soft elements, such as personnel integrity, confidence, organizational culture, motivation, team spirit and quality relationships.
4. Infrastructure

The quality of the outputs is at risk if any of these aspects is deficient in any way.

The approach to quality management given here is therefore not limited to the manufacturing theatre only but can be applied to any business or non-business activity:

- Design work
- Administrative services
- Consulting
- Banking
- Insurance
- Computer software development
- Retailing
- Transportation
- Education

It comprises a quality improvement process, which is generic in the sense it can be applied to any of these activities and it establishes a behaviour pattern, which supports the achievement of quality. This in turn is supported by quality management practices which can include a number of business systems and which are usually specific to the activities of the business unit concerned.

In manufacturing and construction activities, these business practices can be equated to the models for quality assurance defined by the International Standards contained in the ISO 9000 series and the specified Specifications for quality systems.

Still, in the system of Company Quality, the work being carried out was shop floor inspection which did not reveal the major quality problems. This led to quality assurance or total quality control, which has come into being recently.

USING CONTRACTORS AND/OR CONSULTANTS

It has become customary to use consultants and contractors when introducing new quality practices and methods, particularly where the relevant skills and expertise are not available within the organization. In addition, when new initiatives and improvements are required to boost the current quality system, or perhaps improve upon current manufacturing systems, the use of temporary consultants becomes a viable solution when allocating valuable resources.

There are various types of consultants and contractors available in the market; most will have the skills needed to facilitate improvement activities such as Quality Management Systems (QMS) auditing and procedural documentation writing. More

experienced consultants are likely to have knowledge of specialized quality improvement activities such as CMMI, Six Sigma, Measurement Systems Analysis (MSA), Quality Function Deployment (QFD), Failure Mode and Effects Analysis (FMEA), Advance Product Quality Planning (APQP).

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Simply put, quality assurance ensures a product or service is manufactured, implemented, created, or produced in the right way; while quality control evaluates whether or not the end result is satisfactory.

STATISTICAL PROCESS CONTROL

Statistical process control (SPC) is the application of statistical methods to the monitoring and control of a process to ensure that it operates at its full potential to produce conforming product. Under SPC, a process behaves predictably to produce as much conforming product as possible with the least possible waste.

While SPC has been applied most frequently to controlling manufacturing lines, it applies equally well to any process with a measurable output. Key tools in SPC are control charts, a focus on continuous improvement and designed experiments.

Much of the power of SPC lies in the ability to examine a process and the sources of variation in that process using tools that give weight to objective analysis over subjective opinions and that allow the strength of each source to be determined numerically.

Variations in the process that may affect the quality of the end product or service can be detected and corrected, thus reducing waste as well as the likelihood that problems will be passed on to the customer. With its emphasis on early detection and prevention of problems, SPC has a distinct advantage over other quality methods, such as inspection, that apply resources to detecting and correcting problems after they have occurred.

In addition to reducing waste, SPC can lead to a reduction in the time required to produce the product or service from end to end. This is partially due to a diminished likelihood that the final product will have to be reworked, but it may also result from using SPC data to identify bottlenecks, wait times, and other sources of delays within the process.

Process cycle time reductions coupled with improvements in yield have made SPC a valuable tool from both a cost reduction and a customer satisfaction standpoint. Statistical process control was pioneered by Walter A. Shewhart in the early 1920s. W. Edwards Deming later applied SPC methods in the United States during World War II, thereby successfully improving quality in the manufacture of munitions and other strategically important products. Deming was also instrumental in introducing SPC methods to Japanese industry after the war had ended.

Shewhart created the basis for the control chart and the concept of a state of statistical control by carefully designed experiments. While Dr. Shewhart drew from pure mathematical statistical theories, he understood that data from physical processes seldom produces a “normal distribution curve” (a Gaussian distribution, also commonly referred to as a “bell curve”).

He discovered that observed variation in manufacturing data did not always behave the same way as data in nature. Dr. Shewhart concluded that while every process displays variation, some processes display controlled variation that is natural to the process (common causes of variation), while others display uncontrolled variation that is not present in the process causal system at all times (special causes of variation).

In 1989, the Software Engineering Institute introduced the notion that SPC can be usefully applied to non-manufacturing processes, such as software engineering processes,

in the Capability Maturity Model (CMM). This idea exists today within the Level 4 and Level 5 practices of the Capability Maturity Model Integration (CMMI). This notion that SPC is a useful tool when applied to non-repetitive, knowledge-intensive processes such as engineering processes has encountered much scepticism, and remains controversial today.

GENERAL

The following description relates to manufacturing rather than to the service industry, although the principles of SPC can be successfully applied to either. For a description and example of how SPC applies to a service environment, refer to Roberts (2005). SPC has also been successfully applied to detecting changes in organizational behaviour with Social Network Change Detection introduced by McCulloh. Selden describes how to use SPC in the fields of sales, marketing, and customer service, using Deming's famous Red Bead Experiment as an easy to follow demonstration.

In mass-manufacturing, the quality of the finished article was traditionally achieved through post-manufacturing inspection of the product; accepting or rejecting each article based on how well it met its design specifications. In contrast, Statistical Process Control uses statistical tools to observe the performance of the production process in order to predict significant deviations that may later result in rejected product.

Two kinds of variation occur in all manufacturing processes: both these types of process variation cause subsequent variation in the final product. The first is known as natural or common cause variation and consists of the variation inherent in the process as it is designed. Common cause variation may include variations in temperature, properties of raw materials, strength of an electrical current, etc.

The second kind of variation is known as special cause variation, or assignable-cause variation, and happens less frequently than the first. With sufficient investigation, a specific cause, such as abnormal raw material or incorrect set-up parameters, can be found for special cause variations.

For example, a breakfast cereal packaging line may be designed to fill each cereal box with 500 grams of product, but some boxes will have slightly more than 500 grams, and some will have slightly less, in accordance with a distribution of net weights. If the production process, its inputs, or its environment changes this distribution can change. For example, as its cams and pulleys wear out, the cereal filling machine may start putting more cereal into each box than specified. If this change is allowed to continue unchecked, more and more product will be produced that fall outside the tolerances of the manufacturer or consumer, resulting in waste. While in this case, the waste is in the form of "free" product for the consumer, typically waste consists of rework or scrap.

By observing at the right time what happened in the process that led to a change, the quality engineer or any member of the team responsible for the production line can troubleshoot the root cause of the variation that has crept in to the process and correct the problem.

SPC indicates when an action should be taken in a process, but it also indicates when NO action should be taken. An example is a person who would like to maintain a constant body weight and takes weight measurements weekly. A person who does not understand SPC concepts might start dieting every time his or her weight increased, or eat more every time his or her weight decreased. This type of action could be harmful and possibly generate even more variation in body weight. SPC would account for normal weight variation and better indicate when the person is in fact gaining or losing weight.

HOW TO USE SPC?

Statistical Process Control may be broadly broken down into three sets of activities: understanding the process; understanding the causes of variation; and elimination of the sources of special cause variation.

In understanding a process, the process is typically mapped out and the process is monitored using control charts. Control charts are used to identify variation that may be due to special causes, and to free the user from concern over variation due to common causes. By the nature of the control chart, “understanding the process” is a continuous activity. With a stable process that does not trigger any of the detection rules for a control chart, a process capability analysis is also performed to evaluate the ability of the current process to produce conforming product.

When, through the control charts, variation that is due to special causes is identified, or the process capability is found lacking, additional effort is exerted to determine causes of that variance and eliminate it. The tools used include Ishikawa diagrams, designed experiments and Pareto charts. Designed experiments are critical to this phase of SPC, as they are the only means of objectively quantifying the relative importance of the many potential causes of variation.

Once the causes of variation have been quantified, effort is spent in eliminating those causes that are both statistically and practically significant (*i.e.*, a cause that has a only small but statistically significant effect may not be considered cost-effective to fix; conversely, a cause that is not statistically significant cannot be considered practically significant). Generally, this includes development of standard work, error-proofing and training. Additional measures may be required, especially if there is a problem with process capability

PROCESS CAPABILITY IMPROVEMENT

PROCESS CAPABILITY

A process is a unique combination of tools, materials, methods, and people engaged in producing a measurable output; for example a manufacturing line for machine parts.

All processes have inherent statistical variability which can be evaluated by statistical methods. The Process Capability is a measurable property of a process to the specification, expressed as a process capability index (e.g., Cpk or Cpm) or as a process performance index (e.g., Ppk or Ppm). The output of this measurement is usually emphasised by a histogram and calculations that predict how many parts will be produced out of specification (OOS).

Process capability is also defined as the capability of a process to meet its purpose as managed by an organization's management and process definition structures ISO 15504.

Two parts of process capability are:

1. Measure the variability of the output of a process, and
2. Compare that variability with a proposed specification or product tolerance.

Measure the Process

The input of a process usually has at least one or more measurable characteristics that are used to specify outputs. These can be analysed statistically; where the output data shows a normal distribution the process can be described by the process mean (average) and the standard deviation.

A process needs to be established with appropriate process controls in place. A control chart analysis is used to determine whether the process is "in statistical control". If the process is not in statistical control then capability has no meaning. Therefore the process capability involves only common cause variation and not special cause variation.

A batch of data needs to be obtained from the measured output of the process. The more data that is included the more precise the result, however an estimate can be achieved with as few as 17 data points. This should include the normal variety of production conditions, materials, and people in the process. With a manufactured product, it is common to include at least three different production runs, including start-ups.

The process mean (average) and standard deviation are calculated. With a normal distribution, the "tails" can extend well beyond plus and minus three standard deviations, but this interval should contain about 99.73% of production output.

Therefore for a normal distribution of data the process capability is often described as the relationship between six standard deviations and the required specification.

Capability Study

The input of a process is expected to meet customer requirements, specifications, or product tolerances. Engineering can conduct a process capability study to determine the extent to which the process can meet these expectations.

The ability of a process to meet specifications can be expressed as a single number using a process capability index or it can be assessed using control charts. Either case requires running the process to obtain enough measurable output so that engineering is confident that the process is stable and so that the process mean and variability can be

reliably estimated. Statistical process control defines techniques to properly differentiate between stable processes, processes that are drifting (experiencing a long-term change in the mean of the output), and processes that are growing more variable. Process capability indices are only meaningful for processes that are stable.

For Information Technology, ISO 15504 specifies a process capability measurement framework for assessing process capability. This measurement framework consists of 5.5+0.5 levels of process capability from none (Capability Level 0) to optimizing processes. The measurement framework has been generalized so that it can be applied to non IT processes. There are currently two process reference models covering software and systems. The Capability Maturity Model in its latest version also follows this approach.

PROCESS IMPROVEMENT

In organizational development, Process improvement is a series of actions taken by a Process Owner to identify, analyse and improve existing processes within an organization to meet new goals and objectives. These actions often follow a specific methodology or strategy to create successful results.

BENCHMARKING

Benchmarking is a systematic comparison of organizational processes and performance to create new standards or to improve processes. Benchmarking models are used to determine how well a business unit, division, organization or corporation is performing compared with other similar organizations.

A Benchmark is often used for improving communication, professionalizing the organization/processes or for budgetary reasons. Traditionally, performance measures have been compared with previous measures from the same organization at different times. Although this can be a good indication of the rate of improvement within the organization, it could be that although the organization is improving, the competition is improving faster.

There are four types of benchmarking methods:

1. Internal (benchmark within a corporation, for example between business units)
2. Competitive (benchmark performance or processes with competitors)
3. Functional (benchmark similar processes within an industry)
4. Generic (comparing operations between unrelated industries)

Typically, benchmarking models involves the following steps:

- Scope definition
- Choose benchmark partner(s)
- Determine measurement methods, units, indicators and data collection method
- Data collection
- Analysis of the discrepancies

- Present the results and discuss implications/improvement areas and goals
- Make improvement plans or new procedures
- Monitor progress and plan ongoing benchmark.

BUSINESS PROCESS IMPROVEMENT

Business process improvement (BPI) is a systematic approach to help an organization optimize its underlying processes to achieve more efficient results. The organization may be a for-profit business, a non-profit organization, a government agency, or any other ongoing concern. Most BPI techniques were developed and refined in the manufacturing era, though many of the methodologies have been successfully adapted to work in the predominantly service-based economy of today.

While there are differences in the challenges that each type of industry poses, the fact remains that the core principles of BPI and how they apply to business improvement remain portable across industries and functions. It should be noted that BPI focuses on “doing things right” more than it does on “doing the right thing”. In essence, BPI attempts to reduce variation and/or wastage in processes, so that the desired outcome can be achieved with better utilisation of resources.

BPI works by:

- Defining the organization’s strategic goals and purposes
- Determining the organization’s customers
- Aligning the business processes to realise the organization’s goals

The goal of BPI is a radical change in the performance of an organization, rather than a series of incremental changes. Michael Hammer and James Champy popularized this radical model “Reengineering the Corporation: A Manifesto for Business Revolution”. Hammer and Champy stated that the process was not meant to impose trivial changes, such as 10 per cent improvements or 20 per cent cost reductions, but was meant to be revolutionary. Many businesses in the 1990s used the phrase “reengineering” as a euphemism for layoffs. Other organizations did not make radical changes in their business processes and did not make significant gains, and, therefore, wrote the process off as a failure. Yet, others have found that BPI is a valuable tool in a process of gradual change to a business.

Roles

There are four roles within a business Management system: Business Leader, Process Owner, Operational Manager, and Process Operator. The responsibilities of each of these roles are unique, but work together as a system. Some employees in an organization may perform as many as all four of these roles over the course of a day, week, month, or year.

Business Leaders

Business leaders are responsible for creating the business plans and associated resourcing plans necessary to cause the organization to be successful.

Senior leaders are responsible to define the customer and business objectives the organization needs to achieve in order to be successful. This process includes overseeing the development of the organization's mission, vision, and values.

Lower leader levels are responsible for translating senior leaders' business objectives into business objectives that make sense for their level and that support the accomplishment of the senior leaders' business objectives.

The responsibilities of the business leaders follow the PDCA cycle:

- **Plan:** The business leaders create and own the business performance objectives of the organization. Senior leaders need to first understand the requirements of their customers, stockholders, workforce, suppliers, and communities. They need to understand their competition. They need to understand the environmental, economic, technological, social, legal, and political environments that they do business within. Senior leaders need to consider all of these elements as they design a Business model and business Strategy map that will meet the customer and business requirements. Business Leaders then translate these requirements and business environment issues into business performance objectives. Business Leaders then create business plans and associated resourcing plans that will cause the organization to achieve these business objectives. The Business Leaders establish business performance metrics to measure the business's capability to meet these business objectives. Many organizations create a Balanced scorecard to organize and communicate business performance metrics.
- **Do:** The Business Leaders are responsible to communicate to the organization their business plans. As the organization conducts business, the Business Leaders are responsible to build bridges and remove barriers that will allow the business performance objectives to be met. The business performance metric data is produced and collected as business is performed by the organization.
- **Check:** The Business Leaders periodically analyse the business performance data and use it to visualize the business's capability to meet business objectives over time, compare actual performance against performance targets, and identify performance issues.
- **Act:** The Business Leaders are responsible to create improvement actions to address the performance issues that are identified during their analysis of the business performance data. These improvement actions are created to ensure the organization is able to achieve their business plans.

Process Owner

The process owner is the person who is responsible to design the processes necessary to achieve the objectives of the business plans that are created by the Business Leaders. The process owner is responsible for the creation, update and approval of documents to support the process. Many process owners are supported by a process improvement team. The process owner uses this team as a mechanism to help create a high performance process.

The process owner is the only person who has authority to make changes in the process and manages the entire process improvement cycle to ensure performance effectiveness. This person is the contact person for all information related to the process. The responsibilities of the process owner follow the PDCA cycle.

- *Plan:* The process owners create and own the process performance objectives of the organization. The process owner first needs to understand the external and internal customer requirements for the process. This person uses the business plans as a source to help understand the long term and short term customer and business requirements. This person then translates these requirements into process performance objectives and establishes product specifications. This person establishes process performance metrics to measure the process's capability to meet the product specifications and overall process objectives. The set of metrics that are to be reviewed by operational managers and process operators are called key performance indicators. The process owner then designs process steps to describe work that when performed will have the capability to produce product that meets the customer and business requirements.
- *Do:* The process owner is responsible to communicate to the operational managers the details of the processes that the operational managers are responsible to execute. As the operational managers and process operators perform the processes, the process owner is responsible to build bridges and remove barriers that will allow the process performance objectives to be met. The process performance metric data is produced and collected as the process is performed by process operators. The process owner is continually involved with the operational managers and process operators as they use kaizen to continually improve the process as they are performing the work.
- *Check:* The Process Owner periodically analyses the process performance data and use it to visualize the process's capability to operate within control limits over time, compare actual performance against performance targets, and identify performance issues.
- *Act:* The Process Owner is responsible to create improvement actions to address the performance issues that are identified during their analysis of the process performance data. Improvement actions may include the initiation of Lean projects to reduce waste from the process or include the initiation of

Six Sigma projects to reduce variation in the process. Improvement actions may include the use of problem solving tools that would include risk assessment and root-cause analysis. Risk assessment is used to identify and reduce, eliminate, or mitigate risk within the process. This is the proactive approach to avoid problems being created from the process. Root-cause analysis is the reactive way to respond to problems that occur from the process. Root-cause analysis is used to identify the causes of problems within the process and identify and implement improvement actions that will ensure these problems do not occur again.

Operational Manager

The operational Manager is responsible to bring the resources and processes together to achieve the objectives of the business plans that are created by the business leaders.

The responsibilities of the operational manager follow the PDCA cycle:

- **Plan:** The operational managers - in collaboration with each Process Operator, create Process Operator performance objectives for the employees they supervise. The Operational Manager needs to understand the performance requirements of the process. They match employees with the competency and skill requirements of the process to be performed. They ensure that the Process Operators have the budget, facilities, and technology available to them that is necessary to achieve the performance objectives of the processes.
- **Do:** The operational manager is responsible to teach process operators how to perform the processes. Process Operator instruction usually consists of classroom and on-the-job training. The Operational Manager oversees the work and ensures Process Operators receive ongoing informal feedback as to their performance. As the Process Operators perform the processes, the Operational Managers are responsible to build bridges and remove barriers that will allow the process and Process Operator performance objectives to be met. Process and Process Operator performance metric data is produced and collected as the process is performed. The Operational Manager ensures that Process Operators are using Kaizen to continually improve the process as they are performing the work.
- **Check:** The operational manager periodically analyses the key performance indicators during the production cycle to evaluate the work group's ability to achieve the process and process operator performance objectives. This data is used to visualize the process and process operator capability to meet business plan objectives over time, compare actual performance against performance targets, and identify performance issues. They review this performance data and sort out process operator performance issues from process performance issues. Many organizations use a war room concept to post performance data. Within the war room, the operational manager conducts periodic review and analysis of this performance data.

- *Act:* The operational manager is responsible to create improvement actions to address the performance issues that are identified during their analysis of the process and Process Operator performance data. They address Process Operator performance with ongoing feedback to the Process Operator and/or by using an employee performance management review process. They communicate process performance issues to the Process Operator(s) and the Process Owner.

Process Operator

The process operator is responsible to learn and perform the processes (work) necessary to achieve the objectives of the business plans that are created by Business Leaders. The responsibilities of the process operator follow the PDCA cycle.

- *Plan:* The process operators - in collaboration with their Operational Manager, create and own their performance objectives. Process Operators are responsible to understand the performance objectives of the process they are to perform and the specifications of the product they are to produce.
- *Do:* Process operators are responsible to learn the processes (work) that they are to perform. They ensure the processes are performed to meet the process performance objectives and produce product that meets specification. As the Process Operators perform the processes, they are responsible to communicate to their Operational Manager (supervisor) the bridges that need to be built and the barriers that need to be removed to allow the process and Process Operator performance objectives to be met. Process and Process Operator performance metric data is produced and collected as the process is performed.
- *Check:* The process operator periodically reviews the Key performance indicators (KPI's). The Process Operator makes adjustments to their work based on their actual performance compared to KPI targets. The Process Operator is responsible for identifying and reporting any performance issues and stopping production if necessary.
- *Act:* Process operators practice kaizen to continually challenge the process and communicate improvement suggestions to their operational manager (supervisor).

BPI: Key Considerations

Processes need to align to business goals. An organization's strategic goals should provide the key direction for any Business Process Improvement exercise. This alignment can be brought about by integrating programmes like Balanced Scorecard to the BPI initiative. *e.g.*, When deploying Six Sigma, identification of projects can be done on the basis of how they fit into the Balanced Scorecard agenda of the organization.

Customer focus. Fast-changing customer needs underscore the importance of aligning business processes to achieve higher customer satisfaction. It is imperative in any BPI

exercise that the “Voice of Customer” be known, and factored in, when reviewing or redesigning any process. Importance of benchmarks BPI tools place a lot of emphasis on “measurable results”. Accordingly, benchmarks assume an important role in any BPI initiative. Depending on the lifecycle of the process in question, benchmarks may be internal (within the organization), external (from other competing/noncompeting organizations) or dictated by the senior management of the organization as an aspirational target. Establish process owners For any process to be controllable, it is essential that there be clarity on who is the process owners, and what constitutes success/failure of the process. These success/failure levels also help establish “control limits” for the process, and provide a healthy check on whether or not a process is meeting the desired customer objectives.

Methodology of BPI

- Carrying out BPI is a project, so all principles of project management apply. This ensures, for example, that improvement processes do not conflict with each other.
- The first step in BPI is to define the existing structure and process at play.
- Then, the BPI process owners should determine what outcomes would add value to the organization’s objectives and how best to align its processes to achieve those outcomes.
- Once the outcomes are determined, the organization’s workforce needs to be re-organized to meet the new objectives, using the variety of tools available within the BPI methodology.

Implementing BPI

Most resistance to BPI comes from within an organization. Managers often do not wish to change existing structures. The labour force may resist BPI because of fears of layoffs; however, an organization using BPI on a regular basis, argue many proponents, will already have the proper workforce to meet existing business challenges.

Some organizations have implemented BPI on a smaller scale and reported success, by doing the following:

- Start with a small process that can be completed in a short time frame.
- Set clear timelines.
- Do not spread resources thinly and focus on the short term payoff.
- Management and primary stakeholders must be involved, or else even a limited implementation will fail.

BUSINESS PROCESS REENGINEERING

Business process reengineering is the main way in which organizations become more efficient and modernize. Business process reengineering transforms an organization in ways that directly affect performance.

THE IMPACT OF BPR ON ORGANIZATIONAL PERFORMANCE

The two cornerstones of any organization are the people and the processes. If individuals are motivated and working hard, yet the business processes are cumbersome and non-essential activities remain, organizational performance will be poor. Business Process Reengineering is the key to transforming how people work. What appear to be minor changes in processes can have dramatic effects on cash flow, service delivery and customer satisfaction. Even the performance of documenting business processes alone will typically improve organizational efficiency by 10%.

HOW TO IMPLEMENT A BPR PROJECT

The best way to map and improve the organization's procedures is to take a top down approach, and not undertake a project in isolation.

That means:

- Starting with mission statements that define the purpose of the organization and describe what sets it apart from others in its sector or industry.
- Producing vision statements which define where the organization is going, to provide a clear picture of the desired future position.
- Build these into a clear business strategy thereby deriving the project objectives.
- Defining behaviours that will enable the organization to achieve its' aims.
- Producing key performance measures to track progress.
- Relating efficiency improvements to the culture of the organization
- Identifying initiatives that will improve performance.

TOOLS TO SUPPORT BPR

When a BPR project is undertaken across the organization, it can require managing a massive amount of information about the processes, data and systems. If you don't have an excellent tool to support BPR, the management of this information can become an impossible task. The use of a good BPR/documentation tool is vital in any BPR project.

The types of attributes you should look for in BPR software are:

- Graphical interface for fast documentation
- "Object oriented" technology, so that changes to data only need to be made in one place, and the change automatically appears throughout all the organization's procedures and documentation.
- Drag and drop facility so you can easily relate organizational and data objects to each step in the process
- Customizable meta data fields, so that you can include information relating to your industry, business sector or organization in your documentation

- Analysis, such as swim-lanes to show visually how responsibilities in a process are transferred between different roles, or where data items or computer applications are used.
- Support for Value Stream mapping.
- CRUD or RACI reports, to provide evidence for process improvement.
- The ability to assess the processes against agreed international standards
- Simulation software to support ‘what-if’ analyses during the design phase of the project to develop LEAN processes
- The production of word documents or web site versions of the procedures at the touch of a single button, so that the information can be easily maintained and updated.

The software we use by choice is Protos, a very comprehensive Dutch system that has been translated into English.

Protos meets all the above requirements, and many more, and is better than any system originated in English that we have seen.

CAPABILITY MATURITY MODEL INTEGRATION

Capability Maturity Model Integration is a process improvement approach that helps organizations improve their performance. CMMI can be used to guide process improvement across a project, a division, or an entire organization.

CMMI in software engineering and organizational development is a trademarked process improvement approach that provides organizations with the essential elements for effective process improvement.

CMMI helps “integrate traditionally separate organizational functions, set process improvement goals and priorities, provide guidance for quality processes, and provide a point of reference for appraising current processes.”

CMMI currently addresses three areas of interest:

1. *Product and service development:* CMMI for Development,
2. *Service establishment, management, and delivery:* CMMI for Services, and
3. *Product and service acquisition:* CMMI for Acquisition.

CMMI was developed by a group of experts from industry, government, and the Software Engineering Institute at Carnegie Mellon University. CMMI models provide guidance for developing or improving processes that meet the business goals of an organization. A CMMI model may also be used as a framework for appraising the process maturity of the organization.

CMMI originated in software engineering but has been highly generalised over the years to embrace other areas of interest, such as the development of hardware products, the delivery of all kinds of services, and the acquisition of products and services. The word “software” does not appear in definitions of CMMI. This generalization of improvement concepts makes CMMI extremely abstract. It is not as specific to software engineering as its predecessor, the Software CMM.

CMMI was developed by the CMMI project, which aimed to improve the usability of maturity models by integrating many different models into one framework. The project consisted of members of industry, government and the Carnegie Mellon Software Engineering Institute. The main sponsors included the Office of the Secretary of Defence and the National Defence Industrial Association.

CMMI is the successor of the capability maturity model (CMM) or software CMM. The CMM was developed from 1987 until 1997. In 2002, CMMI Version 1.1 was released. Version 1.2 followed in August 2006.

CMMI TOPICS

CMMI Representation

CMMI exists in two representations: continuous and staged. The continuous representation is designed to allow the user to focus on the specific processes that are considered important for the organization's immediate business objectives, or those to which the organization assigns a high degree of risk. The staged representation is designed to provide a standard sequence of improvements, and can serve as a basis for comparing the maturity of different projects and organizations. The staged representation also provides for an easy migration from the SW-CMM to CMMI.

CMMI Model Framework

Depending on the CMMI constellation (acquisition, services, development) used, the process areas it contains will vary. Key process areas are the areas that will be covered by the organization's processes. The table lists the process areas that are present in all CMMI constellations. This collection of eight process areas is called the CMMI Model Framework, or CMF.

**Table. Capability Maturity Model Integration
(CMMI) Model Framework (CMF).**

Abbreviation	Name	Area	Maturity Level
REQM	Requirements Management	Engineering	2
PMC	Project Monitoring and Control	Project Management	2
PP	Project Planning	Project Management	2
CM	Configuration Management	Support	2
MA	Measurement and Analysis	Support	2
PPQA	Process and Product Quality Assurance	Support	2
OPD	Organizational Process Definition	Process Management	3
CAR	Causal Analysis	Support	5

CMMI Models

CMMI best practices are published in documents called models, each of which addresses a different area of interest. The current release of CMMI, version 1.2, provides models for three areas of interest: development, acquisition, and services.

- CMMI for Development (CMMI-DEV), v1.2 was released in August 2006. It addresses product and service development processes.
- CMMI for Acquisition (CMMI-ACQ), v1.2 was released in November 2007. It addresses supply chain management, acquisition, and outsourcing processes in government and industry.
- CMMI for Services (CMMI-SVC), v1.2 was released in February 2009. It addresses guidance for delivering services within an organization and to external customers.
- CMMI Product Suite (includes Development, Acquisition, and Services), v1.3 is expected to be released in 2010. CMMI Version 1.3—Plans for the Next Version

Regardless of which model an organization chooses, CMMI best practices should be adapted by an organization according to its business objectives.

Appraisal

An organization cannot be certified in CMMI; instead, an organization is appraised. Depending on the type of appraisal, the organization can be awarded a maturity level rating (1-5) or a capability level achievement profile.

Many organizations find value in measuring their progress by conducting an appraisal. *Appraisals are typically conducted for one or more of the following reasons:*

- To determine how well the organization's processes compare to CMMI best practices, and to identify areas where improvement can be made
- To inform external customers and suppliers of how well the organization's processes compare to CMMI best practices
- To meet the contractual requirements of one or more customers

Appraisals of organizations using a CMMI model must conform to the requirements defined in the Appraisal Requirements for CMMI (ARC) document. There are three classes of appraisals, A, B and C, which focus on identifying improvement opportunities and comparing the organization's processes to CMMI best practices. Appraisal teams use a CMMI model and ARC-conformant appraisal method to guide their evaluation of the organization and their reporting of conclusions. The appraisal results can then be used to plan improvements for the organization.

The Standard CMMI Appraisal Method for Process Improvement (SCAMPI) is an appraisal method that meets all of the ARC requirements.

A class A appraisal is more formal and is the only one that can result in a level rating. Results of an appraisal may be published on the CMMI Web site of the SEI: Published

SCAMPI Appraisal Results. SCAMPI also supports the conduct of ISO/IEC 15504, also known as SPICE assessments, etc.

Achieving CMMI Compliance

The traditional approach that organizations often adopt to achieve compliance with the CMMI involves the establishment of an Engineering Process Group and Process Action Teams. This approach requires that members of the EPG and PATs be trained in the CMMI, that an informal appraisal be performed, and that process areas be prioritized for improvement.

More modern approaches that involve the deployment of commercially available, CMMI-compliant processes, can significantly reduce the time to achieve compliance. SEI has maintained statistics on the "time to move up" for organizations adopting the earlier Software CMM and primarily using the traditional approach. These statistics indicate that, since 1987, the median times to move from Level 1 to Level 2 is 23 months, and from Level 2 to Level 3 is an additional 20 months. These statistics have not been updated for the CMMI.

The Software Engineering Institute's Team Software Process methodology and the Capability Maturity Modelling framework have been successfully employed to accelerate progress from Maturity Level 1 to Maturity Level 4. They've demonstrated progressing from Level 1 to Level 4 in 30 months, which is less than half of the average time it has taken traditionally.

Applications

The SEI published that 60 organizations measured increases of performance in the categories of cost, schedule, productivity, quality and customer satisfaction. The median increase in performance varied between 14% and 62%. However, the CMMI model mostly deals with what processes should be implemented, and not so much with how they can be implemented.

These results do not guarantee that applying CMMI will increase performance in every organization. A small company with few resources may be less likely to benefit from CMMI; this view is supported by the process maturity profile. Of the small organizations, 70.5% are assessed at level 2: Managed, while 52.8% of the organizations with 1001-2000 employees are rated at the highest level.

Interestingly, Turner and Jain argue that although it is obvious there are large differences between CMMI and agile methods, both approaches have much in common. They believe neither way is the 'right' way to develop software, but that there are phases in a project where one of the two is better suited.

They suggest one should combine the different fragments of the methods into a new hybrid method. Sutherland *et al.* assert that a combination of Scrum and CMMI brings more adaptability and predictability than either one alone. David J. Anderson gives

hints on how to interpret CMMI in an agile manner. Other viewpoints about using CMMI and Agile development are available on the SEI Web site.

The combination of the project management technique earned value management with CMMI has been described. To conclude with a similar use of CMMI, Extreme Programming, a software engineering method, has been evaluated with CMM/CMMI. For example, the XP requirements management approach, was evaluated as not compliant with CMMI.

CMMI can be appraised using two different approaches: staged and continuous. The staged approach yields appraisal results as one of five maturity levels. The continuous approach yields one of six capability levels. The differences in these approaches are felt only in the appraisal; the best practices are equivalent and result in equivalent process improvement results.

INSPECTION

An inspection is, most generally, an organized examination or formal evaluation exercise. It involves the measurements, tests, and gauges applied to certain characteristics in regard to an object or activity. The results are usually compared to specified requirements and standards for determining whether the item or activity is in line with these targets. Inspections are usually non-destructive.

Non-Destructive Examination or Non-Destructive Testing describe a number of technologies used to analyse materials for either inherent flaws or damage from use. Some common methods are visual, Liquid or dye penetrant inspection, magnetic-particle inspection, radiographic testing, ultrasonic testing, eddy-current testing, acoustic emission testing, and thermographic inspection. In addition, many non-destructive inspections can be performed by a precision scale, or when in motion, a checkweigher.

A surprise inspection tends to have different results than an announced inspection. Leaders seeking to discover how well lower echelons in their organization are typically doing sometimes drop in unannounced to see what is going on and what conditions are. When an inspection is scheduled in advance, it gives people a chance to cover up or fix mistakes. A surprise inspection, therefore, gives inspectors a better picture of the typical state of the inspected object than an announced inspection.

TOTAL QUALITY CONTROL

A management approach that seeks to improve quality and performance, and thus meet or exceed customer expectations, by integrating all quality-related functions and processes throughout the organization; takes a holistic approach to managing quality design and development, quality control and maintenance, quality improvement, and quality assurance, at all levels and involving all employees.

QUALITY CIRCLES

The genesis of Quality Circle was in war ravaged Japan in the 1940's, which was struggling to survive in the industrial world. Prof. Ishikawa, who believed in tapping the creative potential of workers, innovated the Quality Circle movement to give Japanese industry that extra edge in creativity.

A Quality Circle is a small group of employees from the same work area who voluntarily meet at regular intervals to identify, analyse, and resolve work related problems. This not only improves the performance of any organisation, it also motivates and enriches the work life of employees. The philosophy behind Quality Circles is building people.

The Quality Circle concept is firmly entrenched in Tata Steel, and the target is to involve all workers in the quality movement through their participation in QC activities. TMH has 57 active Quality Circles in diverse work areas like the Hospital laundry and kitchen, etc. The advantage of this movement has been empowerment of the grass root level.

GENESIS OF QUALITY CIRCLES

After the Second World War Japanese economy was in the doldrums. Seeing this disastrous effect of war, Americans decided to help Japan in improving the quality standards of their products.

General Douglas Mac Arthur who, at that time, was the commander of the occupational forces in Japan took up the task of imparting quality awareness among Japanese to help them improve their products and the reliability of manufacturing systems including men, machine and materials.

Thus, by 1975, they were topping the world in quality and productivity. This astonishing and unique achievement in modern history became an eye – opener to the world. Industrialists and politicians from all over the world started visiting Japan to know how they have achieved such magical results in such a short span.

The answer to this was painstaking and persevering efforts of the Japanese leaders and workers and the development and growth of the philosophy of small working groups. This resulted in the Quality Circle concept being accepted all over the world as a very effective technique to improve the total quality of work life.

DEFINITION

Quality Circle is a small group of 6 to 12 employees doing similar work who voluntarily meet together on a regular basis to identify improvements in their respective work areas using proven techniques for analysing and solving work related problems coming in the way of achieving and sustaining excellence leading to mutual upliftment of employees as well as the organisation. It is “a way of capturing the creative and innovative power that lies within the workforce”.

PHILOSOPHY

Quality Circles is a people – building philosophy, providing self-motivation and happiness in improving environment without any compulsion or monetary benefits. It represents a philosophy of managing people specially those at the grass root level as well as a clearly defined mechanism and methodology for translating this philosophy into practice and a required structure to make it a way of life. It is bound to succeed where people are respected and are involved in decisions, concerning their work life, and in environments where peoples' capabilities are looked upon as assets to solve work-area problems.

The Quality Circle philosophy calls for a progressive attitude on the part of the management and their willingness to make adjustments, if necessary, in their style and culture. If workers are prepared to contribute their ideas, the management must be willing to create a congenial environment to encourage them to do so.

CONCEPT

The concept of Quality Circle is primarily based upon recognition of the value of the worker as a human being, as someone who willingly activates on his job, his wisdom, intelligence, experience, attitude and feelings.

It is based upon the human resource management considered as one of the key factors in the improvement of product quality and productivity.

Quality Circle concept has three major attributes:

1. Quality Circle is a form of participation management.
2. Quality Circle is a human resource development technique.
3. Quality Circle is a problem solving technique.

OBJECTIVE

The objectives of Quality Circles are multi-faced:

- Change in Attitude. From “I don’t care” to “I do care” Continuous improvement in quality of work life through humanisation of work.
- Self Development Bring out ‘Hidden Potential’ of people People get to learn additional skills.
- Development of Team Spirit Individual Vs Team – “I could not do but we did it” Eliminate inter departmental conflicts.
- Improved Organisational Culture Positive working environment. Total involvement of people at all levels. Higher motivational level. Participate Management process.

ORGANISATIONAL STRUCTURE

A Quality Circle has an appropriate organisational structure for its effective and efficient performance. It varies from industry to industry, organisation to organisation.

But it is useful to have a basic framework as a model. *The structure of a Quality Circle consists of the following elements:*

- *A steering committee:* This is at the top of the structure. It is headed by a senior executive and includes representatives from the top management personnel and human resources development people. It establishes policy, plans and directs the programme and meets usually once in a month.
- *Co-ordinator:* He may be a Personnel or Administrative officer who co-ordinates and supervises the work of the facilitators and administers the programme.
- *Facilitator:* He may be a senior supervisory officer. He co-ordinates the works of several quality circles through the Circle leaders.
- *Circle leader:* Leaders may be from lowest level workers or Supervisors. A Circle leader organises and conducts Circle activities.
- *Circle members:* They may be staff workers. Without circle members the programme cannot exist. They are the lifeblood of quality circles. They should attend all meetings as far as possible, offer suggestions and ideas, participate actively in group process, take training seriously with a receptive attitude. The roles of Steering Committee, Co-ordinator, Facilitator, Circle leader and Circle members are well defined.

LAUNCHING QUALITY CIRCLES

The major prerequisite for initiating Quality Circles in any organisation is the total understanding of, as well as complete conviction and faith in the participative philosophy, on the part of the top and senior management. In the absence of a commitment from the Chief Executive to support the Quality Circle movement totally, it would be inadvisable to seriously attempt the starting of Quality Circles. The launching of Quality Circles involves the following steps:

Expose Middle Level Executives to the Concept

Explain the concept to the employees and invite them to volunteer as members of Quality Circles. Nominate senior officers as facilitators.

Form a Steering Committee

Arrange training of co-ordinators, facilitators in basics of Quality Circle approach, implementation, techniques and operation. Later facilitator may provide training to Circle leaders and Circle members.

A meeting should be fixed preferably one hour a week for the Quality Circle to meet. Formally inaugurate the Quality Circle. Arrange the necessary facilities for the Quality Circle meeting and its operation.

TRAINING

Appropriate training for different sections of employees needs to be imparted. Without a proper understanding of the real concept of Quality Circles, both the workers and management might look at this philosophy with suspicion. Each group should know beforehand the commitments and implications involved as well as the benefit that can be obtained from Quality Circles.

Such training comprises of:

- Brief orientation programme for top management.
- Programme for middle level executives.
- Training of facilitators.
- Training for Circle leaders and members.

PROCESS OF OPERATION

The operation of quality circles involves a set of sequential steps as under:

- *Problem identification:* Identify a number of problems.
- *Problem selection:* Decide the priority and select the problem to be taken up first.
- *Problem Analysis:* Problem is clarified and analysed by basic problem solving methods.
- *Generate alternative solutions:* Identify and evaluate causes and generate number of possible alternative solutions.
- *Select the most appropriate solution:* Discuss and evaluate the alternative solutions by comparison in terms of investment and return from the investment. This enables to select the most appropriate solution.
- *Prepare plan of action:* Prepare plan of action for converting the solution into reality which includes the considerations “who, what, when, where, why and how” of solving problems.
- Present solution to management circle members present solution to management for approval.
- *Implementation of solution:* The management evaluates the recommended solution. Then it is tested and if successful, implemented on a full scale.

BASIC PROBLEM SOLVING TECHNIQUES

The following techniques are most commonly used to analyse and solve work related problems:

- Brain storming
- Pareto Diagrams
- Cause and Effect Analysis
- Data Collection
- Data Analysis

The tools used for data analysis are:

- Tables
- Bar Charts
- Histograms
- Circle graphs
- Line graphs
- Scattergrams
- Control Charts

The Quality Circles also are expected to develop internal leadership, reinforce worker morale and motivation, and encourage a strong sense of teamwork in an organisation.

A variety of benefits have been attributed to Quality Circles, including higher quality, improved productivity, greater upward flow of information, broader improved worker attitudes, job enrichment, and greater teamwork. Problem quality circles often suffer from unrealistic expectations for fast results, lack of management commitment and support, resistance by middle management, resentment by non participants, inadequate training, lack of clear objectives and failure to get solutions implemented.

BENEFITS AND LIMITATIONS OF QUALITY CIRCLES

It took more than two decades for the quality control concept to get acceptance in India, after its introduction in Japan. This may be due to the differences in the industrial context in the two countries. Japan needed it for its survival in a competitive market. India had a reasonably protected, sellers market, with consequent lethargy towards efforts to improve quality and productivity. However, with the policy of liberalisation of economy and privatisation of infrastructure development, contexts changed. The concept now needs to be looked upon as a necessity.

APPLICABILITY IN GOVERNMENT ORGANISATIONS

The Public Works Department of Government of Maharashtra is responsible for the construction and maintenance of about 2.00 lakhs km. of road net work in the state and a large number of Government-owned buildings spread all over the state.

The road net work caters for national level traffic to the rural area traffic. The climatic and geographical features also vary from place to place in the state. The ultimate customer of the Department is the common man and it is not generally easy to satisfy the ultimate needs of this user to the fullest extent merely by taking decision at the top management level. The sub-ordinate officers and staff working in the field and who are well-acquainted with the day-to-day problems in their work areas, are the best judges to decide upon the line of action.

The top management in Maharashtra P.W.D. realised this need of the time and introduced the concept of Quality Circle as employee participation method in the organisation. The concept of Quality Circle was launched in the Public Works

Department of Govt. of Maharashtra on 18-11-1997. In a span of about a year 17 Quality Circles have been formed in Mumbai Region wherein 22 problems have been solved so far. 48 Quality Circles have been formed in Pune Region wherein, 8 problems have been solved.

Nashik Region of the P.W.D. has formed 9 Quality Circles and solutions to 3 problems have been found out. Aurangabad Region has formed 21 Quality Circles and solved 7 problems. Nagpur Regions have formed 47 Quality Circles and have found out solution to 57 problems, so far.

The results of a short period of one year are found to be very encouraging so far and many of these Circles have already come up with solutions to the various work related problems as evidenced from the last two state level conventions held at Mumbai and Aurangabad respectively. The presentations by the various Quality Circles at these conventions bear testimony to the fact that the concept of Quality Circle has been received with much enthusiasm at the grass root level.

These presentations also provide concrete evidence fulfilling the expected benefits of the Quality Circle. Taking inspirations from these conventions, more staff of the Department have come forward to form Quality Circles, emphasising the fact that Quality Circles have come to stay in the Department.

Some of the problems solved by the Quality Circles in P.W.D. of Maharashtra are as follows:

- Eliminating delay in issuing observation memos after inspection of site.
- Preventing accidents on highways.
- Reducing electricity bill in office.
- Avoiding duplication of work.
- Removal of encroachment from Govt. land.
- Speeding up pot-hole filling on roads during monsoon.
- Upkeeping of service records of employees.
- Maintaining Govt. offices clean.
- Streamlining reservation system of Govt. rest houses.
- Improvement in maintenance of Govt. Hospital at Solapur.

Some of the problems and the solutions thereof are explained briefly:

- *Delay in issue of observation memos:* The Public Works Department of Maharashtra is divided into six regions each headed by a Chief Engineer. Each regional Chief Engineer is assisted by a Superintending Engineer, Vigilance and Quality Circle who with the help of two Executive Engineer and about 6 Deputy Engineers carries out inspection of the ongoing projects. The samples of the construction material being used on project are collected and independently tested in the Regional Testing Laboratories to ascertain the level of acceptance. The findings of the inspections are communicated to the field engineers in the form of 'observation memo'. It is contemplated that the Execution Engineer in charge of the project ensures immediate compliance

of the remarks raised in the observation memo and carries out corrective measures in the ongoing project as well as preventive measures for the future works. However, very often the observation memos are issued after much delay defeating the very purpose of the observation memo. This problem was identified by the 'OMKAR' Quality Circle formed in the office of Superintending Engineer, Vigilance and Quality Control Circle, Navi Mumbai. The members used various tools and methodology, identified the various root causes that led to this problem and finally solved the problem with requisite foolproofing. By solving this problem, it became possible to improve the quality of work in the field which ultimately brightened the image of the Department. The formation of Quality Circle also benefited employees by providing.

- A systematic style of education that helped them grow.
 - Freedom to modify their work habits.
 - Opportunity to use their brain for development of the Department; and
 - Opportunity to work as a team.
 - The employees were also motivated to improve work culture, self improvement recognition and creativity in work.
- *Preventing accidents on highways:* It was observed that the accident rate on Sion-Panvel Highway passing through New Mumbai had gone up sharply. The Circle members collected the data from the respective police stations and identified the important causes of accidents by using Pareto Analysis. They segregated the causes under Man, Machine, Material and Method. The Circle members found out solutions to the problem such as closing of central verge at all locations except junction, frequent lane marking, installation of Caution boards, Providing lay-byes for buses etc. The data collection of accidents for a period three months after implementation of solutions near Shirvane and Turbhe revealed that the cause of accident had been eliminated and no more accidents occurred at those locations had been eliminated and no more accidents occurred at those locations again. This way the Circle members ensured the effectiveness of the solution implemented. They made presentation to their steering committee for obtaining sanction to other solutions involving more funds.

KAIZEN

Kaizen refers to a philosophy or practices that focus upon continuous improvement of processes in manufacturing, engineering, supporting business processes, and management. It has been applied in health care, government, banking, and many other industries. When used in the business sense and applied to the workplace, kaizen refers to activities that continually improve all functions, and involves all employees from the

CEO to the assembly line workers. It also applies to processes, such as purchasing and logistics, that cross organizational boundaries into the supply chain. By improving standardized activities and processes, kaizen aims to eliminate waste. Kaizen was first implemented in several Japanese businesses after the Second World War, influenced in part by American business and quality management teachers who visited the country. It has since spread throughout the world.

Kaizen is a daily activity, the purpose of which goes beyond simple productivity improvement. It is also a process that, when done correctly, humanizes the workplace, eliminates overly hard work and teaches people how to perform experiments on their work using the scientific method and how to learn to spot and eliminate waste in business processes.

In all, the process suggests a humanized approach to workers and to increasing productivity: "The idea is to nurture the company's human resources as much as it is to praise and encourage participation in kaizen activities." Successful implementation requires "the participation of workers in the improvement."

People at all levels of an organization can participate in kaizen, from the CEO down, as well as external stakeholders when applicable. The format for kaizen can be individual, suggestion system, small group, or large group. At Toyota, it is usually a local improvement within a workstation or local area and involves a small group in improving their own work environment and productivity.

This group is often guided through the kaizen process by a line supervisor; sometimes this is the line supervisor's key role. Kaizen on a broad, cross-departmental scale in companies, generates total quality management, and frees human efforts through improving productivity using machines and computing power.

While kaizen usually delivers small improvements, the culture of continual aligned small improvements and standardization yields large results in the form of compound productivity improvement. This philosophy differs from the "command and control" improvement programmes of the mid-twentieth century.

Kaizen methodology includes making changes and monitoring results, then adjusting. Large-scale pre-planning and extensive project scheduling are replaced by smaller experiments, which can be rapidly adapted as new improvements are suggested.

In modern usage, a focused kaizen that is designed to address a particular issue over the course of a week is referred to as a "kaizen blitz" or "kaizen event". These are limited in scope, and issues that arise from them are typically used in later blitzes.

After World War II, to help restore Japan, American occupation forces brought in American experts to help with the rebuilding of Japanese industry. The Civil Communications Section developed a Management Training Programme that taught statistical control methods as part of the overall material. This course was developed and taught by Homer Sarasohn and Charles Protzman in 1949 and 1950.

Sarasohn recommended William Deming for further training in Statistical Methods. The Economic and Scientific Section group was also tasked with improving Japanese

management skills and Edgar McVoy is instrumental in bringing Lowell Mellen to Japan to properly install the TWI programmes in 1951. Prior to the arrival of Mellen in 1951, the ESS group had a training film done to introduce the three TWI “J” programmes—the film was titled “Improvement in 4 Steps”.

This is the original introduction of “Kaizen” to Japan. For the pioneering, introducing, and implementing Kaizen in Japan, the Emperor of Japan awarded the Second Order Medal of the Sacred Treasure to Dr. Deming in 1960. Consequently, the Union of Japanese Science and Engineering instituted the annual Deming Prizes for achievements in quality and dependability of products in Japan. Reference: US National Archives - SCAP collection

IMPLEMENTATION

The Toyota Production System is known for kaizen, where all line personnel are expected to stop their moving production line in case of any abnormality and, along with their supervisor, suggest an improvement to resolve the abnormality which may initiate a kaizen.

The cycle of kaizen activity can be defined as:

- Standardize an operation
- Measure the standardized operation
- Gauge measurements against requirements
- Innovate to meet requirements and increase productivity
- Standardize the new, improved operations
- Continue cycle ad infinitum

This is also known as the Shewhart cycle, Deming cycle, or PDCA. Apart from business applications of the method, both Anthony Robbins and Robert Maurer have popularized the kaizen principles into personal development principles.

Jeffrey Liker and David Meier discuss the kaizen blitz and kaizen burst approaches to continuous improvement. A kaizen blitz, or rapid improvement, is a focused activity on a particular process or activity. The basic concept is to identify and quickly remove waste. Another approach is that of the kaizen burst, a specific kaizen activity on a particular process in the value stream.

Key elements of kaizen are quality, effort, involvement of all employees, willingness to change, and communication.

QUALITY ASSURANCE

Quality assurance is the process of verifying or determining whether products or services meet or exceed customer expectations. Quality assurance is a process-driven approach with specific steps to help define and attain goals. This process considers

design, development, production, and service. The most popular tool used to determine quality assurance is the Shewhart Cycle, developed by Dr. W. Edwards Deming. This cycle for quality assurance consists of four steps: *Plan*, *Do*, *Check*, and *Act*. These steps are commonly abbreviated as PDCA.

The four quality assurance steps within the PDCA model stand for:

1. *Plan*: Establish objectives and processes required to deliver the desired results.
2. *Do*: Implement the process developed.
3. *Check*: Monitor and evaluate the implemented process by testing the results against the predetermined objectives
4. *Act*: Apply actions necessary for improvement if the results require changes.

PDCA is an effective method for monitoring quality assurance because it analyses existing conditions and methods used to provide the product or service customers. The goal is to ensure that excellence is inherent in every component of the process. Quality assurance also helps determine whether the steps used to provide the product or service are appropriate for the time and conditions. In addition, if the PDCA cycle is repeated throughout the lifetime of the product or service, it helps improve internal company efficiency.

Quality assurance demands a degree of detail in order to be fully implemented at every step. *Planning*, for example, could include investigation into the quality of the raw materials used in manufacturing, the actual assembly, or the inspection processes used. The *Checking* step could include customer feedback, surveys, or other marketing vehicles to determine if customer needs are being exceeded and why they are or are not. *Acting* could mean a total revision in the manufacturing process in order to correct a technical or cosmetic flaw.

Competition to provide specialized products and services results in breakthroughs as well as long-term growth and change. Quality assurance verifies that any customer offering, regardless if it is new or evolved, is produced and offered with the best possible materials, in the most comprehensive way, with the highest standards. The goal to exceed customer expectations in a measurable and accountable process is provided by quality assurance.

EARLY EFFORTS TO CONTROL THE QUALITY OF PRODUCTION

Early civil engineering projects needed to be built from specifications, for example the four sides of the base of the Great Pyramid of Giza were required to be perpendicular to within 3.5 arcseconds. During the Middle Ages, guilds adopted responsibility for quality control of their members, setting and maintaining certain standards for guild membership.

Royal governments purchasing material were interested in quality control as customers. For this reason, King John of England appointed William Wrotham to report about the construction and repair of ships. Centuries later, Samuel Pepys, Secretary to the British Admiralty, appointed multiple such overseers. Prior to the extensive division

of labour and mechanization resulting from the Industrial Revolution, it was possible for workers to control the quality of their own products. Working conditions then were arguably more conducive to professional pride.

The Industrial Revolution led to a system in which large groups of people performing a similar type of work were grouped together under the supervision of a foreman who was appointed to control the quality of work manufactured.

WARTIME PRODUCTION

Around the time of World War I, manufacturing processes typically became more complex with larger numbers of workers being supervised. This period saw the widespread introduction of mass production and piecework, which created problems as workmen could now earn more money by the production of extra products, which in turn led to bad workmanship being passed on to the assembly lines.

To counter bad workmanship, full time inspectors were introduced into the factory to identify, quarantine and ideally correct product quality failures. Quality control by inspection in the 1920s and 1930s led to the growth of quality inspection functions, separately organised from production and big enough to be headed by superintendents.

The systematic approach to quality started in industrial manufacture during the 1930s, mostly in the USA, when some attention was given to the cost of scrap and rework. With the impact of mass production, which was required during the Second World War, it became necessary to introduce a more appropriate form of quality control which can be identified as Statistical Quality Control, or SQC. Some of the initial work for SQC is credited to Walter A. Shewhart of Bell Labs, starting with his famous one-page memorandum of 1924.

SQC came about with the realization that quality cannot be fully inspected into an important batch of items. By extending the inspection phase and making inspection organizations more efficient, it provides inspectors with control tools such as sampling and control charts, even where 100 per cent inspection is not practicable. Standard statistical techniques allow the producer to sample and test a certain proportion of the products for quality to achieve the desired level of confidence in the quality of the entire batch or production run.

POSTWAR

In the period following World War II, many countries' manufacturing capabilities that had been destroyed during the war were rebuilt. The U.S. sent General Douglas MacArthur to oversee the re-building of Japan. During this time, General MacArthur involved two key individuals in the development of modern quality concepts: W. Edwards Deming and Joseph Juran. Both individuals promoted the collaborative concepts of quality to Japanese business and technical groups, and these groups utilized these concepts in the redevelopment of the Japanese economy. Although there were many

individuals trying to lead United States industries towards a more comprehensive approach to quality, the U.S. continued to apply the Quality Control concepts of inspection and sampling to remove defective product from production lines, essentially ignoring advances in QA for decades.

CONCEPT OF QUALITY AND MANUFACTURING

Quality in business, engineering and manufacturing has a pragmatic interpretation as the non-inferiority or superiority of something. Quality is a perceptual, conditional and somewhat subjective attribute and may be understood differently by different people. Consumers may focus on the specification quality of a product/service, or how it compares to competitors in the marketplace.

Producers might measure the conformance quality, or degree to which the product/service was produced correctly. Numerous definitions and methodologies have been created to assist in managing the quality-affecting aspects of business operations.

Many different techniques and concepts have evolved to improve product or service quality. There are two common quality-related functions within a business. One is quality assurance which is the prevention of defects, such as by the deployment of a quality management system and preventative activities like FMEA.

The other is quality control which is the detection of defects, most commonly associated with testing which takes place within a quality management system typically referred to as verification and validation.

DEFINITIONS

The common element of the business definitions is that the quality of a product or service refers to the perception of the degree to which the product or service meets the customer's expectations. Quality has no specific meaning unless related to a specific function and/or object. Quality is a perceptual, conditional and somewhat subjective attribute. The business meanings of quality have developed over time.

Various interpretations are:

- *ISO 9000*: "Degree to which a set of inherent characteristics fulfils requirements." The standard defines requirement as need or expectation.
- *Six Sigma*: "Number of defects per million opportunities."
- *Subir Chowdhury*: "Quality combines people power and process power."
- *Philip B. Crosby*: "Conformance to requirements." The requirements may not fully represent customer expectations; Crosby treats this as a separate problem.
- *Joseph M. Juran*: "Fitness for use." Fitness is defined by the customer.
- *Noriaki Kano and others, present a two-dimensional model of quality*: "must-be quality" and "attractive quality." The former is near to "fitness for use" and the latter is what the customer would love, but has not yet thought about.

Supporters characterize this model more succinctly as: “Products and services that meet or exceed customers’ expectations.”

- *Robert Pirsig*: “The result of care.”
- *Genichi Taguchi*, with two definitions:
 - “Uniformity around a target value.” The idea is to lower the standard deviation in outcomes, and to keep the range of outcomes to a certain number of standard deviations, with rare exceptions.
 - “The loss a product imposes on society after it is shipped.” This definition of quality is based on a more comprehensive view of the production system.
- *American Society for Quality*: “A subjective term for which each person has his or her own definition. In technical usage, quality can have two meanings:
 - The characteristics of a product or service that bear on its ability to satisfy stated or implied needs;
 - A product or service free of deficiencies.”
- *Peter Drucker*: “Quality in a product or service is not what the supplier puts in. It is what the customer gets out and is willing to pay for.”
- *W. Edwards Deming*: concentrating on “the efficient production of the quality that the market expects,” and he linked quality and management: “Costs go down and productivity goes up as improvement of quality is accomplished by better management of design, engineering, testing and by improvement of processes.”
- *Gerald M. Weinberg*: “Value to some person”.

MARKET SECTOR PERSPECTIVES

Operations Management

The dimensions of quality refer to the attributes that quality achieves in Operations Management:

- Quality supports Dependability
- Dependability supports Speed
- Speed supports Flexibility
- Flexibility supports Cost

Quality <-> Dependability <-> Speed <-> Flexibility <-> Cost

Manufacturing

In the manufacturing industry it is commonly stated that “Quality drives productivity.” Improved productivity is a source of greater revenues, employment opportunities and technological advances. However, this has not been the case historically, and in the early 19th century it was recognised that some markets, such as those in Asia, preferred

cheaper products to those of quality. Most discussions of quality refer to a finished part, wherever it is in the process. Inspection, which is what quality insurance usually means, is historical, since the work is done. The best way to think about quality is in process control. If the process is under control, inspection is not necessary.

However, there is one characteristic of modern quality that is universal. In the past, when we tried to improve quality, typically defined as producing fewer defective parts, we did so at the expense of increased cost, increased task time, longer cycle time, etc. We could not get fewer defective parts and lower cost and shorter cycle times, and so on.

However, when modern quality techniques are applied correctly to business, engineering, manufacturing or assembly processes, all aspects of quality - customer satisfaction and fewer defects/errors and cycle time and task time/productivity and total cost, etc.- must all improve or, if one of these aspects does not improve, it must at least stay stable and not decline. So modern quality has the characteristic that it creates AND-based benefits, not OR-based benefits.

Customers

The most progressive view of quality is that it is defined entirely by the customer or end user and is based upon that person's evaluation of his or her entire customer experience. The customer experience is the aggregate of all the touch points that customers have with the company's product and services, and is by definition a combination of these. For example, any time one buys a product one forms an impression based on how it was sold, how it was delivered, how it performed, how well it was supported, etc.

QUALITY MANAGEMENT TECHNIQUES

- Quality Management Systems
- Total Quality Management
- Design of experiments
 - Fractional factorial design
 - Optimal design
 - Response surface methodology
- Continuous improvement
- Six Sigma
- Statistical Process Control
- Quality circles
- Requirements analysis
- Verification and Validation
- Zero Defects
- Theory of Constraints

- Business Process Management
- Business process re-engineering
- Capability Maturity Models

INSPECTION

An inspection is a formal examination or evaluation exercise. It normally involves measurements, gauges and tests to determine whether the characteristics of the object or activity being inspected confirm to pre-decided standards.

Inspections are usually not destructive; rather, inspection is required at all levels in society to ensure that most ordinances are enforced as fairly and efficiently as possible, thus ensuring safety and security for everyone. Automobiles, buildings and homes may be inspected for this purpose. One could also have inspections at various stages of purchase, like pre-purchase and post-purchase of products or when refurbishing one's home. For instance, when a person visits an auto dealer to buy a new car, various aspects like interiors, components and quality are inspected by the buyer. Software inspection refers to the review of the work by peers following standardized testing and inspection procedures.

Normally government officials or private inspectors are employed for assessing the commercial and personal properties. These inspectors are highly trained personnel who are entrusted with the job of assessing and generating reports. These reports in turn help people to abide by the rules and regulations set by the authorities. Thus, a lot of unnecessary expenses are reduced by timely inspection. Nowadays with the use of the Internet, inspections can be performed online, thereby saving time and money. The main purpose of inspection is to ensure that safety standards are adhered to, both at the public and personal level. During the investigation, if the inspector notices low standards, corrective measures are taken immediately.

Inspections such as home inspections and building inspections are highly beneficial to owners as well as to buyers who plan to make huge investments. In all inspections, standards of compliance in accordance with the state are ensured. Therefore a good inspection guarantees public safety by enforcing municipal and state regulations typically for construction activities.

Purchasing Management

Purchasing management directs the flow of goods and services in a company and handles all data relating to contact with suppliers. Effective purchasing management requires knowledge of the supply chain, business and tax laws, invoice and inventory procedures, and transportation and logistics issues. Although a strong knowledge of the products and services to be purchased is essential, purchasing management professionals must also be able to plan, execute, and oversee purchasing strategies that are conducive to company profitability. Sourcing reliable suppliers is a crucial part of purchasing management. Purchasing managers, agents, and buyers usually learn about new products and services from Internet searches, trade shows, and conferences. They meet with potential suppliers in their plants whenever possible. Skills in foreign languages may be helpful for sourcing suppliers in other countries. Purchasing management professionals must always assess potential suppliers in terms of the supplier's ability to deliver quality merchandise at a suitable price on time.

Purchasing management professionals must be good negotiators, understand technical product information, have good mathematical ability, understand spreadsheet software, understand marketing methodology, and be outstanding decision makers. Increased responsibilities in purchasing management require good leadership skills, and higher positions often require a master's degree in a business related subject. Entry level purchasing management positions such as junior buyers, assistant buyers, and purchasing clerks, often require a college degree and some product knowledge.

Larger distributors may require a bachelor's degree for entry level purchasing management positions. Training and learning typically start in sales, then supervision,

and invoice and stock monitoring. Professional purchasing management designations include Certified Purchasing Professional in the United States, and Certified Professional Purchaser in Canada. Purchasing managers, buyers, and materials managers control budgets, manage staff, and may analyse procurement methods as well as negotiate supplier contracts.

Purchasing management professionals must understand tax laws, purchasing trends, ethics, and global outsourcing issues. Buyers and purchasing agents usually deal specifically with purchasing tasks, while purchasing managers usually supervise others, including purchasing agents. However, titles and duties vary greatly between industries and employers. Buyers often have assistants to place orders and keep track of delivery details. Buyers need the ability to select products that consumers will want to buy, so they need to understand trends as well as economic conditions that affect consumer buying decisions. Buyers employed in small stores may be responsible for purchasing the store's entire inventory, while buyers working for larger operations may focus on a few product lines. Merchandise managers may work with an advertising team to create a campaign for the goods they purchase. Follow up is an important part of purchasing management as responsibility does not end in purchasing goods. For example, merchandise managers need to check to be sure the products they buy are being displayed well to attract consumers. Buyers must keep track of customer demands for new products as well as determine how well goods are selling.

OBJECTIVES

To maintain uninterrupted flow of materials to support the development schedules:

- To procure materials economically at a cost consistent with the quality and service required. However, generally all purchases may be attempted at the lowest cost.
- To provide the necessary expertise, advice, information to the Curators and Education Officers with regard to the best quality of material available in the market, supplier's capability and performance, etc.
- To develop and maintain good buyer-seller relationship.
- To promote source development.
- To maintain NCSM's reputation and credibility in the market by fair dealings and prompt payments.

FUNCTIONS

The main functions of the Purchase Department are defined as follows:

- Procurement of stores through indigenous and foreign sources as required in accordance with the rules in force.
- Checking of requisitions/purchase indents.

- Selection of suppliers for issue of enquiries.
- Issuing enquiries/tenders and obtaining quotations.
- Analysing quotations and bids, etc., and preparation of comparative statement.
- Consultation with the Indentor for selection and approval of quotations and with Accounts Officer for pre-audit.
- Negotiating contracts.
- Checking legal conditions of contracts. Consulting Administrative Officer or Secretary, NCSM – where necessary.
- Issue of Purchase Orders.
- Follow-up of purchase orders for delivery in due time
- Verification and passing of suppliers' bills to see that payments are made promptly.
- Correspondence and dealing with suppliers, carriers, etc., regarding shortages, rejections, etc., reported by the Stores Department.
- Maintenance of purchase records.
- Maintenance of progressive expenditure statement, sub-head wise.
- Maintenance of vendor performance records/data.
- Arrangement for Insurance Surveys, as and when necessary.
- Clearance of foreign consignments.
- Keeping various Departments/Divisions informed of the progress of their indents in case of delay in obtaining supplies.
- Serving as an information centre on the materials' knowledge *i.e.*, their prices, source of supply, specification and other allied matters.
- Development of reliable and alternate sources of supply.
 - It shall be particularly ensured:
 - That all purchase are made against properly authorized requisitions, and valid sanctions, showing the correct and detailed end-use.
 - That all the materials requisitioned are duly ordered from the right source after full enquiries on most competitive price taking into account the trade discounts and tax benefits, etc.
 - That the right quantity of the goods consistent with the quality and the specification required is bought.
 - That the deliveries of all goods are received within the stipulated period.
 - That claims due to shortages or any other discrepancies are settled promptly.
 - That cordial and harmonious relations with all parts/divisions are maintained for becoming an efficient service unit in the museum/centre.
- While receiving the planned requirement of the indentors, it should be prevailed upon the Indentors to certify clearly that the quantities indicated are definitely required for the project and specify the period within which the quantities indented will be consumed. It is the duty of the purchase part to check purchase indents against needs and suggest modification of quantities wherever necessary.

METHODS OF PURCHASE

The purchase method of accounting in current standards is similar to the “acquisition accounting” method used in previous accounting standards. Its use provides some useful safeguards.

Previous accounting standards allowed two methods of accounting for mergers and acquisitions; merger accounting and acquisition accounting. The former was generally a more favourable treatment in its impact on the P and L. The purchase method of accounting is now compulsory in the US, the EU and wherever IFRSs have been adopted.

The purchase method of accounting must identify the acquirer. Under IFRSs the acquisition must be valued at fair value. The difference between the purchase price and the fair value should be recognised as goodwill.

A company cannot create a restructuring provision to provide for future losses or restructuring costs as a result of an acquisition. Such costs must be treated as post acquisition costs. This makes the impact of restructuring costs on profits more visible, and prevents the abuse of provisions to take an exaggerated hit to profits on acquisition and thereafter boost reported profits in subsequent years.

The use of the purchase method improves accounts, but investors are likely to disregard the impact of goodwill, which is the largest change.

The elimination of provisions is more useful because of the extra visibility and the prevention of abuses:

- *Full:* The purchase of a receivable in its entirety.
- *Partial:* The purchase of immediate instalments, with the seller retaining remaining installments and/or the balloon payment.
- *Reverse Partial:* The purchase of future installments, allowing the seller to continue receiving immediate installments for a specified period of time.
- *Split Disbursement Partial:* The purchase of a portion of each installment for a specified time and amount. Seller receives the remainder of each installment.
- *Multiple Stage Payout:* The full purchase of a receivable by means of a guaranteed staged payout of proceeds equaling or exceeding the principal balance. Seller receives stated payout regardless of the purchased receivable's performance.
- *Multiple Cash Flow:* The purchase of up to 50% of the cash flow created by multiple receivables held by a single seller.
- *Short Life Yield:* A method of calculating a higher buy price on a long term, Grade A receivable, by calculating the purchase price of the receivable as if there were a 10 year balloon payment.

We realise that you may desire an option other than the quote we provide. Please contact us with any questions you may have for an explanation of how these purchase options can be tailored to meet your needs, now or in the future. We look forward to working with you and providing the best service possible!

PURCHASING PROCEDURE

PURCHASE REQUEST

The specifications and number/quantity and delivery of equipment, devices and materials are determined by the department(s) that will be using the product(s) or materials. The Purchasing Department conducts purchase activities based on purchase requests submitted by the/these department(s).

SELECTION OF COMPANIES FOR ESTIMATE SUBMISSION

The Purchasing Department, at its sole discretion, selects companies from which estimates will be sought. Suppliers are selected from the files of “Companies with Previously Established Business Relationships”, “Companies from Which Estimates Can Be Requested” and “Products and Suppliers”. Selection is made by comprehensively evaluating such factors as the quality and performance of the equipment, device(s) or materials to be purchased, compatibility with existing facilities, degree of reliability, product requirements including safety, delivery time, the scale of the order, after-sale service and the company’s previous business record.

As a rule, Osaka Gas asks several companies to submit estimates. However, only one company may be specified for estimate submission in such special cases as those concerned with industrial property rights, those requiring maximum levels of safety that only one specific supplier can ensure, cases where only one specific supplier can assure compatibility with existing facilities, or in case of urgency.

ESTIMATE REQUEST

As a rule, when requesting an estimate from a company that it has selected, Osaka Gas will set out a specification from listing Osaka Gas’s requirements in respects of quality, performance standard, size, inspection and method of inspection. The selected companies will be asked to submit cost estimates and specifications to Osaka Gas prior to a specified date.

SUBMISSION ESTIMATE

Specification sheets submitted by potential suppliers at their own expense are checked by the Purchasing Department and the department(s) that will be using the product(s), in order to determine whether the required standards are met by the product(s). All products must pass this examination. During this process, Osaka Gas may request additions or changes to the specifications.

NEGOTIATION

After valid cost estimates and specifications have been comprehensively evaluated in respect of price, technical requirements, etc. Osaka Gas will commence negotiation

with the company with the most attractive proposal to discuss the amount of the contract and other terms and conditions. The selection of such a company shall be made by Osaka Gas at its sole discretion. Contract terms and conditions will be decided upon mutual agreement.

CONTRACT CONCLUSION

The business will be established upon conclusion of a contract, in the form of a written document if necessary. The obligations and liabilities of Osaka Gas arise only when such contract is concluded.

DELIVERY AND INSPECTION

Delivery dates specified in the contract must be strictly observed. Precise details of the delivery schedule will be agreed between the supplier and the relevant department(s) of Osaka Gas. Delivered equipment, device(s) or materials must pass inspections conducted by the relevant department(s) of Osaka Gas. When deemed significant, an interim inspection may be conducted during the manufacturing process.

PAYMENT

Payment will be made according to the payment terms specified in the contract.

VALUE ANALYSIS

CONCEPT OF VALUE

The value of a product will be interpreted in different ways by different customers. Its common characteristic is a high level of performance, capability, emotional appeal, style, etc. relative to its cost. This can also be expressed as maximizing the function of a product relative to its cost:

$$\text{Value} = (\text{Performance} + \text{Capability}) / \text{Cost} = \text{Function} / \text{Cost}$$

Value is not a matter of minimizing cost. In some cases the value of a product can be increased by increasing its function and cost as long as the added function increases more than its added cost. The concept of functional worth can be important. Functional worth is the lowest cost to provide a given function. However, there are less tangible “selling” functions involved in a product to make it of value to a customer.

INTRODUCTION TO VALUE ANALYSIS

Lawrence Miles conceived of Value Analysis (VA) in the 1945 based on the application of function analysis to the component parts of a product. Component cost reduction was an effective and popular way to improve “value” when direct labour and material

cost determined the success of a product. The value analysis technique supported cost reduction activities by relating the cost of components to their function contributions.

Value analysis defines a “basic function” as anything that makes the product work or sell. A function that is defined as “basic” cannot change. Secondary functions, also called “supporting functions”, described the manner in which the basic function(s) were implemented. Secondary functions could be modified or eliminated to reduce product cost. As VA progressed to larger and more complex products and systems, emphasis shifted to “upstream” product development activities where VA can be more effectively applied to a product before it reaches the production phase. However, as products have become more complex and sophisticated, the technique needed to be adapted to the “systems” approach that is involved in many products today. As a result, value analysis evolved into the “Function Analysis System Technique” (FAST) which is discussed later.

VALUE ANALYSIS METHOD

In all problem solving techniques, we are trying to change a condition by means of a solution that is unique and relevant. If we describe in detail what we are trying to accomplish, we tend to describe a solution and miss the opportunity to engage in divergent thinking about other alternatives. When trying to describe problems that affect us, we become locked in to a course of action without realizing it, because of our own bias. Conversely, the more abstractly we can define the function of what we are trying to accomplish, the more opportunities as suggested, have for divergent thinking.

This high level of abstraction can be achieved by describing what is to be accomplished with a verb and a noun. In this discipline, the verb answers the question, “What is to be done?” or, “What is it to do?” The verb defines the required action. The noun answers the question, “What is it being done to?” The noun tells what is acted upon. Identifying the function by a verb-noun is not as simple a matter as it appears. Identifying the function in the broadest possible terms provides the greatest potential for divergent thinking because it gives the greatest freedom for creatively developing alternatives. A function should be identified as to what is to be accomplished by a solution and not how it is to be accomplished. How the function is identified determines the scope, or range of solutions that can be considered.

That functions designated as “basic” represent the operative function of the item or product and must be maintained and protected. Determining the basic function of single components can be relatively simple. By definition then, functions designated as “basic” will not change, but the way those functions are implemented is open to innovative speculation. As important as the basic function is to the success of any product, the cost to perform that function is inversely proportional to its importance. This is not an absolute rule, but rather an observation of the consumer products market. Few people purchase

consumer products based on performance or the lowest cost of basic functions alone. When purchasing a product it is assumed that the basic function is operative. The customer's attention is then directed to those visible secondary support functions, or product features, which determine the worth of the product. From a product design point of view, products that are perceived to have high value first address the basic function's performance and stress the achievement of all of the performance attributes. Once the basic functions are satisfied, the designer's then address the secondary functions necessary to attract customers. Secondary functions are incorporated in the product as features to support and enhance the basic function and help sell the product. The elimination of secondary functions that are not very important to the customer will reduce product cost and increase value without detracting from the worth of the product.

The cost contribution of the basic function does not, by itself, establish the value of the product. Few products are sold on the basis of their basic function alone. If this were so, the market for "no name" brands would be more popular than it is today. Although the cost contribution of the basic function is relatively small, its loss will cause the loss of the market value of the product. One objective of value analysis or function analysis, to improve value by reducing the cost-function relationship of a product, is achieved by eliminating or combining as many secondary functions as possible.

VALUE ANALYSIS PROCESS

The first step in the value analysis process is to define the problem and its scope. Once this is done, the functions of the product and its items are derived. These functions are classified into "basic" and "secondary" functions. A Cost Function Matrix or Value Analysis Matrix is prepared to identify the cost of providing each function by associating the function with a mechanism or component part of a product. Product functions with a high cost-function ratio are identified as opportunities for further investigation and improvement. Improvement opportunities are then brainstormed, analysed, and selected.

The objective of the Function Cost Matrix approach is to draw the attention of the analysts away from the cost of components and focus their attention on the cost contribution of the functions. The Function Cost Matrix displays the components of the product, and the cost of those components, along the left vertical side of the graph. The top horizontal legend contains the functions performed by those components. Each component is then examined to determine how many functions that component performs, and the cost contributions of those functions.

Detailed cost estimates become more important following function analysis, when evaluating value improvement proposals. The total cost and per cent contribution of

the functions of the item under study will guide the team, or analyst, in selecting which functions to select for value improvement analysis. A variation of the Function-Cost Matrix is the Value Analysis Matrix. This matrix was derived from the Quality Function Deployment (QFD) methodology.

It is more powerful in two ways. First, it associates functions back to customer needs or requirements. In doing this, it carries forward an importance rating to associate with these functions based on the original customer needs or requirements. Functions are then related to mechanisms, the same as with the Function-Cost Matrix. Mechanisms are related to functions as either strongly, moderately or weakly supporting the given function. This relationship is noted with the standard QFD relationship symbols. The associated weighting factor is multiplied by customer or function importance and each column's value is added. These totals are normalized to calculate each mechanism's relative weight in satisfying the designated functions. This is where the second difference with the Function-Cost Matrix arises. This mechanism weight can then be used as the basis to allocate the overall item or product cost. The mechanism target costs can be compared with the actual or estimated costs to see where costs are out of line with the value of that mechanism as derived from customer requirements and function analysis.

FUNCTION ANALYSIS SYSTEM TECHNIQUE

Function Analysis System Technique is an evolution of the value analysis process created by Charles Bytheway. FAST permits people with different technical backgrounds to effectively communicate and resolve issues that require multi-disciplined considerations. FAST builds upon VA by linking the simply expressed, verb-noun functions to describe complex systems.

FAST is not an end product or result, but rather a beginning. It describes the item or system under study and causes the team to think through the functions that the item or system performs, forming the basis for a wide variety of subsequent approaches and analysis techniques. FAST contributes significantly to perhaps the most important phase of value engineering: function analysis. FAST is a creative stimulus to explore innovative avenues for performing functions.

The FAST diagram or model is an excellent communications vehicle. Using the verb-noun rules in function analysis creates a common language, crossing all disciplines and technologies. It allows multi-disciplined team members to contribute equally and communicate with one another while addressing the problem objectively without bias or preconceived conclusions. With FAST, there are no right or wrong model or result.

The problem should be structured until the product development team members are satisfied that the real problem is identified. After agreeing on the problem statement,

the single most important output of the multi-disciplined team engaged in developing a FAST model is consensus. Since the team has been charged with the responsibility of resolving the assigned problem, it is their interpretation of the FAST model that reflects the problem statement that's important. The team members must discuss and reconfigure the FAST model until consensus is reached and all participating team members are satisfied that their concerns are expressed in the model. Once consensus has been achieved, the FAST model is complete and the team can move on to the next creative phase.

FAST differs from value analysis in the use of intuitive logic to determine and test function dependencies and the graphical display of the system in a function dependency diagram or model. Another major difference is in analyzing a system as a complete unit, rather than analyzing the components of a system. When studying systems it becomes apparent that functions do not operate in a random or independent fashion. A system exists because functions form dependency links with other functions, just as components form a dependency link with other components to make the system work. The importance of the FAST approach is that it graphically displays function dependencies and creates a process to study function links while exploring options to develop improved systems.

There are normally two types of FAST diagrams, the technical FAST diagram and the customer FAST diagram. A technical FAST diagram is used to understand the technical aspects of a specific portion of a total product. A customer FAST diagram focuses on the aspects of a product that the customer cares about and does not delve into the technicalities, mechanics or physics of the product. A customer FAST diagram is usually applied to a total product.

CREATING A FAST MODEL

The FAST model has a horizontal directional orientation described as the HOW-WHY dimension. This dimension is described in this manner because HOW and WHY questions are asked to structure the logic of the system's functions. Starting with a function, we ask HOW that function is performed to develop a more specific approach. This line of questioning and thinking is read from left to right. To abstract the problem to a higher level, we ask WHY is that function performed. This line of logic is read from right to left.

There is essential logic associated with the FAST HOW-WHY directional orientation. First, when undertaking any task it is best to start with the goals of the task, then explore methods to achieve the goals. When addressing any function on the FAST model with the question WHY, the function to its left expresses the goal of that function. The question

HOW, is answered by the function on the right, and is a method to perform that function being addressed. A systems diagram starts at the beginning of the system and ends with its goal. A FAST model, reading from left to right, starts with the goal, and ends at the beginning of the “system” that will achieve that goal.

Second, changing a function on the HOW-WHY path affects all of the functions to the right of that function. This is a domino effect that only goes one way, from left to right. Starting with any place on the FAST model, if a function is changed the goals are still valid but the method to accomplish that function, and all other functions on the right, are affected.

Finally, building the model in the HOW direction, or function justification, will focus the team’s attention on each function element of the model. Whereas, reversing the FAST model and building it in its system orientation will cause the team to leap over individual functions and focus on the system, leaving function “gaps” in the system. A good rule to remember in constructing a FAST Model is to build in the HOW direction and test the logic in the WHY direction. The vertical orientation of the FAST model is described as the WHEN direction. This is not part of the intuitive logic process, but it supplements intuitive thinking. WHEN is not a time orientation, but indicates cause and effect.

Scope lines represent the boundaries of the study and are shown as two vertical lines on the FAST model. The scope lines bound the “scope of the study”, or that aspect of the problem with which the study team is concerned. The left scope line determines the basic function(s) of the study. The basic functions will always be the first function(s) to the immediate right of the left scope line. The right scope line identifies the beginning of the study and separates the input function(s) from the scope of the study.

The objective or goal of the study is called the “Highest Order Function”, located to the left of the basic functions and outside of the left scope line. Any function to the left of another function is a “higher order function”.

Functions to the right and outside of the right scope line represent the input side that “turn on” or initiate the subject under study and are known as lowest order functions. Any function to the right of another function is a “lower order” function and represents a method selected to carry out the function being addressed.

Those functions to the immediate right of the left scope line represent the purpose or mission of the product or process under study and are called Basic Function(s). Once determined, the basic function will not change. If the basic function fails, the product or process will lose its market value. All functions to the right of the basic function(s) portray the conceptual approach selected to satisfy the basic function. The concept describes the method being considered, or elected, to achieve the basic functions. The

concept can represent either the current conditions (as is) or proposed approach (to be). As a general rule, it is best to create a “to be” rather than an “as is” FAST Model, even if the assignment is to improve an existing product.

This approach will give the product development team members an opportunity to compare the “ideal” to the “current” and help resolve how to implement the differences. Working from an “as is” model will restrict the team’s attention to incremental improvement opportunities.

An “as is” model is useful for tracing the symptoms of a problem to its root cause, and exploring ways to resolve the problem, because of the dependent relationship of functions that form the FAST model. Any function on the HOW-WHY logic path is a logic path function. If the functions along the WHY direction lead into the basic function(s), then they are located on the major logic path. If the WHY path does not lead directly to the basic function, it is a minor logic path. Changing a function on the major logic path will alter or destroy the way the basic function is performed.

Changing a function on a minor logic path will disturb an independent function that enhances the basic function. Supporting functions are usually secondary and exist to achieve the performance levels specified in the objectives or specifications of the basic functions or because a particular approach was chosen to implement the basic function(s).

Independent functions describe an enhancement or control of a function located on the logic path. They do not depend on another function or method selected to perform that function. Independent functions are located above the logic path functions, and are considered secondary, with respect to the scope, nature, level of the problem, and its logic path. An example of a FAST Diagram for a pencil is.

The next step in the process is to dimension the FAST model or to associate information to its functions. FAST dimensions include, but are not limited to: responsibility, budgets, allocated target costs, estimated costs, actual costs, subsystem groupings, placing inspection and test points, manufacturing processes, positioning design reviews, and others. There are many ways to dimension a FAST model. The two popular ways are called Clustering Functions and the Sensitivity Matrix.

Clustering functions involves drawing boundaries with dotted lines around groups of functions to configure sub-systems. Clustering functions is a good way to illustrate cost reduction targets and assign design-to-cost targets to new design concepts. For cost reduction, a team would develop an “as is” product FAST model, cluster the functions into subsystems, allocate product cost by clustered functions, and assign target costs. During the process of creating the model, customer sensitivity functions can be identified as well as opportunities for significant cost improvements in design and production.

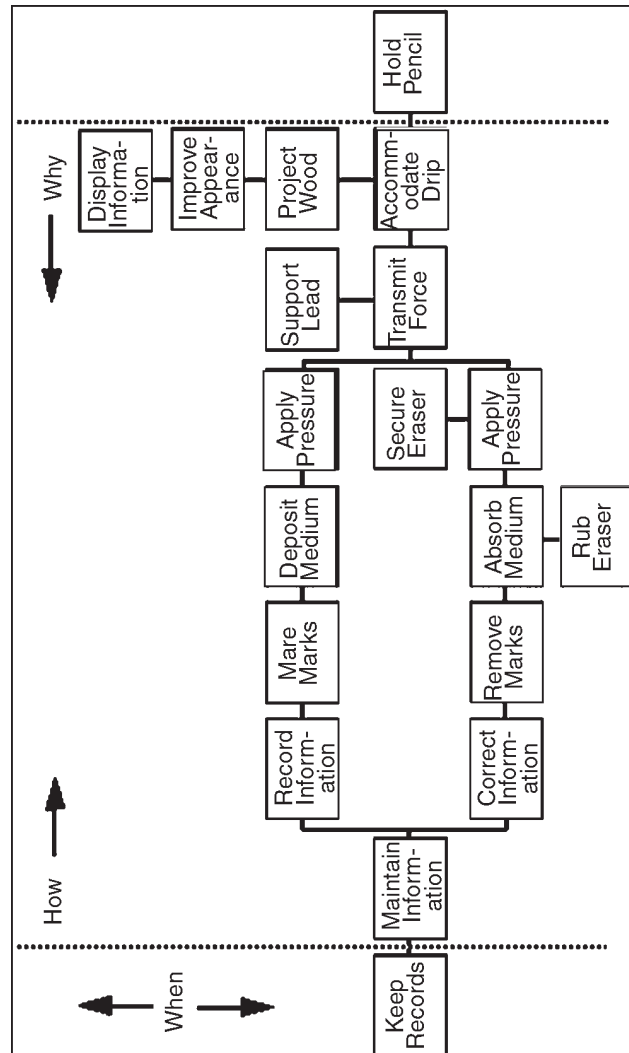


Fig. FAST Diagram- Pencil.

Following the completion of the model, the subsystems can be divided among product development teams assigned to achieve the target cost reductions. The teams can then select cost sensitive sub-systems and expand them by moving that segment of the model to a lower level of abstraction. This exposes the detail components of that assembly and their function/cost contributions.

INTEGRATING QFD WITH FAST

A powerful analysis method is created when FAST is used in conjunction with QFD. QFD enables the uses of the Value Analysis Matrix. An example of a value analysis matrix for the pencil example.

Customer Requirements/ Functions	Importance	Mechanisms				
		Lead	Eraser	Body	Paint	Band
Make Marks	30	⊙ 150				
Remove Marks	20		⊙ 100			
Prevent Smudges	15	⊙ 45		⊙ 45		
Support Lead	5			⊙ 25		
Improve Appearance	10			⊙ 30	⊙ 30	△ 10
Accomodate Grip	20			⊙ 100	△ 20	
Column Weight	555	195	100	200	50	10
Mech. Weight	1.0	.351	.180	.360	.090	.018
Mech. Target Cost	2.80	.98	.51	1.01	.25	.05
Mech. Actual Cost	2.92	1.20	.43	.94	.10	.25

⊙ Strong Correlation
Weight Factor = 5
 ○ Moderate Correlation
Weight Factor = 3
 △ Weak Correlation
Weight Factor = 1

The steps for using these two methodologies are as follows:

1. Capture customer requirements and perform QFD product planning with the product planning matrix. Translate customer needs into directly into verb-noun functions or use a second matrix to translate technical characteristics into verb-noun functions.
2. Prepare a FAST diagram and develop the product concept in conjunction with the QFD concept selection matrix. Review the verb-noun functions in the QFD matrix and assure that they are included in the FAST diagram. Revise verb-noun function descriptions if necessary to assure consistency between the QFD matrix and the FAST diagram.
3. Dimension the system in the FAST diagram into subsystems/assemblies/parts. These are generically referred to as mechanisms.
4. Develop value analysis matrix at system level. The “what’s” or system requirements/function in the value analysis matrix are derived from either a customer FAST diagram or by selecting those function statements that correspond to the customer needs or technical characteristics in the product planning matrix. The importance rating is derived from the product planning matrix as well.
5. Complete the value analysis matrix by relating the mechanisms to the customer requirements/functions and calculate the associated weight. Summarize the column weights and normalize to create mechanism weights. Allocate the target cost based on the mechanism weights. This represents the value to the customer based on the customer importance. Compare with either estimated costs based on the product concept or actual costs if available.
6. Identify high cost to value mechanisms/subsystems by comparing the mechanism target costs to the mechanism estimated/actual costs

A product or system such as an automobile contains a great many components and would result in an extremely complex FAST model. The complexity of the process is

not governed by the number of components in a product, but the level of abstraction selected to perform the analysis. With an automobile, a high level of abstraction could contain the major subsystems as the components under study, such as: the power train, chassis, electrical system, passenger compartment, etc. The result of the FAST model and supporting cost analysis might then focus the team's attention on the power train for further analysis.

Moving to a lower level of abstraction, the power train could then be divided into its components for a more detailed analysis. In other words, the concept of decomposition is applied to a FAST model. The initial FAST model will stay at a high level of abstraction. Starting at a higher level of abstraction allows for uncluttered macro analysis of the overall problem until those key functions can be found, isolated, and the key issues identified. If a function is identified for further study. A supporting FAST diagram is then created for that subsystem function. This process of decomposition or moving to lower levels of abstraction could be carried down several levels if appropriate. Once high cost to value mechanisms are identified in the initial system value analysis matrix, the next step is to focus more attention on those mechanisms and associated functions. Dimensioning groups the functions together into those associated with a particular subsystem, assembly or part. The FAST diagram can be expanded into a lower level of abstraction in the area under investigation.

The steps involved are as follows:

- Use QFD to translate higher-level customer needs to subsystem technical characteristics.
- Create FAST diagram at lower level of abstraction for targeted mechanism/subsystem.
- Prepare a FAST diagram and develop the product concept in conjunction with the QFD concept selection matrix
- Dimension the system in the FAST diagram into assemblies/parts or identify the assemblies/parts needed to perform the given function.
- Develop value analysis matrix at a lower level of abstraction for the targeted subsystem. The "what's" or system requirements/function in the value analysis matrix are derived from either a customer FAST diagram or by selecting those function statements that correspond to the customer needs or technical characteristics in the subsystem planning matrix.
- Complete the value analysis matrix and identify high cost to value mechanisms by comparing the mechanism target costs to the mechanism estimated/actual costs

VALUE IMPROVEMENT PROCESS

Performing value analysis or producing the FAST model and analyzing functions with the value analysis matrix are only the first steps in the process. The real work begins with brainstorming, developing and analyzing potential improvements in the product.

These subsequent steps are supported by:

- The QFD Concept Selection Matrix is a powerful tool to evaluate various concept and design alternatives based on a set of weighted criteria that ultimately tie back to customer needs.
- Benchmarking competitors and other similar products helps to see new ways functions can be performed and breaks down some of the not-invented-here paradigms.
- Product cost and life cycle cost models support the estimating of cost for the Function-Cost and Value Analysis Matrices and aid in the evaluation of various product concepts.
- Technology evaluation leads us to new ways that basic functions can be performed in a better or less costly way. Concept development should involve people with a knowledge of new technology development and an open mind to identify how this technology might relate to product functions that need to be performed. Methods such as the theory of inventive problem solving or TRIZ are useful in this regard.
- Design for Manufacturability/Assembly principles provide guidance on how to better design components and assemblies that are more manufacturable and, as a result, are lower in cost.

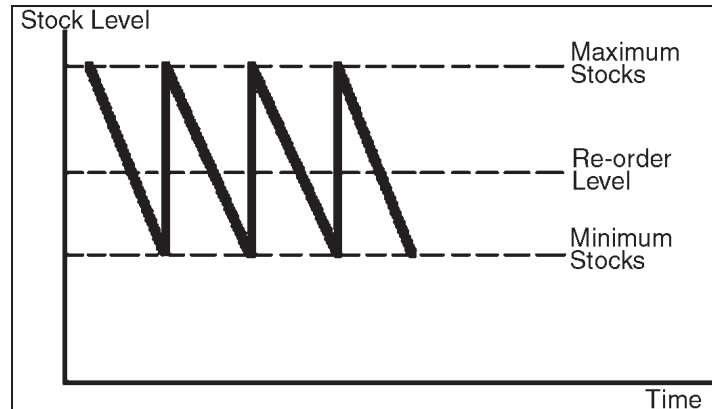
Value Analysis or Function Analysis provide the methods to identify the problem and to begin to define the functions that need to be performed. As we proceed in developing a FAST model, implicit in this process is developing a concept of operation for the product which is represented by all of the lower order functions in a FAST diagram. Concept alternatives will be developed through brainstorming, benchmarking other products performing similar functions, and surveying and applying new technology. Since multiple concepts need to be evaluated, we want to use a higher level of abstraction for the FAST model to provide us with the greatest flexibility and a minimum level of effort. Trade studies and technical analysis will be performed to evaluate various product concepts. A concept selection matrix is a good tool to summarize a variety of different data and support making a decision about the preferred concept.

All of these steps may be iterative as a preferred concept evolves and gets more fully developed. In addition, there should be a thorough evaluation of whether all functions are needed or if there is a different way of accomplishing a function as the concept is developed to a lower level of abstraction. When a Function Cost or Value Analysis Matrix is prepared, functions that are out of balance with their worth are identified, further challenging the team to explore different approaches.

STOCK CONTROL SYSTEMS

FIXED RE-ORDER STOCK LEVEL

This method of stock control is where a business decides the minimum level of stocks it can tolerate, and then re-orders before the stocks reach this level. The exact timing will depend how long the stocks take to arrive. This can be illustrated as follows:



The distance between the re-order line and the minimum stocks level depends how long it may take for the order to arrive - this time is known as the lead time.

FIXED TIME REORDERING

This method is exactly as its title suggests. The firm reorders stocks at a fixed time each month or week. It can offer a good solution as it represents a routine for the firm and ensures that stocks are regularly supplemented. However, it may well mean the level of stocks fluctuating quite a bit depending on the rate they are used up. It is a little inflexible as a system as well unless used very carefully.

ECONOMIC ORDER QUANTITY

For any company there is an optimum level of stocks. The precise level of this will vary in different firms and industries. They have to balance the costs of holding stocks with the costs of ordering stock. The more firms order at once, the better the deal they will usually get.

The level of stocks that strikes the balance between these two things is known as the economic order quantity. If this is taken to be the optimum level of stocks it should help to minimise the firm's costs - an important pre-requisite to maximising profit.

JUST-IN-TIME PRODUCTION

Because stocks cost so much to keep, another method of stock-control was developed in Japan and has now become much more common in the UK. The just-in-time method involves keeping stocks to an absolute minimum, and the raw materials are ordered only when they are needed. In other words just-in-time.

This time period in some cases has been reduced to minutes or hours, and the raw materials arrive on site moments before they are needed. This can be wonderful for helping to reduce the need for working capital, but requires a very high level of organisational skill and a very close relationship with suppliers.

DEFINITION OF PURCHASING

Purchasing refers to a business or organization attempting to acquire goods or services to accomplish the goals of the enterprise. Though there are several organizations that attempt to set standards in the purchasing process, processes can vary greatly between organizations. Typically the word “purchasing” is not used interchangeably with the word “procurement”, since procurement typically includes Expediting, Supplier Quality, and Traffic and Logistics in addition to Purchasing.

Purchasing managers/directors, and procurement managers/directors guide the organization’s acquisition procedures and standards. Most organizations use a three-way check as the foundation of their purchasing programmes. This involves three departments in the organization completing separate parts of the acquisition process.

The three departments do not all report to the same senior manager to prevent unethical practices and lend credibility to the process. These departments can be purchasing, receiving; and accounts payable or engineering, purchasing and accounts payable; or a plant manager, purchasing and accounts payable. Combinations can vary significantly, but a purchasing department and accounts payable are usually two of the three departments involved.

When the receiving department is not involved, it’s typically called a two-way check or two-way purchase order. In this situation, the purchasing department issues the purchase order receipt not required. When an invoice arrives against the order, the accounts payable department will then go directly to the requestor of the purchase order to verify that the goods or services were received. This is typically what is done for goods and services that will bypass the receiving department. A few examples are software delivered electronically, NRE work consulting hours, etc...

Historically, the purchasing department issued Purchase Orders for supplies, services, equipment, and raw materials. Then, in an effort to decrease the administrative costs associated with the repetitive ordering of basic consumable items, “Blanket” or “Master” Agreements were put into place. These types of agreements typically have a longer duration and increased scope to maximize the Quantities of Scale concept. When additional supplies are required, a simple release would be issued to the supplier to provide the goods or services.

Another method of decreasing administrative costs associated with repetitive contracts for common material, is the use of company credit cards, also known as “Purchasing Cards” or simply “P-Cards”. P-card programmes vary, but all of them have internal checks and audits to ensure appropriate use. Purchasing managers realised once contracts for the low dollar value consumables are in place, procurement can take a smaller role in the operation and use of the contracts. There is still oversight in the forms of audits and monthly statement reviews, but most of their time is now available to negotiate major purchases and setting up of other long term contracts. These contracts are typically renewable annually.

This trend away from the daily procurement function resulted in several changes in the industry. The first was the reduction of personnel. Purchasing departments were now smaller. There was no need for the army of clerks processing orders for individual parts as in the past. Another change was the focus on negotiating contracts and procurement of large capital equipment. Both of these functions permitted purchasing departments to make the biggest financial contribution to the organization. A new terms and job title emerged – Strategic sourcing and Sourcing Managers. These professionals not only focused on the bidding process and negotiating with suppliers, but the entire supply function. In these roles they were able to add value and maximize savings for organizations. This value was manifested in lower inventories, less personnel, and getting the end product to the organization's consumer quicker. Purchasing manager's success in these roles resulted in new assignments outside to the traditional purchasing function – logistics, materials management, distribution, and warehousing. More and more purchasing managers were becoming Supply Chain Managers handling additional functions of their organizations operation. Purchasing managers were not the only ones to become Supply Chain Managers. Logistic managers, material managers, distribution managers, etc all rose the broader function and some had responsibility for the purchasing functions now.

In accounting, purchases is the amount of goods a company bought throughout this year. They are added to inventory. Purchases are offset by Purchase Discounts and Purchase Returns and Allowances. When it should be added depends on the Free On Board policy of the trade. For the purchaser, this new inventory is added on shipment if the policy was FOB shipping point, and the seller remove this item from its inventory. On the other hand, the purchaser added this inventory on receipt if the policy was FOB destination, and the seller remove this item from its inventory when it was delivered.

Goods bought for the purpose other than direct selling, such as for Research and Development, are added to inventory and allocated to Research and Development expense as they are used. On a side note, equipments bought for Research and Development are not added to inventory, but are capitalized as assets...

ACQUISITION PROCESS

The revised acquisition process for major systems in industry and defence is shown in figure. The process is defined by a series of phases during which technology is defined and matured into viable concepts, which are subsequently developed and readied for production, after which the systems produced are supported in the field.

The process allows for a given system to enter the process at any of the development phases. For example, a system using unproven technology would enter at the beginning stages of the process and would proceed through a lengthy period of technology maturation, while a system based on mature and proven technologies might enter directly into engineering development or, conceivably, even production.

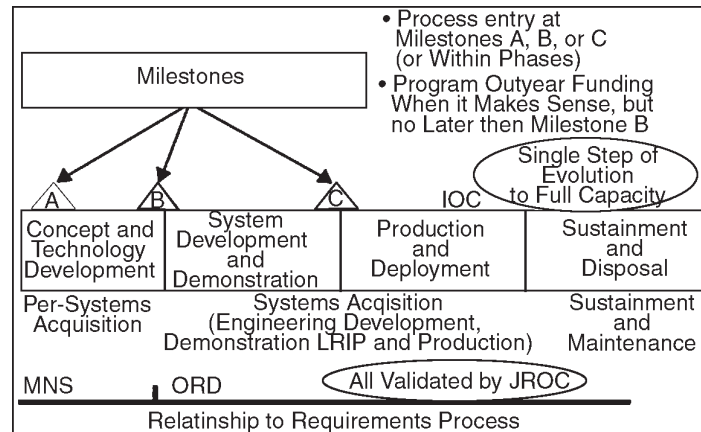


Fig. Model of the Acquisition Process.

The process itself includes four phases of development:

1. Concept and Technology Development: is intended to explore alternative concepts based on assessments of operational needs, technology readiness, risk, and affordability.
2. Concept and Technology Development phase begins with concept exploration. During this stage, concept studies are undertaken to define alternative concepts and to provide information about capability and risk that would permit an objective comparison of competing concepts.
3. System Development and Demonstration phase. This phase could be entered directly as a result of a technological opportunity and urgent user need, as well as having come through concept and technology development.
4. The last, and longest, phase is the Sustainment and Disposal phase of the programme. During this phase all necessary activities are accomplished to maintain and sustain the system in the field in the most cost-effective manner possible.

SELECTION OF BIDDERS

This is the process where the organization identifies potential suppliers for specified supplies, services or equipment. These suppliers' credentials and history are analysed, together with the products or services they offer. The bidder selection process varies from organization to organization, but can include running credit reports, interviewing management, testing products, and touring facilities. This process is not always done in order of importance, but rather in order of expense. Often purchasing managers research potential bidders obtaining information on the organizations and products from media sources and their own industry contacts. Additionally, purchasing might send Request for Information to potential suppliers to help gather information.

Engineering would also inspect sample products to determine if the company can produce products they need. If the bidder passes both of these stages engineering may

decide to do some testing on the materials to further verify quality standards. These tests can be expensive and involve significant time of multiple technicians and engineers. Engineering management must make this decision based on the cost of the products they are likely to procure, the importance of the bidders' product to production, and other factors. Credit checks, interviewing management, touring plants as well as other steps could all be utilized if engineering, manufacturing, and supply chain managers decide they could help their decision and the cost is justifiable. Other organizations might have minority procurement goals to consider in selection of bidders. Organizations identify goals in the use of companies owned and operated by certain ethnicities or women owned business enterprises. Significant utilizing of minority suppliers may qualify the firm as a potential bidder for a contract with a company or governmental entity looking to increase their minority supplier programmes.

This selection process can include or exclude international suppliers depending on organizational goals and criteria. Companies looking to increase their pacific rim supplier base may exclude suppliers from the Americas, Europe, and Australia. Other organizations may be looking to purchase domestically to ensure a quicker response to orders as well as easier collaboration on design and production.

Organizational goals will dictate the criteria for the selection process of bidders. It is also possible that the product or service being procured is so specialized that the number of bidders are limited and the criteria must be very wide to permit competition. If only one firm can meet the specifications for the product then the purchasing managers must consider utilizing a "Sole Source" option or work with engineering to broaden the specifications if the project will permit alteration in the specifications. The sole source option is the part of the selection of bidders that acknowledges there is sometimes only one reasonable supplier for some services or products. This can be because of the limited applications for the product cannot support more than one manufacturer, proximity of the service provided, or the products are newly designed or invented and competition is not yet available.

BIDDING PROCESS

This is the process an organization utilizes to procure goods, services or equipment. Processes vary significantly from the stringent to the very informal. Large corporations and governmental entities are most likely to have stringent and formal processes. These processes can utilize specialized bid forms that require specific procedures and detail. The very stringent procedures require bids to be open by several staff from various departments to ensure fairness and impartiality. Responses are usually very detailed. Bidders not responding exactly as specified and following the published procedures can be disqualified. Smaller private businesses are more likely to have less formal procedures. Bids can be in the form of an e-mail to all of the bidders specifying products or services. Responses by bidders can be detailed or just the proposed dollar amount. Most bid processes are multi-tiered. Acquisitions under a specified dollar amount can

be “user discretion” permitting the requestor to choose who ever they want. This level can be as low as \$100 or as high as \$10,000 depending on the organization. The rationale is the savings realised by processing these request the same as expensive items is minimal and does not justify the time and expense. Purchasing departments watch for abuses of the user discretion privilege. Acquisitions in a mid range can be processed with a slightly more formal process. This process may involve the user providing quotes from three separate suppliers.

Purchasing may be asked or required to obtain the quotes. The formal bid process starts as low as \$10,000 or as high as \$100,000 depending on the organization. The bid usually involves a specific form the bidder fills out and must be returned by a specified deadline. Depending of the commodity being purchased and the organization the bid may specify a weighted evaluation criterion. Other bids would be evaluated at the discretion of purchasing or the end users. Some bids could be evaluated by a cross-functional committee. Other bids may be evaluated by the end user or the buyer in Purchasing. Especially in small, private firms the bidders could be evaluated on criteria or factors that have little if anything to do with the actual bid. Examples of these factors are history of the bidder with the company, history of the bidder with the company’s senior management at other firms, and bidder’s breadth of products.

TECHNICAL EVALUATION

Technical evaluations, evaluations of the technical suitability of the quoted goods or services, if required, are normally performed prior to the commercial evaluation. During this phase of the procurement process, a technical representative of the company will review the proposal and designate each bidder as either technically acceptable or technically unacceptable.

COMMERCIAL EVALUATION

Cost of Money is calculated by multiplying the applicable currency interest rate multiplied by the amount of money paid prior to the receipt of GOODS. If the money were to have remained in the buyer’s account, interest would be drawn. That interest is essentially an additional cost associated with such Progress or Milestone payments.

The manufacturing location is taken into consideration during the evaluation stage primarily to calculate freight costs and regional issues which may be considered. For instance, in Europe it is common for factories to close during the month of August for Summer holiday. Labour agreements may also be taken into consideration and may be drawn into the evaluation if the particular region is known to frequent labour unions. The manufacturing lead-time is the time from the placement of the order until the goods are manufactured and prepared for delivery. Lead-times vary by commodity and can range from several days to years. Transportation time is evaluated while comparing the delivery of goods to the Buyer’s required use-date. If Goods are shipped from a

remote port, with infrequent vessel transportation, the transportation time could exceed the schedule and adjustments would need to be made. Delivery Charges - the charge for the Goods to be delivered to a stated point. Bid Validity Packing Bid Adjustments Terms and Conditions Seller's Services Standards Organizations Financial Review Payment Currency Risk Analysis - market volatility, financial stress within the bidders Testing

NEGOTIATING

Negotiating is a key skillset in the Purchasing field. One of the goals of Purchasing Agents is to acquire goods per the most advantageous terms of the buying entity. Purchasing Agents typically attempt to decrease costs while meeting the Buyer's other requirements such as an on-time delivery, compliance to the commercial terms and conditions.

Good negotiators, those with high levels of documented "cost savings", receive a premium within the industry relative to their compensation. Depending on the employment agreement between the Purchasing Agent and the employer, Buyer's cost savings can result in the creation of value to the business, and may result in a flat-rate bonus, or a percentage payout to the Purchasing Agent of the documented cost savings.

Purchasing Departments, while they can be considered as a support function of the key business, are actually revenue generating departments. For example, if the company needs to buy \$30 million USD of widgets and the Purchasing Department secures the widgets for \$25M USD, the Purchasing Department would have saved the company \$5M USD. That savings could exceed the annual budget of the department, which in effect would pay the department's overhead - the employee's salaries, computers, office space, etc.

POST-AWARD ADMINISTRATION

Post-award administration typically consists of making minor changes, additions or subtractions, that in some way change the terms of the agreement or the Seller's Scope of Supply. Such changes are often minor, but for auditing purposes must be documented into the existing agreement. Examples include increasing the quantity of a Line Item or changing the metallurgy of a particular component.

Risk Management and Business Continuity

Risk management is simply a practice of systematically selecting cost effective approaches for minimising the effect of threat realization to the organization. All risks can never be fully avoided or mitigated simply because of financial and practical limitations. Therefore all organizations have to accept some level of residual risks.

Whereas risk management tends to be preemptive, business continuity planning was invented to deal with the consequences of realised residual risks. The necessity to have BCP in place arises because even very unlikely events will occur if given enough time. Risk management and BCP are often mistakenly seen as rivals or overlapping practices.

In fact these processes are so tightly tied together that such separation seems artificial. For example, the risk management process creates important inputs for the BCP. Risk management also proposes applicable controls for the observed risks. Therefore, risk management covers several areas that are vital for the BCP process. However, the BCP process goes beyond risk management's preemptive approach and assumes that the disaster will happen at some point.

RISK COMMUNICATION

Risk communication is an interactive process of exchange of information and opinion on risk among risk assessors, risk managers, and other interested parties.

Risk communication is an integral and ongoing part of the risk analysis exercise, and ideally all stakeholder groups should be involved from the start. Risk communication makes stakeholders aware of the process at each stage of the Risk Assessment.

This helps to ensure that the logic, outcomes, significance, and limitations of the Risk Assessment are clearly understood by all the stakeholders. Information may be available from the stakeholder.

Industry stakeholders may, for example have unpublished data crucial to the risk assessors, which may be an essential part of the data needed for the risk assessment. There is also information that is typically presented to the stakeholders as an integral part of the risk analysis process.

The identification of particular interest groups and their representatives should comprise a part of an overall risk communication strategy. This risk communication strategy should be discussed and agreed upon between risk assessors and managers early in the process to ensure two-way communication. This strategy should also cover who should present information to the public, and the manner in which it will be done.

Decisions on risk communication, including what, whom and how, should be part of an overall risk communication strategy. Risk communication is most effective if undertaken in a systematic way, and generally starts with the gathering of information on the risk issue of concern.

Therefore the risk manager and risk assessor must be able to briefly and clearly summarize what this issue encompasses, at an early stage, in order to elicit interest and stakeholder input. Communication must then continue throughout the entire process.

Once available information has been used to fully identify the hazards, and decide on and assess the appropriate risks, then the preparation and dissemination of this information is required. This will be followed by further discussion with stakeholders, leading to corrections, amendments, and additions as appropriate, resulting in the final Risk Assessment and risk analysis reports.

COMMUNICATE YOUR IDEAS EFFECTIVELY

If you are not capable of communicating your ideas in an effective manner, then you run the risk of alienating those you are speaking to. On the opposite side of the coin, people will also become agitated if they think you're patronizing them.

The central point of this issue is that you must always have a profound understanding of your audience. It is best to offer good examples and solid information which can assist you in putting the risk into perspective. You must be capable of dealing with uncertainty, and when you talk to people about risks, the results that you get may not be that definitive. You will want to look at the actual sources of uncertainty.

You will want to explain the ways for which the data was gathered, and you will want to study the ways in which it has been analysed. It is also important to show the ways in which the results have been gathered. This shows that after the uncertainties have been showcased, this could lead to a trust increase, and this may further lead to

issues which are related to credibility. At the same time, when you talk about uncertainty, the communicator must take the time to emphasize their knowledge of the subject. This will allow the leader to handle the situation in a way that could block out concerns or fears. There is also the issue of decisions which are related to risk management.

RISK ASSESSMENT

Within the workplace, owners of a business are legally required to assess the risks of injury and ill health affecting employees. Risk assessment is the careful examination of the diverse factors that can bring about these risks. Risk assessment should also make sure that enough precautions are implemented in order to prevent harm coming to an employee.

Ill health and accidents can have a very serious affect on business. They can ruin lives and damage business output. They can also lead to costly court cases and increased insurance costs. Risk assessment procedures should help to dispense with the above and make the workplace a safer environment for employees.

In risk assessment, the most important factor is to decide what is a hazard in the workplace. If the risk is determined to be significant enough, precautions should be put into place so that the risk is minimized or dispensed with altogether. For example, electricity in the workplace is hazardous, but provided that all proper protection is in place, the risk to employees becomes insignificant.

There are five steps that should be undertaken when conducting a thorough risk assessment. The first step is to look for hazards. Take a tour of the workplace and check for potential dangers. Concentrate on anything with the potential to cause serious harm to employees. Also ask for employee opinions on the subject. Accident and ill-health records are a good way of revealing why and how accidents have occurred in the past.

The second step is to decide who might be harmed and how. Decide who might be particularly at risk. Trainees, young workers or expectant mothers may be high on this list. Members of the public who are not familiar with the workplace or anyone who is not in the workplace full time and may not be familiar with the layout may also be at risk.

With the third step, you must calculate whether there have been enough precautions put into place to counter the hazard. For each risk you find, a decision should be made whether you have established enough precautions to reduce the risk. Has your risk assessment procedure taken into account all the health and safety aspects required by law? Is there anything further you can do to reduce the risk, such as issuing protective clothing or preventing access with guardrails?

The next step is to record your findings. Your risk assessment check must show that you have dealt with all the obvious hazard areas. It should also show that you have checked with employees who might be affected and that any hazards that remain have been dealt with and are now reasonably low.

The final step is to review your risk assessment procedures and make revisions if necessary. In the future, new machinery, substances and work procedures could be implemented in the workplace. This is when your revision should take place to ensure that the work environment is risk free and will continue to be so in the future.

EXPLANATION

Risk assessment consists in an objective evaluation of risk in which assumptions and uncertainties are clearly considered and presented. Part of the difficulty of risk management is that measurement of both of the quantities in which risk assessment is concerned - potential loss and probability of occurrence - can be very difficult to measure.

The chance of error in the measurement of these two concepts is large. A risk with a large potential loss and a low probability of occurring is often treated differently from one with a low potential loss and a high likelihood of occurring. In theory, both are of nearly equal priority in dealing with first, but in practice it can be very difficult to manage when faced with the scarcity of resources, especially time, in which to conduct the risk management process. Expressed mathematically,

$$R_i = L_i p(L_i)$$

$$R_{\text{total}} = \sum_i L_i p(L_i)$$

Financial decisions, such as insurance, express loss in terms of dollar amounts. When risk assessment is used for public health or environmental decisions, loss can be quantified in a common metric, such as a country's currency, or some numerical measure of a location's quality of life.

For public health and environmental decisions, loss is simply a verbal description of the outcome, such as increased cancer incidence or incidence of birth defects. In that case, the "risk" is expressed as:

$$R_i = p(L_i)$$

If the risk estimate takes into account information on the number of individuals exposed, it is termed a "population risk" and is in units of expected increased cases per a time period. If the risk estimate does not take into account the number of individuals exposed, it is termed an "individual risk" and is in units of incidence rate per a time period. Population risks are of more use for cost/benefit analysis; individual risks are of more use for evaluating whether risks to individuals are "acceptable"....

RISK ASSESSMENT IN PUBLIC HEALTH

In the context of public health, risk assessment is the process of quantifying the probability of a harmful effect to individuals or populations from certain human activities. In most countries, the use of specific chemicals, or the operations of specific facilities is not allowed unless it can be shown that they do not increase the risk of death or

illness above a specific threshold. For example, the American Food and Drug Administration regulates food safety through risk assessment. The FDA required in 1973 that cancer-causing compounds must not be present in meat at concentrations that would cause a cancer risk greater than 1 in a million lifetimes.

How the Risk is Determined

In the estimation of the risks, three or more steps are involved, requiring the inputs of different disciplines:

1. *Hazard Identification*, aims to determine the qualitative nature of the potential adverse consequences of the contaminant and the strength of the evidence it can have that effect. This is done, for chemical hazards, by drawing from the results of the sciences of toxicology and epidemiology. For other kinds of hazard, engineering or other disciplines are involved.
2. *Dose-Response Analysis*, is determining the relationship between dose and the probability or the incidence of effect. The complexity of this step in many contexts derives mainly from the need to extrapolate results from experimental animals to humans, and/or from high to lower doses. In addition, the differences between individuals due to genetics or other factors mean that the hazard may be higher for particular groups, called susceptible populations. An alternative to dose-response estimation is to determine an effect unlikely to yield observable effects, that is, a no effect concentration. In developing such a dose, to account for the largely unknown effects of animal to human extrapolations, increased variability in humans, or missing data, a prudent approach is often adopted by including safety factors in the estimate of the “safe” dose, typically a factor of 10 for each unknown step.
3. *Exposure Quantification*, aims to determine the amount of a contaminant that individuals and populations will receive. This is done by examining the results of the discipline of exposure assessment. As different location, lifestyles and other factors likely influence the amount of contaminant that is received, a range or distribution of possible values is generated in this step. Particular care is taken to determine the exposure of the susceptible population(s).

Finally, the results of the three steps are then combined to produce an estimate of risk. Because of the different susceptibilities and exposures, this risk will vary within a population.

Small Subpopulations

When risks apply mainly to small subpopulations, there is uncertainty at which point intervention is necessary. What if a risk is very low for everyone but 0.1% of the population? A difference exists whether this 0.1% is represented by *all infants younger than X days or *recreational users of a particular product. If the risk is higher for a particular sub-population because of abnormal exposure rather than susceptibility, there

is a potential to consider strategies to further reduce the exposure of that subgroup. If an identifiable sub-population is more susceptible due to inherent genetic or other factors, there is a policy choice whether to set policies for protecting the general population that are protective of such groups or whether if the group is too small, or the costs too high.

Sometimes, a suitable position is to at least limit the risk of the more susceptible to some risk level above which it seems too inequitable to leave them out of the risk.

RISK ASSESSMENT IN AUDITING

In auditing, risk assessment is a very crucial stage before accepting an audit engagement. ISA315 *Understanding the Entity and its Environment and Assessing the Risks of Material Misstatement*, “the auditor should perform risk assessment procedures to obtain an understanding of the entity and its environment, including its internal control.

Then, auditor obtains initial evidence regarding the classes of transactions at the client and the operating effectiveness of the client’s internal controls. In auditing, audit risk includes inherent risk, control risk and detection risk.

RISK ASSESSMENT IN INFORMATION SECURITY

There are two methods of risk assessment in information security field, qualitative and quantitative. Purely quantitative risk assessment is a mathematical calculation based on security metrics on the asset.

Qualitative risk assessment is performed when the organization requires a risk assessment be performed in a relatively short time or to meet a small budget, a significant quantity of relevant data is not available, or the persons performing the assessment don’t have the sophisticated mathematical, financial, and risk assessment expertise required.

Qualitative risk assessment can be performed in a shorter period of time and with less data. Qualitative risk assessments are typically performed through interviews of a sample of personnel from all relevant groups within an organization charged with the security of the asset being assessed. Qualitative risk assessments are descriptive versus measurable.

Quantitative Risk Assessment

Quantitative risk assessments include a calculation of the single loss expectancy of an asset. The single loss expectancy can be defined as the loss of value to asset based on a single security incident. The team then calculates the annualized rate of occurrence of the threat to the asset.

The ARO is an estimate based on the data of how often a threat would be successful in exploiting a vulnerability. From this information, the annualized loss expectancy can be calculated.

The annualized loss expectancy is a calculation of the single loss expectancy multiplied the annual rate of occurrence, or how much an organization could estimate to lose from an asset based on the risks, threats, and vulnerabilities. It then becomes possible from a financial perspective to justify expenditures to implement countermeasures to protect the asset.

CRITICISMS OF QUANTITATIVE RISK ASSESSMENT

Barry Commoner, Brian Wynne and other critics have expressed concerns that risk assessment tends to be overly quantitative and reductive.

For example, they argue that risk assessments ignore qualitative differences among risks. Some charge that assessments may drop out important non-quantifiable or inaccessible information, such as variations among the classes of people exposed to hazards.

Furthermore, Commoner and O'Brien claim that quantitative approaches divert attention from precautionary or preventative measures.

Others, like Nassim Nicholas Taleb consider risk managers little more than "blind users" of statistical tools and methods.

PRINCIPLES OF RISK MANAGEMENT

- *Organisational Context:* There's no 'one-size-fits-all' when it comes to Risk Management. Each organisation will be affected by different Political, Economic, Societal, Technological, Legal and Environmental factors. It's also worth pointing out that each organisation will have different internal cultures, communication channels, levels of agile-adoption and existing risk management processes.
- *Stakeholder Involvement:* Involve your stakeholders wherever possible. Keep them informed and understand the role they can/could play at each stage in the Risk Management process > Identify, Assess, Respond, Review.
- *Organisational Objectives:* When assessing and responding to a risk, be sure to keep the overall organisational objectives in mind. When considering a Task-level risk, look at the role it plays towards delivering a User Story. If you're concerned about a User Story, consider the impact it has on delivering your Sprint Objective or the relevant Epic. If you're concerned about a particular Epic, then look at the relevant Theme or the Programme of works. Keep things in perspective and don't lose sight of your end-goal.
- *Management of Risk Approach:* This particular principle is less applicable as it refers specifically to the PMBOK Risk Management processes, however the message basically stresses the importance of following best practice guidelines and learning from the mistakes of others.

- *Reporting:* Keep people informed – ensure transparency and visibility. Communication is key!
- *Roles and Responsibilities:* Make sure that everyone understands the role they play at each stage of the Risk Management Life cycle *i.e.*, >Identify, Assess, Respond, Review. Ensure that all bases are covered by someone.
- *Support Structure:* Ensure that everyone understands how risk is managed through the Risk Management Life cycle and who to go to if they have any questions. For example:
 - How are risks identified
 - How and when are risks escalated?
 - Where and in what format are risks documented?
 - How and when are risks reviewed, etc.
- *Early Warning Indicators:* Give yourself the best chance of forecasting/anticipating the transition of a Risk to an active Issue. Ensure that everyone is communicating and that any potential issues are highlighted in the Daily Scrum. It's also important to know how you should react in the event a risk does or is about to be realised *e.g.*, who needs to know and how will you inform them – in the Daily Scrum also? Or, maybe in the Scrum of Scrums? Or, maybe you'll just walk over and tell them.
- *Review Cycle:* Make sure that your Risk Board is visible and that you're regularly reviewing it – you could do this via the Retrospective and as an extension to the Daily Scrum by adding a 4th question:
 - What did you do since the last sprint?
 - What will do you today?
 - Is there anything blocking you at the moment?
 - Any changes to the risk board?
- *Overcoming Barriers to the Management of Risk:* Ensure you're doing everything you can to give you the best chance of successfully assessing the risk and responding to the risk. Some common barriers include:
 - Established roles, responsibilities, accountability and ownership.
 - An appropriate budget for embedding approach and carrying out activities.
 - Adequate and accessible training, tools and techniques.
 - Risk management orientation, induction and training processes.
 - Irregular assessment of Management of Risk approach.
- *Supportive Culture:* Make sure that everyone on the team feels comfortable raising, discussing and managing risks.
- *Continual Improvement:* Use the Retrospective to review the way you manage risk and to assess ongoing risks. Learn from your mistakes.

RISK MANAGEMENT PROCESS

Risk management is a process which could be best described as being systematic. Risk management must never be taken lightly by any organization. It is designed to

deal with risks that may occur in regards to any aspect of a project. While risk management is crucially important, there are a number of additional things that project teams can do to ensure their projects are completed properly and safely. Risk management can be broken down into a number of different steps, and the first of these steps is to take the time to assess the risks that a project faces. By assessing the risks, you are essentially taking the time to think of the things that could go wrong. Once you have an understanding of all the risks that a project faces, you will next need to prioritize them.

By prioritizing the risks, you will essentially take the time to figure out which risks are the most important, in other words, you will rate the risk by how dangerous they are, as well as the probability of the risk actually occurring. Risk priority numbers will generally be used to determine the amount of risk the organization faces.

Once this step has been completed, the next step is to take the time to handle the abatement actions. To do this, you will need to take the time to plan and put in place the actions which are needed to lower the impact or the chance of the risk actually occurring. There are a number of different techniques that can be used within the field of risk management.

The techniques will often differ based on the manner in which the risks have been analysed. For instance, the risk could be evaluated or ranked dependent on the severity, as well as the probability of occurrence. Both effects analysis and failure mode may also be used for the purpose of analyzing and measuring risk.

No matter how you look at it, the fact remains that the risk management process is very important when it comes to managing a project in the proper manner. There are often times when the risk management process must be repeated multiple times throughout the life cycle for a given project. The project team must study the risks and then prioritize them.

HOW TEAMS USE RISK MANAGEMENT?

The team which is working on the project will typically take the time to carry out risk management tasks and record the things that it learned from the project. What this means is that many of the things which have been learned from past projects will need to be used in future projects.

While this takes up an enormous amount of effort, the time and money which is wasted could be saved if a certain process was available which allowed the things learned from any given project to be distributed to other projects. This process is described by a Japanese word called yokoten. It basically translates into “one place to another,” meaning that information from one project is immediately transferred to other projects. This transfer of data can prove to be quite valuable. One way in which knowledge can be transferred from one project to another is simply to make use of documents such as FMEA or even the risk sheets. However, these methods can prove to be tedious and they come with a number of problems. The problem with this method is that it requires an enormous amount of effort, because the new project will need to

analyse the risk management document from all the projects which have been completed. There is also the issue of time. Sharing the information with all the concurrent projects can prove to be very challenging. Then there is the location issue. If the projects are being conducted in different locations, sharing information can be quite challenging.

LIMITATIONS OF RISK ASSESSMENT

Risk assessment is one method in a much broader field of risk management. Risk assessment is a process that does not result in a fixed final answer. It is impossible to determine the true magnitude and extent of any actual contamination at a site.

Research on the effects of chemicals on microbes, plants, animals and humans is patchy at best and confounded by factors such as:

- Variations in individual and species tolerances to the effects of contaminants;
- Environmental conditions and processes affecting the properties of the contaminant such as partitioning, transformation, degradation, temperature, pH, organic material, etc;
- Uncertainty in extrapolating study data between species and within species; and
- Large information gaps about the effects of contaminant mixtures that might have synergistic, magnifying or other effects.
- Large information gaps about the specific mechanisms and processes affecting functions and organs within the body, how these interact, and how they might be affected by a contaminant.

Our understanding of the interactions between affected individual people, plants and animals within a population, populations within a community, and communities within habitats and ecosystems is extremely limited. As a result, generally the best you can hope for is to determine some basic information about the risks posed by the site.

These include:

- Whether there are contaminants present at a site and which contaminants are present;
- Whether these contaminants are likely to have been contained on-site or might be moving off-site;
- What media the contamination has affected including water, soil, air or vegetation;
- Whether these contaminants might possibly be having an effect on microbes, plants, animals or humans; and
- What type of effect that might be.

However, by making informed, careful, and well-documented decisions through a systematic evaluation process, your ability to gauge the relative impact of a contaminant on receptors can be greatly enhanced, and the possibility of underestimating or overlooking adverse effects can be reduced.

AREAS IN RISK MANAGEMENT

As applied to corporate finance, risk management is the technique for measuring, monitoring and controlling the financial or operational risk on a firm's balance sheet..

- *Area 1: ERM implementation:* Many organisations are starting the implementation of their ERM programme. This will require a complete new thinking. Instead of managing different types of risk such as market risk, credit risk and operational risk as separate silos, financial institutions are bringing them together under a single enterprise-wide framework. It will also require a shift from business line aggregation to risk type aggregation of risks and exposures.
- *Area 2: Data quality:* High-quality master and reference data are central to the success of an enterprise risk- management strategy. Each of the components of enterprise risk requires different levels of data input. Other important areas are workflow management and event reporting.
- *Area 3: Systems:* High-quality master and reference data are central to the success of an enterprise. In order to better manage risk, corporations need to bring together data from systems that in the past operated as silos and have seemingly little or no relation to each other. The main challenge in this area is to bring data from different transactional systems, involving different product lines and from different jurisdictions in such a way that data will remain relevant and consistent without duplication or integrity loss.
- *Area 4: Business processes and organisation:* There will be more discussion about 'risk appetite' and 'risk tolerance', which means a greater understanding that companies need to look at their risk culture. Currently they are not yet well embedded as a management concept. There will be a better appreciation that good communication and a learning culture, with openness, lack of blame and analysis of mistakes, is the key to effective risk management. The idea of risk management being embedded in the culture of organisation is central in this context.
- *Area 5: Move from quantitative to qualitative measurement:* One of the reasons of the current financial crisis was cited as over reliance on quantitative approach. The future measurements should combine efficient expert based judgement, risk governance or proper arrangement of roles and responsibilities, stress testing and holistic view of risks. In sound stress testing, management judgement and imagination are essential to determining the scope, likelihood and severity of stress test scenarios.
- *Area 6: Strengthening of the reporting requirements.* Authorities and boards require relevant and timely information concerning key risks that is captured by the risk reporting system to oversee the efficiency of the organisation's risk management approach.

- *Area 7: Convergence of Market and Operational risk.* Risk managers from different disciplines, especially from market and operational risk are working together to develop more forward-looking risk management approaches and techniques.

Market risk is by its nature very data-driven and already using the forward-looking tools such as scenario analysis and stress testing. In contrast to credit risk, which is still mainly driven by historical quantitative data, Operational risk did not produce sufficient historical data. Instead, the Operational risk has focused on using business specialists' knowledge and expertise to generate scenarios. These scenarios have been applied together with the existing data to produce risk measurements or projections of risk probability and severity.

Strategic Planning Knowledge Management

USING KNOWLEDGE MANAGEMENT SYSTEMS TO MAKE STRATEGIC PLANNING MORE EFFECTIVE

A number of leading futurists have predicted that today, as well as into the 21st century, there will be times of great change. Many of the anticipated changes are not minor perturbations, but rather major adjustments in business and social environments. Several of the driving forces behind these changes are global competition, the restructuring of business organizations, the aging of the U.S. population, continued variations in the inflation (deflation) rate, the volatility of the stock market, periodic energy shortages, and accelerating technological changes of all types.

Top-level managers are going to be hard-pressed to deal with this new environment without changing the process of solving problems. Future decisions will involve more complex ones than in the past. To be effective in the future, decisions must merge quantitative and qualitative analyses. Solving the problems of the near future and beyond requires the use of advanced computer systems, like knowledge management systems, to provide top-level managers and their staffs with a more effective approach to strategic planning.

At the outset, it would be helpful to distinguish between strategic thinking and strategic planning. *Strategic thinking* involves intuition and creativity as well as being concerned with a synthesis of what is to come. It is an integral part of executive visioning in a changing world. The outcome of strategic thinking is a forward-looking perspective and an articulated vision of the direction that a company should take at the appropriate time and place.

Knowledge is useful in articulating an executive vision within a strategic thinking framework. In contrast, *strategic planning* is more about analysis—that is, breaking down a company's mission and its objectives into measurable goals. In turn, the expected consequences of these goals are articulated in the form of short- to long-range plans and reports that appear in the form of budgets. Hence, strategic planning from this viewpoint centres on financial measurement and improvement using knowledge that is obtained over time. To assist the typical manager in obtaining a good understanding of this important area relative to knowledge management systems, the need for top-level managers and their staffs to have a real feel for the underlying principles and important factors associated with strategic planning.

For the most part, strategic planning needs for top-level managers and their staffs need to be tied in with executive visioning and problem finding. All of these newer directions set the stage for strategic planning areas that lend themselves to a KMS operating mode—in particular, short-range to long-range strategic planning. In addition, the utilization of knowledge management systems for global best practices provides a means for extending the capabilities of these systems.

SHORT- TO LONG-RANGE STRATEGIC PLANNING IN A CHANGING WORLD

Every organization should recognize that *change can be the engine of growth*. The challenge lies not in embracing this business tenet, but in anticipating, adapting to, and generating fresh ideas that exploit change. Because strategic planning is a logical means for adapting to change, it centres on setting or changing organization objectives and goals as deemed appropriate, obtaining the resources to meet these objectives and goals, and determining the strategies and programmes to govern the use and disposition of these resources. Because it occurs at the highest level and is related directly to top-level executives and their corporate planning staff, strategic planning merges the appropriate external and internal factors that are critical to setting the proper present and future direction for the organization. In turn, it provides input for lower- and middle-level managers.

Strategy considerations relative to future market opportunities and products to fill them are basic to *long-range strategic planning*. A distinctive characteristic of this highest planning level is the use of marketing facts to discover opportunities and then develop effective strategies and programmes to capitalize on these opportunities.

Similarly, the focus is on bringing future problems back to the present for solution. The long-range strategic plans that embrace all aspects of the organization and its environment provide a basis for more detailed medium-range strategic planning.

Medium-range strategic planning, sometimes called *tactical planning*, is concerned primarily with financial planning to place the organization in the best financial position for the next several years. This fiscal planning involves developing the operating programmes and associated budgets for the next several years. On the other hand, *short-range strategic planning*, or detailed operating planning, is related to the financial plans of the current year only.

For an organization that has practiced formal planning on a regular basis, it is normal for every major functional area to prepare annual plans for the coming year. Essentially, these financial plans are brought together from a detailed examination of the key measures of the business for the coming year, such as product-line profitability, variable and fixed costs, inventory turnover, manufacturing capacity, and financial ratios.

UTILIZATION OF KNOWLEDGE IN EXECUTIVE VISIONING

An important element underlying strategic planning from the short range to the long range is *executive visioning*. Many times, executive visioning is tied in with problem finding. An executive view entails farsightedness along with an eagerness to look ahead. But at the same time it must be practical. Effective executive visionaries are not necessarily those who can predict the 21st century and beyond accurately.

Rather, they are the decision makers who can draw a conceptual road map from where the company is now to some imagined future, who can say, “This is how we get there.” Visioning implies a change from the status quo, which helps explain why visionaries are overrepresented in the ranks of entrepreneurs. It also explains why they come in handy to an organization in deep trouble.

Think of Mr. Lee Iacocca saving Chrysler. Vision is not for the complacent. While the executive visionary sees things a bit differently, this individual is no mystic. The person’s sources of information are down to earth—customers and suppliers, for example—and extend beyond his or her gut-level feelings. The most visionary executive can take in large amounts of information and knowledge, not just what is inside himself or herself.

Typically, a broad grounding in a particular industry is almost always a prerequisite to successful direction setting. It is helpful to look at the early career of an executive who comes to be regarded as visionary. Usually, the individual finds an assignment or a series of assignments that enable him or her to see the company from many different perspectives—for example, the individual has served as vice president of marketing, then production, then, finance, or has had a tour as executive assistant to the president.

Jan Carlzon, for instance, held all sorts of positions in the travel business before becoming head of SAS, and Louis Gerstner studied the financial services industry as a McKinsey consultant before taking over American Express’s credit card and traveller’s

check businesses. There can, however, be situations in which the executive does not need to be steeped in an industry to conceive a vision of its future. Upstarts like Steve Jobs dwell on the dangers of falling prey to the “standard world view,” thereby missing opportunities that arise when that standard world begins to show a few cracks. The skeptics concede, though, that someone who has grown up in the business stands a better chance of realizing his or her vision. Executives with vision typically share a couple of other characteristics. They have a high degree of self confidence. It takes a lot of inner strength to imagine a future at variance with the common expectation and to sustain that vision in the face of responses ranging from incredulity to derision.

The visionary may also be a bit of a loner. Ego strength can mean that the individual has less need of other people. But executive vision has a way of exciting others somehow. An executive visionary appeals to the emotions and aspirations of people in a way that goes beyond the usual carrot-and-stick approach.

Most corporate visions come out looking like tarted-up strategies. William Hewlett and David Packard established a company that is dedicated to producing the best instruments for the world’s scientists. That surely counts. Ray Kroc built McDonald’s and Kemmons Wilson founded the Holiday Inns chain—both visionary achievements of 20th-century American capitalism. Vision by itself, however, is not enough. The executive visionary must be able to communicate what he or she has dreamed, and the company must have the required skills to execute the dream.

The leaders of the organization must act consistently with the vision in everything they do. Too often in the past, top-management teams have worked up a statement of corporate vision, promulgated it, and then have thought their work is done. What they overlooked, and what dooms this kind of superficial effort, is the need to plan and manage this vision over time. A knowledge management system, backed up by appropriate information systems, is an excellent vehicle for assisting in the fulfilment of the executive vision.

IMPLEMENTING SUCCESSFUL KNOWLEDGE

Today, knowledge can represent the know-what, the know-how, and know-why of a company’s overall picture as well as its functional areas. Knowledge is acquired by managers and analysts, including their staffs, by examining and investigating areas under study. This is different from the acquisition of specialized skills that are learned by performance over time. Typically, *knowledge work* is performed by company personnel who possess a high level of expertise in their respective functional areas, such as strategic planning, marketing, manufacturing, accounting and finance, and human resources. In addition, there are other company positions that involve some type of knowledge work, directly or indirectly. These include administrative office assistants and factory supervisors who have acquired important knowledge based on their years on the job. Knowledge work can therefore be defined as “the acquisition, creation,

packaging, or application of knowledge.” At the outset, it should be stated that there is no single hardware or software product or combination of the two that can provide a comprehensive approach to knowledge work. If a comprehensive KMS environment is the ultimate goal, hardware and software products cannot be used in isolation. Building a corporate-wide knowledge management infrastructure requires the integration of many different technologies. For a company to unlock successfully its collective knowledge for companywide use, it is necessary not only to implement integrated hardware and software technology but also to integrate a company’s personnel and their related business processes with this knowledge management technology.

If company personnel are not working in a collaborative environment or if no procedures and processes are in place to share the knowledge, no amount of KMS technology can change that. For a KMS environment to work effectively, it must be viewed by company personnel at all levels as a strategic means for the company to become more competitive and ensure success in the long run.

To better understand what needs to be done to develop knowledge management systems successfully, it is best to start with past and current approaches to knowledge acquisition in the form of expert systems and then go on to different levels of knowledge acquisition and use—ranging from operational knowledge to tactical knowledge to strategic knowledge.

ESSENTIAL FOCUS OF KNOWLEDGE MANAGEMENT SYSTEMS

Currently, to achieve good results, a typical organization needs newer methods and procedures to improve its knowledge work. This ties in with the essential focus of KMSs, which is to provide the means for professional and technical workers to capture knowledge and allow it to be accessed easily so that it can be applied effectively to achieve improved results. Knowledge work processes can centre on *finding existing knowledge* (i.e., searching for it, among multiple sources) or centre on *creating new knowledge* from resources unused in the past.

Knowledge can also be created externally—for example, the development of new knowledge about financial reporting by an outside auditor or during the course of the regular annual audit. Similarly, the knowledge work process can focus on the *reuse of knowledge*, such as when adopting object-oriented technology in the computer field.

The task of implementing a successful knowledge management system may seem insurmountable. But in reality, most computer consultants offer a different view. Some have stated that up to 90 percent of the infrastructure required for knowledge management is already in place. This refers in most organizations to the existing structure of their computer networks and servers. Hence, only a small shift in the total computer technology budget is required to make the change to a knowledge management system.

REDESIGN MAY BE NECESSARY FOR CHANGING KNOWLEDGE WORK PROCESSES

To acquire and disseminate narrow- to wide-based knowledge for an organization’s personnel, the present knowledge work processes may have to be improved. From an

overview standpoint, this may involve reengineering in which knowledge work processes have to be revamped so that knowledge flows freely to every functional unit and subpart that needs it. In order to achieve this ambitious knowledge work redesign, several approaches can be undertaken by knowledge management system developers. For one, an organization can *change the content of knowledge* by expanding or contracting what it encompasses in order to better meet desired objectives, such as a one-to-one tie-in with the company's critical success factors. For another, an organization can *reorder the composition of work* so that company personnel may change the focus of their jobs from information to knowledge. This change may require the use of more work teams so that knowledge learned from previous projects, job assignments, and the like can be shared on current work assignments. From another perspective, the change can focus on the *employment of new networking technology that lends itself to groupware*. Employing this technology allows employees to access individual knowledge bases and global knowledge bases in order to bolster the effectiveness of their work. Overall, the above changes are quite consistent with reengineering-oriented approaches to work and, at the same time, bolster the efforts of workers using knowledge to perform their jobs more effectively.

Choose the Appropriate Knowledge Orientation

In order to determine which approach is the proper direction for reengineering knowledge work processes, it is necessary for the knowledge system developers to specify which knowledge orientation is required by the organizational unit and its personnel. Is there a major or minor emphasis on knowledge among organization personnel for the work process under study in the functional unit? Additionally, there are other factors that can affect which direction to take. These include the competitive environment, corporate culture, organizational strategies, problem-finding approach, and the information technology infrastructure—each of which could be study unto itself. By establishing an effective relationship among knowledge work processes, organizational personnel, and the items set forth above, knowledge system developers can sort out the important factors and set forth ones that are useful in the final design.

Although knowledge system developers can sort out which needs should be emphasized or be deemphasized, it is necessary to create a corporate-wide or global knowledge base that brings together the diverse elements of knowledge found throughout the organization. Many of the important elements to consider in developing a corporate-wide knowledge base were included previously in the text, and therefore will not be repeated here. In addition, knowledge system developers need to develop lists of company experts, information maps, and corporate yellow pages that are sources of knowledge. Other possible items for determining an appropriate knowledge orientation include the age of the workforce, educational level, and time on the job. Based on a thorough analysis of these items, an appropriate knowledge orientation can be developed for management, their staffs, and technical workers to help them become more effective in their jobs.

COST JUSTIFICATION

An often asked question by corporate sponsors of knowledge management systems is, “How does one justify the cost in terms of a traditional cost-benefit analysis?” The answer given by those within or outside the computer department who have tried is that one does not. For the most part, the payoff from knowledge management systems are different from other types of information systems in that they are often less tangible, less quantifiable.

For example, in the implementation of a new accounting system for accounts receivable, an information systems manager is able to estimate specific savings from tracking accounts receivable more closely. However, these new knowledge management systems focus on providing managers, their staffs, and technical workers with improved knowledge via some type of information analysis over time. The real payoff is in giving them quick access to analytical knowledge that shows patterns, trends, and projections that may lead to new ways of thinking and better decisions.

On the other hand, there can be instances where savings or gains can be tied directly to knowledge management systems. For the most part, however, no one has been able to come up with a model or system for justifying the cost of these systems. Typically, cost-benefit analysis is not used. In its place, the focus is on the value of these systems to managers, support staffs, and technical workers. Also, it is not easy to predict the total cost of such systems in an accurate manner. The largest expense generally comes from ongoing knowledge acquisition and dissemination to organization personnel as they become more familiar with the system and seek more knowledge about patterns and trends that can be related to future projections to assist them in doing a better job of problem finding.

The difficulty in justifying the cost of these systems keeps many computer departments from proposing them in the first place. If those at the top of the organization are pushing for these systems and have enough clout, however, the cost-justification issue usually fades into the background. Often the cost of a knowledge management system gets justified by one or more managers high up in the organization saying, “Give me this new knowledge management system or else,” which implies that they may look elsewhere for employment. Overall, justifying the cost of knowledge management systems tends to be an evasive compared to the situation for traditional mission-critical information systems.

SUCCESSFUL KNOWLEDGE MANAGEMENT SYSTEMS INCLUDE SELECTING APPROPRIATE SOFTWARE

Knowledge management systems represent a new business intelligence technology that has become popular because it allows decision makers to access and analyse large quantities of information that can be used to distill knowledge concerning current and future patterns and trends. In addition, these systems are attractive because they provide

decision makers with faster access to desired knowledge which can be presented in new ways that do not require extensive custom programming.

Their most appealing feature is that they assist decision makers in making better informed decisions rather than making decisions for them. The rationale is that decision makers have a wide base of resources at their command to use on problems facing them. Since no one software product can serve the full range of user knowledge needs, there are a host of products available today.

Successful knowledge management systems require the cooperation of the vendors, the computer department, and the company's decision makers and their staffs. Most outside vendor products have limitations that will be reached the first time a user (*i.e.*, decision maker) says, "I need ..." What the person then finds is that the package has provided what is thought to be needed. The vendor can do little to adapt it. So instead of simply acquiring this package, it is necessary to look for one or more software packages that provide capabilities for adapting it to the needs of company decision makers.

This naturally leads to the next group that must be involved, the computer department. Working with the computer department rather than treating it as an outsider will result in a knowledge management system much better suited to the users' needs. These computer professionals, after all, have been providing much of the information that managers and their staffs use. Hence, they understand what some of the decision makers' needs are already. They will also be the individuals who will develop and implement the system and adapt it to decision makers' requirements.

Finally, decision makers are the most critical element in a successful knowledge management system. It is important that they know what they really need to guide them to make better decisions. Since it is important that decision makers get involved and support the effort to get the system up and running, they need to take the time to sit down with the computer department and discuss their real needs. Which knowledge-based systems are currently available (*i.e.*, what nuggets of knowledge exist in a company that have been codified and are available now for use by decision makers)?

In addition, does the current knowledge systems approach need to be improved to make the decision makers' jobs easier? In particular, should a broad-based, global knowledge base be installed that encompasses all knowledge elements found throughout the organization? What are the new knowledge requirements of decision makers? Is there something they need that they have not asked for? Overall, decision makers can explore with computer professionals some of the knowledge-based systems currently in place and see how they can be improved using a knowledge management system approach to provide for more effective decision making.

Critical Factors for Evaluating Knowledge Management Software

In order to determine the appropriate software to provide abundant and rich knowledge for decision makers, it is necessary to evaluate a package's critical factors. As a starting point, the package should be easy to use. It should be designed to assist decision makers

in the development and use of knowledge. In other words, the software should be designed to help decision makers make better decisions, not dictate their actions. The software should also provide for analysis of patterns and trends not only in the past but also for the present and future, thereby allowing decision makers to focus on projections. The software should respond readily to decision makers' requests (*i.e.*, their wants here and now).

The software package should allow the decision maker fast access to knowledge that can be presented in very sophisticated ways without custom programming. Similarly, it should allow access to a very large amount of knowledge in a format that allows decision makers to find their own answers as problems and opportunities arise. Because some of the knowledge-based software precalculates and aggregates values, decision makers should be able to slice and dice quickly using a variety of keys to get at important knowledge. Additionally, the software approach, generally a combination of software packages, must be able to handle the organization's size or it will be outgrown in a few years. The software should allow for changes and updates as deemed necessary.

The software should be reasonably priced and flexible to meet the decision makers' needs. It should be flexible to allow for different types of analysis in order to manipulate the knowledge in whatever form is considered to be most useful. As specific knowledge analysis is required, they must be able to be added without losing other functions or requiring changes in the basic software. Other factors come into play that can be asked as questions: "Will the vendor support the installation adequately?" "Will the software run on the organization's hardware?"

"Is the documentation clear?" Needless to say, these questions must be answered to evaluate critically the decision about whether to proceed with a specific software package. It should be noted that it is helpful to visit one other user of the software to get a firsthand look. If multiple software packages are to be acquired, it would be helpful to pay a brief visit to users of each package.

It is highly recommended that the above factors be investigated fully before signing a contract for purchase or lease of the software operating in a networked or mainframe environment. The computer department, working with decision makers, should resist the temptation to sign the vendor's standard agreement. If possible, the matter should be referred to the company's legal department, rather than trying to handle the matter alone. In the final analysis, the key to the longevity of various types of software designed to develop narrow- to broad-based organization knowledge is flexibility to meet the changing needs of users.

CONTINUING SUPPORT OF KNOWLEDGE MANAGEMENT SYSTEMS

If a corporate-wide knowledge management system is developed and implemented correctly, some type of formal training is needed for managers and their staffs. Typically,

managers and their staffs need to know how to use the knowledge that can be displayed on their screens. This knowledge may tend to change over time depending on the industry in which the company operates. If hand-holding of the company's personnel is undertaken, the KMS stands a much better chance of being accepted. Managers and analysts tend not to bother with a user's manual or even a user's pamphlet.

In some cases, a company may supplement the KMS project team with outside consultants. The use of consultants during development, implementation, and support phases does not relieve the company from acquiring expertise in knowledge management. Usually, the consultants have a broader perspective that may help avoid pitfalls and may speed up the learning curve. Outside consultants can give the KMS effort an extra boost and ensure its final success.

For effective continuing support within a KMS environment, essentially the project team becomes the support team. However, it usually takes fewer people to support the knowledge management system than to develop it initially. The KMS support team members should have a continuing interest in the system's success. Needless to say, the implementation of a knowledge management system can cause major changes throughout the organization in terms of altering the flow of information and knowledge to and from the top of the company.

Within this context, managers and their staffs can spend more time analyzing and improving the company's strategic business direction. Whereas most of a manager's time was previously spent gathering data and information, the time has been greatly reduced by requiring less than 20 percent of the manager's effort.

Thus, time savings allow more time for intensive analysis to improve a company's operations. As the world renowned management consultant Peter Drucker has said, "That which is measured ultimately improves." Updating this concept to the present time, a KMS environment must provide a basis for learning and the compounding of learning in order to create relevant knowledge for improving a company's performance over time. In turn, this relevant knowledge becomes an important company asset that needs to be managed effectively.

REAL-WORLD CASE STUDIES OF SUCCESSFUL KNOWLEDGE MANAGEMENT SYSTEMS

An important reason behind the implementation of knowledge management systems is the ability for organizational personnel to share knowledge via some type of online forum. As one person put it, "Why reinvent the wheel if organization personnel can share company experience and knowledge to solve an internal or external problem?" Building systems to manage organizational knowledge or know-how goes beyond the needs of CPAs and consulting firms. Today, information systems professionals in various industries around the globe are engaged in a wide variety of knowledge management system installations.

Although approaches tend to vary, knowledge management systems organize and make available important know-how wherever and whenever it is needed. This centres on “best practices” or guiding principles, projected forecasts, reference sources, proven processes and procedures, patent information, established formulas, corrective fixes, and similar items. Data warehouses, computer networks, company intranets, extranets, groupware, bulletin boards, and video conferencing are the technological tools for storing and distributing appropriate knowledge—as stressed previously in the text. Typical examples of knowledge management systems case studies are given in Part III of this text, in addition to the sample real-world case studies provided below.

Because there is no single approach to knowledge management systems, the aerospace giant, Hughes Space & Communications Company, decided that the best approach was to create its own Hughes Knowledge Highway. Described best as a human network that uses technology, the highway employs video conferencing, Notes, employee home pages, and other technologies. Among other things, the highway is used to spread new management practices, track patents and licenses, and gather competitive intelligence.

Another firm—the Bechtel Corporation—decided the best foundation for its emerging knowledge management effort was an enterprise document management system from Documentum, Inc. (Pleasanton, California). Since authoring, managing, and controlling requisitions, reports, and purchases orders drive its business, the company, aided by EDS Strategic Desktop Services, has installed Digital Equipment Corporation’s VAXstation and Alpha servers in five U.S. offices to serve its 4,000 to 5,000 users worldwide. For example, a civil engineer using the system will be able to access vast amounts of resources on past projects, including plant designs and equipment specifications. Internet links let users download earthquake and weather data directly from the World Wide Web.

Along the same lines, Buckman Labs has created a knowledge web for its 1,200 employees in 80 countries. They have round-the-clock access to K’Netix, which is a rich global web of electronic forums, bulletin boards, virtual conference rooms, online libraries, knowledge bases, and electronic mail that can be accessed via IBM’s ThinkPad 755s. Recently, the company has added Notes and Internet support. It has overhauled its worldwide network to TCP/IP and is employing information and related knowledge from its databases onto internal web servers.

From a different perspective, the primary focus of Monsanto’s efforts in knowledge management is connecting people with people and encapsulating that knowledge so it can be shared. The company has gone from pioneering knowledge management in Monsanto’s Ceregen unit to rolling out the chemical giant’s first enterprise-wide data warehouse for development of pertinent knowledge. Its client/server system uses MicroStrategy’s DSS Agent front end and an Oracle back end on a Digital Alpha workstation. The company’s warehouse contains “a genetic map” that includes charters, organizations, and geographics.

SOFTWARE USEFUL IN KNOWLEDGE MANAGEMENT SYSTEMS

Today, knowledge management systems often integrate a variety of off-the-shelf software products, such as groupware, document management systems, E-mail, relational databases, and workflow, with knowledge extraction tools, knowledge management software, knowledge management intranet search engines, knowledge management alternatives (Lexis-Nexis), and knowledge-discovery or data mining software. In addition, knowledge management systems can employ OLAP software, statistical analysis software, and GUI/4GL software packages to assist decision makers in getting at the knowledge at hand. Depending on the nature and purpose of a knowledge management system, there can be still other software products and hardware needed to form a complete KMS package to gather, organize, collaborate, refine, and distribute knowledge.

Essential to capturing and making individuals' knowledge available to the entire organization is the ability to deal with heterogeneous information. Although data warehouses are useful for collecting and storing structured information in an organization, a large amount of information resides in unstructured text—E-mail messages, documents, presentations, etc. Knowledge extraction tools to get at appropriate organizational information will be covered next. Then the popular software approaches mentioned above will be set forth in some detail to complement the discussion of knowledge management software.

KNOWLEDGE EXTRACTION TOOLS

A starting point within a KMS environment is extracting information initially in the entire knowledge acquisition process. Other than the laborious task of manual mining of information, the preferred way is to make this information structured and readily accessible—that is, turn it into organizational knowledge by identifying and extracting important content from the text automatically.

Knowledge extraction from text involves several underlying phases of document and linguistic analysis: breaking the document into units (titles, paragraphs, sentences, and words), determining the linguistic role of each word in each sentence (*i.e.*, noun, verb, adjective, etc.), normalize different surface representations of the same word (*i.e.*, bought, buys, etc.), and identifying and classifying patterns within the analyzed, normalized text that constitute content-rich structures—noun phrases (“stock market decline”), people (“President Clinton”), and organizations (“Treasury Department). In turn, these extracted objects can be indexed by search engines, for example, in order to support better retrieval, categorization of topic area, and message routing and filtering.

The challenge for providers of knowledge extraction tools is to provide a product that is very fast, accurate, and extendable. Because organizations deal with hundreds of

gigabytes of textual information, system throughput is critical, as are precision (correctness) and recall (completeness). For many companies, knowledge extraction tools must deal with text in several languages. Above all, the tools must be able to be tailored to the customer's domain and to the KMS environment. Due to the complexity of knowledge extraction tools, they have not been developed overnight, but rather over a long period of time.

For example, Inxight's LinguistX product line is the result of 20 years of research and development at Xerox PARC. This product line provides multilingual natural language processing components to extract knowledge at very high speeds. It supports a range of functions, from automatic language and character-set identification to phrase extraction to automatic summarization in 11 European languages and Japanese, with others to follow. Today, LinguistX is incorporated into major applications from companies, such as Infoseek, Excite, Oracle, Textwise, and Verity. These products along with many others are covered in the material to follow on knowledge discovery software.

Knowledge Management Software

Knowledge management software shares its borders with search and knowledge discovery or data mining tools where the differences are somewhat subtle. That is, search tools ferret out information known to be present, but they do not necessarily reveal anything inherent in the data. Data mining focuses on legacy data that is more transactional and financial in nature for extracting patterns, trends, and rules.

In contrast, knowledge management software provides the capability for users to collect and organize information and knowledge, search for what is needed, and share the findings with others. As will be seen in the exposition to follow, knowledge management software varies from one vendor to another. Thus, it is helpful to look at these software tools from this perspective. It should be noted that identifying the intended purpose of knowledge management tools will go a long way towards choosing the appropriate one(s).

GrapeVine for Lotus Domino and Lotus Notes is a groupware management system that helps companies classify and index corporate knowledge using charts and profiles. It is highly scalable and follows earlier release of its client/ server version. If a company has deployed Notes, employees are already using it to exchange information about their projects, sales leads, and the like. GrapeVine attempts to discover and route existing information stored in Notes databases or on local area networks that it thinks company employees have an interest in. With this approach, users might even find the new project they are embarking on overlaps with one already in progress in another part of the company.

Typically, GrapeVine is the best solution for companies that rely heavily on Lotus Notes for messaging, collaboration, and document storage. The application sits on top of the Lotus Domino server and pulls information from Notes databases and other external files. It then distributes the information to users through the Notes messaging

system. Integrating GrapeVine with Notes should be straightforward for a Notes administrator, but GrapeVine does provide a consultant for three days (at no charge) to help with the details. Users familiar with Notes will take to GrapeVine quickly. It is available from GrapeVine Technologies, Inc., Troy, Michigan (248-680-6685).

Wincite is a Windows-based knowledge management system that has fast data retrieval, flexible report distribution, and graphing and support using an Oracle client/server architecture. The Wincite solution tackles the knowledge management system from a conventional approach.

It is aimed at analysts who track market conditions, competitors, and other information. This tool organizes intelligence and distributes it to employees within a business. If analysts are knowledgeable, Wincite will amplify their aptitude; if their insight is less keen, it will distribute those findings as well.

This solution approach to knowledge management requires in-depth customization by the vendor to define the content users want to track, design the data entry screens, and set up the briefing books. Wincite consultants work on site for about six days and then complete the job back at the Wincite office. The entire process takes about one month. Because the Wincite approach to knowledge management is straightforward, it does not profess to find any hidden relationships within the data. However, it organizes what analysts feed into the knowledge base and distributes it throughout the organization.

The knowledge base is easily built by linking Wincite to external files or inserting data directly into the system. Findings are consolidated into tailored reports that can be posted to a web site or sent by E-mail. It can be obtained from Wincite Systems LLC., Chicago, Illinois (312-630-4848).

Another knowledge management software package is KnowledgeX. By feeding it pieces of information, KnowledgeX can discern relationships that the user might otherwise overlook. For example, by feeding it with public documents about management and investment policies, the solution might reveal hidden relationships that might be indicative of how future management decisions will be made. Similarly, KnowledgeX is a useful tool for tracking the career paths of the competition's chief executives and displaying the results graphically. KnowledgeX displays its output either in a hierarchy or in a "spider web" form. Generally, since KnowledgeX is easy to implement for users familiar with SQL databases, it requires little maintenance and lets users create some custom reports.

Other knowledge management tools include the following: Sovereign Hill provides a search and index tool, a web access layer, and a sophisticated database that understands people, places, and things. The software parses free-form data and distills it into the core relationships. For example, after supplying it with industry news, users can query Sovereign Hill about competitors. Its answer will not just incorporate hits on a competitor, but also identifies the relationships the competitor has with employees, customers, or competitors.

It is available from Sovereign Hill Software Inc., Dedham, Massachusetts (781-407-7700). KnowledgeShare (from Cambridge Technology Partners, Inc.) is a client/server-

based system for developing and managing an enterprise-wide repository on intranets. Lastly, Decision Suite 3.0 (from Information Advantage) is an integrated suite that lets a user navigate, filter, visualize, and share live analysis and information. Overall, these knowledge management tools are designed to help typical companies manage corporate knowledge effectively and disseminate it throughout the organization as needed to assist appropriate company employees.

Knowledge Management Intranet Search Engines

Knowledge management software that utilizes intranet search engines is the most basic and widely used tool for finding and accessing information housed on companies' Internet-based computer networks. Many of these products came from earlier ones designed to locate and retrieve data from databases, file systems, and other legacy systems before intranets took hold across companies. Typically, search engines often make up just one part of a broader-based knowledge management product package. Typical intranet search engines are set forth below.

The Knowledge Access Suite from Information Discovery (Hermosa Beach, California) was the first and only set of products to provide business users with a gateway to knowledge that has been predistilled from data and is stored in a Pattern Warehouse. Business users need not perform data analysis but simply query explainable knowledge on the intranet that has been automatically premined. At the heart of the Knowledge Access Suite lies PQL (Pattern Query Language) which is a pattern-oriented query language specially designed to provide business users access to refined information, just as they use SQL to access data.

Originally a supplier of software and services to the CD-ROM market, Dataware Technologies, Inc. developed NetAnswer, knowledge management technology that utilizes its own database. Once information is found, it is placed in the database, thereby eliminating the chance of its disappearing from a non-Dataware data source. The current Dataware NetAnswer release is a query-and-retrieval system featuring relevance ranking and natural language processing. It is compatible with major web browsers and consists of a server and database.

Excalibur Technologies Corporation (Vienna, Virginia) developed Visual RetrievalWare software specifically for finding data in a multimedia age. The software is designed to index and retrieve text, photos, video, audio, and animation via Boolean language as well as statistical, semantic, and pattern recognition. Excalibur also has a product specifically for text searches.

Another intranet search engine is one that is available from Fulcrum Technologies, Inc. (Ottawa, Canada), which pioneered "brokering technology." Brokering technology allows users to index information residing on networked computers and execute simultaneous enquiries across network resources. The key to the company's Knowledge Network product is a feature called Knowledge Map, a folder hierarchy that shows users the information sources available to them through the corporate intranet.

Knowledge Network consists of software that sits on a web server and can be accessed via standard web browsers. Information Dimensions, Inc. (IDI) (Hermosa Beach, California), a spin-off from the Battelle Memorial Institute, first developed a product called Battelle's Automated Search Information System (BASIS). IDI has since developed a BASIS intranet package for UNIX and NT networks that includes a document manager, a web server gateway, and client software. Another important company in this category of knowledge management intranet search engines is Verity, Inc. (Sunnyvale, California).

Verity's flagship product is SEARCH '97, an application platform containing a group of tools designed for accessing information from many data sources across an enterprise. It features agent technology designed to proactively search, filter, categorize, and deliver information to users; database gateways that allow for cross-database searching; and an intranet spider for indexing data. It includes a user interface for the desktop, an information server, and an agent server.

Knowledge Management Alternative—Lexis-Nexis

Setting up a programme with Lexis-Nexis is an alternative to the knowledge management software mentioned above. This company provides businesses with information collected from a wealth of news sources and public records that users can search and retrieve over the Internet. Currently, it houses more than one billion documents in a massive central repository and adds almost 14 million documents to the pool each day. Actually, Lexis-Nexis consists of two separate services. The Lexis is a full-text legal information service that was started in 1973 to help legal professionals more efficiently research the law.

Lexis has evolved into 40 specialized libraries covering all major fields of practice, including tax, securities, banking, environmental, energy, and international law. On the other hand, Nexis news and information service, which began in 1979, is a leading news and business information service. It contains more than 13,500 sources, including regional, national, and international newspapers, news wires, magazines, trade journals, business publications, and public records.

The Lexis-Nexis system is somewhat similar to the Wincite software mentioned previously. It searches for information based on specific criteria and delivers it to the users in a company who need answers. In the early stages of implementation, a Lexis-Nexis representative meets with the knowledge manager to determine the type of information that needs to be collected and the relevant topics that need to be covered. The representative gets a firsthand view of the type of user that the system will serve and the person(s) who will be in control of the information pool. Next, the Lexis-Nexis representative determines which product best serves the needs of the company. Lexis-Nexis offers a number of services, among them Tracker, Info Tailor, and Requester. Tracker is a software programme that lets the knowledge manager search for information and publish the results on the corporate intranet or in other common areas, such as a

Lotus Notes database. Info Tailor is a personal information agent that brings customized news to the user's desktop every morning. Requester is a search tool designed for users who need to track their own information from their desktops. Users can request information on a variety of topics including news, company financial and personnel statistics, industry forecasts, and product data. Among the companies using the Lexis-Nexis system are utility companies, which use it to track their competitors and new regulations that affect the utilities industry. These utilities can also prospect companies that are relocating to other local areas to win new accounts.

Knowledge Discovery or Data Mining Software

Knowledge discovery or data mining software provides an important means for decision makers in a KMS environment to uncover patterns and trends as well as rules for today and tomorrow. Among the major players are: Angoss Software International, Business Objects, Cognos, DataMind, IBM, Information Discovery Inc., Magnify, NeoVista Solutions, Pilot Software, SAS Institute, and Thinking Machines.

The Data Mining Suite is an integrated set of products that provide a powerful, complete, and comprehensive solution for large-scale enterprise-wide decision support and data mining. The system works directly on very large SQL repositories with no need for sample or extract files and delivers results on the web.

Over 90 percent of computations are performed on the server, in parallel if desired. The system directly accesses large volumes of OLAP and relational data, incrementally discovers powerful patterns that are very general and go far beyond decision trees and simple rules. The system delivers automatically generated English text and graphs as explainable documents on the intranet.

Another typical software product is the Intelligent Miner from IBM that runs on AIX using a 5-node RS/6000 SP. Analyses are performed from Intelligent Miner clients on Windows NT. This software package serves as the core technology for Loyalty Consulting's data mining services for clients in finance, retail, telecommunications, and other industries. Intelligent Miner's full range of data mining capabilities help users to acquire more intimate knowledge of their customers, predict customer behaviour, and find the most cost-effective way to influence that behaviour. Most importantly, Intelligent Miner meets users' key criterion for data mining: the ability to deliver measurable results, such as increased revenues and lower selling costs.

It provides a wide range of algorithms for clustering (both demographic and neural), classification (tree induction and neural induction), value prediction, association discovery, sequential pattern discovery, and predictive Modelling. Essentially, Intelligent Miner allows users to build better data mining models than are available with many other tools since it offers a broad range of data mining algorithms that can be combined to provide greater value to users.

Server data mining software licenses typically cost \$150,000 to \$200,000, while desktop or "micro mining" software licenses typically cost \$500 to \$50,000. For

example, IBM's Intelligent Miner can deal with data warehouses containing hundreds of gigabytes or terabytes of information. Its licenses range in price from \$25,000 to \$150,000 for systems from Risc/6000 UNIX machines to System/390 mainframes. Although these costs may appear to be somewhat high, some vendors and users note that data mining offers a good return on investment and an important competitive advantage.

Database and Computer Vendors Incorporate Knowledge Discovery or Data Mining Capabilities: Today, database vendors have integrated their products with the data mining capabilities of other vendors. This decision support strategy pairs the processing horsepower of relational databases with the discovery-driven analysis capability of data mining. This combination improves the scalability and performance of data mining applications, and simplifies their management. For example, Brick Systems (Los Gatos, California) has integrated its RDBMS technology with data mining technology from Data Mind (Redwood City, California). DataMind Professional Edition provides scalable data mining software that builds a model from historical data via an automatic discovery process, thereby allowing business professionals to explore relationships in historical company data. In addition, data mining vendors are cooperating with computer vendors. For example, several vendors, like Information Discovery Inc., NeoVista Solutions, Pilot Software, and SAS Institute, are linked with Hewlett Packard. They are part of HP's OpenWarehouse Alliance Programme which focuses on selecting best-in-class warehouse component providers to ensure compatibility and to maximize functionality on HP computing platforms. Information Discovery System is a collection of client/server data mining tools that work with Oracle, Sybase, and Red Brick relational databases and also perform data mining on the Internet.

This software takes advantage of the parallel processing technology of an HP 9000 Enterprise Parallel Server by decomposing its analysis and sending it to the separate processors automatically. In a similar manner, NeoVista Solutions' Decision Suite Series provides comprehensive, enterprise level, data mining solutions, and associated professional services to Global 2000 companies. Products are based on a comprehensive suite of automated knowledge discovery software—the Decision Suite Series—designed to be highly scalable and to integrate easily with legacy and open systems decision-support environments, such as HP's Enterprise Parallel Server.

The Pilot Discovery Server (a key component in the Pilot Software Decision Support Suite) is a sales and marketing product that uses data mining to help users increase sales and acquire and retain customers. Seamlessly integrated within a large relational data warehouse, Pilot Discovery Server analyzes all available information in customer activity and demographics. The product runs on HP 9000 Enterprise. Similarly, the SAS Institute's software—SAS/Insight and SAS/Spectaview—offers various tools for the exploration of data to uncover anomalies in the data tools for Modelling as well as analytical tools, such as multi-dimensional analysis. SAS software runs on HP computing platforms.

OLAP Software

OLAP software centres on *multidimensional analysis*, which is an architecture for performing computer analysis in new and different ways. OLAP software packages can be used to uncover new patterns and knowledge about a company's operations. Typically, this analysis divides data into the important dimensions used to manage the business.

For example, dimensions for marketing applications might include products, market areas, distribution channels, and time periods. In turn, these dimensions can be used to describe each data point in the database, such as selling price, units sold, total revenue, and market share. Dimensions can be further described by attributes, such as location, size, or year. Attributes can also describe hierarchies within a dimension, even overlapping and inconsistent hierarchies.

From an end user's perspective, multidimensional analysis provides for selection, analysis, summarization, and reporting by dimensions and attributes within dimensions. Multidimensional analysis can support virtually any time-series application. These include forecasting and budgeting. Most current spreadsheet, database, and reporting tool vendors offer simplistic multidimensional tools.

However, some of these vendors and others are offering more sophisticated tools. That is, all multidimensional engines are not created equal. There is a need to look for engines that can work with any data warehouse, use meta data, and use any PC development tools for the user interface. Not only should the engine be robust to support an unlimited number of dimensions, custom groupings, and automatic and custom calculations but the engine also needs to reside on a server large enough to support hundreds of users, large volumes of data, and intensive processing.

The major OLAP software products include Essbase from Arbor Software, Inc. (Sunnyvale, California); Commander from Comshare, Inc. (Ann Arbor, Michigan); Express from Oracle Corporation (Redwood Shores, California); KnowledgePoint from Knosys Inc.

(Boise, Idaho); and LightShip Suite, from Pilot Software, Inc. (Cambridge, Massachusetts). They exhibit some interesting and helpful features to users. For example, some software products handle up to six dimensions with ease. In addition to this software, some users are turning to alternative products because they are considered easier and faster to implement. For example, Business Objects from Business Objects, Inc. (San Jose, California) gives managers the ability to perform complex multidimensional analysis, thereby allowing them to follow lines of questioning on their own. As another example, SAS/EIS from SAS Institute (Cary, North Carolina) uses fourth generation language-based packages that are designed to provide multidimensional views and analyses.

Statistical Analysis Software

Although many of the current software packages have statistical analysis capabilities, there are those that are designed primarily as statistical analysis systems for uncovering

important relationships and trends. For a typical business situation, a series of data is used to see what the results look like when plotted under a certain set of conditions. Interactively, the set of conditions can be changed and new output plotted.

Needless to say, there are a whole series of business problems that benefit from graphic solutions: project planning, break-even analysis, learning curves, and so forth. Since a picture is often worth a thousand words, a thousand lines of printout, or ten minutes of discussion, decision makers are able to absorb a tremendous amount of information when it is shown graphically. Because knowledge management systems make great use of computer graphics, they allow organizational personnel to build models and speculate with their computers using graphic output concerning how hypothetical decisions might translate into reality.

A number of statistical analysis packages currently available include SAS (Statistical Analysis System) from the SAS Institute, Inc., SAM (Strategic Analysis Model) from Decision Sciences Corporation, SPSS from SPSS, Inc., STATSII from CompuServe, and DISSPLA and Tell-A-Graf from Integrated Software Systems Corporation. SAS is used for most statistical applications that are widely found in business today.

SAM has been used for expediting strategic planning by generating alternative future scenarios. SPSS is useful in market-research studies; STATSII centres on business statistics, forecasting, and econometric Modelling. DISSPLA is a graphics software system with a high-level subrouting plotting language for producing graphs, surfaces, and maps. Tell-A-Graf enables users to call up graphs, charts, and plots using simple English-like statements.

GUI/4GL Software

Currently, there are a large number of GUI (graphical user interface)/4GL (fourth-generation language) packages that epitomize the term “client/server development tools.” At the outset, it should be noted that although today’s more advanced GUI languages have replaced the fourth generation in many situations, 4GL remains the tool of choice for many straightforward and basic administrative jobs, large reporting tasks, and screen setups that require page breaks, margins, and specific page formatting.

These GUI/4GL packages offer a broad range of features and functionality. Ease of use and RAD (rapid application development) via iterative prototyping are important hallmarks of this group. While CASE (computer-aided software engineering) and “pure” 4GLs provide integrated support for many of the functions required for building complex systems (data Modelling, version control, software distribution, interapplication middleware), GUI/4GL tools typically, but not exclusively, provide this same functionality by way of vendor alliances.

Generally, products in this group can be differentiated according to their functionality. Some products are strongly object oriented (OO), fully supporting multilevel and multiple inheritance, while others are more object based, limiting OO support to encapsulation and single-level inheritance. Also, GUI/4GLs can be distinguished by whether they

generate compatible code or interpreted code. Since many of the 4GLs lack the highly optimized “pure” 4GLs, this issue becomes important as systems are scaled up to serve more users. All tools in this class use a high-level 4GL to specify applications.

A class of tools that is very similar to the GUI/4GLs is the visual 3GLs. Originally, these products were not considered client/server tools *per se*, but this has changed recently. The explosive growth of Microsoft’s Visual Basic (VB), coupled with VB 4’s greatly improved database access capabilities and Borland’s phoenix-like rise on the strength of its Delphis tool, now positions these products as front-rank client/server tools. Typical GUI/4GL tools include Powersoft Corporation’s PowerBuilder, Forte Software’s Forte, Uniface Corporation’s Uniface, and Oracle Corporation’s Developer 2000. Rapid application development is an integral part of PowerBuilder. Within the boundaries of its client-side OOP (object-oriented programming) paradigm, the applications are quite usable and maintainable. The developer can build screens, input some data, and thereby allow users to give feedback. In total, PowerBuilder is an object-oriented client/server development system that is quite useful for mission-critical Windows applications.

A specialized form of fourth generation languages include financial planning languages. When users find that their models are too large for electronic spreadsheets and need advanced features such as risk analysis, linear optimization, and goal seeking, financial planning languages that operate in conjunction with microcomputers and computer mainframes are used. They are high-level financial generators designed to excel in a certain class of applications that includes financial planning and analysis, strategic planning, budgeting, Modelling, and business simulations.

Actually, they are quite useful in handling any problem whose answer is likely to end up on a row-and-column format or whose complex nature reflects logic and data manipulation rather than support for managers. Their statements must resemble a typical person’s vocabulary and must track unstructured and continually changing thought processes. Furthermore, these generators must react in sufficient time to allow a person’s attention to remain focused on the problem. Decision-making support is further promoted by permitting the language statements to appear in any order.

Financial planning languages, such as Comshare’s Commander and Information Builders’s Focus, enable managers and their staffs to become competent in building and solving models in an interactive, exploratory manner. Typically, users of financial models are found in manufacturing as well as banking and finance where corporate planning and finance are the largest user groups. Modelling applications most frequently used centre on cash-flow analysis, short-term investments and debt structure, accounts payable, and inventories. Other typical uses include pro forma financial reports and financial forecasts, investment analysis, profit planning, sales forecasts, budgeting, merger-and-acquisition planning, and project models.

METHODOLOGIES FOR DEVELOPING SUCCESSFUL KNOWLEDGE MANAGEMENT SYSTEMS

Just as there are a number of software packages and languages that are useful within a KMS environment, there are also methods for developing knowledge management systems successfully. Popular ones include joint and rapid application development (JAD and RAD), windows-based application development, visual programming development, and object-oriented (OO) technology development. These packages are discussed below, as is their tie-in with the World Wide Web.

It is essential that knowledge system developers get a firsthand look at what is currently going on in the organization's operations (*i.e.*, learn how people actually work). The result will be usable knowledge about the organization's internal operations that can be applied to functional units and their subunits. Similarly, knowledge system developers should learn something about their competitors. Perhaps, a tie-in with their auditors or an outside consulting firm who maintain knowledge bases about specific industries can be of assistance here. The essential message is that knowledge system developers should understand rudimentary methods and procedures involved in how people work. In some cases, it may be useful to move into different areas of the company for several weeks so that real, usable knowledge and not theoretical knowledge is gathered.

Experienced knowledge system developers emphasize the need to start with knowledge work processes. Knowledge work processes drive a successful knowledge management system, not the technological issues. The rationale is that these newer systems are helping decision makers make better decisions in terms of their work activities. The technology is incidental to this critical issue. Examples given in this text have focused on knowledge work processes and how they can be developed further to assist decision makers.

For example, Cigna's Property and Casualty Division has created a new desktop knowledge system for underwriters. There was a two-day "innovation workshop" to encourage creative suggestions from users. Half-day feedback sessions at dozens of field offices also were held. Now, with a few mouse clicks, underwriters can call up dozens of worksheets, forms, motor vehicle and financial reports, and lists of company experts to improve their knowledge work processes. Substantial user input and speedy development using Microsoft Corporation's Visual Basic was crucial in improving these work processes.

Joint and Rapid Application Development (JAD and RAD)

Joint application development (JAD), which is basically a systems design technique, has as its goal to improve the quality of project specifications by engaging users in the requirements definition process. JAD is performed in structured workshops driven by a JAD leader and documented by a scribe. Over time, a quality systems specification

evolves and becomes a blueprint for project development. Typically, computer projects are prone to delays, errors, and failure for a number of reasons. Poor communications, a protracted systems development process, a “we-they” environment, and massive, incomprehensible, and inconsistent documentation are some of the symptoms.

To be responsive to organizational needs, computer projects must be completed before the problem parameters change. By having computer professionals and a group of users work together in a nonthreatening series of analysis and design sessions, the total time to understand and document the requirements for a new system can be greatly reduced. The goals of JAD are to accelerate the development process, increase productivity, improve quality, and build better developer-client relationships.

Overall, joint application development facilitates the interaction between systems analysts and users. Much of the value in this approach is that it requires consensus. Conflict can be used to energize the group so that they accept differences, focus conflict on substantive (rather than personal or political) issues, promote deeper communication and understanding and, in the end, achieve a superior quality product. It has been said that the value of JAD is not so much in the end product but in the discovery process.

Rapid application development (RAD), on the other hand, represents the body of development approaches that use JAD, CASE tools, and 4GLs to build systems faster. RAD is a systems development approach that has gained acceptance in the computer development community. It combines elements of JAD and automated prototyping techniques with a stringent time limit to create an environment that fosters rapid systems development and maintenance. RAD employs small teams with specialized skills. Unlike other development approaches, it has no interim deliverables. Each RAD team simply builds a piece of larger prototype and refines it until it becomes a working system. Some computer development experts have noted that using JAD and RAD together represent a good way to achieve internal customer needs for quality and speed.

Currently, many software vendors are offering their own RAD products and services. Rapid application development tools include Microsoft Corporation’s Visual Basic and Powersoft Corporation’s PowerBuilder. While vendors offer widely differing products, their approaches to RAD are characterized by some key common elements: an interactive development approach; great emphasis on enterprise and data Modelling; use of automated tools for rapid prototyping and code generation; active end user involvement throughout systems development; and reliance on reusable code, forms, and modules.

Since most businesses revolve around a few critical success factors, the reason for centring on data Modelling is to identify just what those success factors are and how the pieces fit together. In turn, this foundation can be utilized for building and prototyping system applications.

A most important distinguishing factor of RAD is its time limit. Projects typically run from 60 to 120 days. While this short time limit can be a motivator, it can also become the single greatest reason for employee dissatisfaction and turnover. Careful planning and execution are needed to make RAD projects work. Rapid application

development calls for leaders who are forward thinking and confident enough in their staffs to empower them to get the job done. Overall, most system developers agree about the benefits of RAD tools: development is faster and easier and code can be reused and ported across platforms. Because end users are involved in prototyping, chances are that they will be satisfied with the final product.

Going one step further, RAD technology is making use of Modelling that allows all information about a complex application to be stored in a repository. This allows the application to change as business rules change, with new application code being reflected in the model. Many Modelling tools are being combined with CASE tools, which allow users to create complex data models for applications used in enterprise environments. Tight integration between these tool sets is important to ensure that changes in an application are reflected in the CASE data model and vice versa.

Current RAD tool vendors have decided to offer CASE-like Modelling. For example, Powersoft acquired S-Designor, a database Modelling tool that has been integrated with PowerBuilder, and it has added process Modelling as well. Similarly, Forte, Oracle, and others have integrated analysis-and-design tools into their own development environments. Analysis-and-design tool vendors have also formed alliances with RAD tool vendors.

For example, Visible Systems has coupled with Unify Vision and provides bidirectional integration with PowerBuilder, Uniface, and SQLWindows. Bidirectional integration automatically reflects changes during the coding that affect the design back to the model, ensuring that the design as captured by the model and the application code remain synchronized. Without such synchronization, the organization loses the benefits of having a model when the need for maintenance arises. Generally, experienced RAD programmers do not like the discipline imposed by model-based analysis and design. However, given the complexity and risk of enterprise-scale client/server computing, this is a good alternative to system development.

Windows-Based Application Development

There is currently a new set of development tools available to systems analysts and programmers: windows-based application development. The most popular representatives of this new breed of high-level development languages and/or frameworks include Borland Turbo C for Windows, Microsoft Visual Basic with Professional Toolkit, Asymetrix Toolbook, the Whitewater Group's Actor, and Microsoft Visual C/C. These products and many more have sprung up to assist system developers in making the change to developing for, and sometimes within, a graphical environment.

Many Windows programmes are graphically based versions of existing text-based applications. But as Windows 3.X has gained unprecedented market penetration along with the next generation of Windows (NT), there has been and continues to be more interest in developing new types of applications geared to the operating environment. Because Windows is event driven, Windows programmers are freed from thinking in

the traditional logical and sequential manner of DOS (disk operating system) programming. Windows programmers construct applications in an environment that allows them to think in a much less linear fashion. Since events can be invoked from all parts of the screen (menu, toolbar, dialog, box, etc.), a Windows programmer needs to anticipate all the actions a user might like to perform and take into account any order that the user decides.

To this must be added the fact that other programmes may be running and interacting with the programme under development. One of Windows' greatest strengths is its multitasking capability. Typically, windows-based application development gives programmers an easy way to get up to speed.

The development of Internet applications, including intranets, has become a high priority for many development teams. For example, Sun has created much excitement with Java, which has the ability to embed Java applets within HTML-based documents and support dynamic content and access to other applications.

Java-specific tools are now available in the Windows NT market, but most lack integration with applications on computer mainframes or other servers. Microsoft has added to the momentum by developing the ActiveX control strategy, which allows authors to integrate BackOffice data, Java applets, and older OLE-based applications. Microsoft has moved to leverage its visual development tools by releasing a Java development environment called J. Several other vendors have moved to provide Internet/intranet controls for the Windows NT development environment.

Visual Programming Development

A more current approach is visual programming development that is tied in with the World Wide Web. Visual programming tools simplify the overall development process and create a synergy between end users and developers. They let developers create applications using point-and-click and drag-and-drop techniques instead of text-oriented methods. Code is generated on behalf of developers, and visual techniques let them more easily organize development tasks. Simplifying the application development process is considered to be a big advantage of visual programming tools. In the text-based world, the difference between textual and visual would be similar to that between character-based and graphical. Programming with visual tools is faster and requires fewer skills and less training.

Currently, data visualization software—initially found in scientific research laboratories—helps improve comprehension of complex business data. That could mean displaying the information on a map, a chart, or even an animated, virtual-reality scene that shows changes over time. For example, Silicon Graphics uses its internally developed Data Navigator to monitor sales activity in geographic regions. Using a mouse, analysts can “fly” through sales territories and manipulate the level of detail shown.

Sales targets are portrayed as hovering disks that are broken when the target is exceeded. From this view, data visualization is a pictorial representation of data. It can

be sliced, diced, or plotted to create a “landscape” of the data and information. For improved clarity, colour and time-elapsd motion can be used. The bottom line is that data visualization software can turn large, complex data sets into patterns that are easily recognized.

In the World Wide Web universe, visual programming tools are quite useful. As Web applications become a necessary component in enterprise development, users will expect more from developers in terms of support. The Web has brought developers and end users closer. Netscape Communications Corporation, for example, is driving its development effort off the Internet. The company puts beta software on its web site, lets users work with it to discover bugs and suggest changes, and then modifies it for general release. As developers of web applications focus on the client, they will demand some of the traditional backend services, such as data integrity, protection, recovery, and joining, to be delivered to them.

Object-Oriented (OO) Technology Development

A widely used methodology for any type of system development, including a KMS operating mode, is object-oriented (OO) technology. It is useful to increase the productivity of systems analysts and programmers by giving them a single construct with which to address analysis, design, and related activities. The fundamental construct is the object, which combines both data structure and behaviour in a single entity. Object models are particularly valuable for communication and documentation. And they serve well as the framework for programme code and database schemas. With object-oriented methodology, the result is a much better understanding of requirements, cleaner designs, and more maintainable and reliable systems.

Object technology is sometimes considered to be a subset of CASE because the two share the characteristic of reusable programming modules. However, there are important differences. While CASE is a semi automated form of structured programming in which each function is broken down into sets of instructions, object-oriented programming is based on Modelling the physical aspects of the business rather than the procedures themselves. Objects contain programming and data so that the object knows what it is and how it behaves.

While a CASE module could be reused in a similar application with a little extra coding, a well-designed object could be used in any application, regardless of its type. Object modularity also means that a designer need only make changes to a single object in order to modify a programme, rather than having to modify hundreds or even thousands of lines of code, because one change causes a chain reaction. Such modularity is a primary reason why many organizations are moving to this important direction in systems development and maintenance.

Currently, the World Wide Web has kick-started objects into broad acceptance. Next-generation web applications, such as electronic commerce, will depend on distributed-object technology, particularly CORBA-compliant ORBs, to become a reality. Object

Request Brokers (ORBs) are a type of middleware that enables objects to pass messages among themselves. Objects will be able to interact with each other over the Internet, providing services such as transaction processing, replication, fallback and recovery, and multilevel security, that are difficult if not impossible to build into applications created using other programming technologies of HTML.

The transition to object technology is currently being undertaken by companies in a evolutionary fashion. Because there is a steep learning curve, thinking in terms of objects may take some time. Typically, it takes close to a year to make the shift. It should be recognized that class construction or the building of objects is a complex business. Every object can have hundreds of relationships with other objects, and its creation requires strong analysis and design skills. If applications are being downsized or moved to a client/server environment, systems analysts and programmers are likely to be more open to object technology. If deemed appropriate, external training and consulting organizations should be used for the transition to object technology.

STEPS IN IMPLEMENTING SUCCESSFUL KNOWLEDGE MANAGEMENT SYSTEMS

Knowledge management systems were described earlier as a group of related technologies that assist in the collection, organization, and transfer of knowledge among company employees. Sometimes, a company's customers can be involved in the whole knowledge process.

A knowledge management system should function as a knowledge warehouse where users have easy access to the knowledge contained in disparate information systems as well as have the ability to contribute valuable knowledge. Because there is no comprehensive KMS software approach available today (or anticipated in the near future), a number of steps for implementing a successful knowledge management system are suggested. Underlying all of these steps is the empowerment of employees (and customers) to have more control over their operations. Finally, it should be noted that the ultimate goal of a knowledge management system is not to capture all possible knowledge in people's minds, but rather to do a better job of using the individual's knowledge to run the company more effectively in the short to long run. Typically, these steps are taken in the following order:

- Get support by starting at the top of the organization.
- Appoint a chief knowledge officer (CKO) to manage knowledge.
- Select an experienced team to implement the system.
- Select an appropriate methodology to develop the system.
- Select appropriate knowledge software that meets users' needs.
- Determine a proper organization to acquire, renew, and disseminate knowledge.
- Develop initial knowledge applications.

Get Support by Starting at the Top of the Organization

The first step in implementing a knowledge management system is to obtain executive sponsorship (*i.e.*, to identify a *corporate sponsor*). No KMS system will succeed without a strong advocate at the very highest level. Ideally, it should be the number one or two executive in the organization. If top management support is not there, the system is not going to go very far. This is like trying to undertake a knowledge management system from a grassroots effort. Overall, getting started centres on obtaining the support of the president of a company or, at least, the support of the company's executive vice president.

Appoint a Chief Knowledge Officer (CKO) to Manage Knowledge

Additionally, sponsorship must go beyond a corporate executive sponsor and include a day-to-day operating sponsor. Today, this person is the *chief knowledge officer* (CKO), who makes the initial request for the system and oversees its continuing development over time.

Sponsorship means more than sitting back and saying, "I'm for this project and will be its champion throughout the development process," and then delegating all of the work to others. A successful KMS implementation requires a total commitment from the company's chief knowledge officer.

This person must be capable of managing old and present knowledge as well as putting in motion what it takes to acquire new knowledge.

The individual must equip the organization to respond to as-yet-unknown forces for change. That is, the CKO must question assumptions, learn new technologies, and thrive in the face of uncertainty. The CKO must therefore be capable of operating comfortably in either the business or technological areas of an organization.

Select an Experienced Team to Implement the System

After the appropriate chief knowledge officer has been recruited and has his or her job under control, there is a need to establish a project team that is capable of carrying out the implementation of the knowledge management system on a corporate-wide basis. The head of the KMS project team should be knowledgeable and receptive to the new knowledge technology set forth earlier. Ideally, this person should be able to work on the project full time, have an excellent understanding of both the organization's and management's knowledge needs, and have priority access to the chief knowledge officer. Overall, the KMS manager should have the proper credentials to head the project team.

Because knowledge is a level higher than information and uses information as its raw material, the recruitment of a KMS project team must take this important fact into consideration. A balanced mix of technology specialists and business analysts is needed.

Technical specialists from the information systems department who are knowledgeable about data mining, data warehousing, network computing, capacity planning, and security issues need to be recruited. Similarly, business analysts who are

knowledgeable in identifying, locating, and utilizing knowledge required by the company's management at all levels need to be recruited in order to complement the information systems professionals. Ideally, these business analysts, representing the various business/financial areas, should be providing information and current knowledge to the company's managers that focus on the accomplishment of a company's objectives, goals, and strategies.

Select an Appropriate Methodology to Develop the System

Underlying the appropriate methodology to develop a successful knowledge management system is the consideration of whether to go with a "push" or "pull" philosophy for delivering knowledge. If the *push approach* is utilized, the knowledge necessary for company employees, managers, and others to do an effective job is relayed to their desktops.

The knowledge produced by the knowledge management system is pushed forward to users on a routine basis to assist them in their day-to-day tasks. In contrast, if the *pull approach* is employed, users are provided the tools needed to find the knowledge they need. A wide range of knowledge management tools and related information software were discussed as possible candidates to assist in this pull approach. Needless to say, two different system methodologies have a decided impact on the design architecture.

The design methodology set forth below focuses on the push design approach since the pull design approach is very open ended and varies from one installation to another. An important part of this first approach is the development of a corporate-wide knowledge base. Typically, such a knowledge base consists of various kinds of knowledge, such as established facts, rules, heuristics, strategies, procedures, and standards, along with an explanation of this knowledge.

The capability of Modelling knowledge and using fuzzy knowledge can also be included under the heading of the corporate-wide knowledge base. A well-designed, corporate-wide knowledge base assists a company in such processes as knowledge acquisition, knowledge inference, and explanation, including validation of the software quality assurance process.

Concurrent with the development of the corporate-wide knowledge base by the KMS project team is the design of a user-friendly interface for knowledge base construction and validation as well as implementation. A user-friendly interface is also helpful for knowledge retrieval and updating.

For example, this can take the direction of a graphical user interface (GUI). Related to these design activities is the selection of a suitable underlying methodology for knowledge representation. An object-oriented method, or one of the other methods set forth earlier, should mesh with the objective of establishing a knowledge management system and meeting users' needs.

In total, the appropriate KMS design methodology for a push or a pull approach is not a product or a total solution that the project team can buy off the shelf. Rather, it

focuses on the capability to link important structured and unstructured factors over time with the changing rules by which people apply knowledge. The methodology must provide the basis for learning and the compounding of learning—that is, the ability to create longer-term intelligence. Essentially, an effective KMS design methodology is a process that is, in actuality, implemented over years and focuses on the effective interfacing of human minds with the best of business practices.

Select Appropriate Knowledge Software that Meets Users' Needs

An integral part of the above KMS design methodology by the project team is the selection of the appropriate knowledge tools that meet users' needs. The result of using these tools is an efficient and effective mechanism for knowledge acquisition, knowledge inference, and knowledge explanation. Some of the knowledge tools (*i.e.*, knowledge extraction tools, knowledge management software, knowledge management intranet search engines, knowledge management alternatives [Lexis-Nexis], and knowledge discovery or data mining software).

KnowledgeSEEKER will be employed as a typical knowledge tool to demonstrate its usefulness for a typical company's functional areas. For the most part, it is prudent to build flexibility into the knowledge management system as users' needs multiply over time.

Appropriate knowledge tools today include using the company's intranets, extranets, and the Internet along with some type of groupware, such as Lotus Notes. By adopting an easy-to-use platform plus standardizing on a single-technology platform, there is easy access to knowledge as well as the ability to connect everyone in the company. They are useful for indicating trends and identifying opportunities for improvement. From this view, the "best practices" for a company in a specific industry can be shared throughout an organization.

Additionally, appropriate knowledge tools include those that create rich meta data structures for capturing information about the knowledge assets. For example, was a proposal successful or not? What did the individual learn? This enables the capture of tacit knowledge that can be more useful than merely reading an archived document. It allows the capturing of knowledge in a more structured form, which allows users to perform fielded searches on classes of knowledge assets and enables uniform presentation of the resulting information.

However, organizational knowledge goes beyond mere documents, web pages, and other explicit forms. Finding out who knows what in an organization has always been a time-intensive process. An effective knowledge management system should provide the capability to find people based on their skills and areas of expertise so that company members can quickly find the experts for whom they are looking.

The selection of appropriate knowledge tools by the KMS project team centres around their capability to meet users' needs—that is, their contribution to assist managers and analysts, including their staffs. Obviously, a knowledge management system will not

provide all possible knowledge, nor should it attempt to be a total solution, because people are the problem solvers. The system is simply a tool to support company personnel.

Ultimately, they will have to ask “why,” and they will have to come up with the solution. Overall, not only do knowledge tools need to be selected carefully so that they provide *flexibility* for users, but also the output from these tools should provide the capability to improve their *effectiveness* in everyday operations.

Determine a Proper Organization to Acquire, Renew, and Disseminate Knowledge

Once the appropriate knowledge tools have been selected, it cannot be assumed that data and information exist that can be converted into knowledge. If they do, it cannot be assumed that the data and information are compatible with a KMS environment.

Because a knowledge management system represents a key tool for leveraging the contents of an organization’s current databases plus data and information stored in its data warehouse, the system must be properly organized to acquire, renew or refresh, and disseminate a company’s knowledge. In working through these issues, the KMS project team must give users the capability to acquire (knowledge capture), renew (knowledge inventory), and disseminate (knowledge transfer) knowledge in new and different ways.

In terms of *acquiring knowledge*, the capture and collection of tacit knowledge is related to an organization’s corporate-wide knowledge base (as noted previously). No matter what the size of the organization, knowledge is not a static source. True knowledge is found, to a large degree, in the sophistication of the methods and processes by which that knowledge can be consistently renewed.

From this perspective, knowledge cannot be preserved for very long without losing its inherent value, which is *timeliness*. While the accounting knowledge base changes little in a five-year period, the knowledge base for engineering and designing integrated circuits changes every few months. Effective knowledge capture is based on its relevance to today and tomorrow. Having instant access to yesterday’s best practices does nothing for a consulting firm if the knowledge is outdated. Hence, replacing knowledge is much more difficult than simple knowledge capture.

Once procedures have been established by the KMS project team to capture relevant knowledge for the times, there must be procedures put in place to *catalog this knowledge*. In the past, cataloging information meant searching and finding a set of documents. However, with knowledge links, any single document can lead to an indeterminate number of other links, making navigations almost impossible. To resolve this difficulty, there is a need for intelligent inventory systems that catalog knowledge as it is needed rather than in advance. This approach recognizes that knowledge is changing constantly. Today’s knowledge extraction tools, such as Inxight’s LinguistX product line, support a wide range of functions from automatic language and character set identification to phrase extraction to automatic summarization.

The *dissemination of knowledge* or knowledge transfer means transferring the implicit nature of both the “what” and the “how” of what should be done. While prior information systems, such as OLAP systems, placed the burden of asking the right questions on the users, knowledge management systems shifted the burden to a large degree, to the system. At the same time, a knowledge management system provides users with the ability to expand their understanding of the company’s operations over time as they discover new areas that they did not think to ask about in the first place. From this view, the KMS project team can pinpoint new directions for transferring knowledge to system users that were not obtainable with past information systems.

Develop Initial Knowledge Applications

The KMS project team should start with one or more important knowledge applications that have wide appeal across the organization. Typical applications could focus on a company’s critical success factors (CSFs). CSFs centre on performing those factors that are critical to the survival of a company. If, for example, customer satisfaction or quality are two important CSFs, what information as well as knowledge should be retrievable by managers to evaluate and improve these areas. In terms of an OLAP system, for example, new sales to existing customers might be the measure of customer satisfaction, or quality might be measured by the volume of product defects.

In contrast, knowledge of customer satisfaction would go a step further by starting with the customer as the focus for improvement. This would entail utilizing in-depth customer surveys, examining point-of-purchase information, forming customer focus groups, and employing knowledge discovery techniques. In reference to knowledge discovery or data mining techniques, a store could collect useful information about their customers by asking them for their ZIP codes and correlating this with bar-coded data collected at the cash register.

This approach could also be applied to the quality of goods and services. The focus would initially be on how customers perceive the quality of the company’s offerings—good, bad, or indifferent. In turn, this could lead to internal improvements that impact the goods or services offered to customers. The linkage between a company’s customers and the employment of knowledge within a KMS environment is essential if the project team wants to show important results for initial knowledge applications. Overall, companies that utilize knowledge management systems are capable of adding to their top and bottom lines through the greater use of superior solutions across the organization.

STRATEGIC PLANNING PRINCIPLES BASED ON KNOWLEDGE OF TOTAL BUSINESS OPERATIONS

One of these principles serves as a starting point. It is the recognition that *change is a constant*. Generally, fundamental changes to an organization can take from five to ten

years to complete, while superficial improvement can be achieved in a much shorter time frame. The first step in successfully managing any change is analyzing the gap between the present situation and the intended goal.

It is vital for management and employees to work together towards the final goal. Secondly, there must be a plan in place to reach the final destination. The focus is on short-range to long-range corporate plans. It is necessary to break down these plans into manageable pieces so that relatively simple steps, accompanied by progress reviews at each stage, make the change less difficult. Finally, it is necessary for managers at all levels to manage the transition to the intended goal. Even the most detailed planning will not fully anticipate myriad unexpected events. It is best to manage the process of change by responding with needed training and support.

Because change is considered to be a constant in today's business world, there is a need to consider this factor in determining

- A company's mission,
- Identifiable objectives,
- Measurable goals and strategies. A company's mission gives overall direction to its entire organizational efforts. For example, say that a company's mission is to be the dominant supplier of widgets to the most profitable domestic market segments and to provide a high-quality employee work environment. Identifiable objectives, regardless of how general they may be in their conception, become a measure of success in very real terms. They should be looked upon not as a reactive approach to setting direction for a company, but rather as a proactive force to realise organizational goals successfully. These objectives, such as to compete in profitable market segments and be the low-cost producer in the company's market segments, provide the basis for determining measurable goals, such as increasing market share by 12 percent in high-profit market segments while reducing manufacturing costs by 8 percent per unit and achieving a 22 percent return on investment. In turn, these goals provide a basis for developing appropriate strategies that realise the company's mission. For example, strategies could include expanding market research activities to identify high-profit market segments and developing one or more product lines that fit the requirements and needs of these high-profit market segments. The successful linkage of a company's mission to its objectives, goals, and strategies requires knowledge of a company's total business operations. The principle of effective use of resources applies here and underlies the entire strategic planning function.

Another important strategic planning principle is one that relates to a company's long-term strategy. Competitive strategy is about being somewhat or completely different in the eyes of customers. Essentially, a company can outperform its rivals only if it can establish a difference that is perceived by its customers. This difference must also be preservable over the long haul.

This approach contrasts with concentrating on operational effectiveness, where techniques can be duplicated over time by competitors. Creating a successful longterm strategy involves choosing activities that are desired by customers and that are somewhat difficult for the competition to match. The essence of an effective long-term company strategy, then, is choosing to market goods and services differently from rivals. Otherwise, such a strategy is nothing more than a marketing ploy that will not withstand competition over time.

Related to the previous principle is one that allows a company to transform new technologies of products and services into an important advantage, not only for itself but also for its customers. Too often, companies experience a technological breakthrough but are not capable of capitalizing on it. It cannot succeed without capable management to define or redefine the company's goals and its core competence. Without the proper managerial guidance, these pioneering efforts can result in disastrous forays into unknown territories. On the other hand, development of carefully prepared corporate short- to long-range plans that are supported by an appropriate organizational structure and effective marketing can thrust a company into a position of market leadership. Over time, it is necessary that technological pioneering efforts be maintained by continually bringing out important customer-oriented features ahead of competition.

For companies that are not engaged in technological pioneering but are operating in established industries, there is a need for top management to step back and take a hard look at its operations. This ties back to the principle of strategic thinking—that is, thinking in terms of strategies that are linked to today and tomorrow. Thinking involves identifying and mapping the driving forces that the company needs to evaluate.

The mapping power provides top management and the corporate planning staff with visual representations of possible solutions and results, thereby helping them to achieve consensus in determining the driving forces. Once top management and its staff has formulated a new strategy, it must align the strategy with the company's resources to make sure that implementation is possible. In turn, this strategy can be translated into action by using aggregate project planning, as discussed below.

An important guiding knowledge principle of strategic planning is one that recognizes the need to develop project plans that focus on the development of new products and services. To map out and manage a group of strategic development projects effectively, there must be a total project plan in place. Getting to the point of an aggregate project plan is not easy, but working through the detailed process is a critical part of developing a sustainable development strategy. While the specific plan is extremely important, the planning process itself is even more so since the plan will change as events unfold and managers make adjustments.

But choosing the right mix and determining the number of projects the resources can support, as well as picking the right projects, raise essential questions about how product and process development ought to be linked to the company's competitive opportunities. Overall, creating an aggregate project plan not only gives direction and

clarity to the overall development effort but also provides a framework for outstanding performance. In the development of a company's strategic plans, the principle of examining the best- and worst-case scenarios comes into play. The best-case scenario is based on an economy that is experiencing a 5 to 6 percent growth rate where profits are very high. In contrast, the worst-case scenario considers the financial results of natural disasters like fires or floods. These are usually short lived and can be solved quickly, though at high cost.

The types of disasters referred to here are conditions that prevent a company from fulfilling its long-term mission, objectives, and specific goals. Also, a worse-case scenario can refer to a reduction in business volume. Knowledge of these unfavourable conditions is necessary if an effective job of short- to long-range corporate planning is to be undertaken. While most corporate planners think in terms of good times, realistic projections should also include provision for bad times, especially when global operations are used in developing realistic short- to long-range plans.

For effective corporate planning under any scenario, a most important principle recognizes the importance of short- to long-range strategic planning. Essentially, these strategic plans, recognize that the company knows what its customers want and need before they do.

This proactive stance will help a company gain a competitive edge in the marketplace. At this point, it is safe to say that financial plans need to incorporate this important fact of competitive advantage. A company's short- to long-range corporate plans should also be related to its strategies (as discussed in a previous principle) so that the right hand knows what the left hand is doing.

They should not only be realistic in financial terms but also should challenge management and employees to meet financial targets that go beyond what a company has done in the past. Knowledge of past operations coupled with the challenge of the future should provide the company with opportunities to develop new ideas and concepts in order to meet these future projections. Essentially, long-range strategic plans provide a basis for action which, in turn, drives medium-range and then short-range strategic plans.

A principle that ties in with the need for short- to long-range strategic planning is one that cultivates a new style of corporate behaviour that encourages changing established views of the company's products and services. There is also a need to encourage creativity in strategy and business processes. From this view, innovation is a style of corporate behaviour that is comfortable with, even aggressive about, new ideas, change, and even failure.

It must permeate an organization. Although it is not necessary to abandon a healthy business just for the sake of trying to be innovative, it is important to make sure there are many good ideas in various stages of fermentation. When the time comes, the company needs to be willing to take risks. If a company waits to move until everything is certain, competition will have been there first. Thus, innovation can be the lifeblood

of a company that uses vast amounts of internal and external knowledge to keep ahead of competition. To assist management and its staff in getting a handle on important factors affecting a company's total operations, including innovative directions, the principle of spending time with customers goes a long way towards understanding their point of view. This recognition about the customer's importance suggests that the customer is more than the next step in the distribution chain.

Senior managers and their staffs need to spend several days in the lives of their key customers. There is no substitute for senior executives' instincts, imagination, and personal knowledge of the marketplace. Creativity on their part can bring insights and expertise to their customers' problems. Hence, the strategic planning process for all of a company's functional areas can be improved by making the customer the focus of investigation. In turn, knowledge that is learned about a company's customers can help provide an underlying framework for a KMS operating mode. An integral part of the preceding principles is the element of risk (covered briefly in one principle above).

Essentially, a company's risk-management strategy needs to be integrated into its overall corporate strategy. A small company, for example, can ill afford to have "all of its eggs in one basket," since a disaster in the one area could bankrupt the company. In contrast, a large company can take on risks in one or more areas of its business.

If this area(s) fails, it can be written off against other profitable areas of operations. However, a large company has to be cautious, since many times the level of risk is affected by its competitors' strategies. Overall, an effective strategic planning principle states that the level of risk needs to be balanced with the company's ability to withstand significant losses and survive in the long run.

Underlying all of the above strategic planning principles (based on knowledge of total business operations) is the fact that management and its staff have the authority and responsibility over a company's operations. Not only does management assisted by its staff have the *authority* over developing and implementing short- to long-range strategic plans, but it also has the *responsibility* to carry out these strategic plans.

Knowledge gained from within and outside the company provides management with the means to discharge its duties in an effective manner. The fact that authority and responsibility are linked to *accountability* is often overlooked. Accountability means that management is held accountable for operating results, financial or otherwise.

Thus, a most important strategic planning principle ties in authority and responsibility over operations with accountability for results. Needless to say, a knowledge management system can assist management in tying together these three items successfully as well as maintaining them over the long term. Strategic planning principles based on knowledge of total business operations.

IMPORTANT FACTORS

Today, the changing world, should be regarded as a *constant*. Change creates important opportunities but, if ignored or unanticipated, it can be devastating. The business

community showcases change and its consequences like no other. The \$30 billion-a-year package delivery business, for example, would be significantly smaller had Federal Express and UPS not anticipated a surge in worldwide demand for time-guaranteed delivery.

To assist corporate management in adapting to all types of changes, including changing computer technology, important factors that are useful in discovering strategic planning knowledge within a KMS environment are set forth below. These centre on the following:

- Problem finding in discovering strategic planning knowledge,
- Enlarged view of core competence to develop new opportunities,
- Importance of a company's critical success factors (CSFs) and their measurement,
- Benchmarking the company to detect the best business practices, and
- Software useful in discovering strategic planning knowledge.

Related to the foregoing factors is the need for a company's managers to translate the executive vision that revolves around a company's strategies and express it in terms that guide action at the local level. Needless to say, it is helpful that this executive vision be based on knowledge of a company's operations. To help get a grip on the knowledge base, companies today are employing a chief knowledge officer (CKO) to harvest and distribute professional expertise via information technology.

As indicated previously in the text, the chief knowledge officer, also called the chief learning officer or VP of information strategies, identifies what corporate knowledge needs to be developed and retained as well as what knowledge is useful in strategic planning. The focus is on seeing patterns and trends that enable decision makers to make the transition to insight and prediction. This focus is particularly useful in short- to long-range strategic planning.

Strategic Planning Principles Based on Knowledge of Total Business Operations

- Recognizing that change is a constant, fundamental changes to an organization can take from five to ten years to complete, while superficial improvements can be achieved in a much shorter time frame.
- Change is a most important factor affecting a company's mission, identifiable objectives, measurable goals, and corporate strategies. As such, there is a need for the effective utilization of a company's resources for changing times.
- A company can outperform its rivals only if it can establish a difference that is perceivable by its customers. A long-term company strategy is to market goods and services differently from its rivals.
- Technological breakthroughs should provide an important advantage not only for a company but also for its customers. This approach can propel a company into a position of market leadership.
- The need for strategic thinking is paramount in order for a company to identify and map its driving forces. The mapping process provides management with visual representations of possible solutions and end results.

- To map and manage a group of strategic development projects effectively, there must be a total project plan in place. An aggregate project plan not only gives direction and clarity to the overall development effort but also provides a framework for outstanding performance.
- While most corporate planners think in terms of good times, realistic projections should also include provision for bad times. The development of a company's strategic plans, then, should include examining the best- and worst-case scenarios.
- A company's short- to long-range strategic plans should challenge its management and employees to meet financial targets that go beyond what has been done in the past.
- Short- to long-range strategic planning should include cultivating a new style of corporate behaviour that encourages changing established views of a company's products and services so that innovation is the order of the day.
- There is no substitute for senior management and its staff spending several days in the lives of their key customers. This approach can bring insights and expertise to their customer problems.
- The level of a company's risk is related to its size and needs to be balanced with the company's ability to withstand significant losses and survive in the long run.
- Although authority and responsibility are widely discussed in describing a manager's limits, an often overlooked fact is their linkage to accountability for operating results under the manager's jurisdiction.

Problem Finding in Discovering Strategic Planning Knowledge

For effective problem finding, strategic planning must be undertaken from a proactive standpoint where external and internal environmental factors that affect the organization, from the short range to the long range, are taken into consideration. Many times, such information is found in the company's data warehouses.

The most important *external environmental factors* focus on customers, suppliers, the government, investors, the public, financial institutions, and competition. The most important *internal environmental factors* centre on organizational strengths and weaknesses, objectives and goals, functional areas, and personnel. Obtaining and using this information and knowledge is an integral part of the strategic planning process. Overall, the raw material of strategic planning is needed by top management and its corporate planning staff to initiate problem finding. Likewise, it is the means by which strategic decisions are made to further organization's objectives and goals.

To have an effective planning process, top-level managers and corporate planners need to identify potential problems in the future and bring them back to the present for resolution. Problem finding tends to centre on finding solutions to problems that many deem impossible to solve. Going one step further, there is the need to *identify*

opportunities that are related to future problems. As in the past, top-level managers and corporate planners must also be involved in problem solving. Thus, they need to go beyond problem solving and be actively involved in problem finding to allocate and use the organization's resources effectively.

To assist this highest-level group in problem finding, knowledge discovery tools along with mathematical and statistical models can be employed to develop appropriate analyses. Because the highest-level problems tend to be semistructured and unstructured, a series of "what if" questions can be asked and answered to get a feeling for future problems that are poorly defined. Hence, the needs of top-level decision makers, when they are oriented towards the near future, usually centre on using multidimensional analysis but, more importantly, on knowledge discovery as a means to answer pressing questions.

However, as the time frame is extended further into the future, the problems become even more unstructured and may require heuristics (*i.e.*, rules of thumb) based on past strategic planning principles (such as noted previously). Due to the vast number of uncertainties over time, the reliability of answers to these "what if" questions are, needless to say, less accurate. Nevertheless, top-level executive needs are being met as well as possible based on a magnitude of unknowns.

An important benefit from problem finding is that a manager can prevent a molehill from becoming a mountain if the manager is warned early enough to take corrective action. What is it worth to a manager to be warned of a business problem sometime sooner? What it is worth to the manager and to the company is the avoidance of a crisis. Often, it is too late to react to problems that are already out of control. This strategic advantage from using problem finding within a KMS environment, while difficult to quantify, is very significant and real for the typical company today.

Enlarged View of Core Competence to Develop New Opportunities

In order to create new markets for products and services, early and consistent investment in what is called "core competencies" is one important factor. In turn, corporate imagination and expeditionary marketing are the keys that unlock these new markets. A company that under invests in its core competencies, or inadvertently surrenders them through alliances and outsourcing, generally robs its own future. But to realise the potential that core competencies create, a company must also have the imagination to envision markets that do not yet exist and the ability to stake them out ahead of the competition. In a few words, a company must anticipate what the customers want before they are aware of what they want.

A company will strive to create new competitive space only if it possesses an opportunity perspective that goes far beyond the boundaries of its present businesses. This perspective identifies, in broad terms, the marketing territory top management hopes to stake out over the coming years, a terrain that is unlikely to be captured in anything as precise as a five-year marketing plan.

The initial enthusiasm that several Japanese companies brought to developing high-definition television (HDTV) grew out of just such a vision. Creative considerations of the many new opportunities that might emerge if HDTV could be made a reality led them beyond the traditional boundaries of colour television business to identify potential markets in cinema production, video photography, video magazines, electronic museums, product demonstrations, and training simulations, among others. As this example demonstrates, a company's opportunity perspective represents its collective imagination of the ways in which an important new benefit might be harnessed to create new competitive space or reshape existing space. Commitment to an opportunity perspective does not rest on ROI (return on investment) calculations but on an almost visceral sense of the benefits that customers will finally derive should pioneering work prove successful. The more fundamental the envisioned benefits and the more widely shared the enthusiasm for the opportunity, the greater the company's perseverance will be.

Although there is a need for core competencies to create new markets, there is also the need for an enlarged view of these competencies. Conceiving a company as a group of core competencies rather than as a group of products and services is one way to extend the opportunity perspective considerably. Because Motorola sees itself as a leader in wireless communications, it is not just a producer of paging devices and mobile telephones.

Instead, the company's charter permits it to explore markets as diverse as wireless local area computer networks and global positioning satellite receivers. Ajinomoto, a giant grocery products company, is not only in the food business but also applies the skills it has mastered in fermentation technology to produce an elastic paper for Sony's top-end headphones. The point to be made from these examples is that if company managers are not able to think outside current business boundaries, they will miss important new opportunities that depend on the combination of skills from several sources. Core competence with the proper application of knowledge should prosper and flourish over the years to the betterment of all parties concerned. Dominance in core products and services, then, allows companies to shape this evolution to keep one step ahead of competition.

Importance of a Company's Critical Success Factors (CSFs) and Their Measurement. For any business, critical success factors (CSFs) are the limited number of areas in which results, if they are satisfactory, will ensure successful competitive performance. They are the key areas where things must go right if the organization is to flourish. If results in these key areas are not adequate, the organization's efforts for the period will be less than desired. As a result, the critical success factors are areas of activity that should receive constant and careful attention from management. The current status of performance in each area should be continually measured, and that information should be made available to the higher levels of management.

Typically, critical success factors support the attainment of measurable organizational goals. These goals represent the end points that an organization hopes to reach. Critical

success factors are the few areas in which good performance is necessary to ensure attainment of those goals. For example, the automobile industry has four industry-based CSFs: having the right styling for the times, having a good-quality dealer system, having effective cost control over selling and manufacturing operations, and having the capability to meet current energy standards in terms of average mileage per gallon for all cars produced. Though the automobile manufacturers must pay attention to many other factors, these four areas represent the underpinnings of successful operations.

From a broad viewpoint, the principal sources of critical success factors have been identified by John F. Rockhart at MIT's Sloan School of Management. They are as follows:

- *Structure of a particular industry.* Each industry by its very nature has a set of CSFs that are determined by the characteristics of the industry itself.
- *Competitive strategy, industry position, and geographic location.* For smaller organizations in an industry that is dominated by one or two large companies, the actions of the major companies will often produce new problems for the small companies. The competitive strategy for the latter may mean establishing a new market niche, getting out of a product line completely, or redistribution of resources among various product lines.
- *Environmental factors.* As the economy changes as potential factors change, as energy problems become more acute, and the like, critical success factors can change for an organization.
- *Temporal factors.* Internal organization considerations often lead to temporal critical success factors. Inventory, for example, which is rarely a CSF for top management, might become a high-level CSF if there is far too much or too little stock.

Although these four sources for identifying CSFs are determinable, critical success factors are different for an individual industry and even for companies within that industry.

Going one step further, the concept of *key performance indicators* (KPIs) is a way of formalizing and describing critical success factors. Fundamentally, a comprehensive CSF system has four basic components which are shown in Figure 6-2. *First*, executive visioning that centres on broad-based knowledge of a company sets the company's overall direction and is linked to its corporate objectives and goals. *Second*, general company objectives are related to overall corporate goals, which are then broken down into appropriate measurable goals for divisions and business units. These goals provide the means for developing appropriate corporate strategies and programmes.

Third, each business unit identifies a number of critical success factors that must be performed well to achieve its goals and strategies as well as carry out its programmes. These activities are then assigned to the people responsible for their completion. *Fourth*, each business unit establishes a measurement system to quantify success. These measures are the key performance indicators, which normally include a number of financial ratios. Reference to KPIs and financial ratios.

For example, a company has decided that one of its corporate goals is to improve customer satisfaction by 10 percent this year. At the corporate level, a periodic survey of customers would be a key performance indicator, since corporate managers are responsible for the overall corporation effort. Measurable activities that contribute to improvements might be improved product quality, improved customer service and support, improved delivery times, and more customer-suggested product improvements.

Each of these activities suggests its own set of key performance indicators. The most important part of this process is that the KPIs are measures that the people responsible for them can actually control and for which they can be held accountable. And because of this combination of responsibility, control, and accountability, these KPIs, including financial ratios, are certain to be relevant and important to the managers assigned to them.

Benchmarking the Company to Detect the Best Business Practices

An important way to judge the performance of a company is to compare it not only with other units within the company but also with outsiders that represent the best industry practices. Commonly, this technique is called *benchmarking*. When the Xerox Corporation started using benchmarking not too long ago, management's aim was to analyse unit production costs in manufacturing operations. Uncomfortably aware of the extremely low prices of Japanese plain-paper copiers, the manufacturing staff at Xerox wanted to determine whether their Japanese counterparts' relative costs were as low as their relative prices. The staff compared the operating capabilities and features of the Japanese machines, including those made by Fuji-Xerox and tore down their mechanical components for examination.

As somewhat expected, the investigation revealed that production costs in the United States were much higher. Discarding their standard budgeting processes, U.S. manufacturing operations thereby adopted the lower Japanese costs as targets for driving their own business plans. Top management, gratified with the results, directed that all units and cost centres in the corporation use benchmarking.

In contrast, distribution, administration, service, and other support functions of Xerox found it difficult to arrive at a convenient analogue to a product. These non-manufacturing units began to make internal comparisons, including worker productivity at different regional distribution centres and per pound transportation costs between regions. Next, they looked at competitors' processes. In logistics that meant comparing the transportation, warehousing, and inventory management of Xerox's distribution function with those of the competition.

The initial step in the process is to identify what will be benchmarked (*i.e.*, measured): expense-to-revenue ratios, inventory turns, service calls, customer satisfaction—whatever the “product” of the particular function is. Then it is necessary to pinpoint the areas that need improvement. In Xerox's experience, managers tend to concentrate first on comparative costs. But as they become more knowledgeable about benchmarking,

managers discover that understanding practices, processes, and methods is more important because these define the changes necessary to reach the benchmark costs. Moreover, as managers become more confident about benchmarking, they can readily extend it beyond cost reduction to profit-producing factors, like service levels and customer satisfaction.

Annual reports and other easily available publications are important sources for the purpose of comparison. They can uncover gross indicators of efficient operation. Universally recognized measures like return on assets, revenue per employee, inventory turns, and general and administrative expenses will help identify well-managed companies. To identify superior performance in specific functions, a company can utilize trade journals, consultants, annual reports, other company publications in which “statements of pride” appear, and presentations at professional and other forums. The same well-run organizations keep turning up.

Software Useful in Discovering Strategic Planning Knowledge

To assist small businesses in enhancing their planning efforts, software packages provide an effective starting point and can lead the planner through the process of generating an effective business plan. The software planning packages are not only easy to install in standard hardware but also easy to use. They also provide some capabilities to carry out “what if” analysis. In addition, they offer the capability to utilize historical and current knowledge in the preparation of a business plan.

While there are variations in business planning software, typically a formal business plan consists of several common elements: executive summary, company overview, products/services, market analysis, marketing plan, and financial plan. A snapshot of the whole plan is represented by the *executive summary*, which highlights the company’s mission, measurable objectives and goals, product and marketing strategies, and the financial plan. When preparing the executive summary, keep it short and simple.

From one to several pages is ideal. Following the executive summary is the rest of the business plan. The *company overview* sets the stage for the venture and provides details of its history and ownership, operating structure, management team, and business purpose. *Products/services* describes what the company sells and how it develops, produces, and distributes products and services. This section also offers an analysis of competition and future plans. This is followed by the *market analysis*, which describes current and potential customers and the demographics of the marketplace.

In addition, the *marketing plan* reveals the company’s strategy for reaching its markets through advertising, alliances, distribution channels, promotions, and a sales force. Finally, the *financial plan* sets the words to numbers and shows the recent operating history. It projects the next three to five years’ worth of cash flow, profits, and asset and liability balances. It also typically contains a break-even analysis, financial ratios, capital requirements, and the timing and amount of the return on investment.

Typical business planning software packages include the following: Avantos ManagePro helps managers plan and track goals and progress while fostering success through focused management activities. It includes goal planning and tracking tools, such as the Top Level Goal Planner, for planning and delegating key business objectives, strategies, and tactics. The Goal and People Status Boards enable managers to monitor the status of primary business goals and obtain at-a-glance reinforcement of where to focus attention. Jian BizPlan Builder is a complete business plan template on disk with more than 90 typed pages of example text that are formulated into word processing files.

Templates use standard spreadsheet applications like Excel or Lotus 1-2-3 to calculate financials and generate graphs. The step-by-step format guides the user through, explains issues, and gives clear and sensible advice.

Once a company's business plans have been developed (from the short-range to the long-range), software can be employed that is useful in determining the company's overall financial performance. Such software goes beyond the monthly actual reports and budget reports that are routinely produced by today's information systems. Currently, companies are employing corporate scorecard software, which is a sophisticated business model that helps a company understand what is really driving its success. In effect, it acts like the control panel on an airplane—that is, the business equivalent of a flight speedometer, odometer, and temperature gauge all rolled into one. A corporate scorecard keeps track of a company's financial progress as well as its softer measurements.

Ranging from customer satisfaction to return on investment, these areas need to be managed to reach a company's final destination (*i.e.*, profitable growth). A scorecard, for example, might graph customer service to determine if it is improving or deteriorating and, at the same time, tally product defects to determine if they are rising or falling—and where. Going a step further, scorecard software, which is usually distributed throughout a company's computer network, lets company employees across the entire organization be certain they are talking about like items when they get together. If customer satisfaction is declining, sales, manufacturing, and research and development will all be reading the same score and thus will be able to tackle the problem from a common perspective.

Today, there are two views on the utilization of scorecard software. One is that a company's yardsticks should be purely financial. Managers should employ indicators like revenue growth and return on investment to guide a business. The other is that a corporate scorecard should be balanced. A company, for instance, should not only keep close watch on performance numbers like gross profit margins on new products but should also use softer measurements. Other softer measures include the number of new products, product development cycle times, and the rate of on-time deliveries. The bottom line from this view is that a company is forced to evaluate critically those drivers that really determine its performance over the short run to the long run. Although business planning software and scorecard software is useful in a typical information system

mode, no technique or software can replace experience and knowledge about a business or its markets, which is necessary for effective strategic planning. In contrast, knowledge discovery and data mining can make business knowledge and experience more valuable than ever. They provide a means for conducting complex data analysis to discover patterns, relationships, and associations within a set of data and information. Such patterns and the rules inferred from them can be used to guide decision making and even predict the outcome of those decisions by corporate planners.

An example of knowledge discovery software that can be of assistance to management and its corporate planning staff is KnowledgeX. Essentially, this software can discern hidden relationships among people, organizations, and positions. It is able to take information, organize it, and extract from that material the connections and webs of relationships. Using Intelligent Knowledge Exchange, KnowledgeX can perform a text processing task called “SmartParse” to select and extract textual information. If it encounters terms it already knows, it will automatically assign the correct set of connections to that term. If it does not know the term, the user can define it once and all future uses of that term will be connected according to rules. KnowledgeX is object oriented. It allows multimedia information to be accessible from within the programme.

Once the relationships are established, KnowledgeX has the ability to display these connections in a graphical way or as a text outline. The graphical display is easy to grasp and often reveals relationships that have been buried within the text and not obvious to the researcher.

By creating maps showing relationships among customers, companies, individuals, and organizations, KnowledgeX makes it easy to grasp how these people and organizations are interconnected. The more information the knowledge base contains, the better KnowledgeX is at finding unknown and unnoticed relationships.

In addition, KnowledgeX can combine information entered by many people on the network and find the connections within that information. Fragments coming from other people can therefore be converted into something like a centralized knowledge source. That is, KnowledgeX takes information and knowledge coming from many sources and reveals relationships that had never been known before because the pieces were too fragmented and dispersed among too many people. From this perspective, this knowledge discovery software is quite useful in bringing together diverse elements needed for resolving critical strategic planning issues.

KNOWLEDGE MANAGEMENT SYSTEMS IN STRATEGIC PLANNING

In order to understand knowledge management systems better in corporate strategic planning, marketing, manufacturing, finance, and human resources and the following ones, reference can be made to either a manufacturing-oriented company or a service-oriented company. In either company, the corporate strategic planning system is linked

directly with other systems. That is, planning activities are related to external environmental factors through a company's mission to its organization's objectives and goals. In turn, these important relationships are related to a company's internal environmental factors (*i.e.*, marketing, manufacturing, finance, and human resources). All of these external and internal environmental factors form the basis for short- to long-range strategic planning within a KMS operating mode.

The capability of the knowledge management system to extract online knowledge, facts, information, and the like now makes this system a forward-looking one for answering questions about the future. From this viewpoint, an effective strategic planning system makes great use not only of multidimensional analysis but also of whatever software is useful for answering tomorrow's questions in view of today's projections.

AREAS THAT LEND THEMSELVES TO DISCOVERING STRATEGIC PLANNING KNOWLEDGE

In view of this current approach using a KMS operating mode, an effective strategic planning operating mode needs to translate visioning down to a practical level. That is, *executive visioning* can be used to help managers build a consensus around its strategies and programmes so that it guides action at the lower or local levels. This necessitates letting managers *communicate* these strategies and programmes up and down the organization and, at the same time, linking them to each business unit and individual objectives and goals. This planning model enables the company to integrate its *business and financial plans* at the lower levels.

This model also allows for *feedback*, since it gives the company the capability of studying alternatives for strategic improvement. Strategic improvement consists of continuous feedback—that is, testing the environment on which the strategies are built and making the necessary adjustments where deemed necessary.

Since short- to long-range strategic planning in a typical company makes use of this broad-based approach, this will be the essential focus of the areas that lend themselves to discovering strategic planning knowledge. Typically, a company's corporate planning staff, which undertakes strategic planning activities, is responsible directly to the president or chief executive officer (CEO), has relatively few members, exists at the corporate level, and operates on a full-time basis. This corporate planning staff operates in an environment characterized by product and technological change.

Also, the character of a company's departmentalization seems to influence the organization structure for strategic planning. An important part of the corporate planning staff's time is spent determining and analyzing a company's critical success factors as well as related key performance indicators, including financial ratios. For example, say that a staff has determined CSFs to be:

- Prices (responsiveness to competitive pricing),
- Cost control (reducing the cost of plant and office operations),
- Inventory turnover (improving the times the inventory turns over yearly),
- Product mix (having the right products for the times). These CSFs are in line with those set forth earlier for other industries and can be related to key

performance indicators and financial ratios. Knowledge about a company CSFs, which is obtained from its current operations via its KPIs and financial ratios, may indicate that there are subtle changes occurring currently. The end result is that there may well be a shift in what constitutes an effective way to evaluate a company's performance.

LONG-RANGE STRATEGIC PLANNING

Within a KMS environment, corporate planners view decision making as a *continuous decision process*. It centres on setting appropriate corporate objectives and measurable goals, employing strategies and programmes to achieve these desired objectives and goals, allocating a company's resources in an optimum manner, specifying the critical success factors, and determining the key performance indicators and financial ratios to measure the CSFs.

All of these planning and related activities are influenced by visioning that is based on a comprehensive knowledge of a company's operations. The implemented plans are then monitored to keep them on course and to correct any deviations from the original or revised plans. Revisions are usually necessary since change is integral to the nature of strategic planning and its decision process.

Since there is a need to evaluate planning alternatives continuously for achieving stated objectives and measurable goals, the capability for studying various aspects of the long-range plans in its various stages becomes critical. In practical terms, this means producing many runs or variations of the same plan, using different sets of assumptions and perhaps different logic. In *evaluating resources*, for example, varying combinations of resources can be applied, and numerous questions concerning their nature and size can be posed. "Do we have the necessary resources to achieve our objectives, or must they be acquired? If resources are to be acquired, at what cost will they be acquired and in what time span? Will they be operational in time to achieve the corporation's objectives within a given schedule?"

What are the optimum mixes of resources to satisfy different objectives under different sets of conditions? What are the likely effects of miscalculating availability and cost of equipment, personnel, or funds to acquire the needed resources? What can happen for every one percent of error in the forecast and, therefore, how critical is it that we hold strictly to the plan?" Similarly, in *evaluating strategies*, for example, various questions can be asked concerning improved revenues, lower costs, higher reliability, meeting customer requirements, relationship to other products, enhanced image in customer eyes, better information, and relationship to organizational plans and derivative programmes. The anticipation of consequences of given assumptions and the evaluation of alternatives is a major activity of a company's managers at the highest level and of the corporate planning staff. The utilization of appropriate information and its resulting knowledge is helpful because it provides the means whereby the final strategic plan mirrors the real world as closely as possible.

Knowledge discovery can help corporate planners evaluate alternatives and (usually) choose the best one. Such evaluation has come to be known as the “what” is likely to happen “if” the company takes or does not take a certain course of action. As will be seen in the materials to follow, a KMS operating mode can prove to be extremely useful for a company’s corporate planners.

Integrate Changing Times with Long-Range Strategies and Resources

An integral part of the entire strategic planning process is the integration of changing times with long-range strategies and resources. A Labour Department study, for example, forecasts a slowdown in future economic growth. It estimates that the country’s gross domestic product, adjusted for price change, will grow at annual rates of just over 3 percent. That would be about 25 percent below the average growth rate of about 4 percent that has prevailed during much of the time since World War II. In addition to the external long-range factors, a typical company’s executives and their corporate planning staff should consider the internal factors that affect the long run.

Essentially, they include the factors mentioned above (marketing, manufacturing, financial, and human resources). The interplay of these internal environmental factors and the external environmental factors can have a dramatic impact on a company’s strategic plans. In fact, incorrect information concerning the internal and external environmental factors may lead a typical company down the wrong planning path.

To get a better idea of how to adapt to changing times as reflected in a company’s long-range strategies, reference can be made to Figure 6-3 for three typical situations. First, a company can invest available resources in order for the business to *grow* and thereby strengthen its profitable position. Second, it can *protect* business strengths in order to maintain a strong position in a moderately attractive and mature market. Third, it can *divest* the business in order to exit from a weak position in a relatively unattractive market.

Similarly, measurement and analysis using KPIs and financial ratios also differ based upon the situation. In all three situations, the discovery of new knowledge can be useful to management in getting a complete understanding of a grow, protect, or divest long-range strategy.

Once a company defines its current state and its long-range strategy, it can and should match its people to that strategy. These three typical strategies require distinct sets of managerial characteristics and management styles. Broadly speaking, an entrepreneur is needed for the *grow* situation, a competent manager for the *protect* situation, and a critical manager for the *divest* situation. Not only do these three types of organizations differ in the optimum management style of their executives, but also in the way they are organized and the climate that exists within each organization as well as the current economic climate.

For example, in the *grow* situation, corporate planning for problem solving and problem finding is longer term in its outlook and there is a great deal of delegation of authority. There is considerably less freedom to act independently in the protect organization and planning is more medium-term in nature. The successful manager in either situation is one who thrives in the company's culture and feels most comfortable within its organizational structure.

Table. Relationship of Company Long-Range Strategies to Allocation of Company Resources,

Company LongRange Strategies	Allocation of Company Resources	Measurement and Analysis Using Key Performance Indicators and Financial Ratios
Grow	Improve market share Develop new products Increase manufacturing capacity Expand sales force	Market share Sales volume growth Capitalized ROI Relative quality
Protect	Maintain market share Optimize margins Differentiate products Increase efficiency	Marketing expenses/sales R&D expenses/sales Investment/sales Capacity utilization
Divest	Forego market share for profits Hold prices and margins Prune product lines Shrink sales force	ROI Net cash flow/investment Pretax profit/sales Sales/assets

Using Knowledge to Improve Long-Range Strategies

Related to a company's long-range strategies are a number of underlying concepts. A popular concept that embodies many of the strategic planning principles set forth previously is this: The essence of strategic unity between business and technology is manifested in changing roles and structures, high expectations, and a collaborative spirit. From this view, long-range strategies are broad visions of the future that are adaptable to a changing environment. They encourage corporate planners to think about the future in creative and productive ways.

Such planners see their job as getting others to question conventional wisdom. To perform their jobs, corporate planners have to resort to provocation or shock tactics like raising difficult questions and challenging conventional assumptions. A number of

the strategic planning principles set forth previously may be of help in changing the status quo. The knowledge contained in these principles may well provide the impetus for improving a company's long-range strategies. Overall, development of long-range strategies is the process of discovering and understanding broad visions of the future as opposed to the strategic planning process, which is about turning these visions into action.

A very important long-range strategy that is widely used today to improve customer service and lower operating costs is the Internet and its corporate equivalent intranets. All elements of a company's supply chain, for example, can be linked by electronic networks. If a person buys a pair of Nike running shoes, that information can be transmitted back to the plant in Taiwan where the shoes are made and the shoes can be express mailed directly to the person's home.

Similarly, items purchased electronically from several suppliers in the same region could be shipped in a batch using electronic interfaces. A whole host of intermediaries in the chain who used to make their living coordinating and moving information can be eliminated.

Intranet-to-intranet communication is blurring the lines between companies. The notion of where a corporation starts and stops is very different today. As another example, a company's expertise might be harvesting timber or processing lumber, but the company also needs to move its products to the construction industry. Traditionally, all those steps were brought together in a soup-to-nuts operation. Now, given interconnectivity, someone else can run a company's truck fleet, but it will still operate like one's own fleet.

For some companies, the growth of network computing offers new business opportunities. Quick Response Services Inc. (based in Richmond, California) provides electronic links between manufacturers and retailers, thereby becoming an information switch for several large apparel manufacturers, including Liz Claiborne and Perry Ellis. By collecting information from the retailers about which products are selling, the company then works with manufacturers to restock their inventories.

The Microsoft Corporation has handed off troubleshooting responsibilities for its in-house computers to an outside vendor that uses the software maker's intranet to get at faulty computers. Another example of network computing is the Chicago-based Navistar Inc., which no longer maintains a tire and rim inventory at its Springfield, Ohio, truck-assembly plant. That responsibility is now handled electronically by Goodyear Tire & Rubber Company, one of Navistar's tire suppliers.

Basically, a Goodyear office in New York receives Navistar's manufacturing schedule and tire-and-rim requirements by electronic data interchange (a proprietary electronic network). The information is then sent to a Goodyear plant in Ohio where tires are mounted on rims. The completed assemblies are shipped to the Springfield plant, arriving just eight hours ahead of when they are needed. The tire-and-rim assemblies come in the same order as the trucks that are coming down the assembly line.

To assist its clients in using knowledge to improve their operations and assist in developing appropriate long-range strategies, Andersen Consulting has developed Knowledge Xchange. Essentially, Knowledge Xchange creates *virtual communities of practice*:

Lotus Notes groupware and Microsoft Office Suite run on a Novell NetWare-based LAN. The LAN is connected to a proprietary client server-based WAN. Because the value of this consulting firm is in its knowledge capital, Knowledge Xchange makes all of the firm's accumulated knowledge and skills immediately available to each of its consultants, and, therefore, to every client.

It is a virtual place where all of the firm's 27,000 consultants can share their knowledge and expertise in global communities of practice. Knowledge Xchange is available to each consultant on desktop and laptop computers and affects the work done on a daily basis.

In fact, it lets a consultant find out who has skills out there that he or she does not have so that the consultant can quickly put together an online team to help solve a client's problem. This way, the proper skills can be brought to bear on developing a company's strategies and meeting its strategic planning needs. The knowledge obtained from Knowledge Xchange can make an important difference between success or failure.

Long-Range (Five-Year) Strategic Planning

Typically, a company's strategies that need to be translated into action take the form of long-range (*i.e.*, five-year) strategic plans. To assist in the development of these long-range plans, corporate planners need to begin with a knowledge and understanding of existing products, divisions, margins, profits, return on investment, cash flow, availability of capital, research and development capabilities, skills and capacities of personnel, etc.

For a manufacturing-oriented company, this understanding needs to be extended to an in-depth analysis of manufacturing operations that is tied in with centralized operations at the corporate level. There is a need to examine performance over the past few years as well as the current year as part of an initial overall review process.

Evaluating significant aspects of past and current operations is the basis for determining how well the organization objectives and goals are being met. In like manner, plans for the coming five years, based on short- to medium-range plans for improving operations, become an essential part of getting started on long-range strategic planning.

A typical five-year plan for a manufacturing-oriented company includes the *external environmental factors* shown in Figure 6-4. These are generally not controllable, such as customers, government, public, competitors, suppliers, investors, and financial institutions. On the other hand, four *internal environmental factors* that are controllable by the company centre on the following:

- *Marketing planning*: Focuses on expanding current product lines and entering new product markets. This involves increasing use of selling outlets and/or distribution to sell the company's products, changes in pricing policy and

pricing practices to effect higher sales, and consideration of new advertising media for more effective penetration of the company's markets.

- *Manufacturing planning*: Centres on major facilities contemplated and improvements in processing efficiency, including the percent of capacity that is now and will be employed with current facilities and machinery as well as the steps that are being taken to use any excess capacity.
- *Financial planning*: Relates to projected sales by product lines, contribution (sales less direct manufacturing costs) by product lines, indirect manufacturing costs plus sales and general and administrative expenses, net profits before federal income taxes by product lines, fixed and working capital needs, return on investment by product lines, and comparable financial ratios and analyses.
- *Human resources planning*: Centres on projected requirements for key management personnel and production labour when considering turnover and future growth.

To develop projected strategic plans, the corporate planning staff employs the company's corporate databases and data warehouses to analyse meaningful longrange information as well as to discover new knowledge about a company's operations over time. In turn, this output is used to finalize the company's five-year strategies and programmes.

Problem-Centred Approach to Long-Range Strategic Planning

The knowledge obtained to improve a company's long-range strategies can be integrated with problem finding. Because company executives and their corporate planning staffs must combine knowledge of today's marketplace, including economic and political analysis, with a vision of the future, strategic planning does not involve forecasting of the future in the conventional sense, but rather a projection of a variety of possible scenarios under good, average, and poor economic conditions.

Against these scenarios, corporate planners analyse the strengths and weaknesses of the company and evaluate its resources under these conditions. This includes a rigorous study of the finances. Market forecasts and financial Modelling can provide the foundation for a company's strategic plans. To illustrate the problem-finding process, the problem-Centred approach is applied to strategic marketing plans in the discussion to follow.

In this illustration, the corporation is engaged in selling products to the home market. Its president recognizes that its future markets may well change drastically over time. In fact, the individual is convinced that about half of the present products will not be sold five years hence. In view of this potential major upheaval in the marketplace, he has enlisted the services of a national consulting firm.

The consultants who were assigned to this marketing project formed a group of high-level management personnel, consisting of the executive vice president, the

marketing vice president, four regional sales managers, and themselves. As a starting point, the group met for two days at corporate headquarters and initially Centred on *problem generation* (first step) regarding present and future products using decision conferencing. Under the direction of a senior consultant, the group examined marketing problems by using a brainstorming approach.

The major discussion focused not on the products being marketed under good, average, and poor economic conditions, but rather on the changing nature of customer buying habits. Due to the continuing trend towards smaller living quarters and smaller families caused by inflationary times and other factors, the corporation is faced with developing new products to reflect the changes that affect the home market. The group also brainstormed the trend towards more people living alone.

Based on the information displayed on a public screen, *evaluation* (second step) consisted of reviewing the marketing areas of the future and their related problems. As it turned out, all of the areas displayed on the public screen were determined to be valid for managerial concern in the *validation* stage (third step). The trend towards smaller living quarters represents a major problem for the corporation.

In contrast, its current products are oriented towards medium- to large-sized homes where there is ample space to accommodate them. In light of this major problem and its related problems in the future, the group spent some time *establishing boundaries* (fourth step). Although the home market in foreign countries was considered to be a viable one, it was decided to put first things first by channelling the corporation's resources into the development of new products for the domestic market.

Although a number of problems needed to be solved, the major one was the first one solved—that is, developing over the next five years a wide range of new products that would be more compact and, at the same time, capable of performing a wide range of services to meet homeowner needs. The *solution* took the form of having the group employ knowledge discovery software within a KMS environment.

In turn, this collected knowledge was useful in passing judgement on the profitability of the proposed products and comparing one to another for implementation. The profitability of the products (*i.e.*, net profit on sales and return on investment) were determined in terms of Modelling the anticipated revenues and costs over their life. Overall, the employment of the problem-Centred approach resulted in identifying future product problems. In turn, these problems were brought back to the present time for solution. As a result, management by perception was put into practice by this high-level executive task force for determining future strategic marketing plans.

SHORT-RANGE AND MEDIUM-RANGE STRATEGIC PLANNING

Short-range strategic planning, sometimes called operational planning, is a derivative of medium-range strategic planning which, in turn, comes from long-range strategic

planning. For a typical company, short-range strategic planning is concerned with the efficient use of available capacities. It is a detailed financial plan that specifies both how the company's objectives and goals for the coming year will be attained and the operational procedures for managing daily operations. As such, the short-range plans outline the specific steps for accomplishing the medium-range plans. Also, they play a major role in implementing business strategies by translating long-range plans into action.

Additionally, short-range plans centre on detailed objectives and specific measurable goals and strategies of the company and the means for achieving them—usually in the form of flexible budgets or profit plans by product groups. Management needs cost and revenue margin information so that it can identify areas of strength and weakness.

Knowledge about a company's product margin (contribution information) and its competitors over time is also needed to measure the profit impact of alternative courses of action. Hence, approved short-range strategic plans become budgets so that actual results can be measured and compared with them monthly for more effective control.

Using Knowledge to Prepare Short-Range and Medium-Range Profit Plans

For a typical company's products, an annual profit plan is an integral part of corporation-wide strategic planning. In a similar manner, overall profit plans are determined for two, three, and four years hence. As information about changing times becomes available, profit plans for the coming year must be revised to reflect the changes and expected changes in the business environment.

Effective profit planning, therefore, must be a continuous effort rather than a periodic one, within a KMS operating mode. Although budgets for profit planning are generally prepared by the accounting department, the responsibility rests with the corporate planning staff, who must not only select the appropriate financial information for specific planning decisions but must also combine this information from corporate databases and data warehouses in useful and meaningful ways.

Many times, there is a need to employ knowledge discovery techniques to extract financial trends and patterns. Staff members must also review and coordinate the estimates provided by the functional managers involved in a particular decision. They must measure the profit impact of alternative courses of action and offer advice on the meaning and significance of financial analysis. In summary, long-range strategic plans within a KMS environment are translated into medium-range plans for the next several years and finally into short-range strategic plans, or annual profit plans (including budgets) for the coming year.

Flexible budgets or detailed profit plans for the coming year are developed to take into account planning for marketing, manufacturing, human resources, and finance that have an impact on the current year. Similarly, overall profit plans for medium-range strategic plans can be developed. To assist the corporate planning staff, appropriate

software, such as an OLAP software package, can be used to answer a series of “what if” questions about planning and to perform sensitivity analysis to determine the impact the change of one or more variables might have on the final profit plans. Managers and their staffs interact with the company’s information to answer “what if” questions and to undertake sensitivity analysis.

In this case, there is a six-month analysis of six new products in terms of profitability. Profits increase over the months due to a number of factors, including a reduction in production times, larger sales volumes due to the acceptance of the product, and a reduction in scrappage as the learning curve improves.

A variety of answers, then, can be obtained to assist in the development of profit plans—the corporate strategic plans for current and future operations. The development of a company’s profit plans can benefit from the use of knowledge management techniques, particularly knowledge discovery (data mining) software.

Going beyond the preparation of short- and medium-range profit plans, it is helpful to measure actual performance against these profit plans. In terms of the short range of the current time period, can help management at this point. Besides getting an overview of how a company is doing, specific areas of a company can be measured and evaluated for performance. For example, the scorecard software can check on defect rates plant by plant and see how each plant’s quality is improving. It can link measurements like on-time deliveries to certain financial indicators.

The software can measure the percentage of sales due to new product introductions, gross margins on new products, and corporate-wide indicators like revenues and return on investment. It should be noted that different companies at different times have different needs and aims.

For example, a service company that has just merged would not build a strictly financial model that focuses, say, on productivity. Output per employee is not a driver for that business. The real value of scorecard software is that it forces a company to reexamine its assumptions about what really drives performance. It forces a company to focus and become much more explicit about what matters to its customers and, ultimately, what matters to its stockholders.

Using Information and Knowledge to Finance an Expansion

To assist in short-range strategic planning, information and knowledge can be helpful in the review of quarterly financial forecasts to finance an expansion. For example, top-level managers can apply for a loan to finance an expansion although they lack the historical financial data necessary for a statistical analysis of earnings. The solution is to construct a four-quarter earnings forecast combining present market conditions with their best knowledge about the future. The gathering process is started by polling the managers for number estimates and market assumptions.

For this type of Modelling process, there is almost complete consensus among all managers on all assumptions other than unit sales and selling price. The importance of

unit sales and selling price growth, coupled with the range of estimates received, makes the use of multiple projections prudent. Consequently, the model provides the best-case and worst-case scenario forecasts for review by top-level managers and their staffs. Basically, the model incorporates assumptions for several variables, including growth factors for unit sales and selling prices for the next four quarters. Also, current sales along with the actual cost of goods sold, sales commissions, administrative expenses, and federal income taxes are needed for the current quarters. Using the estimates based on assumptions, projected quarterly income over four periods can be displayed for high-growth and low-growth projections in unit sales and selling prices along with income, representing the best-case and worst-case forecasts.

Additionally, sensitivity analysis allows the company's managers to examine the impact of a change in one variable. A more rigorous multidimensional analysis can be performed by varying a number of variables to see what impact each change has on net profit on sales. In this way, managers can evaluate how sensitive the model is to changes in given variables.

Another forecasting approach is to utilize one of the popular software packages available today. For example, Forecast Pro for Windows allows users to let the package—via its expert system—determine the best forecasting model based on the software's observation of the data. The software package recommends a forecasting model and will even explain why a particular model was selected. This feature provides an accurate model as well as teaches the user what to look for when analyzing a data series.

Comparison of Company Against Competitors and Noncompetitors

Going beyond assistance in the preparation of final flexible budgets and review of quarterly financial forecasts, a KMS approach is also helpful in measuring a company's operations against competitors as well as noncompetitors. To enable corporation executives to accomplish such analysis, there may be a need for information and knowledge from external databases. That is, specific information and knowledge for such analysis must be stored online for later use by executives. When executives develop plans that centre on improving productivity for manufacturing operations, adequate information and knowledge are available to set the wheels in motion for developing short- to long-range strategic plans for beating or, at least, meeting competition.

Similarly, external financial information and knowledge about competition can be stored online so that executives can measure their own corporation's performance against the competition. In this manner, executives can see where the corporation is experiencing good, average, or poor performance in its major areas of operation against the best of its competitors.

It should be noted that benchmarking against the competition poses problems. For one thing, comparisons with competitors may uncover practices that are unworthy of emulation. For another, while competitive benchmarking may help a corporation meet its competitors' performance, it is unlikely to reveal practices for beating them. Moreover,

getting information about competitors is obviously difficult. Finally, it has been observed that people are more receptive to new ideas that come from outside their own industry. A noncompetitive investigation can give executives information about the best functional practices in any industry. These may include technological advances unrecognized in their own industry (like bar coding, which originated in the grocery industry but has since been widely applied). Adoption of these practices can help a corporation achieve a competitive advantage.

To place the preceding discussion in its proper perspective, one of the Xerox Corporation's most valuable and memorable benchmarking experiences was with L. L. Bean, Inc., the outdoor sporting goods retailer and mail order house. It was carried out by Xerox Logistics and Distribution (L&D), which is responsible for inventory management, warehousing, and transport of machines, parts, and supplies. Historically, productivity increases of Xerox's L&D had been 3 to 5 percent per year. Recently, it became clear that improvement was necessary to maintain profit margins in the face of industry price cuts. The inventory control area had recently installed a new planning system, and the transportation function was capitalizing on opportunities presented by deregulation. Warehousing was next in line for improvement, and the distribution centre managers wanted a change. They identified the picking area as the worst bottleneck in the receiving-through-shipping sequence.

A new technology, automated storage and retrieval systems (ASRSs) for materials handling, had appeared on the scene and was the subject of hot debate in Xerox's distribution function. The corporation had just erected a high-rise ASRS warehouse for raw materials and assembly parts in Webster, New York, in the same complex as a large finished goods distribution centre. Internal benchmarking evaluations by L&D showed that heavy investment in capital equipment for ASRS could not be cost justified for finished goods. They needed a different way to boost warehousing and materials handling productivity. But what?

The L&D unit assigned a staff member half-time to come up with a suitable noncompetitor to benchmark in the warehousing and materials handling areas. The staff member combed trade journals and conferred with professional associations and consultants to find the companies with the best reputations in the distribution business. He then targeted those companies with generic product characteristics and service levels similar to Xerox reprographic parts and supplies. The staff member had singled out L. L. Bean as the best candidate for benchmarking in terms of their warehouse operations.

The staff member was particularly struck with the L. L. Bean warehouse system design. Although extremely manual in nature, the design minimized the labour content, among other benefits. The operation also did not lend itself to automation of handling and picking. The design therefore relied on very basic handling techniques, but it was carefully thought out and implemented. In addition, the design was selected with the full participation of the hourly workforce. It was the first warehouse operations designed by quality circles. To the layperson, L. L. Bean products may bear no resemblance to Xerox parts and supplies. To the distribution professional, however, the analysis was

striking; both companies had to develop warehousing and distribution systems to handle products diverse in size, shape, and weight. This diversity precluded the use of ASRS. Later, a Xerox team visited Bean's operations in Freeport, Maine. Besides the person in charge of benchmarking in L&D, the team consisted of a headquarters operations executive and a field distribution centre manager.

These two people represented the line employees who would ultimately make any changes. The findings resulted in L&D incorporating some of L. L. Bean's practices in a programme to modernize Xerox's warehouses. These practices included materials location arranged by velocity, to speed the flow of materials and minimize picker travel distance as well as enhanced computer involvement in the picking operation. Xerox plans to put together a totally computer-managed warehouse.

GLOBAL BEST PRACTICES

Today, an effective KMS operating mode extends the flow of knowledge beyond national boundaries. Sales executives, for example, need to share vital statistics about customers, products, services, competitors, and selling techniques around the world. In order to do this, it is necessary to develop a KMS that encompasses the entire business world. Such an approach was taken by Arthur Andersen. Going beyond Knowledge Xchange—an in-house version of the kind of consulting help it sells its clients—Arthur Andersen in the early 1990s began creating the Global Best Practices (GBP) knowledge base.

Basically, the GBP is a central repository of knowledge about world-class business practices. The goal was to capture qualitative and quantitative information and knowledge about how companies achieve best-in-the world standards of performance in activities that are common to most companies regardless of their industry. This would then be used by Arthur Andersen's professionals to provide clients with an ever-growing body of knowledge that could serve as a reference point for performance improvement.

The contents of the knowledge base include the following: best practices information, best company policies, top ten studies and articles, world-class performance measures, diagnostic tools, customizable presentations, process definitions and directory of internal experts, best control practices, tax implications, and relevant Arthur Andersen engagement experience.

Arthur Andersen gathers knowledge in different ways from a number of sources. The Global Best Practices hot line receives more than 250 calls each month. Calls are tracked on a Lotus Notes database and sorted into business process categories. This knowledge, combined with ongoing research into emerging areas, is used to determine which process areas will be developed or enhanced for an upcoming release. Research analysts team with content experts who serve clients to develop best practices content and diagnostic tools, about 250 pages per process.

Quantitative and qualitative information and tools are released on a CD that can be accessed by all Arthur Andersen professionals worldwide and used in the field with clients. Best practices information is also generated through suggestions received by

the hot line or captured through formal and informal surveys. The impetus for building GBP came from the outside. Arthur Andersen needed to leapfrog competitors in the then-emerging area of best practices, so it set a three-month timetable to start advertising a resource tool to give clients access to best-practice insights. Although the tool did not exist at that time, the firm began to create, develop, and release the GBP knowledge base.

The four-year initiative has taught the project's development team some unexpected lessons about implementing changes that fundamentally shift a company's culture and the way it does business. Many of these lessons contradict generally accepted ideas about what constitutes good planning and execution.

Elements of Strategic Management

MISSION

Mission is the description of an organization's reasons for existence, its fundamental purpose. It is the guiding principle that drives the processes of goal and action plan formulation, "a pervasive, although general, expression of the philosophical objectives of the enterprise."

Mission should focus on "long-range economic potentials, attitudes towards customers, product and service quality, employee relations, and attitudes towards owners."

It provides identity, continuity of purpose, and overall definition, and should convey the following categories of information.

- Precisely why the organization exists, its purpose, in terms of
 - Its basic product or service,
 - Its primary markets,
 - Its major production technology.
- The moral and ethical principles that will shape the philosophy and character of the organization.
- The ethical climate within the organization.

Thus mission outlines the firm's identity and provides a guide for shaping strategies at all organizational levels. The role played by mission in guiding the organization is an important one.

Specifically it:

- Serves as a basis for consolidation around the organization's purpose.
- Provides impetus to and guidelines for resource allocation.
- Defines the internal atmosphere of the organization, its climate.
- Serves as a set of guidelines for the assignment of job responsibilities.
- Facilitates the design of key variables for a control system.

Deal and Kennedy claim that a strong culture is the key to long-term corporate success and that culture has five elements: business environment, values, heroes (people who personify values), rites and rituals (routines of day-to-day corporate life), and the cultural network (communication systems).

The mission statement describes primarily the second of these cultural factors, corporate values. The strong cultural companies studied by Deal and Kennedy all had "a rich and complex system of values that were shared by the employees."

The mission statement must be believable in that the company's behaviour should correspond to it over both the short and long term. In this way it can serve as the foundation for the development of respect for and pride in the firm by management, employees, owners, customers, suppliers, and others who interact with it. Broad-based acceptance of the values represented by mission can lead to three characteristics of firms that accomplish this acceptance:

- They stand for something—the way in which business is to be conducted is widely understood.
- From the topmost levels of management down through the firm's organization structure to the lowest level of production jobs, the values are accepted by all employees.
- "Employees feel special because of a sense of identity which distinguishes the firm from other firms."

Many examples of firms that have these characteristics as a result of a finely honed sense of cooperation and value acceptance are presented by Deal and Kennedy. A few of these are listed here, along with the slogans that have come to represent their value systems.

- *Dupont*: "Better things for better living through chemistry—a belief that product innovation, arising out of chemical engineering..."
- *Sears, Roebuck*: "Quality at a good price—the mass merchandiser from Middle America.
- *Dana Corporation*: "Productivity Through people— enlisting the ideas and commitment of employees at every level in support of Dana's strategy of competing largely on cost and dependability rather than product differentiation..."
- *Chubb Insurance Company*: "Underwriting excellence—an overriding commitment to excellence in a critical function.
- *Price Waterhouse and Company*: "Strive for technical perfection" (in accounting).

Mission typically is not considered a part of a firm's strategy set. It reflects the essential preferences of owners and managers for what the firm will do. Strategy will accomplish the task of reducing mission to operational terms.

As such mission is somewhat a personal choice of a firm's dominant group of actors and is an *input* to the strategy formulation process.

LEVELS OF STRATEGY

There is wide diversity in the strategic management literature of labels attached to the different levels of strategy that may exist in a firm. For example, Thompson and Strickland propose four levels: corporate strategy, business strategy, functional area support strategy, and operating-level strategy.

They go on to say, "Each layer is... progressively more detailed to provide strategic guidance of the next level of subordinate managers." Lorange defines three levels for a typical divisionalized corporation: Portfolio strategy (corporate level), business strategy (division level), and strategic programmes (functional level).

He defines the focus of each as follows:

- Portfolio strategy: Developing the desired risk/return balance among the businesses of the firm.
- Business strategy: Source of competitive advantage of a particular business relative to its competition.
- Strategic programmes: Bringing to bear functional managers' specialized skills on the development of programmes.

He notes that smaller firms may involve only the last two of these, but in any firm there rarely would be more than three. Hofer, *et al.* list four levels of strategy for business organizations. First, strategy at the societal level is concerned with the definition of a firm's role in society.

It would specify the nature of corporate governance, political involvement of the firm, and trade-offs sought between economic and social objectives.

The second strategy level is corporate strategy which addresses:

- The nature of the firm's business
- Management of the set of businesses necessary to achieve its goals.

Third, business strategy addresses how the firm should be positioned and managed so as to compete in a given business or industry. Finally, functional area strategy is the lowest level of corporate strategy. It is concerned with their respective functional area environments.

Newman and Logan present two levels—business strategy and functional policy—for non-diversified firms, and a total of three (with the addition of corporate strategy) for diversified firms.

Higgins identifies four levels of strategy: Societal response strategy (enterprise strategy), mission determination strategy (corporate level), primary mission strategy (business level), and mission supportive strategy (functional level). He defines their contents as follows:

- *Societal response strategy*: How the firm relates to its societal constituents.
- *Mission determination strategy*: The organization's field of endeavour.
- *Primary mission strategy*: How the organization will achieve its primary mission.
- *Mission supportive strategies*: How primary mission strategy will be supported.

Another model proposes five levels of strategy but the levels are not tied to organizational structure. Gluck, et al suggest that the levels of planning activity consist of corporate, sector, shared resource unit (SRU), natural business unit (NBU), and product market unit (PMU).

The advantages of this system are:

- It separates the strategic management process from organization structure to a large degree
- Pushes it farther down the organization than traditional systems do.

These characteristics stem from focusing planning level selection on strategic issues or problems shared by the organization's activities rather than on the organization levels of its business activities.

Corporate level planning is that which involves identifying trends and formulating strategy in global, technical, and market arenas, responsibility for which rests with corporate headquarters in most cases.

Sector level planning, where sectors represent national and technological boundaries, may involve several SBU's, product categories, or even product/service-based divisions of an organization. Shared resource unit planning calls for the development of strategies for activities of the business that are shared by SBU's or the various product-market focuses which the company might have.

Natural business units, "...are largely self-contained businesses with control over the key factors that govern their success in the marketplace—their market position and cost structure". Finally, product-market unit planning is the lowest level at which planning takes place and those activities that directly relate the company's output to its markets.

There are many other interpretations of the levels of strategy. They differ primarily in terms of the organizational levels to which they apply. Those discussed above and most of the others have a number of commonalities.

First, the uppermost levels in each scheme tend to concern the problem of fitting the organization to its environment; lower levels address the problem of integrating functional areas in ways consistent with upper-level strategy. Second, the topmost level tends to involve structuring the set of acquisitions of divisionalized firms and is usually called corporate-level strategy. Third, they contain a business or strategic business unit (SBU) level of strategy that applies almost equally to a firm comprised of only one line of business and to the individual subsidiaries of multibusiness corporations.

Finally, the various schemes include a functional level of strategy that represents the ways in which functional departments are expected to respond to business-level and, in turn, corporate-level goals and action plans.

During the mid-1980s some authors began to include the fourth level: enterprise or societal goals and action plans. Societal strategy was intended to capture the essential ways in which the firm was expected to respond to goals related to the major social issues confronting it.

Interpreted fundamentally, then, there are four primary levels of strategy: societal-level, corporate-level, business-level, and functional-level. The concerns of societal-, corporate-, and business-level strategy are clearly cross-functional. That is, they contain implications for each of a firm's functional areas (although more distantly removed in the case of societal- and corporate-level strategy), whatever they may be and regardless of the type of firm.

By contrast, functional area strategies are more operationally focused than the others. The process of determining how each functional area should be managed is a more specialized problem, defined largely by the practice and theory applicable to each functional (or operational) area.

That is, the content of marketing strategy is the subject of marketing texts and courses, finance strategy can be found in finance texts and courses, personal strategy in personal texts and courses, and so on.

SOCIETAL STRATEGY

Societal strategy consists of goals and action plans of which the overall purpose is to guide the ways in which management intends the organization to respond to the major social demands placed on it.

It is an explicit definition of the organization's social responsibilities: how it is expected to react to the demands of particular groups of external constituents. The recent emergence of this level deserves further explanation.

The idea of isolating social responsibility in a separate (from corporate, business, and functional) strategy level was introduced in 1979 by Ansoff and modified by Schendel and Hofer. Ansoff presented it as follows.

The original concept of product/market strategy appears as only one component of a much broader concept. We might call it *enterprise strategy*, which is needed to integrate and relate the new dimensions of the strategic problems.

The development of societal legitimacy (or enterprise) strategy is Ansoff's proposed solution to the increasing importance of... Sociopolitical variables in the life of the firm. Included in these variables are "new consumer attitudes, new dimensions of social control and, above all, a questioning of the firm's role in society."

In other words, Ansoff suggests concentrating managers' attempts to deal with new liberal views about the social role of business, growing egalitarianism, and the proliferation of constrictive regulation, among other social demands, in a societal legitimacy strategy that specializes in matters of social responsibility. Actually Ansoff's proposal for enterprise strategy goes beyond social responsibility issues to include other substrategy areas. Schendel and Hofer limit its contents to social-legitimacy concerns

because of their belief that the other strategy areas are effectively addressed within prevailing definitions of corporate-business-and functional-level strategies.

Thus they recommend, Enterprise strategy attempts to integrate the firm with its broader non-controllable environment, not in terms of product/market matches in a narrower economic sense, but in the sense of the overall role that business, as one of society's important institutions, should play in the everyday affairs of society... More explicit attention will have to be given to (societal legitimacy strategy) in the future, just as Ansoff suggests, just as it has been necessary to separate corporate and business strategy considerations from the problems of functional area interpretation, over the past two decades.

However, some strategists may prefer not to institute this new strategy level, for reasons explained by Dill: "There has been less turbulence in the environment for companies which have spent time over the years listening to and assessing what various internal and external constituencies have to say about the powers they are and are not willing to delegate to corporations and corporate management."

Stated differently, some managers may prefer to address *all* social legitimacy issues within the rubric of existing strategy levels.

For them social responsibility concerns would be addressed within the fabric of goals and action plans at the corporate, business, and functional levels.

Our conclusion about the usefulness of enterprise strategy is that it should be employed by organizations that have particularly sensitive relationships with the environment.

Firms such as utility companies, toy manufacturers, and educational institutions, for which an especially sharply focused public scrutiny has devolved, would be well suited for development of separate social-legitimacy goals and action plans.

For others, whose society interaction is less controversial, matters of social responsibility might be handled more efficiently, and even effectively, within goals and action plans at the corporate, business, and functional levels alone.

However, the advisability of their formulation for a given firm as a distinct activity is intended to be a matter of judgement depending upon whether or not sufficiently poignant social issues either confront the firm at present, or are expected to do so in the future.

Corporate-Level Strategy

A corporation's portfolio of business units (SBUs) can vary in its desirability just as a financial manager's portfolio of investments can.

The acceptability of a particular *business portfolio* is determined by assessing whether it reflects mission and expected strengths, weaknesses, threats, and opportunities (determined through environmental, industry, and internal analyses of each business unit) as well as values and political exigencies.

Additions to and deletions from the present portfolio would then be expressed in merger and/or divestment strategy. Finally, these strategies would be implemented by

the appropriate organizational units and the machinery would thus be activated to adjust the present portfolio. The content of corporate-level strategy is a set of action plans and corresponding corporate-level goals expressed in terms of the firm's collection of subsidiaries. By comparison, business strategy focuses on expected operational results of a business unit. However, it also consists of action plans that relate to goals (at the business level).

The substance of corporate-level goals and action plans can be summarized as follows:

- *Corporate-level goals*: The desired portfolio of business units by the end of the firm's planning period.
- *Corporate-level action plans*: How to get from the present portfolio to that targeted in terms of:
 - Business unit strategies to be retained.
 - Additions of business units expressed as either growth strategy parameters (for "home grown" units) or merger and acquisition strategy.
 - Business units to be divested.

Business-Level Strategy

The content of business-level strategy is also a combination of goals and action plans. Business-level goals should be stated in such a way that for each one, the following four components are evident:

- *Identifying, characteristics* of each objective or target.
- Each target's *substantive threshold value*; that is, the hurdle value upon reaching which the target will be considered attained (often stated as a specific value of the evaluative criteria).
- *Evaluative criteria* to be monitored in order to measure the firm's progress towards the target.
- *Time frame threshold value*; that is, the date by which, or the elapsed time during which, the substantive threshold value would be reached.

Business-level action specifications should be devolved so that collectively they define the following elements:

- The firm's *product market scope*; that is, the nature of the products or services provided and the markets served by the firm.
- *Input-output transformations* in which the firm is engaged; that is, the nature of the activities with which the firm adds value to its labour and material inputs and what those inputs are.
- What *synergies* are sought in the operation of the firm.

Whether a strategy is being analyzed from within or outside the firm, these characteristics should be present. The outside analyst may use them as dimensions along which a firm's strategy can be described. The inside analyst or strategist may use them as guidelines for determining the comprehensiveness of strategy.

When taken together, these elements of business-level strategy should combine uniquely to:

- Define the business of the SBU or firm,
- Describe its competitive edge.

Thus strategy aligns the strategic business unit or firm relative to its competitors, distinguishes it from them, and hopefully propels it beyond them.

Functional-Level Strategy

In contrast with the other levels of strategy, functional strategies serve as guidelines for the employees of each of the firm's subdivisions. Which ones of these segments or functional areas are included in a firm's functional strategy set is itself a matter of strategy. For example, whether to have an R&D department or not in the first place is a strategic decision.

Functional goals and action plans are developed for each of the functional parts of the firm to guide the behaviour of people in a way that would put the other strategies into motion. If part of a firm's business-level strategy were a target of a 10 per cent increase in sales to be brought about by market penetration, for example, marketing strategy might include a change in compensation policy for salespersons and a specified increase in the advertising budget.

In that way marketing strategy would provide some detail about how the marketing aspects of the market penetration action plan would be implemented. Similarly, financial strategy would consist of a set of guidelines on how the financial elements of the firm would be put into effect. Personal strategy, production strategy, research and development strategy, and appropriate other functional strategy areas would do the same.

Nature of Strategic Decisions and Strategic Thinking

Although the process of creating strategy is often discussed as if it were an unconstrained design process, keep in mind that while strategists evaluate strategy, the firm is operating.

This evaluation involves assessing the extent to which present strategy is meeting expectations. It may be the case that only a small part of, say, marketing strategy would have to be changed to correct a problem. In effect, then, such a change would constitute an acceptance of corporate- and business-level strategy, and also of the firm's functional strategy set. Marketing strategy would be all that was rejected. When a firm's performance is less than satisfactory, the reason often is a functional strategy shortcoming. One might say that a "good" business-level strategy would have been poorly implemented by part of its functional strategy set. For this simple example, a change in marketing strategy could improve performance while other levels of strategy would remain unchanged. Alternatively, a problem with the nature of a firm's or SBU's business brought about by a major environmental opportunity or threat, a change in that level's

goals set, or the development of some internal capability or weakness could necessitate a business-level strategy change. The new strategy would probably include vestiges of the old along with some unfamiliar elements. In most cases a whole new functional strategy set would likely have to be designed and put into effect to implement the new business-level strategy.

More generally, one could conceivably change parts of a firm's functional strategy set without changing business-level strategy. However, rarely would one expect to encounter the case in which a change in business-level strategy did not trigger the necessity to alter functional-level strategy in some way, at least not in a successfully managed business.

There is a risk of incorrectly identifying the strategy level at which a problem exists. A tendency exists in business to change functional-level strategies or organizational structure in an attempt to remedy any problem.

Of course, if the problem existed within the firm's corporate-or business-level strategy, for example, changing functional-level strategy would not correct it. In fact, this move would most likely aggravate the situation. The reason for this tendency is probably that functional strategy changes are potentially less disruptive than changes in the other levels. They certainly would affect fewer people than modifications at the corporate or business levels. The results of trying to solve a business-level strategic problem with a functional-level solution is well illustrated by the "big four" U.S. automobile companies.

With overseas competitors exporting fuel-efficient automobiles to the United States, and with widely acknowledged shrinkages of fossil fuel supplies, they still stubbornly tried to retain their old business and corporate strategies, well into the 1970s, by changing market strategy only. A set of major environmental threats, particularly at the business level, was met by minor model changes and increased efforts to convince the car-buying public that big autos were what we all really wanted. During the 1970s and early 1980s, these companies finally began to respond to their business strategy problems with business strategy changes—major philosophical alterations in the nature of their businesses.

If U.S. auto firms were to rely on small cars instead of traditional American big cars as their primary sources of growth, far-reaching changes in their functional strategy sets were also necessary.

New growth rates, profit targets, market-share goals, and the like, along with related broad action plans that centered around contraction in many forms, were implemented along with new functional strategy.

They included such things as cash rebates, Chrysler's government-guaranteed loan, employee layoffs, close cooperation with UAW officials, R&D efforts focused on producing smaller, less powerful engines as a primary effort rather than as an inconvenience.

The U.S. auto firms' strategy changes were finally implemented during a period of chaos. However, by then they were all in the red, and the doomsayers were having a

heyday. South claims that the key to successful strategic decisions is the creation of competitive advantage—selection of competitive areas within which success is clearly achievable. The Japanese have taught us that there are really two very different forms of competition: Reciprocal and strategic. Reciprocal competition is the traditional form in which companies in mature industries compete with other firms having similar strategic positions while attempting to distinguish themselves on the basis of operations.

Many colleges and universities offer similar programmes, with traditional facilities and are significantly different only in terms of the geographical areas within which they operate. Strategic competition, on the other hand, involves competing primarily by establishing superior strategic position.

That is, the main concern of strategists is selection of a strategically advantageous posture which affords their organization a position of strength from which to do battle. Lincoln Electric Company has successfully differentiated itself on the basis of price and quality.

Their industrial welding equipment is regarded as the top quality product of its kind and the firm has consistently passed along cost savings to customers in the form of low prices. As a result, Lincoln is very difficult to compete with because of its strong price position.

Following this set of ideas, then, strategic decisions are those which involve “clear and favourable differentiation from competitors,” so that one’s competitive advantage is tangible, measurable, and preservable.

Defining strategic thinking is made especially difficult by the absence of a consensus on what strategic management is. On the latter point, Gluck, et al commented, “hardly anyone yet fully understands what strategic management is, knows exactly how to do it, or can clearly define its benefits.” Their explanation of strategic management is that it is an approach to management which fuses strategic planning (plan development through analysis of environment, competition, and strategy choices) and the firm’s system of operational decision making. This process has five dimensions.

First, strategic management requires a widely understood strategic planning process tightly interwoven with a strategically Centred organization structure. Second, it requires widely discussed, analyzed and negotiated goal choices from which are culled out the ones that are not adoptable. Third, strategic management relies on the shared ability to think strategically by managers throughout the company.

Fourth, a performance evaluation system is necessary that focuses top level evaluators’ attention on critical positive and negative strategic factors. Finally, it requires a strategy supportive motivational system and value orientation among managers.

These qualities, they say, define a strategically managed organization. Four of these qualities are readily understandable. But still subject to question is what is meant by the ability to think strategically.

Strategic thinking is that which generates:

- Creative, environmentally relevant ideas
- Concepts about how to turn them into systematically managed action plans.

It has the following prerequisites:

- Input information is based on facts and logical data.
- Previously unquestioned assumptions are sought out and examined.
- A burning desire for resource conservation, and
- Indirect, spontaneous, and unexpected thought processes which are hard for competitors to predict.

These requirements of strategic thinking set it apart from other kinds of decision making. First, strategic thinking requires factual and logical input data because it is competitively dangerous to base strategy formulation on erroneous information.

The stakes are too high to “hoof it.” Second, long-held assumptions should be identified and analyzed to make sure they still apply. This is especially true about goals which are in force, understandings about the environment and competitors, market acceptance and image, etc.

Third, the manager attempting to think strategically should be committed to conservation of the organization’s resources rather than expecting that a good idea will precipitate a cornucopia of funds, people, and support. It is easy to be creative while assuming that most resource requirements can be taken care of.

A greater degree of creativity is required when one must conserve resources. Finally, strategic thinking must be done without setting patterns which competitors can identify and anticipate. The problems of predictable strategic thinking are analogous to the football coach sending in plays to his quarterback using hand signals that are understood by the opposing team’s coaches.

EVOLUTION OF STRATEGIC MANAGEMENT

Gluck, Kaufman, and Walleck propose a model of the evolution of strategic management within organizations whose managers have a propensity to plan. The model is based on a study undertaken by McKinsey and Company. Part of the reason for including a summary of their model here is to help students recognize the strategic management stage in which a particular organization might be.

Such information may provide additional insight into the operation of the firm and help understand the degree of strategic management progress made by managers. More importantly, however, it may be useful in helping students understand strategic management as the highest level of a hierarchy of management activities.

The McKinsey analysis discovered four quite distinct phases of strategic management evolution. In Phase 1, financial planning, management focuses on the preparation of budgets with an emphasis on functional operations. Most organizations have a budgeting process, in at least rudimentary form, as a way of allocating resources among functional units, subsidiaries, or projects.

The second, forecast-based planning, follows naturally from the first as managers project budget requirements beyond the one-year cycle.

This phase represents an effort to extend managers' attention beyond the immediate future as scenarios are developed which describe their expectations about future time periods.

Budgets are often constructed for several years at a time and are rolled over annually so that the appropriateness of a budgeted amount can be reviewed several times before it is operationalized. Phase 2 planning is very "now" oriented.

Current operations and characteristics are stressed in analyses of the firm and there is little attention to or patience for considering operational options or development of strategic changes. The business portfolio of a Phase 2 firm is often viewed as the final expression of strategy rather than as an input to the strategy formulation process. Current structure and business activities may be considered fixed, not as strategic variables.

Planners are required to adopt an external orientation and tools and procedures for environmental and internal assessment.

Concern centres on understanding the organization's environment and competitive position and generating ideas about how the company might better fit its environment. Several choices, contingency plans, are often devised for how the company might fit its environment. Lower level planners and managers are often involved in the process of generating choices, an activity that soon puts top management in the position of choosing a plan in which it had little involvement in developing.

Strategic management, evolves as top management senses the need to more heavily invest in the planning process because of its lack of understanding of or involvement in the details of earlier plan development. Strategic management is the meshing of Phase 3 planning and operational management into one process.

It is analysis and conclusion that takes place year-round and ties performance evaluation and motivational programmes to strategy.

DELIBERATENESS OF STRATEGY

Sometimes outsiders impute strategy to the behaviour of firms. Obviously, students analyzing case studies are placed in this position when they impute strategy from the data they are able to generate on the firm's operations. Similarly, journalists and the managers of competing firms may impute strategy to a firm's behaviour; and it may or may not accurately reflect the real strategy in place.

Outsiders may also imply intent to an imputed strategy. That is, they assume not only that the strategy they imputed from the firm's behaviour is the real strategy its employees are implementing, but they imply that this strategy is the one intended for the firm by its management. Seldom is this the case.

Mintzberg developed a taxonomy which is useful for discussing the realism and deliberateness of strategy. First, he distinguished between strategy that is the result of a plan, and of a pattern of behaviour.

He referred to them as "strategy as plan" and "strategy as pattern," respectively. Strategy as plan is a chosen course of action; it could be a real strategy (one intended

for implementation) or a ploy (a tactical move whereby a competitor may be influenced into making a mistake). Some people think that Coca-Cola's rumored change in Coke's formula in the mid-1980s was such a ploy.

The implication is that Coca-Cola had not intended to really change the formula. Instead, it is thought, they publicized their plan to change formula, introduced a new product with a different formula that tasted a lot like a competitor's product, and finally graciously conceded to continue producing the old formula product when the public demonstrated a preference for it over the new—"similar to a competitor's—" formula.

Incidentally, if this was in fact a ploy, it has to rank among the top marketing moves ever attempted by any business. Coca-Cola reaped an immediate increase in market share of about 15 per cent that thrust them once again into unquestioned dominance in the huge U.S. softdrink market.

Strategy as plan, when implemented, may or may not be what the firm ends up with. That is, the planned strategy could ultimately be either realized or unrealized.

If it is realized, then the entire process would be a textbook case of strategy formulation and implementation in the sense that the firm successfully implemented what was intended.

But what happens if the planned strategy is implemented and, for some reason, the strategy that is realized is not the intended one? We might say that the planned strategy was unrealized, and the realized strategy (the one that seems to describe what the company is actually doing) arises out of some consistency in the behaviour of the company.

Mintzberg and Waters call this unintended realized strategy, "strategy as pattern," or a pattern in a series of actions by the organization. Strategy as pattern is what you will end up with when you impute strategy to the behaviour of a company you are analyzing in a case study, or what journalists produce when they attribute a strategy to a company based only on its actions.

Thus, a realized strategy could be either a deliberate strategy as plan, or an "undeliberate" strategy as pattern. If the realized strategy was planned and also accurately described the firm's actions, then strategy as pattern and strategy as plan would be synonymous. However, when realized strategy is not intended strategy (that is, it was either not what was intended by management when they drafted a planned strategy, or they drafted no strategy at all), then it simply "grew" out of the activities of the company. In Mintzberg's terms, it "emerged" as a pattern of behaviour in the absence of intention, or despite unrealized intentions.

A realized strategy is what a company is actually doing. If it is the one intended by management, then it is deliberate. If not, then the intended strategy was unrealized, and the realized strategy is emergent. An emergent strategy is, by definition, not deliberate.

However, a manager may choose not to consciously formulate strategy and, instead, "go with" the emergent one. But even here, the resultant emergent strategy would not have been deliberate in the same way an intended strategy would have been.

Often it is convenient to distinguish between intended and emergent strategies. When management performs no strategic management at all, they still will have a realized strategy that is emergent.

This emergent strategy could be recognized by outsiders (and insiders for that matter) even though it may not have been intended by management.

Contingency Strategies

Managers may develop several versions of strategy, each of which is intended to fit a different environmental scenario. An environmental scenario consists of a set of environmental characteristics that describe a firm's external circumstances. By structuring external variables into sets of related variables, different scenarios result.

Thus several scenarios (usually three) can be arrayed on a probability distribution for which the horizontal axis is a measure of the extent to which they are viewed as "good" or "bad"—supportive or not supportive—to the firm.

In the scenarios that fall on the left side of the distribution are considered pessimistic (undesirable but only a small probability that they might have to be dealt with); those on the right as optimistic (desirable but probably will not materialize either); and those in the middle as most likely to occur and having some good and bad features.

Next a strategy set is established for each scenario. These could be entirely different strategies or variations of one. Evaluative criteria then are formulated for each scenario. These are variables to be monitored in order to figure out which of the scenarios is going to apply in the future.

Actually this is a matter of deciding whether the present scenario will continue to be the best description of the firm's environment or, instead, if one of the others seems to be emerging as most correct. By identifying critical values for evaluative criteria, the strategist knows beforehand under what set of conditions to switch from the present strategy set to the pessimistic or optimistic set.

Summary

This chapter serves to familiarize the reader with the underlying principles of strategic management and to outline the path through that field taken by the chapters. First, the history of strategic management as an academic discipline is reviewed to provide perspective on its current stage of development. Next, summarizes the results of some of the major studies conducted to answer the question "Does strategic management pay off?" Also discussed in this section are the primary methodological problems encountered in doing such research.

The mission statement is considered an input to the strategy formulation process, which is outlined in the next section. Strategy at four organizational levels comprise the focus of this section and the book's text. They are the enterprise, corporate, business, and functional levels.

STRATEGIC THINKING

When a roomful of executives is asked, is it fair to say that your business is getting more competitive? A resounding Yes! From each one echoes in response. This happens regardless of industry or service sector, the size of the executive's organization, whether it is domestic or global, or even whether it is from the public or private sector.

All organizations face competition—for resources, people, cash, or customers. Likewise, all organizations face uncertain environments, because both product and technology life cycles are shortening relentlessly.

Never has it been more important for businesses and their leaders to position their organizations strategically for competing successfully in the future.

This requires that managers thoroughly understand the dynamics of their industries, the trends in their external environments, and the basic economics of their firms.

They also must think creatively enough to craft strategies that make their companies unique and inspire their organizations to perform to ever increasing standards. Finally, they must balance the multiple activities of the various functions in their organizations and galvanize them towards unified action. To accomplish all these activities simultaneously requires that managers engage in *strategic thinking* and *strategic management*.

Strategic management encompasses the activities of senior-level executives in their attempts to influence the overall direction of their corporations. This textbook covers many of the approaches that help executives increase that influence by developing successful business and corporate strategies.

Strategic thinking, on the other hand, can occur at all levels of the organization—from brand managers to production supervisors, from sales executives to financial planners and analysts. Strategic thinking at these levels involves understanding the role of one's firm in the marketplace and how one's individual function contributes to that role.

Just as a guitar player or drummer needs to know the bandleader's intent with a piece of music in order to provide masterful support to the whole, every individual in a corporation can provide better customer service, better information back to the home office, and make better personnel decisions if she or he has a good understanding of both the company's strategy and some basic strategic management principles.

This textbook provides concepts and tools to aspiring managers who wish to add value to their companies by making strategically sound decisions, whatever their functions or responsibilities may be.

EVOLUTION OF STRATEGIC THINKING

Strategic thinking evolved from the different approaches that companies utilize to plan their business activities. Prior to the 1920s, few firms did any of what today might be called "long-term planning." Most businesses were run more or less entrepreneurially,

and business decisions were relatively short term in focus. Manufacturing produced products, sales sold them, and accounting counted the money. Many businesses were driven by a focus on survival.

With the advent of the modern corporation, such as General Motors and DuPont in the 1920s, larger and more far-flung enterprises needed mechanisms to plan and allocate resources. Financial planning provided senior managers with a sophisticated set of tools with which to plan annual production, expenses, and capital investments.

The methods typically involved translating sales forecasts into production schedules, estimating the costs associated with the planned volume, and deriving profit forecasts. Once capital appropriations, research, development, advertising, and administrative overhead were taken into account, projecting cash flows available for dividends was a simple and straightforward matter.

This approach, however, was based on a fundamental assumption that sales projections could be made. In a stable world, this assumption held. Sales forecasters had only to extrapolate from the past.

But when times changed—the economy slowed, markets got saturated, or other events affected demand—the limitations of this method became obvious. Also, this approach tended to treat departments within the business as separate functions (sales forecast? manufacturing costs? financial results) and tended to have a one-year horizon.

As a consequence, changes occurring over several years, such as shifting consumer preferences or market saturation, would go unnoticed. Finally, the driver of business behaviour under financial planning systems tended to have short-term goals, which were often oriented towards meeting the budget for the year.

To deal with longer-horizon changes, forecast-based planning was developed in the 1950s and 1960s. Here, the rigour of analyses jumped dramatically, as planners began using such tools and concepts as product life cycles, discounted cash flows and net present value analyses, and statistically valid market research of consumer preferences and buying behaviour.

Because the level of sophistication and time required by these tools was significantly greater than called for previously, managers needed assistance from a new breed of technical experts—strategic planners, business school graduates, and consulting firms. Business schools grew at unprecedented rates in the 1960s and 1970s, and the consulting profession boomed as new strategic planning firms, such as the Boston Consulting Group and Bain and Company, were founded and flourished. The behaviour driver under forecast-based planning was to predict the future and to plan strategies that would adapt the firm to that future.

In other words, it was reactive. At the same time, staff planners and consultants found that many of their strategic recommendations were not getting implemented. This occurred for two reasons.

One reason was that often the analyses and recommendations were performed by people who were not responsible for executing the strategy. Without the ownership that

comes with participation, many line executives either rejected plans or, at best, were lukewarm in their compliance. The second reason was that plans that were based on analyzing consumer trends and product acceptance often failed to take *competitor* behaviour into account. The strategic management approach was developed in response to these limitations.

Strategic *management*, in contrast to strategic *planning*, is the function of line executives—the people on the firing line who deal with the real world of customers, machines, products, and people.

The philosophy is simple: the people closest to the action are in the best position to know what is going on in the business, what works, what trends are developing, and what competitors are doing. In other words, those who must execute must be involved in developing strategy.

A second part of the philosophy is also simple: it is not sufficient to look forward (at potential consumer demand for a new product, for example). Managers must also look to the right and to the left—at what competitors are doing. In other words, competitive analysis becomes a key feature in strategic thinking.

The dilemma, however, is twofold:

- Usually, line managers do not have the time to do strategic analyses. As you will see in the Astral Records case, a general manager's life is cluttered with a multitude of conflicting demands on his or her time. So, managers must either take time out—such as at a planning retreat, for example—or they must learn so much about strategy and competition that they are thinking strategically on the job.
- Even if managers create the time for strategic planning, they often lack the tools. Their training may have been thorough in the *functional* areas of marketing, accounting, engineering, or human resources, but it may have been insufficient in *strategic* concepts, frameworks, and tools. As stated earlier, the purpose of this book is to provide these tools.

In a firm equipped with strategic management tools and strategic thinking managers and executives, the behaviour driver shifts from a mentality of survival, meeting the budget, or merely forecasting the future, to one of *creating* the future by shaping one's industry and markets rather than reacting to them.

A firm that is truly thinking strategically can be identified in a number of ways. First, its executives know and understand what the firm's mission and strategy are. Second, they know their core competencies and capabilities as an organization.

Third, they can identify their competitors and predict their reactions to the firm's strategic moves. Fourth, they have developed a culture that supports their strategy.

Finally, they are constantly developing new capabilities for renewing their strategies and for competing successfully in the future.

BECOMING A STRATEGIC THINKER

So, how do managers and their companies develop strategic thinking? Must they go through the stages listed above, one at a time? Fortunately, no. It is essential that managers know how to do financial planning, market research, and other fundamental business analyses. But they need not wait to master these before progressing to strategic management and thinking.

ELEMENTS OF STRATEGIC THINKING

Strategic *planning* has a fairly well-established history. The first book on strategy was written in 1965 by Igor Ansoff, an executive at Lockheed Aerospace. Titled *Corporate Strategy*, it presented an extraordinarily logical, step-by-step set of concepts and approaches to making diversification decisions.

When presented with this model as a how-to-do-strategy template, most practicing managers run for cover. The thoroughness of the model forces the analyst to think through the logic and economics of the business. But this thoroughness comes at the price of appearing overwhelming to busy line executives.

All strategic planning approaches attempt to find an optimal match between the resources and capabilities available within the firm (Strengths and Weaknesses) and the external market conditions and environmental trends (Opportunities and Threats).

This match or coalignment (often called a SWOT analysis) results in a strategy, whose efficacy translates into some level of corporate performance.

A key feature in the basic strategy model is the reciprocal influence of all of the variables. For example, in the late 1960s a major science-based consulting firm, Sigma Consultants, observed that:

- The level of expenditures on health care in the United States was accelerating, a trend likely to continue,
- The firm had several competent professionals on its staff who had carried out health care related assignments such as designing hospitals and evaluating medical group acquisitions.

By responding to the market opportunity and articulating a health care strategy or consulting practice, Sigma would define a new domain.

Having chosen the new domain, the firm would now have to respond to the dictates of the marketplace (arrow from environment to strategy) by, for example, designing products and services that were in demand or adapting to whatever governmental regulation might be relevant. To implement the new strategy, Sigma might reorganize its staff so that all professionals working on health care related assignments were grouped together.

Over time, this group might increase its proficiency by sharing experience, knowledge, and the like. The degree of success would be manifested in some level of performance

(arrow from strategy to performance), and, in turn, the performance level would serve as feedback, indicating that further adaptations of strategy might be in order.

In essence, then, the model represents a system of interacting parts. But the picture painted thus far is not complete. A “perfect” strategy may be found that violates managers’ predispositions. Conversely, management’s values may suggest strategies that are not optimal in an economic sense. For example, President Jay Monroe of the Tensor Corporation refused to change his very “functional” high-intensity Tensor lamps when new competitors started introducing style variations.

As a result, his company lost 60 per cent of the market, which it probably could have retained had Monroe’s values been more economic than aesthetic.

In our Sigma Consultants example, there was strong support for the health care venture among senior management and some of the professional staff. That is, the venture fit the values of top management. However, Sigma was originally organized on the model of a research lab, in which professionals were unimpeded by bureaucratic hierarchy and were free to choose their projects.

Assignment to projects was by invitation, and, assuming the invited person was billing sufficient time to avoid the pressure to join projects indiscriminately, he or she could turn these projects down.

The health care strategy, in contrast, suggested an organizational unit in which consultants would be expected to dedicate themselves to one major type of activity. This ran smack against the prevailing culture. The strategy was abandoned.

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Index

A

Accomplished 9, 16, 24, 51, 55, 64, 87, 194, 203, 216
Algorithm 77, 91
Allocation 37, 58, 59, 103, 116, 117, 120, 121, 136, 292
Ambitious 103, 133
Assembled 141
Assumptions 22, 104, 128, 129, 131, 132

B

Behavioralist 17
Benchmarking Methods 169
Brainstorming 147, 211, 212
Bureaucratic 16, 20, 51
Business-level 294, 295, 297, 298, 299

C

Cad 31, 63, 71, 84, 129, 193
Cam 66, 72, 146, 147, 192, 198
Capabilities 183
Cheapest Method 141

Collaboration 217
Collaborative 192
Comprehensive 43, 44, 45, 59, 64, 69, 75, 128, 135, 177, 191, 193, 194, 201
Contingency 302, 304
Culture 292, 307, 309

D

Decisional 7, 9, 10, 11
Demonstrated 47, 158
Diligence 149
Dimensions 295, 297, 300

E

Elements 292, 297, 298, 299
Emphasize 10, 160, 222
Endeavour 42, 103
Engage 110, 203, 206
Eventually 123

F

Finance Strategy 295
Formulation 291, 293, 296, 301, 302, 303, 304
Functional-level 295, 296, 298, 299

G

Goals 293, 294, 295, 296, 297, 298, 299, 301, 306

H

Hazard 221, 222, 224, 226
Hierarchical 51, 59
Hierarchy 301, 309

I

Illustrated 212
Implications 6
Influencers 110, 111, 112, 113
Insignificant 126
Instantaneous 50

K

Kaizen 172, 173, 174, 188, 189, 190

L

Loophole 149

M

Mergers 200

Mission 291, 292, 293, 294, 296, 304, 307

N

NDA 40

O

OECD 158, 159

P

Parallel 69, 79, 88, 100, 115, 123, 124, 127, 136, 137

Participation Method 186

Permissible 51, 125

Pert 32, 42, 47, 49, 53, 81, 83, 84, 105, 139, 141, 158, 161, 163, 166, 168, 177, 189, 196,
198, 201, 225, 229, 230, 231

Precautions 222

Predetermined 50, 67, 191

Procurement Methods 198

Q

Quantifiable 50, 51, 53, 65, 67

R

Responsibility 6, 8, 10, 11, 13, 14, 15, 20

Rigorous 63, 69, 128

Roethlisberger 16, 23

S

Scalar Chain 15, 16

Scheduling 29, 30, 74, 75, 76, 77, 79, 189

Scientific Methods 12
Sequential 114, 116, 123, 124, 136, 137, 138, 185
Simultaneous 60, 92, 114, 116
Strategic Management 293, 294, 300, 301, 302, 304, 305, 307, 308

T

Troublesome 52

U

Unequivocal 65, 69
Unequivocally 65
Universality 3, 4